

PRACTICAL SUSTAINABILITY STRATEGIES

HOW TO GAIN A COMPETITIVE ADVANTAGE

SECOND EDITION

GEORGE P. NASSOS | NIKOS AVLONAS



WILEY

**PRACTICAL
SUSTAINABILITY
STRATEGIES**

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HOW TO GAIN A COMPETITIVE ADVANTAGE

Second Edition

George P. Nassos

Principal

George P. Nassos & Associates, Inc.

Nikos Avlonas

Founder and President

Center for Sustainability & Excellence

WILEY

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About 40 years ago, I had the fortune to work for a subsidiary of an American international chemical company that was headquartered in Cologne, Germany. While experiencing many great times during the three years I lived in Europe, there were a few that stood out because of their impact on the natural environment and energy efficiency. Here are three that I will never forget as they planted the seed in my interest in sustainability.

1. I was walking to the subway and had to take the escalator to the lower level. When I arrived for the first time, I was disappointed that the escalator was not operating. So I did what I would normally do under such a circumstance—walk down the stationary escalator. As soon as I took the first step to walk down, the escalator started and continued to operate until I reached my destination. With no one else on the escalator, it stopped when I got off. There was a pressure switch under the first step that activated the motor. That really made sense. Why should it be operating when no one was using the escalator?
2. I was participating in a business meeting at a high-rise office building in the La Defense suburb of Paris. Sitting in a conference room on about the 40th floor with nothing but windows on the outside wall, I couldn't help but notice that from time to time the lights would go off and later come back on. After witnessing this occurrence two or three times, I asked the hosts if they were having problems with their electrical supply. I was embarrassed to learn that light sensors controlled the need for artificial light when the sunlight was insufficient. The turning on and off of the lights on that day was due to a partly cloudy day.
3. I lived in an apartment building two blocks from the Rhine River and about six or seven kilometers south of the city center where the company office was located. My commute to the office was by driving on a four-lane boulevard along the river. About every kilometer or less were intersections with stoplights to allow traffic to enter or exit the boulevard. In addition to these traffic lights at the intersections, digital signals existed on the boulevard about midway between the

intersections. These digital signals would indicate what speed the cars should travel in order to arrive at the next intersection with a green light. This system provided several benefits like reduced fuel consumption, reduced emissions, reduced congestion, and reduced stress. I am not sure how many years this system was in operation prior to my experiencing it, but it surely made sense.

After working for the chemical company for 16 years, I returned to the United States and joined an environmental company for the next 15 years. A combination of my experiences in Europe and Germany, in particular, along with the environmental company convinced me that this was a field of great interest. I then decided to enter the academic field and combine my chemical and environmental background with my business background (as I also have an MBA) and teach at a graduate business school. I wanted to offer students a combination of teaching theory as well as my business experiences in the real world. I was fortunate to learn that the Illinois Institute of Technology—Stuart School of Business had just developed a new program—MS in Environmental Management, and the dean was looking for a director of the program. This was a great fit for both the school and me.

During my first year at the school in 1997, I attended a conference sponsored by the World Resources Institute focusing on environmental and social sustainability. About 150 people attended this annual conference with about 70% from academia and the balance from the business and government sectors. I had never heard of this topic, but I found it extremely interesting. The following year I attended the conference again, and after this second experience I was convinced that sustainability was very, very important. During the next academic year, I introduced sustainability into the capstone course I was teaching and renamed it “Business Strategy: The Sustainable Enterprise.” Over the next few years, I introduced the sustainability concept in other courses and shortly thereafter changed the program name to MS in Environmental Management and Sustainability. This program was eventually ranked No. 11 in the world by the Beyond Grey Pinstripes biennial survey.

I became so interested in sustainable strategies that I researched the topic on a continuous basis and added new strategies and case studies to the course almost every year. Since about 80% of the students were working professionals—part-time students—the final exam for the course was actually a project. Each student was challenged to apply one of the sustainability strategies to their workplace and show how it would enhance the company, a subsidiary or a strategic business unit (SBU) in terms of environmental integrity, social equity, and economic viability. If it didn’t,

I didn't want to hear about it. The full-time students that were not working for a company had the option to select a publicly held company or create a new one based on the sustainability strategy.

In addition to researching for new strategies, I was also interested in books about sustainability. Over the past 10 years, I did not come across any book that dealt with the various sustainability strategies I was teaching. I am convinced that sustainability should not be a discipline in a business school like marketing, accounting, finance, or organizational behavior, but rather should be imbedded in all the appropriate courses. All graduates of a business school should have an understanding of the sustainability concept and its benefits. This book was written to enhance business programs at any or all business schools. It can also be used as the basis for a course on sustainability or as a reference to cover the topic in one or two modules of any other business course.

This does not mean that the book has been written only for business schools. Small- and medium-sized companies as well as large corporations will certainly benefit from the contents of this book. In addition, government agencies and non-governmental organizations (NGOs) will also benefit by adopting selective strategies. Just as I am convinced that sustainability should not be a discipline in a business school, it should not be the responsibility of an individual or department in a company or organization. Sustainability should be imbedded in the culture of the organization so everyone can work together to achieve their goal of operating as a truly sustainable company.

A number of years ago, I had the pleasure of meeting Nikos Avlonas, CEO and Founder of the Center for Sustainability and Excellence, an organization that trains and consults in the sustainability field. His expertise is in the measurement of sustainability metrics that are used to determine how well the organization is performing and in the reporting of its activities. Consequently, it made sense to join forces and write a book about the strategies, measurements, reporting, and communicating.

Although I retired from IIT—Stuart School of Business in 2011, I joined the Driehaus College of Business at DePaul University in 2017 as Director of the MS in Sustainability Management program. So I am continuing to do what I love best in training more disciples in the field of environmental and social sustainability.

George P. Nassos
Glenview, IL, USA
November 2019

PREFACE OF NIKOS AVLONAS

*As you set out for Ithaka
hope your road is a long one,
full of adventure, full of discovery.*

My journey to sustainability was anything but linear. It started off with an interest, and then a job in Quality Assurance, passed through the challenging environment of Total Quality Management to finally reach its destination. Every step of the journey added to my knowledge of the field, offered insight, understanding, and experience.

Coming from a small town of fewer than 20,000 inhabitants and a small country of less than 12,000,000, I am still in awe of the journey so far. I founded the Centre for Sustainability and Excellence (CSE), a global Sustainability Strategic Advisory and Training organization with activities in North America, Europe, MENA, and Asia regions and had the honor of receiving recognition for my work (one of the “Top 100 Thought Leaders in Trustworthy Business Behavior” by Trust Across America in 2010, CSR Professional of the Year by PR News in 2017, and Sustainability Professional of the Year Award by the Silicon Valley Community Foundation, in 2018). I have enthusiastically passed the flame of Sustainability on to others through teaching (among others DePaul University, University of San Francisco, Athens University of Economy and Business), speaking at conferences and, in keeping with the times, through the creation of the online Sustainability Academy platform.

On this journey I have had the opportunity to meet and work with people from different walks of life, from a variety of sectors, from a multitude of countries. What lies behind the different backgrounds and interests, however, is the ardent wish to make this planet sustainable, which eventually requires that we practice what we teach and what we are taught. In this revised edition, George Nassos and I are pooling together our experience and knowledge to promote the practical side of sustainability. I hope this

book will accompany students, practitioners, and enthusiasts well on their own journey.

*And if you find her poor, Ithaka won't have fooled you.
Wise as you will have become, so full of experience,
you'll have understood by then what these Ithaka mean.*

Nikos Avlonas
November 2019

REFERENCE

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ABOUT THE COMPANION WEBSITE

This book is accompanied by a companion website:



www.wiley.com/go/Nassos/PracticalSustainabilityStrategies_2e

The website includes:

1. Case Studies
2. PPT slides

PART I

INTRODUCTION TO SUSTAINABILITY

Urgency to Adopt Sustainability

It has been close to 60 years since we started reading books or articles about the environment with “Silent Spring” by Rachel Carson being one of the first important books published in 1962. Many other outstanding books have been written about the environment since then such as “The Ecology of Commerce” by Paul Hawken in 1993 and “Natural Capitalism” by Paul Hawken, Amory Lovins, and L. Hunter Lovins in 2008. The number and frequency of new books have increased as more and more people are concerned about the state of the environment.

Very few people question the decline in the state of our environment, only the degree to which it has deteriorated or the rate at which it is continuing to deteriorate. Regardless of the current status of our environment, it is important to put in perspective what has happened to our earth since its creation. Historians estimate that the earth is about 4.5 billion years old, but it is really difficult to understand exactly what this means. What does 1 billion really mean? Let’s consider a situation where a 21-year-old girl is given US\$1 billion as a gift, and she places the money in a noninterest bearing account. She will be able to spend US\$60,000 every day of her life until she retires at the age of, say, 65 and still have US\$36 million left over for retirement. This gives someone a better understanding of 1 billion really means.

So how can we put 4.5 billion years in perspective so we can understand what has happened to the earth since its creation. As suggested by David Brower [1], former Executive Director of the Sierra Club, let us compress

the geologic time, from the initial formation of the earth until now, into the six days of biblical creation [2], from Monday to Saturday.

CREATION OF THE ENVIRONMENT

Using the compressed time scale, the earth was formed at midnight, the beginning of the first day, Monday. There is no life until Tuesday about 8:00 A.M., and millions of species begin to appear and disappear throughout the week. Photosynthesis begins and it gets into high gear by Thursday morning, just after midnight. By Saturday, the sixth and last day of creation, there is sufficient oxygen in the atmosphere that amphibians can come onto land, and enough chlorophyll manufactured for the vegetation to begin to form coal deposits. The giant reptiles appear around 4:00 P.M. and primates show up at 10:00 P.M. on this last day, but *Homo sapiens* don't appear until 11:59:54—just six seconds ago. In other words, if we compress the age of the earth to six days, or 144 hours, “man” is not created until the last six seconds. A quarter of a second to midnight, Jesus Christ appears. One-fortieth of a second ago is the beginning of the industrial age, and one-eightieth of a second ago, we discover oil, thus accelerating the carbon blowout started by the industrial revolution.

Scientists have predicted that this 4.5 billion year old earth will be around for another “week.” But look at the damage that has been done in just the past one-fortieth of a second. Almost 90% of the major fisheries have been depleted or are at their biological limit [3]. It is estimated that the forest cover has been reduced by as much as 40% worldwide [4]; 50% of the wetlands [5], and more than 95% of the US grasslands have been lost [6]. By 2025, two-thirds of the world's population may face water shortages [7]. The big question now is how long will we last, another one-fortieth of a second—about five generations? Or will we be able to survive for another quarter of a second—about 2000 years? Or can we make a difference to extend a healthy world to some indefinite period of time? Or is it too late, and are we in the midst of a period of overshooting the carrying capacity of the earth, followed by a rapid collapse?

God did not create the natural environment for the benefit of the people so they can use and misuse it. The environment *can* be used indefinitely as long as it is replenished. It has the capacity to support the needs of living creatures—plants and animals, including humans—but only a finite number. If this carrying capacity is exceeded to the degree that it cannot be replenished, the population that it is supporting will decrease significantly. This can be demonstrated by a real experiment conducted by scientists a number of years ago.

EXCEEDING THE ECOLOGICAL FOOTPRINT

Near the end of World War II in 1944, the United States Coast Guard placed 29 reindeer on St. Matthew Island in the Bering Sea as an emergency food supply for the US military. This island consisted primarily of vegetation and was void of any predators. Specialists had calculated that the island could support between 13 and 18 reindeer per square mile, or a total population of between 1600 and 2300 animals.

In 1957, the population was 1350; but just six years later in 1963, the population had exploded to 6000. Were the scientists wrong in their calculations of how many reindeer the island could support?

Eventually it was determined that the original calculations had been correct. The 6000 reindeer vastly exceeded the carrying capacity of the island and they were soon decimated by disease, starvation, and extreme weather conditions. Such a drastic overshoot, however, did not lead to restabilization at a lower level with some of the reindeer dying off. Instead, the entire habitat was so damaged by the overshoot of reindeer that the number of animals fell far short of the original carry capacity. In 1966, just three years later, there were only 42 reindeer living on St. Matthew Island rather than the expected 1600–2300. The reduction was primarily due to disease and starvation.

This is an example of what could happen to the earth. In the case of St. Matthew Island, the resources used by the reindeer were grasses, trees, and shrubs, all renewable resources that can be replenished. Many of the resources necessary for human survival, however, are not renewable. There is only a finite quantity of resources such as minerals, oil, and coal. We must be cognizant of the over utilization of both renewable and non-renewable resources.

To examine this over-utilization of the earth's resources, we must look at a concept called the ecological footprint. This is a tool for measuring and analyzing human natural resource consumption and waste output within the context of nature's renewable and regenerative capacity (or bio-capacity). It represents a quantitative assessment of the biologically productive area required to produce the resources (food, energy, and materials) and to absorb the wastes of an individual or region. In terms of resources, it includes cropland, grazing land, forest, fishing grounds, and built-up land. The footprint to handle waste output includes the forests required to absorb all the carbon dioxide emissions resulting from the individual's energy consumption.

In order to be sure we don't exceed the carry capacity of the earth, the footprint for humanity must be within the annual regenerative capability of

nature. Similarly, we must not exceed the absorptive capacity of the planet for the handling of the waste that is produced. A *sustainable* environment will exist if we live within the earth’s regenerative and absorptive capacity. If we remove more from nature than can be provided indefinitely, we are on an unsustainable track.

An organization called Global Footprint Network [8] has been calculating and analyzing the ecological footprint of over 200 countries. The footprint refers to the amount of the earth’s carrying capacity it takes to sustain humanity’s consumption of goods and services, basically the need for food, clothing, shelter, energy, and disposal of waste. According to its calculations, in the late 1970s, humanity’s collective ecological footprint breached the sustainability mark for the first time, and it has remained unsustainable ever since. In fact, the deficit for maintaining sustainability has grown every year since then, and it appears that this deficit is on a path to grow further in the foreseeable future. Currently, it is estimated that we need 1.7 earths to ensure that future generations are as well off as we are today.

It is interesting to note the variation in the ecological footprint by region or nation as seen in Table 1.1. Surprisingly, the largest footprint belongs to Qatar where it is 35.6 acres per capita. This means that for each individual living Qatar, about 36 acres are necessary to provide the consumptive and disposal needs for that person. By comparison, the footprint for the United States is 20 acres. Two additional questions that might be asked are the following: (i) is the footprint increasing with time, and (ii) how does this

TABLE 1.1 Ecological footprint for 2016

Country	Ecological footprint (acres per capita)
Qatar	35.6
Luxembourg	31.9
United States	20.0
Germany	12.0
Japan	11.1
China	8.9
World	6.8
Mexico	6.4
Egypt	4.5
Bio-capacity	4.0
India	2.9
Pakistan	2.1

The value shown in bold shows what the earth generates in one year.
Source: Global Footprint Network.

footprint compare with the available capacity? Growth in the ecological footprint can be attributed to an increase in population, an increase in consumption, or both. Of the Western European countries, Sweden, Belgium, Portugal, Spain, and Switzerland have increasing footprints, while Denmark and the Netherlands are making concerted efforts to reduce their footprints. The most striking result of this Ecological Footprint analysis is that if the entire world lived like the people of the United States, it would take over five planet earths to support the present world population.

THE LIMITS TO GROWTH

In 1972, a team of MIT experts wrote a report titled *The Limits to Growth* and presented it to scientists, journalists, and others and shortly published it as a book. It was the first time that computer modeling was used to answer the question as to whether the population would outgrow the planet and the resources available. The purpose of the study was to show the interrelationship between global growth factors like population, resources, persistent pollution, food production, and industrial activity. Based on this study, they predicted that if human beings continued to consume more than the environment was capable of providing, there would be an economic collapse and a sharp decline in population by 2030, which is not too far away.

This topic of overshoot and collapse was addressed again in “Limits to Growth: A 30-Year Update” [9], which stated that “overshoot can lead to two different outcomes. One is a crash of some kind. Another is a deliberate turnaround, a correction, a careful easing down. We explore these two possibilities as they apply to human society and the planet that supports it. We believe that a correction is possible and that it could lead to a desirable, sustainable, sufficient future for all the world’s peoples. We also believe that if a profound correction is not made soon, a crash of some sort is certain. And it will occur within the lifetimes of many who are alive today.”

Although the 1972 report seemed to focus on a very negative scenario, they looked at various changes that could avert a collapse. One positive variable was looking at technological changes that increased agricultural productivity, reduced pollution, and provided an increase in the available supply of natural resources. Technological advancements would have a positive impact, but this alone could not avert a collapse. Social and cultural changes would also be necessary to reduce consumption and stabilize population growth. Since it had been 40 years since the report, data were collected and compared with the predictions. To mark the 40th anniversary of the report, experts gathered to discuss the challenges ahead to into a sustainable

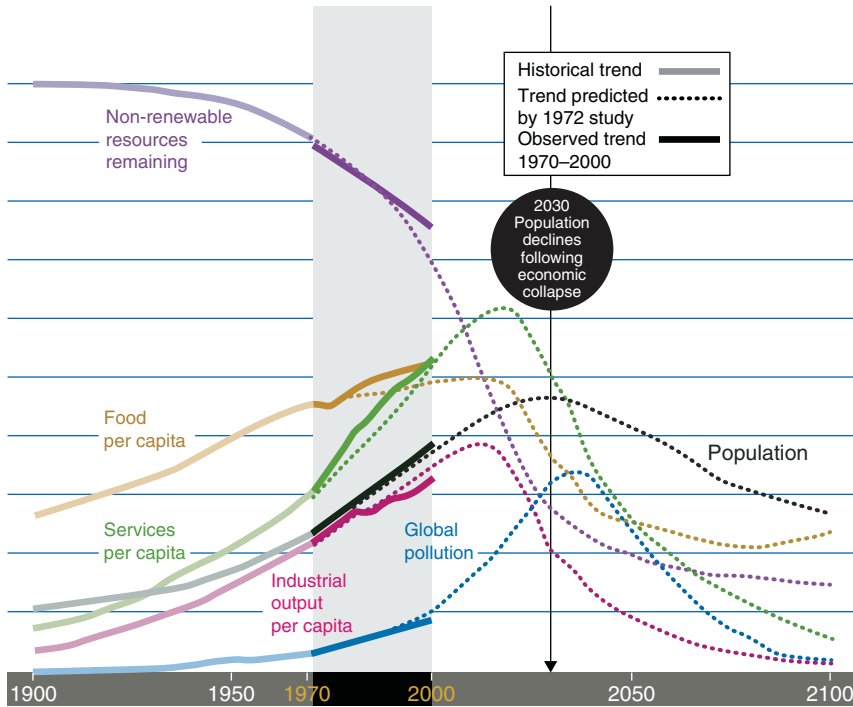


FIGURE 1.1 Limits of growth.

future. Their concern was depicted in Figure 1.1 [10], which shows that the world is following the predictions of the study.

You can see that with 30 years of data, pollution, industrial output, population, and services per capita are all increasing as expected. At the same time, the remaining non-renewable resources are decreasing a little slower (good), but food per capita is increasing a little faster than expected (bad).

The study was also concerned with sustainable development, which was defined by the notion that the developed nations can keep what they have while the poor people try to catch up. Or perhaps, keep on doing what we are all doing but through technological advances, we can expect less pollution and use fewer resources. Unfortunately, we are not succeeding with this expectation. We are currently consuming 70% more than what the earth is able to provide as explained earlier by the ecological footprint.

What we are consuming can be described as the different forms of industrial capital. This capital really refers to the machines and factories that produce the manufactured goods. These products manufactured by the industrial capital can be defined as industrial output. This industrial output derived from industrial capital can be used to generate service capital for the

service industry, like banks, schools, and hospitals, which provide services for the people. Industrial capital is also converted to agricultural capital to generate agricultural output. Likewise, it is converted to resource-obtaining capital to generate resource output. In addition, industrial capital is used to manufacture consumer goods. As each of these industrial outputs continues to grow, there is a need for more capital to be invested in the factories and machines that serve each of the outputs. Consequently, there may be an exponentially growing requirement for industrial output to expand the capacity for production in the future. This leads to more and more consumption.

CONSUMPTION FACTOR

Another way of looking at overuse of the earth's resources is to talk about consumption. There is a great variation today in consumption of the many nations in the world. Consumption is defined as the needs of people for survival in terms of food, energy, materials, and the disposal of waste. The disparity in the consumption rate is that it is 32 times greater in the United States, Canada, Western Europe, Japan, and Australia than in the developing countries [11].

In 2019, the world population is estimated to be around 7.7 billion people of whom only about 1 billion live in the fully developed countries listed earlier. By the middle of this century, it is estimated that the world population could grow to 9 billion people, and there are questions as to whether the earth can support this number of people or will it collapse. It is not really a question of how many people are on this earth, but what is the consumption rate of these people?

People in third-world countries are aware of a major difference in the consumption rate per capita, although they probably don't know the magnitude of the difference. In general, their goal is to catch up to the developed countries, but if they believe their chances of catching up are hopeless, they could get frustrated, angry, or even participate in terrorist activities. Another option is to immigrate to a first world country like the United States and Western Europe, but then they would contribute to the consumption rate of that country.

If one considers the fastest growing economy in China, these people are already aspiring to increase their consumption to the 32 factor. If the Chinese were to succeed, it would be equivalent of doubling the world's consumption rate. If India were to do the same thing, the consumption rate would then triple. If the entire world had the same consumption rate as these first world countries, it would be the same as having 72 billion people on this

planet at the current consumption rates—and there is no way the earth could handle this.

Since we are in no position to restrict the rest of world from improving their quality of life, the only answer is that the high consuming countries mentioned earlier must lower their consumption rate. But will they do it for the benefit of the rest of the world? Whether they want to or not, they must reduce their consumption rate because what they are doing today is not sustainable.

If these countries reluctantly agree to reduce their consumption rate, does it mean that they will have to reduce their quality of life? Definitely not! For example, the people in Western Europe consume half as much oil (gasoline) per capita than the people in the United States. But the Western European standard of living is considered higher than that in the United States in terms of life expectancy, healthcare, infant mortality, vacation time, quality of public schools, and several other criteria. Does a large gas-guzzling automobile really contribute positively to any of these quality-of-life factors? Probably not!

The current state of the environment can also be presented by looking at four major environmental issues: (i) water scarcity, (ii) energy sources, (iii) climate change, and (iv) population growth.

CONSERVATION OF WATER

Water is a natural resource with a finite quantity. The amount of water on this planet 2000 years ago is the same as it is today, but the population during this time interval has gone from approximately 150 million to over 7 billion. But of all the water on the earth, how much is readily available to all of the living creatures? Figure 1.2 [12] provides a summary of the current situation.

The earth's surface is about 71% water; however, 97% of all the water on earth is saline. Of the remaining 3%, 68.7% is in the form of icecaps and glaciers, 30.1% is ground water, and 0.9% is in some other unavailable form. This leaves only 0.3% of the fresh water on earth available to us on the surface with 87% in lakes, 11% in swamps, and 2% in rivers. This means that only 0.1% of all the water on the earth is available for industrial, agricultural, and human use. And of these three general uses, 70% is for agricultural use, 20% for industrial use, and only 10% for human consumption. Going further with the calculations results in only 0.01% of all the water on the earth being consumed by humans, and as the population grows, that leaves less for everyone.

According to the United Nations, two-thirds of the world's population is projected to face water scarcity by 2025. In the United States, a federal

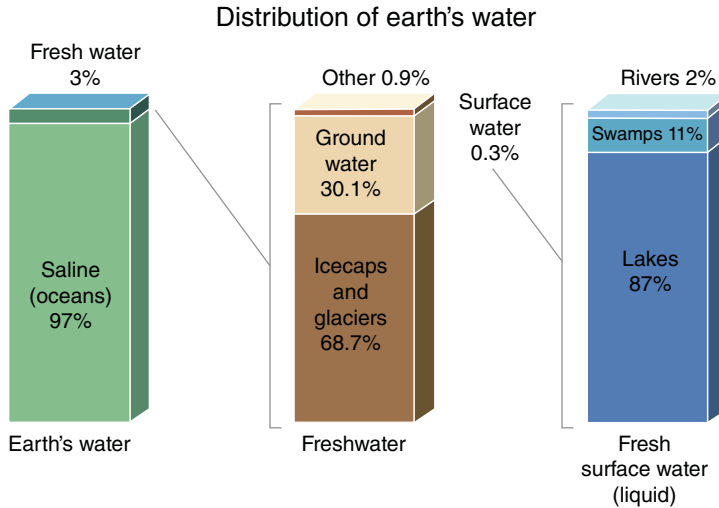


FIGURE 1.2 Distribution of earth's water.

report [12] by the General Accounting Office shows that 40 of the 50 states are anticipating water shortages by 2023. In 2008, the state of Georgia tried, unsuccessfully, to move the state's border north in order to claim part of the Tennessee River.

The concern for this water shortage is partly due to the companies that require so much for their processes. It takes roughly 20 gal of water to make a pint of beer, about 130 gal of water to make a 21 bottle of pop, and about 2000 gal of water to make a pair of Levi's stonewashed jeans. Why so much? For the pop, it includes the water used to grow the ingredients such as sugar cane. For the jeans, it includes the water used to grow, dye, and process the cotton.

Companies are now calculating the "water footprint" in order to manage better the water consumption. This is not dissimilar to the carbon footprint that organizations and individuals have been calculating for some years. The water-footprint concept was first developed in 2002 by A.Y. Hoekstra at the University of Twente in the Netherlands [13]. Following the water-footprint concept, studies were conducted to calculate the embedded, or virtual, water required for a product, which was then added to what is consumed directly. Embedded water includes everything from raising beef in South America, growing oranges in Spain, or growing cotton in Asia. By calculating the embedded water, you would learn that a typical hamburger takes 630 gal of water to produce. Most of the water is used to grow the grain to feed the cattle. This represents more than three times the amount the average American uses every day for drinking, bathing, washing dishes, and flushing toilets.

At first glance, these large numbers representing water footprints for certain products seem very alarming. However, they are not necessarily bad if there is available water and it is well managed. Since most of the water is used for crops, it becomes part of the water cycle where it is eventually evaporated or it is runoff. This water becomes temporarily unavailable for other uses, but that is not really a problem in an area that has plentiful water. If it doesn't return to the same aquifer or it returns as rainfall in another region, this could be a problem.

THE DEPLETION OF FOSSIL FUELS

In 1956, M. King Hubbert, a scientist with Shell Oil, proposed that fossil fuel production in a given region over time would follow a roughly bell-shaped curve without giving a precise formula [14]. Hubbert assumed that after fossil fuel reserves are discovered, production at first increases approximately exponentially, as more extraction commences and more efficient facilities are installed. At some point, a peak output is reached, and production begins declining until it approximates an exponential decline as shown in Figure 1.3.

The Hubbert curve suggests that the oil production rate increases as more reserves are discovered, and the rate peaks when half the estimated ultimately recoverable oil is produced. This is followed by a falling production

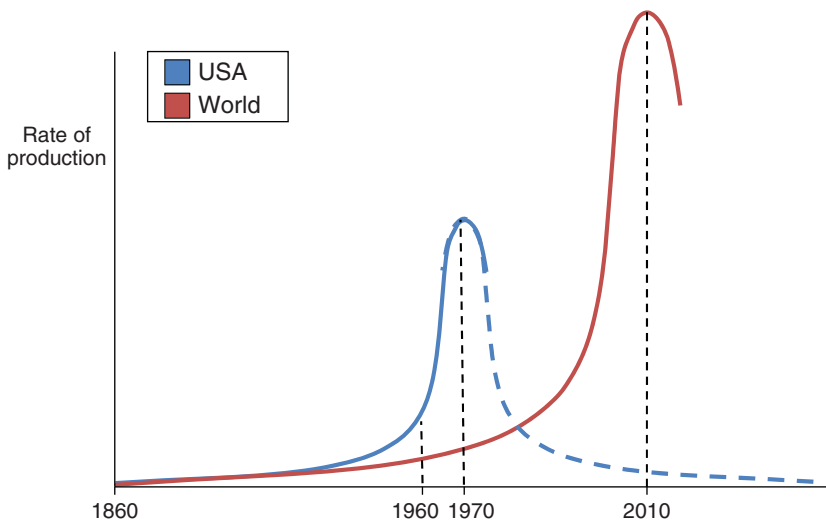


FIGURE 1.3 Hubbert's peak.

rate, all along a classic bell curve. This same analysis has shown that it took 110 years to produce about 225 billion barrels of crude oil in the United States, but half of that oil was produced in the first 100 years and the second half in the next 10 years. In the United States, Hubbert predicted that this production rate peak would be achieved in 1970, the year when half of the estimated ultimately recoverable oil was utilized and then production would start its steady decline. That in fact is what happened when the United States lost its preeminence as the world's leading producer of oil and caused a spike in gasoline prices and long lines at the pumps. His prediction that the United States would peak in oil production in 1970 actually came true although it peaked 17% higher than he projected, and its pathway since has not followed the bell-shaped curve he predicted. On a global basis, this milestone was expected to occur around the year 2010. What has happened since Hubbert's predictions is that more reserves have been discovered and new extraction technologies have been applied? It is now expected that the current oil reserves will last another 50 years [15]. Based on a similar analysis, natural gas reserves are good for 53 more years and coal reserves for at least 100 years.

CLIMATE CHANGE

In 1988, the United Nations General Assembly created the Intergovernmental Panel on Climate Change (IPCC) with the task of reviewing and assessing the most recent scientific, technical, and socio-economic information produced worldwide relevant to the understanding of climate change. Further, it would provide the world with a clear scientific view on the then current state of climate change and its potential environmental and socio-economic consequences, notably the risk of climate change caused by human activity.

The first assessment report of the IPCC was presented in 1990 and along with subsequent reports led scientists to conclude that the earth cannot tolerate more than a 3–5 °F increase in temperature. In order not to exceed this level, the carbon dioxide emissions must be reduced 60–80% of the 1990 levels by the year 2050. The first in a series of international meetings took place in 1992 in Rio de Janeiro, Brazil, called the Rio Earth Summit. As a result of that meeting, five years later the Kyoto Protocol was adopted. It recognized that climate change was a result of greenhouse gases (GHGs) created by human industrial activity. The idea was that rich nations, which had already benefited from industrialization, would reduce their GHG emissions in the first part of the treaty and developing nations would join in later. Milestones were created in various intervals through the year 2050. One of the milestones was to reduce GHG 5% below 1990 levels by 2012.

Instead, the world increased GHG by 58% above 1990 levels as the Kyoto Protocol came to an end.

At the next international meeting, which took place in Doha, Qatar, at the end of 2012, the developing countries once again demanded, as they did in Kyoto in 1997, for the rich countries to make a commitment to set real targets for reducing their GHG output. The rich nations then agreed to make some commitment toward a stronger legal agreement when they meet in Paris in 2015.

The primary goals of the Paris Agreement were to hold the increase in the global average temperature to less than 2°C from pre-industrial levels, but also achieve net zero GHG emissions beginning in 2050. Each country made a pledge to reduce carbon emissions commensurate with its rate of emissions along with its technical ability. For instance, the United States committed to reducing emissions 26–28% below 2005 levels by 2025. To achieve these goals, the public and private sectors must act boldly and quickly to reduce the use of fossil fuels and increase renewables as quickly as possible. In addition, if the technology can be developed, sequestering more carbon from the atmosphere would really enhance the goals of the agreement.

For the Paris Agreement to be successful, all of these countries, particularly the developed countries, must make the necessary investments in educating the masses, adopting renewable energy, reducing the use of fossil fuels, and developing carbon capture technology. But this will take a major financial commitment. For that reason, the next meeting of this Conference of Parties (COP22) took place in November 2016 in Marrakech, Morocco, to take action on the Paris Agreement including obtaining financial commitments from each of the countries and to establish plans for tracking progress by each country. This is all going in the right direction but none of these laws is binding.

Another measure of the carbon dioxide in our atmosphere is the actual average concentration in parts per million (ppm). Since the beginning of human civilization up until about 200 years ago, our atmosphere contained about 275 ppm [16] of carbon dioxide. A concentration of 275 ppm CO₂ is a useful amount—without some CO₂ and other GHGs that trap heat in our atmosphere, our planet would be too cold for humans to inhabit. In 2008, James Hansen, the top NASA climatologist indicated: “If humanity wishes to preserve a planet similar to that on which civilization developed and to which life on earth is adapted, paleoclimate evidence and ongoing climate change suggest that CO₂ will need to be reduced from its current 385 ppm to at most 350 ppm [17].” This level of 350 ppm has been the target of many environmentalists and even Bill McKibben, who founded a worldwide organization to curb global warming, named his organization

“<https://350.org>” [18]. Unfortunately, we are heading in the wrong direction as scientific instruments have shown that the earth’s average carbon dioxide concentration reached 400 ppm in 2013 [19].

There are some scientists, while in the minority, who believe that global warming may exist and/or it is not anthropological. Regardless, reducing carbon dioxide emissions is like an insurance policy. If one assumes that global warming does exist and therefore takes the necessary action, the downside risk is minimal. If we learn in 20 years that global warming never really existed, it would have resulted in unnecessary development of renewable energy sources and possible introduction of a carbon tax. If on the other hand, one assumes that global warming does not exist and therefore takes no action at all, what would happen if this assumption is eventually determined to be incorrect. The result to the world population could be catastrophic with rising sea levels leading to flooding and droughts leading to dwindling food production.

POPULATION GROWTH

Each of the environmental issues described earlier, consumption, fossil fuel reserves, water scarcity, and climate change are all related to the world population. Figure 1.4 [20] provides a summary of the historical as well as the projected growth in population.

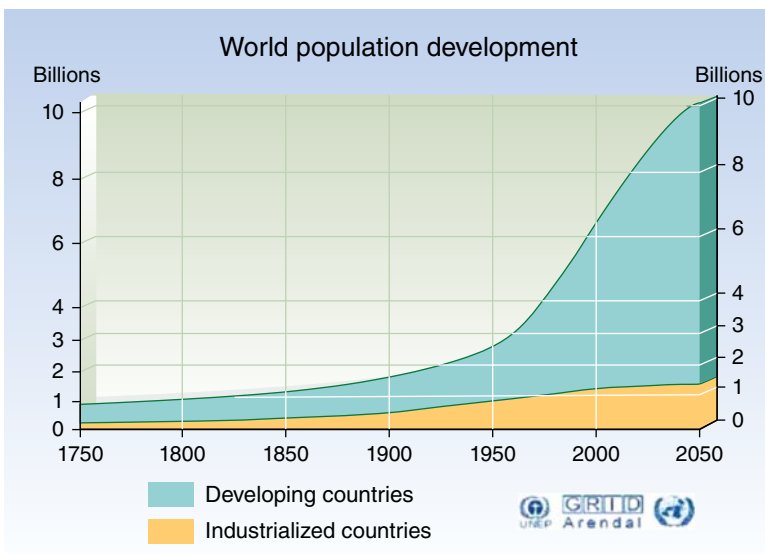


FIGURE 1.4 World population growth.

The world population reached 1 billion people in 1804, 2 billion in 1927, 5 billion in 1987, 6 billion in 1999, and 7 billion in 2011 [21]. In 2019, the world population is 7.7 billion and growing at a rate of 10 million people every 6.3 weeks [22]. Most of this growth is in the developing countries, which may not be a major problem because of the lower consumption rate. However, as indicated earlier some of these economies are increasing as are their consumption rates. China and India are examples of such growing economies.

THE ENVIRONMENT'S BIG FOUR

Today these are the four major environmental concerns in the world, specifically:

- Water quality and quantity
- Depletion of natural resources
- Climate change resulting primarily from fossil fuels
- Population growth—eventually exceeding the earth's capacity

Mitigating the impact of these four major environmental issues leads to an urgency for sustainable development.

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Development of the Sustainability Concept and CSR

In 1983, the United Nations convened the World Commission on Environment and Development and was chaired by Dr. Gro Harlem Brundtland who was the Prime Minister of Norway at the time. The agenda of this commission was the concern with the accelerating deterioration of the human environment and natural resources and the consequences of that deterioration for economic and social development. The final report was published in 1987 as “Our Common Future” [1] and resulted in the definition of sustainable development, or sustainability for short:

Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs.

This has become now well-known as the Brundtland definition of sustainability, but there are many other definitions that really all mean the same—protecting the environment so it will be available for the people forever. A definition developed for the Center for Sustainable Enterprise at the Illinois Institute of Technology—Stuart School of Business [2] is:

The practice of sustainable development assures that the natural resources and energy we use to provide today’s products and services will not deny future generations the resources necessary to meet their needs while building and preserving communities that are economically, socially and environmentally healthy.

Practical Sustainability Strategies: How to Gain a Competitive Advantage,
Second Edition. George P. Nassos and Nikos Avlonas.
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Companion website: www.wiley.com/go/Nassos/PracticalSustainabilityStrategies_2e

The impact on the environment can also be presented as a mathematical relationship. For instance,

$$EB = P \times A \times T_1$$

where EB is environmental burden; P is population; A is affluence; and T_1 is technology.

Environmental burden refers to the negative impact on the environment, which is a function of (i) the population, (ii) affluence or consumption by this population, and (iii) technology that enhances or mitigates this impact. This formula can be restated as follows:

$$EB = \frac{P \times A}{T_2}$$

where T_2 is the inverse of technology.

But now, the technology is what creates products and services sustainable. Population and affluence are really inversely proportional, as the population increases, consumption tends to decrease. As the people become more educated and consumption increases, the population tends to decrease. However, if the population was to double and the consumption was to, say, increase by 10-fold, technology would have to increase by 20-fold to keep the environmental burden constant.

The bottom line to this discussion is that there is an urgent need for corporations, government agencies, non-governmental organizations (NGOs), and individuals to find ways to be sustainable and to implement them. Chapters 3–17 in Part 2 will provide strategies for everyone, but for corporations in particular to develop, maintain, or extend a competitive advantage **without** having a negative impact on the environment. Being more efficient in energy usage, water usage, natural resource usage, and creating less waste are all very, very important. These can all be classified as being less bad. However, being “LESS BAD IS NOT ENOUGH.”

While most corporations are very concerned with profitability, implementing sustainability does not necessarily mean an opposition to profitability. In fact, these two terms are complementary. By striving for corporate sustainability, the company will achieve long-term profitability. This is the goal for all **corporate stakeholders**, a term first used by multinational companies in the late 1960s and early 1970s, referring to all people and entities upon which the company has an impact. At the same time, the term Corporate Social Responsibility (CSR) was used to describe the process with an aim to embrace responsibility for the company’s actions and encourage a positive impact through its activities on the environment, consumers, employees, communities, stakeholders, and anyone else in the

public sphere who may also be considered as stakeholders. While this term was not widely used during the next three decades, it is currently used often to describe a company's integration of sustainability in its operations. Perhaps a more accurate name for the process would be Corporate Social Responsibility and Sustainability, or CSR and Sustainability as used by McDonald's Corporation [3].

In 2011, the European Commission defined CSR as "a concept whereby companies integrate social and environmental concerns in their business operations and in their interaction with their stakeholders on a voluntary basis" [4].

CSR concerns actions by companies over and above their legal obligations toward society and the environment. Certain regulatory measures create an environment more conducive to enterprises voluntarily meeting their social responsibility.

According to European Union (EU) policy, a strategic approach to CSR is increasingly important to the competitiveness of enterprises. It can bring benefits in terms of risk management, cost savings, access to capital, customer relationships, human resource management, and innovation capacity.

Because CSR requires engagement with internal and external stakeholders, it enables enterprises to anticipate better and take advantage of fast changing societal expectations and operating conditions. It can therefore drive the development of new markets and create opportunities for growth.

By addressing their social responsibility, enterprises can build long-term employee, consumer, and citizen trust as a basis for sustainable business models. Higher levels of trust in turn help to create an environment in which enterprises can innovate and grow. Through CSR, enterprises can significantly contribute to the European Union's treaty objectives of sustainable development and a highly competitive social market economy. CSR underpins the objectives of the Europe 2020 strategy for smart, sustainable, and inclusive growth, including the 75% employment target. Responsible business conduct is especially important when private sector operators provide public services. Helping to mitigate the social effects of an economic crisis, including job losses, is part of the social responsibility of enterprises. CSR offers a set of values on which to build a more cohesive society and on which to base the transition to a sustainable economic system.

CSR TODAY: FROM SHAREHOLDER VALUE TO STAKEHOLDER VALUE

Corporations today are being met by greater pressure from consumer and stakeholder groups to act upon societal and environment obligations.

In essence, their actions are to mitigate the negative impact their activities are having upon the wider world. More and more, business leaders are themselves coming to the realization that their activities reach beyond the sphere of the business marketplace, impacting social, environmental, and political realms as well, and often negatively, and have implemented various measures, such as voluntary CSR reporting and recycling programs, to counteract the social and environmental tolls of global business.

With the rise of consumer education and activism, global social consciousness has become a hot topic of discussion, with business leaders jumping at the chance to exert themselves as champions of the people, and promote their businesses as “green” or “socially responsible.” And in the wake of a continually evolving set of ethical business guidelines comes the redefinition of the very concept of *business*. No longer is the “shareholder value prospect,” that is, maximum profit, the only player in the game, but its rival, CSR, a largely stakeholder and consumer concern, has emerged as the new MVP, or Most Valuable Priority. This reshuffle of priorities reflects the ability of globally responsible and creative business leaders to remain profitable while expending company resources toward the betterment of society, the transparency of business, and the protection of the environment.

Stakeholders—customers, employees of all ranks, suppliers, partners, affiliated institutes and governments, shareholders, and “owners”—represent the most fertile and diverse source of easily procured information available to a business or an organization. It should, therefore, come as no surprise that stakeholder engagement and dialogue is crucial for the profitability and continued existence of a company, as stakeholder opinions can be utilized to guide its vision and mission, and reflect on its performance. To simultaneously realize the potential of stakeholder groups, and to retain their valuable insights, a company must be willing to engage stakeholders in frequent dialogue, weigh their opinions, and allow them to participate in various decision-making processes. If this is not done, the company runs the risk of losing stakeholder capital—whether this is monetary, intellectual, or labor capital—which could prove potentially catastrophic to the life of the company in the long term.

Follow-up and feedback with stakeholders provide a cycle of opportunity for the growth and improvement of a company. Diverse stakeholder groups, when engaged in frequent dialog, can steer a company in a profitable and productive direction through the provision of perspectives on a variety of issues, from within a range of industries, geographic locals, demographic groups, and cultural backgrounds. For efficient and effective stakeholder dialog to be established, constant, clarifying communication between parties is essential, such as the identification of common goals among stakeholder groups and the company, as is the engagement and participation

of stakeholders in relevant fora. Building relationships that transcend traditional business boundaries is a proven way to involve stakeholders in the mission of the company on a personal and emotional level, and evokes a genuine desire to learn from partnerships.

CSR MEASURING AND REPORTING

In addition to CSR, another term being used by business community is Environmental, Social and Corporate Governance (ESG). This refers to the three main areas of concern that have developed as the central factors in measuring the sustainability and ethical impact of an investment in a company or business. Whether a company is concerned about CSR or ESG, how does it know whether it is meeting its goals and objectives? And how does the investment community know which companies are achieving the CSR and ESG objectives?

The best way to really understand if your organization is meeting any goals and objectives is to take appropriate measurements of the key performance indicators (KPIs). But it is important to measure both the environmental/social indicators and the financial indicators because a balanced scorecard is important. In other words, the balanced scorecard consists of financial and non-financial indicators.

Another approach at obtaining this balance is to consider the environmental, social, and economic aspects of the company, or the triple bottom line (TBL) as coined by John Elkington [5]. It is also known as P³ or planet, people and profit, or E³ for earth, equity, and economy. This is often shown as a triangle (Fig. 2.1) depicting a three-legged chair as all three legs are necessary. A more realistic depiction can be seen in Figure 2.2 that shows the planet and environment as a large circle. Within the environment, in a smaller circle, are the people that depend on the environment. And what is

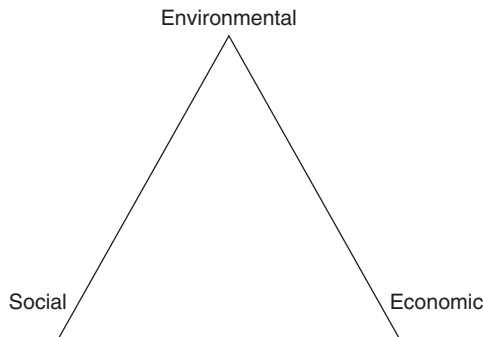


FIGURE 2.1 Triple bottom line.

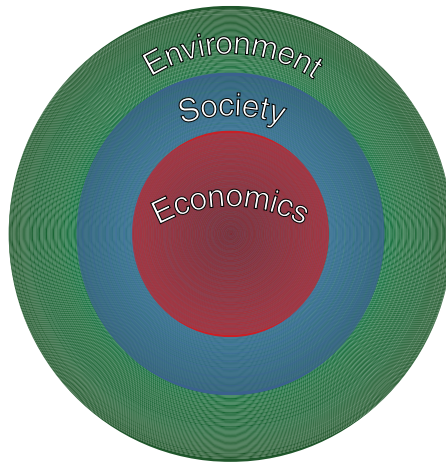


FIGURE 2.2 Realistic triple bottom line.

dependent on the environment and society is the economy. But there needs to be a measurement to understand achieving the TBL. The Global Reporting Initiative (GRI), developed in 1997, is the internationally accepted standard for TBL reporting. The GRI was created to bring consistency to the TBL reporting process by enhancing the quality, thoroughness, and value of sustainability reporting.

Chapters 18–24 in Part 3 will provide readers with various measuring standards as well as reporting standards. The benefits to the organization will also be outlined.

THE SUSTAINABLE DEVELOPMENT CONCEPT THOUSANDS YEARS AGO

Although the term CSR is used today, concepts analogous to social responsibility have existed for centuries. The idea that those in an advantageous position, either in terms of money or power, should conduct in a socially responsible way can be traced back to classical Athens of 500 BC.

This view supports that from the early days of civilization it was understood, in some cases by the state or in others by wealthy individuals of the time, that it would only be fairer for the rest of society if such individuals shared a portion of their amassed wealth with their fellow citizens. This reality came into being from the realization of the fact that has it not been

for the rest of society, wealth making would probably be an impossibility. Thus, bearing this realization in mind, certain laws were voted by the state which at the time reflected precisely and directly the opinion of the people. Namely, these were the liturgies, “a wide range of public service paid for out of the pockets of the people owning substantial property” [6]. There were two different kinds of liturgies, the regular liturgies which had a religious character and the extraordinary liturgies which had a military character.

Aristotle’s “magnificent man,” who bestows collective benefits upon his community out of his own wealth, is depicting the exact definition of “euergetism” in ancient Greece. The nobles, who because of their birth and wealth controlled access to all essential services, were expected to provide various services to their cities in exchange for the public bestowal of honor from the inhabitants. In the *Art of Rhetoric*, Aristotle states that true wealth consists in doing good; that is, in monetary handouts, giving of scarce and costly gifts, and helping others to maintain an existence [7].

A last notion connecting CSR and ancient Greece is that of heroism. Nowadays, the term “hero” has many different meanings. It addresses to people whose work is extremely risky like firemen, or to people whose work is challenging like teachers. “Hero” originates from the Greek word *hêrôs*. Although in ancient Greece this word was referring only to warriors, later on it was introduced for the people who have offered in favor to the local community but they are now dead. In order for someone to obtain the heroic status, he should deal with local sponsorship. There were no exact rules that someone had to follow in order to attain a heroic status. Someone who was involved in the matters of public interest and his goal was the welfare of his community could be characterized as a hero.

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SUSTAINABILITY STRATEGIES

Imbedding the UN Sustainable Development Goals to Achieve Sustainability

It is important to note that there are many activities that an organization can undertake for the benefit of the environment. Some of these activities may be sustainable, but many of them are not truly sustainable yet very beneficial as they reduce the negative impact on the environment. These environmentally beneficial activities embraced by corporations become the guidelines that are followed, better known as “principles.”

As a result of the Exxon Valdez oil spill in 1989, a group of investors launched the Coalition for Environmentally Responsible Economies (CERES). This group, which subsequently went by the name CERES, consisted of over 130 institutional and socially responsible investors, environmental and social advocacy groups, and other public interest organizations. This organization developed a set of 10 principles that organizations should adopt in order to achieve sustainability. In addition to the CERES principles, other organizations adopted their own set of principles with associated goals. Some of these other principles include the following:

- The Hannover Principles
- Precautionary Principle
- Community
 - Principles of Sustainable Development for Minnesota
 - Ontario Round Table on Environment and Economy (ORTEE)
 - The Netherlands National Environmental Policy Plan (NEPP)

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- The Earth Charter
- International Council of Local Environment Initiatives (ICLEI)
- Natural resources
 - American Petroleum Institute (API) Environmental, Health, and Safety Principles
 - Forest Stewardship Council (FSC) Principles and Criteria for Forest Stewardship
 - Marine Stewardship Council (MSC) Principles and Criteria for Sustainable Fishing
 - The Asilomar Declaration for Sustainable Agriculture
- Ecological design
 - The Five Principles of Ecological Design
 - Todds' Principles of Ecological Design
 - The Sanborn Principles
 - US Green Building Council and Leadership in Energy and Environmental Design (LEED)
- Biosphere
 - Deep Ecology's Basic Principles
 - Charter of Rights and Responsibilities for the Environment
 - Biomimicry Principles
 - Mollisonian Permaculture Principles

UN 2030 AGENDA

In September 2015, the United Nations passed a resolution titled *Transforming our world: The 2030 Agenda for Sustainable Development*. Included in this agenda were 17 sustainable development goals (SDGs) along with 169 targets. The purpose of the goals and targets was to stimulate action over the next 15 years in areas of critical importance for humanity and the planet. More specifically, the agenda focused on the following five areas:

- (1) *People*—We are determined to end poverty and hunger, in all their forms and dimensions, and to ensure that all human beings can fulfill their potential in dignity and equality and in a healthy environment.
- (2) *Planet*—We are determined to protect the planet from degradation, including through sustainable consumption and production,

sustainably managing its natural resources and taking urgent action on climate change, so that it can support the needs of the present and future generations.

- (3) *Prosperity*—We are determined to ensure that all human beings can enjoy prosperous and fulfilling lives and that economic, social, and technological progress occurs in harmony with nature.
- (4) *Peace*—We are determined to foster peaceful, just, and inclusive societies that are free from fear and violence. There can be no sustainable development without peace and no peace without sustainable development.
- (5) *Partnership*—We are determined to mobilize the means required to implement this Agenda through a revitalized global partnership for sustainable development, based on a spirit of strengthened global solidarity, focused in particular on the needs of the poorest and most vulnerable and with the participation of all countries, all stakeholders and all people.

THE 17 SUSTAINABLE DEVELOPMENT GOALS

The 17 SDGs adopted by the United Nations in 2015 are most often depicted as a table such as Figure 3.1.

The SDGs were written as a blueprint to achieve a better and more sustainable future for everyone on the planet. They address the global



FIGURE 3.1 2015–2030 Sustainable development goals.

challenges that are before everyone, including those related to poverty, inequality, climate, environmental degradation, prosperity, and peace and justice. The SDGs interconnect with a target to be successful by 2030. The 17 SDGs can be summarized as follows:

- (1) *No poverty*—End poverty in all its forms everywhere.

Extreme poverty has been cut by more than half since 1990, but more than one in five people still live on less than the target figure of US\$1.25 per day. That target may not be adequate for human subsistence, however, as there is always an effort to increase it. People also live in poverty if they lack basic services such as healthcare, security, and education. They also experience hunger, social discrimination, and exclusion from decision-making processes.

Gender inequality plays a large role in perpetuating poverty and its risks. Women face potentially life-threatening risks from early pregnancy and frequent pregnancies. This can result in lost hope for an education and for a better income. Poverty affects age groups differently, with the most devastating effects experienced by children. It affects their education, health, nutrition, and security, impacting emotional and spiritual development.

Achieving SDG 1 is hampered by growing inequality, increasingly fragile statehood, and the impacts of climate change.

- (2) *Zero hunger*—End hunger, achieve food security and improved nutrition, and promote sustainable agriculture.

SDG 2 states that by 2030 we should end hunger and all forms of malnutrition. This would be accomplished by doubling agricultural productivity and incomes of small-scale food producers (especially women and indigenous peoples), by ensuring sustainable food production systems and by progressively improving land and soil quality.

Other targets deal with maintaining genetic diversity of seeds, increasing access to land, preventing trade restriction and distortions in world agricultural markets to limit extreme food price volatility, eliminating waste with help from the International Food Waste Coalition, and ending malnutrition and undernutrition of children.

A report by the International Food Policy Research Institute (IFPRI) of 2013 stated that the emphasis of the SDGs should not be on ending poverty by 2030, but on eliminating hunger and undernutrition by 2025. The assertion is based on an analysis of experiences in China, Vietnam, Brazil, and Thailand. Three pathways to achieve this were identified: (i) agriculture-led; (ii) social

protection- and nutrition- intervention-led; or (iii) a combination of both of these approaches [1].

- (3) *Good health and well-being*—Ensure healthy lives and promote well-being for all at all ages.

Significant strides have been made in increasing life expectancy and reducing some of the common killers associated with child and maternal mortality; however, the number of children dying under age five is still extremely high at 41 deaths per 1000 live births [2]. Additional key strategies for meeting SDG 3 will be to reduce adolescent pregnancy (which is strongly linked to gender equality), provide better data for all women and girls, and achieve universal coverage of skilled birth attendants.

Continued improvement on access to clean water and sanitation will reduce the spread of malaria, tuberculosis, polio, and HIV/AIDS. SDG 3 also aims to achieve universal health coverage, including access to essential medicines and vaccines. Attention to health and well-being also includes targets related to the prevention and treatment of substance abuse, deaths, and injuries from traffic accidents and from hazardous chemicals and air, water, and soil pollution and contamination.

- (4) *Quality education*—Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

Major progress has been made in access to education, specifically at the primary school level, for both boys and girls, but there is still a ways to go. Target 1 of SDG 4 is to ensure that, by 2030, all girls and boys complete free, equitable, and quality primary and secondary education. But the problem with tracking SDG 4 is that most of the countries do not have the necessary data to track progress.

- (5) *Gender equality*—Achieve gender equality and empower all women and girls.

According to the United Nations, “gender equality is not only a fundamental human right, but a necessary foundation for a peaceful, prosperous, and sustainable world [3].” However, about one-fourth of the countries have not taken the step to provide the same. Though child marriages are four times higher among the poorest than the wealthiest in the world, most countries need to accelerate progress among both groups in order to reach the SDG 5 target to eliminate child marriage by 2030 [2].

This goal should also refer to the increase in women’s roles in the corporate world as well as in the governments.

- (6) *Clean water and sanitation*—Ensure availability and sustainable management of water and sanitation for all.

The SDG 6 has 8 targets and 11 indicators that will be used to monitor progress toward the targets. The targets relate primarily to drinking water supply and sanitation. Safe drinking water and hygienic toilets protect people from disease and enable societies to be more productive economically.

- (7) *Affordable and clean energy*—Ensure access to affordable, reliable, sustainable, and modern energy for all.

Targets for 2030 include access to affordable and reliable energy while increasing the share of renewable energy in the global energy mix. This would involve improving energy efficiency and enhancing international cooperation to facilitate more open access to clean energy technology and more investment in clean energy infrastructure. As of 2017, only 57% of the global population relies primarily on clean fuels and technology, falling short of the 95% target [2].

- (8) *Decent work and economic growth*—Promote sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all.

For the least developed countries, the economic target is to attain at least a 7% annual growth in gross domestic product (GDP). Achieving higher productivity will require diversification and upgraded technology along with innovation, entrepreneurship, and the growth of small- and medium-sized enterprises (SMEs). While most targets are for 2030, there is one for 2020 to reduce youth unemployment and operationalize a global strategy for youth employment.

- (9) *Industry, innovation, and infrastructure*—Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation.

It has been known for much time that manufacturing is a major source of employment. In 2016, the least developed countries had less “manufacturing value added per capita.” The figure for Europe and North America amounted to US\$4621, compared with about US\$100 in the least developed countries [4]. The manufacturing of high products contributes 80% to total manufacturing output in industrialized economies but barely 10% in the least developed countries. SDG 9 aims to increase manufacturing jobs in the less developed countries.

- (10) *Reducing inequalities*—Reduce income inequality within and among countries.

One target is to reduce the cost of exporting goods from least developed countries. This would help increase the sales, productions, and work force in the least developed countries. Also, reducing the cost of transferring money in or out as well as mobile phone charges should be targets.

- (11) *Sustainable cities and communities*—Make cities and human settlements inclusive, safe, resilient, and sustainable.

The target for 2030 is to ensure access to safe and affordable housing. The indicator named to measure progress toward this target is the proportion of urban population living in slums or informal settlements. The cities and communities should also be designed to include as much renewable energy as possible as well as sustainable mobility.

- (12) *Responsible consumption and production*—Ensure sustainable consumption and production patterns.

The targets of SDG 12 include using eco-friendly production methods and reducing the amount of waste. By 2030, national recycling rates should increase, as measured in tons of material recycled. Further, companies should adopt sustainable practices and publish sustainability reports. Recommendations from the Global Footprint Network [5] should be incorporated in order to move Earth Overshoot Day later in the year. This is the day in the year that the world will have consumed all of the natural resources generated by the earth in that calendar year.

- (13) *Climate action*—Take urgent action to combat climate change and its impacts by regulating emissions and promoting developments in renewable energy.

In May 2015, a report concluded that only a very ambitious climate deal in Paris in 2015 could enable countries to reach the SDGs and targets [6]. The report also states that tackling climate change will only be possible if the SDGs are met. Further, economic development and climate change are inextricably linked, particularly around poverty, gender equality, and energy.

In October 2018, the Intergovernmental Panel on Climate Change released a report [7] on the risks and benefits of limiting global warming to 1.5°C (2.7°F) above pre-industrial levels, along with recommended pathways for limiting temperature increases to that level. Staying within 1.5°C, says IPCC, would require “rapid and far-reaching” transitions in land, energy, industry, buildings, transport, and cities, in short, changing almost everything.

According to IPCC, staying within 1.5°C warming would require, in effect, a radical transformation of unprecedented scale to avoid climate catastrophe. The report says that based on current forecasts, we are likely to reach the 1.5°C threshold sometime between 2030 and 2050.

One of the most comprehensive reports on mitigating climate change is *Drawdown*, edited by Paul Hawken and discussed in Chapter 11.

- (14) *Life below water*—Conserve and sustainably use the oceans, seas, and marine resources for sustainable development.

Oceans cover 71% of the earth's surface, and this results in about 97% of all the water on the Earth being saline. But these oceans are essential for making the planet livable.

The oceans contain more than 200,000 identified species, and there might be thousands of species that are yet to be discovered. Oceans are the world's largest sources of protein. However, there has been a 26% increase in acidification since the industrial revolution. A full 30% of marine habitats have been destroyed, and 30% of the world's fish stocks are over-exploited [8]. Marine pollution has reached shocking levels; each minute, 15 tons of plastic are released into the oceans [9], 20% of all coral reefs have been destroyed irreversibly, and another 35% are in immediate risk of collapse [10].

Individuals can help the oceans by reducing their energy consumption and their use of plastics. Nations can also take action. In Norway, for instance, citizens, working through a web page called *finn.no*, can earn money for picking up plastic on the beach [11]. Improving the oceans contributes to poverty reduction, as it gives low-income families a source of income and healthy food. Keeping beaches and ocean water clean in less developed countries can attract tourism, as stated in SDG 8, and reduce poverty by providing more employment.

The targets include preventing and reducing marine pollution and acidification, protecting marine and coastal ecosystems, and regulating fishing. The targets also call for an increase in scientific knowledge of the oceans.

- (15) *Life on land*—Protect, restore, and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss.

This goal relates to targets for preserving biodiversity of forest, desert, and mountain eco-systems, as a percentage of total land mass. It is important to restore degraded forests and land lost to

drought and flood. SDG 15 calls for more attention to preventing invasion of introduced species and more protection of endangered species.

- (16) *Peace, justice, and strong institutions*—Promote peaceful and inclusive societies for sustainable development, provide access to justice for all, and build effective, accountable, and inclusive institutions at all levels.

Reducing violent crime, sex trafficking, forced labor, and child abuse are clear global goals that come under SDG 16. The international community values peace and justice and calls for stronger judicial systems that will enforce laws and work toward a more peaceful and just society. One target is to see the end to sex trafficking, forced labor, and all forms of violence against and torture of children. SDG 16 also targets universal legal identity and birth registration, ensuring the right to a name and nationality, civil rights, recognition before the law, and access to justice and social services. With more than a quarter of children under five unregistered worldwide as of 2015, about one in five countries will need to accelerate progress to achieve universal birth registration by 2030 [2].

- (17) *Partnerships for the goals*—Strengthen the means of implementation and revitalize the global partnership for sustainable development.

Increasing international cooperation is seen as vital to achieving each of the 16 previous goals. SDG 17 is included to assure that countries and organizations cooperate instead of compete. Developing multi-stakeholder partnerships to share knowledge, expertise, technology, and financial support is seen as critical to overall success of the SDGs.

A REAL POSITIVE VIEW OF THE SDGS

While Figure 3.1 lists the 17 SDGs in numerical order and tabular form, another view of the goals is depicted as a pyramid in reverse order as shown in Figure 3.2.

A FURTHER LOOK AT THE SDGS

A careful look at the 17 SDGs will reveal that there may be some conflicts between the goals. Achieving a target for one goal may affect another goal



FIGURE 3.2 Sustainable development goals as a pyramid.

negatively. For example, SDG 8: Decent Work and Economic Growth calls for an annual increase in GDP of 7%. To achieve this goal could have a negative effect on SDG 12: Responsible Consumption and Production.

Another potential conflict can be between SDG 2: Zero Hunger and SDG 7: Affordable and Clean Energy. There are some clean energy options that are produced from vegetation such as a low carbon emitting vehicle fuel made from corn.

The 17 SDGs include a total of 169 targets [12]. This large number of goals and associated targets has been estimated to cost about US\$2–3 trillion/year for the next 15 years [13]. A big portion of this cost is to provide clean water to entire nations. Implementation of the SDGs started worldwide in 2016, but each nation, organization, and corporation within each nation has started or will start at different times.

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The Natural Step

In Chapter 3, we discussed the development of the UN Sustainable Development Goals (UNSDGs) and various other principles that can be considered as guides to achieving environmental and social sustainability. One of the principles not mentioned is The Natural Step (TNS). This one is somewhat unique and deserves special mention.

In the late 1980s, Karl-Henrik Robèrt, a Swedish medical doctor and cancer-treatment researcher could not help but be confused between two groups of people. On one hand, there were some people that were more concerned with increasing their personal wealth than a concern with the environment and its potential impact on children. And on the other hand, he was seeing concerned parents bringing to the hospital cancer-diseased children.

Dr. Robèrt became obsessed about the potential destruction of the environment. He saw humankind running into a funnel of declining life-sustaining resources and increasing demands [1]. He was aware of the various environmental issues like climate change, increased pollution, water scarcity, and all of which would be compounded with population growth. At the same time, he was not sure whether businesses would focus beyond short-term profits to long-term growth for the good of the environment, or whether there would be any legislation to make it happen. He then began to think: “My epiphany came one day when I was studying cells from cancer patients. It hit me that cells are the unifying unit of all living things. The difference between our cells and the cells of plants are so minor that it’s almost embarrassing; the makeup is almost identical all the way down to the molecular level.” [2].

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Dr. Robèrt drafted a first version of such a “consensus document” and sent this draft to a broad cross-section of scientists, including over 50 ecologists, chemists, physicists, and medical doctors in Sweden and asked for their input. Twenty-one drafts later, there was at last consensus about what in principle needed to sustain the human civilization on earth. With the support of His Majesty the King of Sweden, Carl XVI Gustaf, this “consensus document” and accompanying audio tape were sent to every household and school in Sweden.

Based on this document, in 1989 Dr. Robèrt started TNS, an environmental not-for-profit organization in Sweden. He wanted to address these troubling environmental problems and seek advice and expertise of a growing circle of scientists, economists, business leaders, and other stakeholders in society. This dialogue with key participants spread throughout the world, and, as a result, TNS organizations were launched in the United States and other countries like Canada, the United Kingdom, Australia, and New Zealand.

Robèrt and his advisors developed a simple framework to look at the impact of the environment on businesses and how to provide a means for businesses to become sustainable. The framework was based on systems thinking, which is looking at the whole system, but in such a way that it can be made simple, thus allowing it to apply to almost any organization. The framework includes the following four core processes [1]:

- Perceiving the nature of the unsustainable direction of business and society and the self-interest implicit in shifting to a sustainable direction
- Understanding the first-order principles for sustainability, that is, the four System Conditions
- Strategic visioning through “back-casting” from a desired sustainable future
- Identifying strategic steps to move the company from its current reality toward its desired vision

He then applied the funnel effect to show how the supply and demand of environmental resources are converging. This impact on the environment can be best described as follows [3]:

Global society is currently unsustainable. Because we only have one planet Earth, and the laws of thermodynamics are such that matter and energy cannot be created or destroyed, there are limits to how much we can grow—in terms of both sources (the resources we use) and *sinks* (the natural systems where we deposit our waste). We are currently surpassing those limits because on the one hand population, consumption, competitiveness, and pollution are *systematically* increasing, while on the other

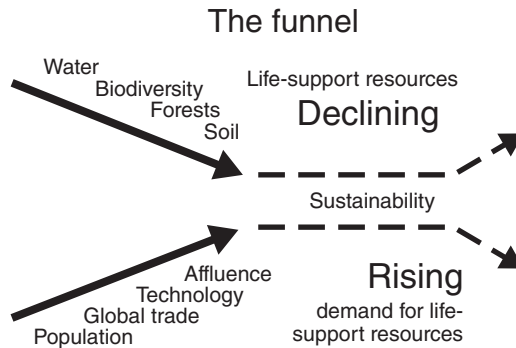


FIGURE 4.1 The funnel.

hand resources, purity, ecosystems, and social equity are *systematically* decreasing.

A “whole systems” perspective is necessary in sustainable development, as the various problems (greenhouse gases, extreme poverty, deforestation, illiteracy, etc.) are inherently interrelated and complex.

The funnel (Fig. 4.1) metaphor is a way of thinking about the unsustainable path that global society is on, where our space for deciding on options is becoming narrower and narrower per capita. This is very different from the illusion that limits to growth are represented by a “cylinder” where isolated social and ecological impacts can be addressed and “solved” separately.

The negative effects of society’s unsustainable path can be described as “hitting the funnel walls.” Organizations (communities, businesses, project teams, etc.) can feel the impact of hitting the funnel walls in a variety of ways, including the following:

- Increased costs for resources and waste management
- Lost investment in projects that quickly become obsolete
- Stricter legislation
- Litigation
- Loss of market share to more cutting edge
- Insurance costs
- Consumer and shareholder activism
- Loss of good reputation

Organizations that have an understanding of the funnel will be better able to act strategically, communicate more effectively to internal and external stakeholders, and lead the shift toward a sustainable society.

THE FOUR SYSTEM CONDITIONS FOR SUSTAINABILITY

In order to build this framework for sustainability, society must adopt the four system conditions in order to be ecologically sustainable. TNS summarizes the Four System Conditions in the following text: In order for a society to be sustainable, nature's functions and diversity are not systematically:

1. Subject to increasing concentrations of substances extracted from the earth's crust.
2. Subject to increasing concentrations of substances produced by society.
3. Impoverished by overharvesting or other forms of ecosystem manipulation.
4. Resources are used fairly and efficiently in order to meet basic human needs worldwide.

System Condition 1—Substances from the Earth's Crust Must Not Systematically Increase in the Ecosystem

This means that materials like fossil fuels, metals, minerals, and other natural resources must not be extracted from the earth at a faster rate than they are reentering or depositing into the earth's crust or even its atmosphere. A good example of violating this condition is the extraction of fossil fuels like oil or coal. When the fossil fuel is combusted, the carbon in the coal, for instance, is converted to carbon dioxide. For most of the human existence, the carbon dioxide emitted would eventually be reabsorbed by trees and other vegetation. The rate of emitting carbon dioxide was equal to the rate of absorption. Consequently, the carbon cycle was in balance and there was no violation of System Condition 1. The concentration of carbon dioxide was consistently about 280 ppm. However, we are now emitting carbon dioxide at a faster rate than it is being absorbed, thus causing an increase in the carbon dioxide concentration in the atmosphere—exceeding 400 ppm. This cover of carbon dioxide is preventing the release of reflected sun rays back into the atmosphere, thus causing an increase in the temperature of the Earth—global warming, which is also known as climate change.

If this condition is not met, the concentrations of substances in the ecosystem will increase and eventually reach some limits beyond which it will be almost impossible to reverse. In the case of carbon dioxide, many scientists believe that it will be impossible to reverse, even if we stopped emitting carbon dioxide. In a report by the National Oceanic and Atmospheric Administration, it stated that “changes in surface temperature, rainfall, and

sea level are largely irreversible for more than a thousand years after carbon dioxide emissions are completely stopped.” In an interview with the lead author, Susan Solomon, she said, “People have imagined that if we stopped emitting carbon dioxide that the climate would be back to normal in 100 or 200 years. What we’re showing here is that that’s not right” [4].

System Condition 2—Substances Produced by Society Must Not Systematically Increase in the Ecosphere

This means that substances must not be produced at a faster rate than the rate at which they can be broken down and integrated back into nature or redeposited into the earth’s crust. If this condition is not met, the concentration of substances in the ecosphere will continue to increase and eventually reach undesirable limits beyond which it will be difficult reverse. In most cases, this upper limit is not known.

It is critical that we eliminate our contribution to the progressive buildup of chemicals and compounds produced by society such as dioxins, polychlorinated biphenyls (PCBs), and dichlorodiphenyltrichloroethane (DDT). In addition to these very toxic chemicals, there are many other substances produced by society that are accumulating in nature and are detrimental to the environment.

A very interesting example of the violation of System Condition 2 is that of ozone. The chemical name for ozone is tri-oxygen as it consists of three oxygen atoms as opposed to two atoms for the oxygen molecule. Electromagnetic radiation or the sparking from high voltage applications will cause the oxygen molecule in the air to disassociate into two oxygen atoms, and then each atom will combine with another oxygen molecule to form ozone. Devices that require high voltages and as a result produce ozone are arc welders, ionic air purifiers, laser printers, and photocopiers. Ozone is also created by precursors such as nitrogen oxides, carbon monoxide, and various volatile organic compounds (VOCs).

Ozone in the lower atmosphere is a pollutant and may cause respiratory problems. In the upper atmosphere, however, ozone is beneficial as it prevents potentially damaging electromagnetic radiation from reaching the earth’s surface. Other man-made substances like chlorinated fluorocarbons (CFCs) break up in the atmosphere and the free chlorine or fluorine atoms become catalysts to break up the ozone molecule. This has caused the formation of an “ozone hole” in the atmosphere that allows electromagnetic radiation to reach the earth. So this is an example of the formation and destruction of ozone being the result of violating System Condition 2.

As a result of the damage created by the CFCs, the Montreal Protocol of 1987 called for the reduction and the eventual ban of CFCs chemicals for

use as refrigerants and other applications. The major chemical companies then started producing hydrogenated chlorinated fluorocarbons (HCFCs), which are considerably less destructive to the ozone layer [5]. Electrolux, the refrigerator and appliance company, decided not to replace their CFC chemicals with a less toxic HCFC because this still went against System Condition 2. Instead, they opted to research for a totally biologically harmless substitute.

System Condition 3—The Physical Basis for Productivity and Diversity of Nature Must Not Be Systematically Diminished

This means that we cannot harvest or manipulate ecosystems in such a way that productive capacity, ecosystem services, and diversity systematically diminish. While the first two system conditions refer to generation of substances into the ecosphere, this condition pertains to the destruction of the natural resources in such a way or to an extent that they cannot regenerate and be useful. An example is the over harvesting of forests to meet the current demand for wood, or for its conversion to paper. This also has a detrimental effect in reducing the sequestration of carbon dioxide. Human activities need to work in harmony with the cyclic principle of nature. Carbon dioxide emitted as a result of the combustion of fossil fuels is absorbed by vegetation and returned to the earth's ecosphere. Overharvesting to meet the current demand may make it impossible for future generations to meet their requirements.

Other examples include the destruction of forests to convert the land for agricultural use. The increase in demand for food has forced some nations to trade off forests for agricultural land. This condition can also be considered in violation for the production of ethanol from corn. The land needed for this industrial crop could also be used for food crops. Overharvesting of fish may eventually deplete the fish population to a level below which regeneration to its original level may be impossible.

System Condition 4—There Must Be Fair and Efficient Use of Resources with Respect to Meeting Human Needs

Basic human needs must be met with the most resource-efficient methods possible. Unless basic human needs are met worldwide through fair and efficient use of resources, it will be difficult to meet conditions 1, 2, and 3 on a global scale. As the human population continues to grow, it becomes more and more critical for the efficient use of the natural resources.

In 2019, the human population reached 7.7 billion and this will require even more consumption of the earth's natural resources. This increase in

population will lead to an increase in energy consumption, and with a finite quantity of available fossil fuels more efficient use of coal, oil, and natural gas will be necessary. More efficient use also means an increase in renewable energy development and consumption. More efficient use of these energy resources will satisfy System Condition 4. At the same time, by adopting as much renewable energy as possible, less carbon dioxide will be emitted into the ecosphere and this will satisfy System Condition 1.

The use of plastic beverage containers is continuing to increase as the population grows and as more people drink bottled water. Unfortunately, in 2016 over 480 billion beverage [6] containers were sold and about 79% accumulated in landfills [7]. It is critical that more efficient use of plastic beverage containers is adopted in order to use less of the natural resources in their manufacturing and satisfy System Condition 4. In addition, there will be less plastic containers in the ecosphere and thus satisfy System Condition 2.

As the population continues to grow, another negative impact is the continued increase in fish consumption. Fishing is central to the livelihood and food security of 200 million people, especially in the developing world, while one of five people on this planet depends on fish as the primary source of protein. According to the UN Food and Agricultural Organization [8] in 2015, two-thirds of the world's fish species are either fully exploited or depleted. The dramatic increase of destructive fishing techniques worldwide destroys marine mammals and entire ecosystems. More efficient harvesting of the fish population is critical in order to meet the needs of future generations and thus satisfy System Condition 4. By decreasing the fish population, the ability to increase the population will also diminish thus violating System Condition 3. This is another example of showing that meeting System Condition 4 is critical to meeting the other three conditions.

SCIENTIFIC RATIONALE FOR THE NATURAL STEP

There are four basic scientific principles that form the foundation for TNS framework [1]:

1. Matter and energy cannot be created or destroyed (according to the first law of thermodynamics and the principle of matter conservation). This means that the overall mass of the earth remains constant. We have the same volume of matter now as we did 4.5 billion years ago. Therefore, matter—or the earth's resources—only changes its form.

2. Matter and energy tend to disperse (according to the second law of thermodynamics). This means that sooner or later matter that is introduced into society will be released into natural systems. This is the underlying mechanism behind our experience that energy and material transformation operate to reduce the available energy in the system and increase the dissipation of matter throughout the system.
3. Material quality can be characterized by the concentration and structure of matter. What we consume are the qualities of matter and energy—the concentration, purity and structure of matter, and the ability of energy to perform work. Because nothing disappears and everything tends to disperse, a carpet turns to dust and a car turns to rust, and not the reverse. Dust does not reassemble into a carpet or rust into a car. As matter disperses, it loses its concentration, purity and structure.
4. The net increase in material quality on earth is produced by sun-driven processes. Photosynthesis is the only large-scale producer of material quality. While the earth is a closed system with regard to matter, it is an open system with respect to energy. This is the reason why the system has not already run down with all of its resources being converted to waste. The earth receives light from the sun and emits heat into space.

In addition to these scientific principles, TNS conditions are influenced by the cyclic principle that can be summarized as follows: (i) waste must not systematically accumulate in nature and (ii) the reconstitution of material quality must be at least as large as its dissipation.

THE NATURAL STEP RECENT PROJECTS

Interface—The First Company to Adopt the Natural Step

Interface became the first US company to adopt TNS framework and methodologies [9]. Under the guidance of TNS and other sustainability visionaries, a new vision for Interface was later crafted—Mission Zero™; to eliminate any negative impact Interface has on the environment by 2020, showing it is possible to create a better world, and being restorative by the power of our influence.

Interface included aspects from TNS's Framework for Strategic Sustainable Development (FSSD) in the company's strategic planning process and all employees training. Gradually the FSSD became a vital and visible part in the company's operations and culture, and it was consistently positioned as providing the most rigorous definition of sustainability.

From 1996 to 2013, Interface reduced the manufacturing waste it sends to landfill by 84%. About 49% of all raw materials are now either recycled or bio-based, including 36% of yarn and 51% of carpet backing. Savings from avoided waste costs amounted to US\$450 million. In terms of energy, total energy use at the global factories has been reduced by 39% per unit. Also, 36% of energy consumed globally is from 100% renewable sources (with seven out of nine facilities operating with 100% renewable energy), and actual greenhouse gas emissions have been reduced by 41%. Interface, Inc. today uses only 20% as much water to make products as they did in 1996. This is an impressive accomplishment, especially for a materials manufacturer, and shows the benefits of looking at a business through the lens of sustainability.

Twenty-two years ago, Interface Inc. started on a journey led by their founder, Ray Anderson. The company adopted a bold new vision—"To be the first company that, by its deeds, shows the entire industrial world what sustainability is in all its dimensions: people, process, product, place and profits—and in doing so—become restorative through the power of influence." Reaching this vision meant that Interface would need to evolve into an entirely different kind of organization, but there was no blueprint for this kind of organization in business.

Interface started by asking how they could translate the operations of nature into a model for business. They relied heavily on TNS Framework to guide their thinking as they mapped out how to change their business. Interface ultimately developed the Seven Fronts of sustainability—seven key areas where Interface would focus to remake their company. Over the last 22 years, Interface has followed these Fronts and made progress reducing the impacts of the company and its reliance on natural resources. This has included a strong focus on the manufacturing operations where they traditionally used large amounts of raw materials, energy, and water.

As Interface approached the management of their manufacturing operations through a sustainability lens, they were able to identify numerous opportunities to not only conserve resources, but also cut costs.

Some savings have been the result of making fundamental shifts in how Interface manufactures its flooring products. For instance, engineers at Interface's Bentley Prince Street factory in California reconsidered the way they made carpet that resulted in a reduction of the water used to manufacture each unit by 47%. The company simply moved away from the chemical-intensive dye-injection and yarn dye methods for adding colors and patterns to their products. Dye-injection involved using a large machine, much like a big ink-jet printer, to apply colors to long lengths of carpet. The yarn-dye method involved soaking fibers in dye solution before

tufting it into carpet. In pursuit of their sustainability objectives, Interface Bentley Prince Street totally phased out the use of dye-injection carpet printing in 1999 and reduced carpet made with the yarn-dye process from 45% of its output in 2001 to 1% in 2009.

As a substitute, the company shifted its focus to two alternative processes—piece dyeing and solution dyed yarns. The piece dyeing process involves manufacturing a blank slate of white carpet, and then adding patterns and colors on a made-to-order basis. This process of customization is far more energy and resource efficient than the sweeping dye-injection method.

The company also reconsidered the process of creating the carpet yarn itself. Instead of soaking fibers in dyes, the switch was made to the use of yarn filaments that are extruded from solutions already impregnated with pigment. This dyeing method drastically reduces the amount of water, energy, and chemicals required in the process and at InterfaceFLOR's modular carpet factory in Georgia. This change in the manufacturing process resulted in a savings of over 88% in water per unit of production.

Interface has also taken advantage of the low hanging fruit available to them by making easy changes in building operations. At InterfaceFLOR in Thailand, they have reduced total water usage by 30% through the installation of water efficient toilets, showers, and sinks. Outside, they selected plants for their landscape plan that only required one year of temporary irrigation.

Interface's significant reduction in water use is just one example of how managing the environmental impacts of business can bring about meaningful change. By relentlessly pursuing sustainability focused innovation, Interface is racing to fulfill its long-term vision of becoming a "restorative company" and is continuing to set the pace as a world leader in next generation manufacturing.

VinylPlus—The European PVC Industry's Voluntary Commitment to Sustainable Development

In 2000, together with the Environment Agency in the United Kingdom, TNS published a report on the challenges the polyvinyl chloride (PVC) industry must address to align its activities with the four system conditions for sustainability [10]. The report highlighted five key sustainability challenges and followed an extensive study of PVC including consensus-building workshops with industry, NGOs, retailers, and regulators. Over the years these challenges have provided a reference point for the industry, initially in the United Kingdom and subsequently across

Europe. The most proactive organizations in the sector have been working to address them.

On June 22, 2011, the European PVC industry launched VinylPlus—a new voluntary commitment to enhance the sustainable production and use of PVC by 2020. It is no secret that PVC is a controversial material that has both critics and supporters, so when an entire industry consortium mobilizes around a voluntary commitment such as this, it is surely a positive sign that sustainable development is being taken seriously in the board rooms of major corporations.

On top of this, the VinylPlus initiative represents an expanded scope and level of ambition over prior efforts. Targets are derived from TNS System Conditions for sustainability and build on previous work in the sector.

Over the last eight years, through a series of activities, TNS has worked with the board of the initiative to review and map out the challenges for PVC, offer capacity building workshops, gather stakeholder views and provide recommendations on the formulation of the new industry charter.

They used the ABCD planning process to help industry leaders apply the framework and arrive at the priorities. They did this by first beginning with a vision of where the industry wants to head, identifying challenges that must be overcome, generating ideas on what needs to be done, and finally arriving at priorities and targets for the period to 2020.

Before deciding on the priorities and actions to be undertaken, they contacted 113 stakeholders across Europe on behalf of the board of Vinyl2010 with a briefing on the challenges they had identified together with industry representatives. Information from these stakeholders were used in the formulation of exact goals and targets.

They were pleased to see an increased level of ambition in VinylPlus and believe it sets out a clear roadmap for the industry. Proof will nevertheless be in the action, and it will be judged by whether targets are met and stakeholders feel their views are being heard.

Moving forward TNS's role will be to follow the new initiative in the following capacity:

1. *Critical friend*—Challenging and advising the industry to make progress while ensuring the initiative retains the direct link to TNS system conditions for sustainability.
2. *Stakeholder intermediary*—Encouraging constant external monitoring and communication with external stakeholders
3. *Capacity builder*—Supporting industry to integrate sustainability principles into their operations in order to achieve the goals set out in the initiative

Dow Measures Up

By any measure, Dow is a sustainability leader.

Dow has been listed in the Dow Jones Sustainability Index nine times. They have won “green” awards from Michigan to China for their wide array of products from insecticides to solar panels. And in 2006, CEO Andrew Liveris enthusiastically announced seven new sustainability goals for 2015 even *after* the company had already met or exceeded many of its 1996 sustainability goals. In that initial 10 years span, they saved more than US\$5 billion in bottom-line costs.

Yet probably their most impressive feat to date might be their array of clear and simple bar graphs that dot the end of each of their quarterly sustainability reports. These simple beige and green charts aren’t flashy—they probably were produced right in an Excel spreadsheet. But it is the function of these graphs that make them so meaningful and compelling, based on what they represent.

First, whether one particular graph is showing their product safety leadership or the intensity of greenhouse gas emissions, you can immediately see the upward or downward trajectory of the company. Their goals are always marked with a simple black line, and for the most part, the green and orange bars show that they are hitting their intended targets. *Dow’s progress is clear.*

Second, Dow may have some of the most complete, public, and long-range sustainability data available. It is immediately evident that the company keeps great records of the data that they produce. This meticulous study certainly seems central to the corporate culture, with good reason—Dow is driven by the exacting art and science of chemistry, which demands precision from heaps of data. *Dow’s progress is credible.*

The data are so complete that their sustainability managers are able to make sense of specific anomalies when they occur. For instance, in 2009, the company’s intensity of greenhouse gas emissions (the amount of energy used to create a single pound of finished product) increased 5%, which went in the opposite direction after four years of downward progress. They were able to attribute this slide to lower operating efficiencies of their production facilities during the worldwide recession. In short, they could see exactly what the effect underutilizing their facilities had on their overall emissions levels—a fantastic feat for any company as large as Dow. *Dow’s progress can be meticulously tracked.*

There is not a company in the world that can claim to be operating in a “sustainable” way, and Dow most certainly has a long way to go to get beyond doing “less bad” and claim that they have no negative impacts on social and ecological systems. But one thing is clear: with the help of their

simple bar graphs, Dow will probably know it the exact moment when they do get to that point!

Nike's Core Values

Nike credits TNS, Business for Social Responsibility (BSR), Deloitte & Touche, and the World Economic Forum as collaborators that helped Nike define its strategic path toward sustainability, and they are very proud to have played such a larger part in Nike's ongoing efforts [11].

This step is a big one and it reflects a cultural shift coming from the core values of the organization. CEO Mark Parker explained:

And for all the athletic and cultural and financial successes of the company, I believe our work in sustainable business and innovation has equal potential to shape our legacy. For that to happen, we have to focus on the lessons we've learned:

- *Transparency is an asset, not a risk*
- *Collaboration enables systemic change*
- *Every challenge and risk is an opportunity*
- *Design allows you to prototype the future, rather than retrofit the past*
- *To make real change, you have to be a catalyst*

As part of this work, Nike tested the real business impact of the changing world on NIKE, Inc. and explored how the consumer brand could thrive in a sustainable economy. They concluded that it needs to continue to refine its existing business model while simultaneously looking at new ways of doing business.

In 2008, Nike partnered with TNS again to help assess and further develop its approach to product innovation by defining a long-term vision for sustainable products. The resulting North Star vision and innovation goals position Nike to become a leader in sustainable product innovation and navigate toward a sustainable future.

As Nike entered 2010 the work continued, with a number of ambitious goals:

1. Put investing in sustainability as a key innovation/R+D priority on consumer brands' agendas
2. Fast track innovation through investment and collaboration
3. Launch the GreenXchange as a platform for enabling the sharing of intellectual property to fast track changes efficiently
4. Build an advocacy agenda to push for large-scale policies and investments in sustainable innovation

Nike's commitment did not come about overnight. It had been transforming how the organization saw its place in the market and the world for 10 years. They are proud to be part of that ongoing transformation and hope other organizations would recognize and rise to the challenges they face with similar optimism and enthusiasm. One of the company's major goals is to create "zero waste" and operate with "100% renewable energy."

Pratt and Whitney Canada's Sustainability Journey

Aerospace manufacturer Pratt & Whitney Canada (P&WC) wanted to explore how building on existing sustainability initiatives, toward a proactive approach to sustainability, could help capture additional market share and drive innovation for the company.

In early 2011, P&WC engaged with TNS Canada to undertake an organizational review, create a vision and goals for the organization, and develop a sustainability roadmap to guide their progress toward minimizing their environmental impact while capturing value for the organization.

To begin, TNS Canada undertook an organizational review of P&WC's operations and culture. This helped determine the business's ability to capture the sustainability value proposition and identify opportunities and threats that might exist. Information was collected over several months through document review, interviews, and surveys and was then analyzed to determine internal attitudes toward sustainability and how staff felt about the organization taking a sustainability-driven leadership role within the aerospace industry. In addition, an external scan conducted by TNS Canada also provided P&WC with a benchmark report about the aerospace industry's sustainability opportunities.

Next, during a planning workshop, a group of executives, senior managers, and members of the internal sustainability team at P&WC were introduced to TNS Framework. Armed with a common language and understanding of sustainability, the group worked to identify some of P&WC's key sustainability challenges as well as strategic goals to address them. This process enabled the development of a series of transition strategies and key milestones to guide P&WC's sustainability efforts over the next few years.

A few months later, in alignment with the roadmap from the workshop, P&WC had a more-defined approach, sustainability goals, and strategy teams in place. One of the organization's priorities became to embed all dimensions of sustainability into its corporate culture while also working toward more sustainable products and operations.

The feedback from the company resulted in: "At Pratt & Whitney Canada, we are firmly committed to ensuring that our products are

designed, produced and operated to enable minimization of environmental impacts throughout their entire life cycle. TNS helped us define a clear sustainability strategy and key priorities as we continue to strive to become ‘the best aerospace company FOR the world’.”

As a result of the work at P&WC, the parent company established goals in 2013 to improve further the sustainability of its factories, suppliers, and products by 2025 and increase efficiencies, create added value for its customers and employees, and lead industry efforts to reduce aviation’s environmental impact. Pratt & Whitney’s global sustainability goals for 2025—the company’s 100-year anniversary—focus on Waste, Energy, Water, Safety and Wellness, Materials, Suppliers, and Products. The company’s sustainability improvements to date demonstrate a record of leadership, achievement and commitment [12].

The Circular Economy

TNS was one of the first sustainability strategies that changed the focus from the typical linear model of “take, make, dispose” to the circular model of “take, make, dispose.” This relatively new Circular Economy will be discussed in detail in Chapter 10.

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Eco-Effective Versus Eco-Efficient: Sustainability Versus Being “Less Bad”

In 1973, the Organization of the Petroleum Export Countries (OPEC) proclaimed an oil embargo in response to the US decision to assist Israel in its war with Egypt. This embargo affected not only the United States but also its Western European allies. Although the United States was energy self-sufficient through 1950, it was now importing about 35% of its oil. Although this embargo was lifted the following year, the US administration was now concerned that something similar could happen again.

In 1975, the US Congress passed the Corporate Average Fuel Economy (CAFE) standards to double the average then fuel economy of 13.5–27.5 miles per gallon (mpg) by 1985. These standards started at 20.0 mpg in 1980 and increased by 2.0 mpg each year until they reached 27.5 mpg in 1985. These standards remained the same for the next 25 years, after which they were increased in order to make the auto industry more efficient.

Making automobiles more efficient with respect to gasoline consumption is the way to go. It is critical that less oil be consumed as it is a non-renewable energy source with depleting inventory worldwide. To become more eco-efficient, we can increase the automobile fuel economy by, say, 50%, which will reduce the consumption of oil. But with growing economies in China and India, what if the number of automobiles in the world doubles. The total consumption of oil will continue to increase.

Practical Sustainability Strategies: How to Gain a Competitive Advantage,
Second Edition. George P. Nassos and Nikos Avlonas.

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About the same time, in 1987 the UN World Commission on Environment and Development summarized its meeting in a report titled *Our Common Future* and emphasized that industrial operations should be encouraged to reduce the consumption of non-renewable natural resources, generate less waste and pollution, and minimize irreversible negative impacts on human health and the environment. It was five years later that the Business Council for Sustainable Development (BCSD), a group of 48 industrial companies, officially adopted the word “eco-efficient.” At the 1992 Earth Summit in Rio de Janeiro, the BCSD brought this concept of eco-efficiency to the representatives of 167 countries. Industrial machines would now become cleaner, faster, quieter, and thus more efficient.

Another version of this same concept is *lean manufacturing*, which focuses on more efficient operations while generating less waste. Lean manufacturing is actually based on the Toyota Production System developed by Toyota Motor Corporation [1]. Eco-efficiency also led to the initiative of three Rs, *reduce*, *reuse*, and *recycle*. Since then, this initiative has grown to five Rs with the addition of *redesign* and, to a lesser extent, *regulate*.

While eco-efficiency is extremely important in leading to a more healthy and sustainable environment, it is a partial solution. Eco-efficiency really means being *less bad*. We must strive for a total solution and be *eco-effective*. Reducing toxic emissions like dioxins from a power plant by adding new pollution control equipment is eco-efficient. However, preventing any toxic emissions is eco-effective. Recycling of plastic beverage bottles is eco-efficient, but banning plastic beverage bottles and replacing them with reusable bottles is eco-effective. Let us consider a few examples of how eco-efficiency can be improved by eco-effectiveness.

FUEL EFFICIENCY

Going back to the beginning of this chapter, there is a major effort to increase the fuel economy of automobiles worldwide. In the United States, there are new regulations that will require the CAFÉ standards to increase to 44.2 mpg by 2020 and 55.3 mpg by 2025. If the number of cars on the road in 2025 in the United States is the same as 2011, the consumption of oil will be about 55% in 2025 as compared with 2011. However, let us look at the rest of the world. The number of cars in the world increased by 3.5% to over 1 billion in 2011 and is expected to exceed 2.5 billion by 2050 [2]. Half of this growth was from China, which increased the number of cars on the road by 27.5%. Most of the growth in cars on the road by 2050 will be from China and India.

The fuel economy standards in China were already at 36.5 in 2009, went to 47 mpg in 2020, and then to 59 mpg by 2025 [3]. In India, the fuel

economy standards increased from 42.7 mpg in 2017 to 51.4 mpg in 2022 [4]. If we can assume that the overall fuel economy will double by 2050, the consumption of oil will still increase by 25%. Despite being eco-efficient, more oil will be consumed and more carbon dioxide will be emitted. We need to be eco-effective.

The route to eco-effectiveness is to eliminate fossil fuels altogether. One way of accomplishing this task is to switch to manufacturing electric automobiles and eventually remove all the gasoline powered cars off the roads. Battery-operated cars, like gasoline cars, have a limited range, which today is much less than the gasoline cars. Consequently, the batteries must be charged at shorter intervals, but the charging system is critical to eco-effectiveness. If the power required to charge the batteries is generated from a coal-fired or natural gas plant, fossil fuels are being consumed and carbon dioxide is being emitted. So even if less fuel is consumed as compared with an eco-efficient, gasoline-driven vehicle, this is still being eco-efficient.

An electric car charging network like Better Place [5] possesses several features that make this system eco-effective. This system consists of a network of charging stations geographically located so that the battery range will allow the electric car to travel between stations. Once the car arrives at the station, the batteries are changed out for fully charged batteries in a matter of minutes, and the driver continues the trip. The key to making this system eco-effective is that the batteries are charged by employing renewable energy like solar or wind. The first system deployed by Better Place is in Israel where the batteries are charged with solar energy. In a small country like Israel or Denmark, it is possible to develop an infrastructure of charging stations that makes it possible for a driver to travel throughout the country and find conveniently located charging stations. This is truly being eco-effective. However, better place went out of business due to lack of funding.

COMPUTING EFFICIENCY

Since the beginning of transistor powered computers, the major manufacturer of the transistors has been Intel, a company founded by Gordon Moore. The business model of this company and its competitors was to double the number of transistors in a computer every 24 months, and along with making them faster, the computer's processing capacity would double. This was eventually known as Moore's Law [6], although an Intel executive modified the doubling period to 18 months and this number is quoted more often.

Based on Moore's Law, every 18 or 20 months the computers become more efficient and thus more can be accomplished with less. This is another

example of eco-efficiency. Thus businesses that require the most efficient computers, like financial services companies, will replace their computers every 18–20 months, primarily to be more efficient. But what happens to the older, slower computers. Many of them are transferred to another organization that does not require the latest computer efficiency, and in turn theirs are sent to less proficient users. For example, some universities always provide the most efficient computers for the student computer labs or IT classes. The replaced computers would then be given to the faculty and staff, and their old ones would be sold for home use or to local schools. In any event, while eco-efficiency is being exercised, there are more and more computers being placed in use.

In order to be eco-effective, a different business model must be introduced so that additional computers are not introduced into society if not needed. An alternative would be to manufacture computers in which the processor can be replaced with a newer, faster version. There is no need to replace the keyboard, the monitor, and other components if no improvement in these components is anticipated. This model would require less resources like metal and plastic and would create less waste. This model would be eco-effective.

MORE DURABLE BRAKE PADS

Brake pads for automobiles were originally made from materials like carbon black, graphite, and asbestos. As the brake pads are employed for their function, particles of the composite material are emitted into the atmosphere and the landscape. In the 1980s, the brake pads were made smaller, which required the composition to be more durable. So the pads were made from carbon black, bronze, and resins. More recently, the brake pad composition has changed to fiberglass, Kevlar, ceramic, and some asbestos [7]. In each of these types of brake pads, the material of composition is eventually dispersed in the atmosphere. The focus has been to be more eco-efficient and manufacture more durable brake pads so that less material is used in the manufacturing process and less toxic material is emitted into the atmosphere and landscape. The US government is even trying to legislate regulations for the prevention of the emission of toxic substances like asbestos [8].

In order to be eco-effective, the brake pads should be manufactured using materials that are not only non-toxic, but also renewable and/or compostable. The disintegrated brake pad material is found primarily in the landscape along roads. With a durable and compostable brake pad material, the process would become eco-effective.

INCREASE POLYMER RECYCLING

The amount of plastic goods being manufactured is continuing to increase, mostly made from oil. The continued increase in production is due to increased demand and the deficiencies in adequate recycling. What recycling of plastic is accomplished, some people call it “down-cycling” because in most cases the recycled plastic is then used for some lesser application. Some uses for plastic products have specifications that require the product to be manufactured from petroleum rather than recycled material.

In order to be more eco-efficient, efforts are being made to recycle more plastic so that less plastic is manufactured from petroleum. In some cases, the product can be made from a combination of virgin plastic and recycled plastic. Some beverage bottles are currently manufactured from that combination with the recycled content as high as 30%. This is going in the right direction, but it is not enough. It is being “less bad.”

In order to be eco-effective, we need to think about fulfilling the demand for the plastic and not manufacturing any additional plastic. Rather than “down-cycling,” we need to find ways of reusing the plastic material. Some products manufactured from plastics might be disassembled at the end of their useful life and reassembled for some other application. Another possibility is to develop a process that allows for continuous recycling of the same plastic material. Approaching that concept is PepsiCo, which is constantly looking for ways to be eco-effective. In 2011 PepsiCo introduced a plastic bottle made from 100% polyethylene-terephthalate (PET) [9]. How many times that bottle could be recycled has not yet been determined.

Another way to become eco-effective is to manufacture the bottle from renewable resources rather than non-renewable. Both PepsiCo and Coca Cola are developing a beverage bottle made entirely from plants. In 2009, Coca Cola announced that it is manufacturing a beverage bottle made from 30% plant-based plastic. The two companies are racing to see which will be first with a 100% plant-based plastic bottle [10]. This is truly being eco-effective.

REDUCED SEWAGE EFFLUENT

One of the industries that has been of concern for some time relative to waste effluent is that of textile dyeing and finishing. The whole cycle of finishing consists of mechanical and chemical processes, which are used depending on the kinds and end uses of the fabric. Mechanical processes

include drying, calendaring, schreinerizing, embossing, sueding, and chemical processes include the application of special substances on the fabric, impregnation with size, starch, dextrin, and other polymeric substances. The effluent can include chemicals like halogenated organic compounds, heavy metals (chromium, copper, zinc), surfactants, and salts.

In order to be eco-efficient, it is critical to minimize the volume of the effluent and/or reduce the concentration of the pollutants in the effluent. When assessing options to minimize wastewater streams, one environmental agency [11] suggests the following methods:

- Minimize machine cleaning through better
- Maintenance and production planning
- Optimize production to reduce liquor ratio
- Optimize and reduce the number of rinses
- Optimize cycle times and job turn-around
- Use lower liquor ratio machinery
- Reduce reprocessing through better quality controls
- Combine rinses with scours
- Scours may be done in dye-bath
- Recycle steam condensate back to boilers
- Recycle cooling waste to use as hot/mixed hot-cold fill
- Recycle rinses as feeds for dye-baths and scour baths
- Recycle "clean rinses"

These recommendations should be implemented in order to be eco-efficient, but this, again, means being "less bad."

What should be done to make the textile dyeing and finishing industry eco-effective? Industry engineers and chemists should find additives, such as vegetable dyes, that are compostable so the residue from the wastewater treatment system can be returned to the earth. Employing compostable dyeing and finishing chemicals may also make the final product compostable at the end of its useful life if the fabric is a natural material like wool or cotton.

MORE EFFICIENT CLOCKS

Here is another example that was first presented by Denis Hayes, President and CEO of the Bullitt Foundation. We have been using electric analog clocks for many years, and they are not very efficient. A typical electric clock requires about four watts of power, which translates to 96 Wh/day.

In one year, the clock will consume about 35 kWh of energy. Assuming there is one electric analog clock per person in the United States, a total of about 1000 MW of power would be required, and this much power consumes about 100 railroad cars per day. How can we make these clocks more efficient?

Electric wrist watches operate with small batteries, which possess minimal power. The same number of these watches would require about 3 kW in total. This amount of power from a coal-fired plant would need one railroad car of coal every five years. This is an excellent example of eco-efficiency, but how do we make these clocks eco-effectiveness?

In a two-step process, the operation of electric clocks can be made eco-effective. The first step would be to make the electric clock as energy efficient as a battery-operated wrist watch. The second step would be to connect the clock to a small solar/battery system so that all the power required for the clock would be provided by the solar cell. However, a battery charged by the solar cell would be needed to operate the clock during the evening. Now this system is eco-effective.

CRADLE TO CRADLE

Most systems designed for eco-efficiency are based on getting more output from less input. Some of these systems may be based on dematerialization—or servicizing (see Chapter 6), extended life of the product, multiple recycling, reducing pollution or toxicity, or obtaining more output from the resources. These strategies are all based on a linear flow of materials from their source to the landfill, or from cradle to grave. McDonough and Braumgart [12] proposed a five-step system that leads to eco-effectiveness, or cradle to cradle.

Step 1—Get free of known undesirable substances:

There are many substances that are known to be harmful and thus undesirable in products utilized by businesses or consumers. Most companies are not aware of the toxicological impact of substances in products through their supply chain. However, they should have a general knowledge of the most dangerous substances in their products. These are referred to as X-substances when referred to eco-effectiveness. For companies like this, a first step in moving toward eco-effectiveness is to find replacements for the X-substances in their products. This includes substances like mercury, cadmium, and lead that are known or suspected carcinogens, polyvinylchloride (PVC), mutagens, or endocrine disruptors. Removal of X-substances is almost always a step in the right direction, but such an approach

has to be applied carefully to ensure that replacement substances are indeed better than those that are replaced.

Step 2—Follow informed personal preferences:

Once the most undesirable substances have been removed from a product, the next step is to begin to make educated choices about those substances that should be included in the product. The best way to do this may be to have a detailed knowledge about the impacts of a particular substance on ecological and human systems throughout its life cycle. But this is often impractical or even impossible. Furthermore, different substances may have different types of impacts in different end products. Some products may require an additive that could be one of several different materials. Each of these may have a different impact such as persistency in the environment, contribute to climate change, or harm plant life. Without a detailed scientific knowledge of a substance’s toxicological profile and its fate throughout the life cycle of a product, these decisions can be difficult to make. Sometimes it may be necessary to bring a product to the market without really knowing the eventual impact. Without complete knowledge of these substances, the best way to make decisions about which chemicals and materials to include in a product comes down to personal preferences based upon the best available information. Though decisions guided by personal preferences may not always result in the most eco-effective design choices, they generally will result in a product that is at least less bad than its predecessors. Although it is “less bad,” it is on its way to become eco-effective.

Step 3—Creating a “Passive-Positive” List:

This step includes a systematic assessment of each ingredient in a product to classify it according to its toxicological and eco-toxicological characteristics, especially its capability to flow within biological and technical metabolisms. For products of consumption, the criteria to examine should include impacts like toxicity to humans (acute, delayed, developmental, reproductive), aquatic toxicity, persistence and bioaccumulation in nature, sensitization potential, mutagenicity, carcinogenicity, and endocrine disruption potential. Based upon the assessment of a material or chemical according to these criteria, a passive positive list can be generated, which classifies each substance according to its suitability for the biological metabolism. This list can be used to determine the degree of additional optimization necessary for a particular product to be a true product of consumption. One can also use this process for service-type products although the impacts

may be different. Cadmium, for instance, is a highly toxic heavy metal and is often applied in photovoltaics in the form of cadmium telluride which may not necessarily be bad. Although cadmium telluride is not an ideal substance from an ecological perspective, its application in photovoltaics as a product of service may be considered acceptable until something better is found and employed.

Step 4—Activate the positive list:

In this step, the objective is to determine which component materials provide a positive technical or biological nutrient. In Step 3, the objective is to optimize the amount of the material to minimize its impact, whereas in Step 4 the objective is optimize the component to its fullest extent. An example is an upholstery fabric whose constituent materials are positively defined as biological nutrients. This was designed by McDonough and Braumgart and others for a German company and is a completely biodegradable and compostable fabric. Each component was selected according to the manufacturer's positive listing methodology for its positive environmental and human health characteristics and its suitability as a biological nutrient.

Step 5—Reinvention:

The concept of reinvention addresses the interconnected nature of ecological, social, and economic systems by considering the biological and technical impacts of existing product and service forms. Strategies for reinvention view products from the perspective of the services they provide and the needs they fulfill for customers. They also consider the broader context of social and ecological systems. While most companies try to design a product with minimal negative impacts, reinvention considers the design of a product with maximum positive impacts. An example may be to design a car whose emissions can be considered nutritious for nature or industries. Perhaps it could burn a fuel that emits water vapor that can be turned back to water and used for some end purpose. Or perhaps the carbon dioxide released from the exhaust can be captured and converted to carbon, which can be used in the manufacture of rubber tires. As more cars are manufactured and driven in the world, more nutrition will become available.

This five-step process to develop an eco-effective product, system, or service will take a trial-and-error process along with time, effort, money, and creativity.

DO NOT TAKE IT TO THE EXTREME

If one gives considerable thought to this sustainable strategy, it should be possible to design a system that is extremely eco-effective, but not very desirable. One might design a pill that provides all the nutrients and vitamins needed for sustaining a healthy life. You could call it a dinner pill. But should we go to that extent and eliminate the pleasure of eating?

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Servicizing and the Sharing Economy

Over the past generation, the economies of the United States and other wealthy industrialized states have undergone significant structural changes. Services have attained new prominence, and the relative contribution of traditional manufacturing to these economies has diminished. These changes have created enormous opportunities for entrepreneurs and new national wealth on the one hand—and huge social costs attendant to the decline of traditional industries and challenges for public policy on the other.

Because the human impact on the environment is intimately linked to economic activity, these changes present both a challenge and an opportunity for environmental policy. The structural changes producing a service and information-led economy is often presented in environmental terms as gradually divorcing economic growth from material and energy throughput and environmental burden. The idea of a *functional* economy—in which the focus of consumption is not goods per se, but the services which those goods deliver—has been associated with the idea of eco-efficiency. In a functional economy, commercial and domestic consumers buy cleaning services instead of washing machines, document services rather than photocopiers, and mobility services rather than cars.

Systematic analysis of the environmental implications of a service and information-led economy has only recently begun. *It is clear that the simplest and most optimistic view—a service economy is inherently a clean economy—is insufficient and incorrect.* Instead, the service economy is better characterized as a value-added layer resting upon a material-intensive, industrial economy. All else equal, economic growth in services may be less environmentally problematic than growth in manufacturing. But that is not sufficient when society already exceeds environmental limits in a number of

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crucial ways. *If services are to produce a greener economy, it will be because they change the ways in which products are made, used, and disposed of—or because services, in some cases, supplant products altogether.* This recommendation was presented in a report to the US Environmental Protection Agency (EPA) about 20 years ago [1].

The concept of services *supplanting* products has been considered for some time. Two examples are the “dematerialization” potential of information technology in reduced travel and transport of physical goods. However, not all products can be dematerialized. For example, the provision of basic needs such as food, shelter, and clothing requires the mobilization of an irreducible physical minimum of material and energy, but the information technology, which may make dematerialization possible, rests upon an extensive, sophisticated manufacturing and maintenance infrastructure. While most examples of dematerialization were pursued for economic reasons, more recent efforts are for environmental reasons.

When one talks about dematerialization and providing a service, it should be made clear that this type of service is different from the traditional services like banking, accounting, hair dressing, and reservations. Consequently, the term that was created to make this differentiation is “servicizing” and the process can be called “servitization.” Rather than selling and buying a product, the property rights can be some form of a diversification of leasing, pooling, sharing, or take-back.

Servicizing can also be a driving force for extended product responsibility (EPR). This is the principle whereby the process participants along the product chain or lifecycle share responsibility for the lifecycle environmental impacts of the whole product system, including upstream, production, and downstream impacts. This should result in lower lifecycle environmental impacts for products or product systems. The link between servicing and EPR is that both require manufacturers or service providers to extend their involvement with, and responsibility for, the product to phases of the lifecycle outside the traditional seller/buyer relationship. If servicing contains within it potential environmental benefits, it is because this altered relationship with the product drives superior environmental performance; in short, because servicing drives EPR.

When companies manufacture a product for a particular application or function, it transfers ownership of the product to the customer who takes over all responsibilities. The manufacturer may provide a warranty for its operation over a particular period of time, but the ultimate responsibility for recycling, reuse, or disposal belongs to the customer. There may be, however, certain advantages of extending the product responsibility along the supply chain or the product’s lifecycle.

For many of the manufacturing businesses today, the business model is based on utilizing the natural resources and energy to manufacture a product that provides some useful function to the consumer. With the exception of some warranties, the manufacturer usually has no other obligations for that product. The consumer uses it throughout its useful life and then disposes of it, with the product eventually finding its way to the local landfill, even after some recycling.

Employing this servicizing paradigm, companies can sell the function of the product rather than the product itself, and maintain ownership of the product throughout its useful life. Under such a scenario, the manufacturer would be more inclined not to consider planned obsolescence and manufacture a more durable product. This would result in fewer products manufactured, less resources employed, and less waste created. Let's look at some examples of products that can be utilized through servicizing.

SELL ILLUMINATION

A simple example of this business model is in the manufacturing and use of incandescent light bulbs—even though this product is not manufactured anymore but it serves as a great example to understand the process. Manufacturers like Philips and GE sell light bulbs, through a supply chain, to an end user, such as a school. But the school doesn't really need light bulbs; it needs the function of light bulbs—illumination. So why not sell illumination? This could be accomplished by providing illumination in terms of watts per hour or lumens per hour, or some other metric with the ultimate cost to the school being the same or less than purchasing the light bulbs.

In the original business model, the sale of light bulbs is a revenue item, and it shows up on the company's profit and loss statement as a sales. In business terminology, it is considered a profit center for the company. By selling illumination, this service becomes the profit center and the light bulb itself becomes a cost center, and thus providing an incentive for the company to reduce that cost. But how? Currently, when a light bulb burns out, it is because the tiny metal filament in the bulb is broken. The owner of the light bulb throws it out and replaces it with a new one, even though the only thing wrong with it is the tiny filament inside. When the bulb is disposed, the metal, ceramic, and glass are all discarded even though it is only the filament that went bad.

What if GE or Philips could manufacture a light bulb where the glass bulb itself could be safely removed and the broken filament replaced? Now the only resource necessary to produce a new light bulb is the tiny filament and the only waste that would be created is the broken filament. Actually, this

tiny filament need not be wasted if it is eventually melted and reformed into a new filament. In theory, once all the light fixtures are fitted with this new light bulb, there would be no need to ever manufacture another light bulb. Of course, this new reusable bulb would cost a little more to produce, but the ultimate use of resources and creation of waste would approach zero.

SELL A PAINTED CAR

There are some real examples of how this new business model can be implemented. The Ford Motor Company has an assembly plant in England where the cars are painted using product supplied by DuPont. DuPont's objective is to sell as much paint as possible to Ford, as selling paint is how it makes money, that is, its profit center.

A number of years ago, Ford management approached the DuPont sales people and asked them to sell Ford a painted car instead of the paint. Upon agreement with this suggestion, DuPont assumed the responsibility for painting the cars, and in providing this service, paint was no longer the profit center as it became a cost item. DuPont would now be paid on a per-car basis rather than on a per-gallon-of-paint basis. DuPont then worked on a new formulation to use less paint while meeting Ford's specifications and improved the efficiency of the spray guns in order to create less over-spray and thus less waste. The end result was an improved system, using less paint and at a lower cost, with the savings shared by the two companies.

In addition to the environmental and cost savings, this new program enhanced DuPont's position as a supplier to Ford by creating a partnership. It would be more difficult to change a supplier of painted cars than a supplier of paint.

SELL FLOOR COMFORT AND ESTHETICS

Interface, one of the largest manufacturer and supplier of commercial carpeting, introduced a new model for providing carpeting to its customers. Instead of selling carpeting, it sold a floor covering service called "Ever-green Lease." Its customers pay for the comfort and esthetics of a floor covering and pay for it on a monthly or yearly basis. Interface manufactures the carpeting in the form of tiles, usually 2 ft × 2 ft, and after it is installed, the company provides a service to maintain it. After all, the carpeting itself belongs to interface, not the customer. Since the carpeting wears out in the major traffic areas and not along walls or under tables, only the worn carpet tiles need to be replaced. The old carpeting is returned to the manufacturing

facility, separated into its original component and recycled to produce new carpeting. With this business model, less carpeting is produced, and much less is disposed of in the landfills. Again, less natural resources and energy are required for this business model.

While this was an excellent example of servicizing, it never really took off. The primary reason was the inability for Interface to convince its customers to switch floor coverings from a capital item to an operating expense.

SELL WATER TREATMENT SERVICES

Water treatment chemical suppliers, like Nalco, Calgon, or Dearborn, sell their products to a variety of applications that require a special treatment of, usually, waste water. One such example is the treatment of paint waste from an automobile assembly plant.

During the painting of new automobiles, the overspray from the paint guns is captured by a vertical wall of flowing water. This water flows into a large basin that resembles a 6 ft deep swimming pool. Here the water is treated with chemicals that enhance the coagulation of the paint particles. Various methods are used to separate the paint from the water, but for each system, having large paint particles makes the process considerably more efficient. One system may employ a horizontal blade at the surface of the water that sweeps the large paint particles to one end of the pool for removal. While the removal of the paint includes a certain volume of water, the quantity of water is reduced for the next step, which could be filtering. Regardless of the process employed, the water is continuously circulated throughout the system.

The chemical supplier normally provides the customer with the recommended quantity of chemical addition for the system. This recommended quantity gives assurance that the paint coagulation will occur at a maximum level. In some cases, the recommended addition may include a safety factor to be sure the system works at its maximum efficiency. At the same time, the chemical supplier is also interested in maximizing its sales while keeping its relationship with the customer.

What has occurred in many situations, the customer has now hired the chemical supplier to operate the water treatment system. The supplier now has an employee working full time at the auto assembly plant with the responsibility of maximizing the efficiency of the paint separation system while utilizing the minimum quantity of the water treatment chemicals. The treatment chemicals are now a cost center for the chemical supplier, and not a profit center. It has been shown that fewer chemicals are employed for this kind of system and savings are accrued to the two parties.

THE EVOLUTION OF THE SHARING ECONOMY

While servicing is an excellent strategy for the reduction in consuming natural resources as well as reducing waste, the sharing economy can also be applied to the service sector. This can best be done by looking at “underutilized assets.” If you think about some of your own assets that are not used as often as the manufacturer intended, they can lead, or have led, to opportunities. Of course, the manufacturer would prefer the limited use of the product as it can then sell more of it.

When movies became available on video tape, and later on discs (DVDs), they were purchased to watch at the owner’s convenience and as often as they wished. However, movies are not viewed very often, usually no more than once or twice. So companies like Blockbuster bought the recorded movies and rented them out so the product was used considerably more often. With the sharing economy still evolving, Blockbuster closed its doors as this type of business went a step further so that a DVD is not even required. Netflix allows the customer to stream the movie through the Internet and onto the television, so a physical product is not even required.

Another example of an underutilized asset is the family automobile. If a car is driven 12,000 miles per year at an average speed of 25 miles per hour, which assumes both highway driving and neighborhood driving, it amounts to 1 hour and 20 minutes per day. This means that the car is used for a little over an hour per day and sits idle for almost 23 hours per day. What a waste! This thinking was the impetus for companies like Zipcar (www.zipcar.com) and I-Go (<http://igo.moosebeta.com/>) to purchase automobiles and allow members of their organization to use them when they need transportation. Of course, the members must reserve the car in advance, pick it up at a specific location, and return it to the same or another specific location. Most of the members that might have been two-car families usually sell one of their cars within a year after becoming a member.

This service business has gone to the next step where automobile owners provide this service themselves. In the United States, RelayRides (www.relayrides.com) is a relatively new company consisting of people who rent their **own** car, usually to neighbors. The company provides the technology to connect the car owner with the car renter and to provide the necessary insurance. In Europe, similar companies like Buzz Car (<https://fr.getaround.com/>) and Whip Car (www.whipcar.com) allow people to rent their cars to other drivers.

The sharing economy for automobiles has been extended a step further where the owner of his/her underutilized asset drives it more by providing a personal “taxi” service. This has led to the development and growth of

ride sharing companies like Uber, Lyft, and Didi (China). These companies are very profitable as their only asset is the app for the drivers and the customers. The major asset being the automobile is owned by the driver. Uber currently operates in about 800 major metropolitan areas worldwide with over 100 million worldwide users [2].

The transportation service business can be extended to bicycles, the most common form of transportation in the world, other than walking. Many cities have developed bicycle rental programs that allow a person to rent a bicycle at a location, ride it to the destination, and then return it to another location within the network. However, this service has been extended to the rent-your-own concept developed with automobiles. An example is a company called “Spinlister.” In early 2012, it opened up access to anyone who wanted to rent a bike, anywhere in the United States. But there was just one catch: to rent a bike, you also had to list a bike. It did that to boost inventory, but has since taken away that sort of listing pay wall to new users. Shortly after its startup, the company came out with a whole new brand—although the service would more or less stay the same. The company changed its name “Liquid” to highlight the whole idea of creating a “liquid marketplace” for sharing people’s assets—like bikes!

In 2007, two roommates decided to rent out their spare bedroom to people visiting the city for a design conference [3]. An idea was born, and the company, Airbnb (www.airbnb.com), took off in 2008 during the Democratic and Republican conventions in Denver, Colorado, and St. Paul, Minnesota. When Denver ran out of hotel beds, more than 300 residents used Airbnb to offer spare rooms. Unwittingly, they had opened a new tier in the holiday accommodation industry. Similar companies have been created since then with names like Crash Padder in the United Kingdom, Couch Share (www.couchshare.com), and I Stop Over (www.istopover.com).

To develop some new businesses, one only needs to think of some underutilized assets like church buildings, which are used primarily on weekends. Another example could be something as simple as a pizza box for carry outs. The customer may use it for as little as 10 minutes to bring the food from the restaurant to the home. The pizza is removed and the box is thrown in the trash unless it is still clean which allows it to be recycled—still a waste due to the energy required for recycling.

COLLABORATIVE CONSUMPTION

One can also extend the concept of the sharing economy to something called “collaborative consumption.” Fast company calls it the reinvention of traditional market behaviors—renting, lending, swapping, sharing, bartering,

gifting—through technology [4]. It is not really new but goes one step further. People have rented out rooms in their homes (boarding) or provided freelance skills in the past. What's different now is that the Internet has expanded the market and made it easier for people who need something to find someone who has it.

Here are a few places you can rent your items or sell your skill:

Sell your skills:

Gigbucks

FittyTown

Fiverr

Freelancer

Mechanical Turk

TaskRabbit

Tenrr

TwentyVille

Upwork

Room rental:

Airbnb

Vrbo

Rent your car:

GetAround

JustShareIt

PeerRenters

Zilock

Sell stuff:

ArtPal

Rent your parking spot:

ParkEasier

Rent your clothes:

Borrowing Magnolia (wedding dresses)

Date My Wardrobe

Rent yourself:

Bridesmaid for Hire

Rent a Friend

Rover (pet sitting)

Wag (pet sitting and walking)

Care (child, senior and pet care, house-keeping, tutoring, and more)

Miscellaneous items to rent:

GoBaby (baby items)

Kitsplit (camera gear)

Neighborgoods

BigStockPhoto

Cash4Books

Decluttr

Gazelle

CLASS FINAL PROJECTS

During my academic career, I have taught sustainable strategies for about 20 years. The final project for each student required the application of one of the strategies to the student's place of work, for a public company, or for a new start-up. The student was asked to develop the concept but not necessarily be responsible for the technical design of the system. As many of the final projects employed the servicizing strategy, the following is a sample of their presentations.

Medication Delivery

One of the large medical equipment companies provides the delivery of medicine to a patient via an intravenous system [5]. This allows for the introduction of the medicine directly into the blood stream for a more effective impact on the patient. These intravenous fluids are provided in one-use plastic bags, which are eventually disposed and possibly recycled. The student, who was working for a large medical supply company, presented a system where the plastic bags could be sanitized and reused. The presentation included a life cycle analysis of the reuse system versus the disposal/recycle system, and it confirmed the justification for a reuse system.

Refill Perfume Shop

The global market for perfumes, colognes, and other fragrances is about US\$3 billion, and the consumer's bottle has not been employed efficiently [6]. People dispose the container before the bottle completes its life cycle or it just sits there in peoples' shelves. A new shop will sell only the perfume (liquid) requiring the customers to bring their old empty perfume bottles, which must be the original ones, to the shop and where it would be refilled with the same scent and original brand of the old perfume bottle. This system would have economic benefit as very few new bottles would have to be produced, would have social benefits as the consumer would achieve savings and create more jobs with more shops, and would have environmental benefits by using less resources and creating less waste.

Luggage

If we can assume that the average person takes 2 weeks of vacation each year and replaces the luggage every 10 years, the luggage is used about 20 weeks before it is disposed or given away. People may have the desire to replace their luggage every 10 years as new and improved models are introduced. In the meantime, it is stored in the home for the other 50 weeks/year. This leads to the manufacture and eventual disposal of luggage that may not have been use throughout its useful life. A new company would purchase the latest design of luggage and rent them to vacationers for, say, 2 weeks at a time. In theory, the luggage would be used 52 weeks/year and disposed after its useful use. The company will always be offering the latest and most cost-effective luggage. There will be less luggage manufactured meaning that fewer raw materials will be consumed. In addition, there will be less material disposed in landfills.

Home Improvement Paint

Many home owners purchase a gallon of paint when they really only need one to two quarts [7]. However, since a gallon of paint usually costs less than two quarts and there may be a need for a little more than two quarts, paint is usually purchased by the gallon. This usually results in excess paint stored in the basement of the home. To reduce and possibly eliminate this problem, a new paint store sells paint by a volume probably greater than is needed. The consumer can then return the excess paint to the company, and the company refills the container for the next customer. Under this scenario, there is never any leftover paint that is wasted. Of course, if the consumer wishes to retain some paint for a later touch-up job, a small amount can be retained in a separate container. A challenge for the company would be how to handle special order paint colors.

Nutrient Services

An agricultural chemical company sells fertilizer to farmers through its supply chain [8]. The farmer applies the fertilizer in such quantities to be sure that the plant receives the proper amount. However, a large portion of the applied fertilizer does not reach the roots of the plant, but rather is dispersed in the soil and groundwater. Perhaps it would be more effective if the agricultural chemical company provided a nutrient service whereby it would develop a system for the nutrients to go directly to the plant roots. The quantity of fertilizer chemical consumed would be considerable less and no unnecessary chemicals would migrate into the groundwater.

Baby Mattresses

With the fertility rate in developed countries decreasing to less than a zero growth rate, there are many families with only one or two children. This means that baby crib mattresses are probably used for considerably less than their useful life. After the baby outgrows the baby crib, the crib and mattress are stored away and possibly never used again. This proposal suggests a rental service for mattresses that can be used for, say, 10 years rather than the typical 1 year. By improving the quality of the mattress for a small incremental cost, the useful life could be extended to, say, 14 years for a total of seven different uses. After the mattress is returned to the company, it could be recovered for sanitary purposes and the next user would have a like-new baby mattress. Applying the number earlier, there would be one-seventh as many mattresses manufactures and one-seventh as many mattresses eventually disposed.

Seat-Go-Round

Instead of selling car seats, Seat-Go-Round will lease *child transportation safety* [9]. Parents will not have to purchase a new seat for every stage of a child's growth. Instead they will lease *child transportation safety* services for a nominal fee of less than US\$10 per month. As the child grows or the seat becomes worn, the parents turn in the seat for a newly refurbished one. The used seat will be inspected, cleaned, and repaired and will go back into circulation for another family to use. This will ensure that the product gets maximum usage instead of sitting idle until needed again or being disposed of after one use. This process will continue until the seat is no longer able to be safely used upon which time the seats will be disassembled and the materials will be recycled. By shifting the focus from buying the car seat to renting *child transportation safety*, this will result in a more effective use of each car seat produced.

Bike Helmets

In certain developing countries where motorized bikes are the most common form of transportation, helmets are required when riding on a main road or highway [10]. While riding on other surfaces, not only are helmets not required, but the drivers do not even wear them. However, for that possibility of riding on a main road, the driver must have a helmet in his or her possession. That leads to every rider having a helmet in their possession even though the use of the helmet may be limited. The proposal suggests establishing kiosks at the entrance and exit of the main roads so that bike riders can obtain a helmet the when entering main road and returning it to a similar kiosk when exiting the main road. Less helmets would be manufactured and less helmets would go unused.

Q Card for Better Transportation

The word "transportation" elicits many images: highways, traffic, smog, parked cars, and gas stations at every corner. This is the old transportation set-up [11]. The Q Card is an all-inclusive transportation smart card that brings an indispensable tool to urban travel. Bike sharing and storage, busses, trains, car sharing, and taxicabs are how people get around. All on the same smart card and smart phone app, and priced to ensure maximum savings, the Q Card is "Transportation 2.0." The Q Card goes beyond traversing through turnstiles and opening car doors. The Q Card links to a checking and credit account, automatically reloads, and keeps you on the go. The Q Card offers variable pricing to ensure value for the customer and a full transit

system. With multiple transportation choices using smart card and smart phone technology, it is now easier to not own a car in the digital age.

The Q Card offers real solutions to real transportation problems. The focus is on impediments to ridership, with an entrepreneurial eye on the serving customers' needs. Public transportation has been run according to outdated models of success and failure. Currently, over US\$6 is lost with every transaction on the Chicago Transit Authority (CTA). With government funds not guaranteed forever, it's obvious there is an urgent business reason to adapt and close this gap. The modus operandi of the Q Card is that sustainability, cost effectiveness, and ergonomics all go hand in hand.

The first action of the Q Card will be a Metra Station Bike Storage Facility. This facility will prove when biking is integrated into rail transit; it becomes the fastest, cleanest, and healthiest commute possible. This will force rail operators to adapt their business models regarding bicycle integration.

Applying the Octopus card model used in Hong Kong, the Q Card will combine all major transportation systems on the same card. The upfront cost and increased revenue will be shared between transportation and financing partners.

The Q Card will introduce variable pricing for advanced purchase of fare rides. By separating pricing from a "per ride" basis into an "overall use" basis, the combined transit system will allow for a single transportation ride to be cost competitive with an impulse car use. Creating and sustaining the captive customer base will be as important as any per ride fare collection increases.

The Q Card will incorporate successful trip planning and transportation tracking software to optimize customer decision making. The smart phone accessible software will make transportation planning, purchasing, and billing on the cutting edge of easy. These are not your grandparents' tokens. This is the Q Card, which may have been the inspiration for the Ventra card currently used in Chicago for many forms of transportation.

Other examples submitted by students can provide an idea of all the possibilities of more sustainable businesses. These include eco-toys shared by children, dry cleaning hangers and bags, wedding services, and a neighborhood share system.

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Adopting Systems Thinking

A simple way to introduce systems thinking is with an ancient story [1].

Beyond Ghor, there was a city. All its inhabitants were blind. A king with his entourage arrived nearby; he brought his army and camped in the desert. He had a mighty elephant, which he used to increase the people's awe.

The populace became anxious to see the elephant, and some sightless from among this blind community ran like fools to find it.

As they did not even know the form or shape of the elephant, they groped sightlessly, gathering information by touching some part of it.

Each thought that he knew something, because he could feel a part.

The man whose hand had reached an ear said: "It is a large, rough thing, wide and broad, like a rug."

And the one who had felt the trunk said: "I have the real facts about it. It is like a straight and hollow pipe, awful and destructive."

The one who had felt its feet and legs said: "It is mighty and firm, like a pillar."

Each had felt one part out of many. Each had perceived it wrongly.

This ancient Sufi story was told to teach a simple lesson but one that we often ignore. The behavior of a system cannot be known just by knowing the element of which the system is made.

So how do you really know whether you have a system or a group of different parts? You need to ask yourself if you can identify the parts and whether these parts have an impact on each other. In addition, you need to ask if these parts act differently on their own as compared with when they

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are with each other. Further, does the effect over time persist in different ways?

Too often, people look at the result of a particular outcome and come to a conclusion as what one single event caused that outcome. In most cases, it is not a single cause but a set of things that are responsible. This set of things, or objects, can be called a system, and the things can be people, vehicles, cells, molecules, plants, or whatever. To form this system, these things are interconnected in such a way that over time they produce their own patterns of behavior. The system can be moved, activated, constrained, or driven by some outside forces but the result is dependent on the characteristics of the system itself. The system must consist of three kinds of things: elements, interconnections, and a function or purpose.

SYSTEM ELEMENTS

The elements of a system are the individual objects, or things, or parts that make up the system. In an automobile, the elements are the engine, the wheels, the brakes, the steering wheel, the gas tank, and others. For a tree, the elements can be the trunk, the roots, the branches, and the leaves. For a house, the elements can be the foundation, walls, roof, water pipes, a furnace, and others.

While elements are easily recognized as being tangible objects, the elements can also be intangible. If one considers a church as a system, some of the elements can be the faith of the members or service provided by the members to the church.

Each of the elements of a system can be described as a system itself. The engine of an automobile consists of the engine block, cylinders, spark plugs, and wires. These are all elements of the engine. A leaf that is an element of the tree can be considered a system consisting of a stem, veins, tip, and blade. A furnace, an element of a house, can be described as a system consisting of a gas burner, a blower, and a thermostat. Each of these sub-elements can be broken down to sub-sub-elements.

SYSTEM INTERCONNECTIONS

The interconnections of a system describe the relationships between the elements. In an automobile, the engine operates in order to turn the wheels, while the brakes are needed in order to stop the movement of the wheels. The gas tank contains the necessary fuel that is required for the engine to operate, while the steering wheel is necessary to guide the directions of the wheels.

If one looks at the engine element as a system, it consists of the various sub-elements. The engine block is the physical part that contains the cylinders, spark plugs, and cooling water. The cylinders are needed to move in two directions and are able to transfer the movement to the wheels so they can turn. The sparkplugs are made to spark at a certain frequency to ignite the gasoline, thus making the cylinders move in the two directions.

In a tree, nutrients are absorbed from the soil by the roots and are transferred up through the trunk of the tree. The nutrients are then transferred to the branches, which in turn provide the nutrients to the leaves. Thus the leaves become larger, provide cover for the ground and objects beneath it, and absorb carbon dioxide.

In a church, the faith of the members means that they will attend the church services, extend their faith to other potential parishioners, and provide some services to the church. The service provided may enhance the faith of other parishioners and possibly provide assistance to those in need.

SYSTEM FUNCTION OR PURPOSE

Once the elements are identified and the interconnections are described, there must be a function or purpose for the system. In an automobile, the elements and their interconnections have one primary purpose, and that is to move one or more passengers from one location to another location. This is the primary purpose, but a secondary purpose might be to attach a trailer to the rear of the automobile in order to transport a boat.

The interconnections between the elements of a tree can be for multiple purposes or functions. In this system, the purpose could be to grow the tree as large as possible in order to generate wood for construction purposes. Another purpose for the tree is to provide shade to a house and thus reduce the cost of cooling. A third purpose might be to have the tree leaves absorb carbon dioxide in order to reduce the concentration of this gaseous emission.

For a house, there are many elements and many interconnections. These are all necessary to provide shelter for one or more people. In addition to providing shelter, the purpose also includes a place to sleep, a place to eat, a place to read and write, and a place to relax.

When describing intangible elements along with the interconnections, a function or purpose is still necessary. In the case of the faith and service elements for parishioners of a church, the purpose could be to provide a place for people to worship and obtain comfort. The purpose could also be to spread the word of the religion and thus increase the number of its faithful.

While the terms “function” and “purpose” are used interchangeably, there may be fine distinction between the two. Purpose usually pertains to a

human system such as the faith of a religion. Function usually pertains to a non-human system such as the movement of an automobile. However, these two words can be used interchangeably without much loss in the discussion of systems.

The description of a system along with its components elements, inter-connections, and functions is described here in very basic terms. A more thorough presentation of systems thinking has been provided by Donella Meadows [1].

TUNNELING THROUGH THE COST BARRIER

Many times when people evaluate an operation to determine how to make it more efficient, the first task is to find that improvement that makes the most impact at the least cost [2]. In other words, look for the best return on the investment.

Once that task is accomplished, people then look for another modification that could result in additional improvements. Most likely, this one will cost a little more and the result may be a little less than the first one. This process is likely to be continued until the improvements become less and less while the cost of implementation continues to increase. You reach the point of diminishing returns, and that is where you stop on trying to improve the efficiency as seen in Figure 7.1.

However, if one looks at this project as a system, there are opportunities to save even more energy which will allow to “tunnel through the cost barrier,” making the cost come down and the return on investment go up. Basically,

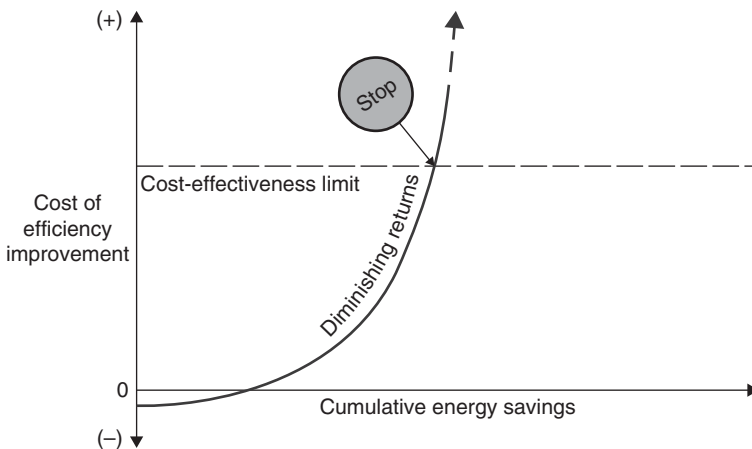


FIGURE 7.1 Diminishing returns. *Source:* Natural Capitalism.

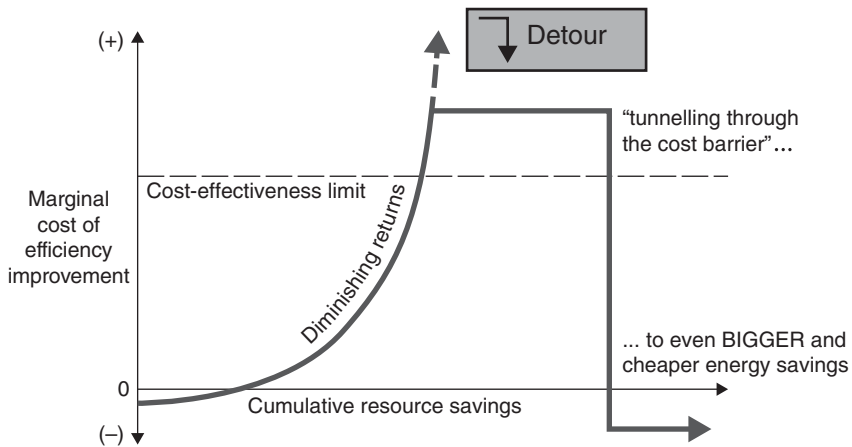


FIGURE 7.2 Tunneling through the cost barrier. *Source:* Natural Capitalism.

one can make a detour to a considerably higher return on investment by considering the entire system as shown in Figure 7.2.

An example of this is the design of the heating, ventilation, and air conditioning (HVAC) of a home. A builder will tell you that more efficient materials will cost much more than the normal materials which are usually less efficient. This can refer to the wall thickness, the insulation, the quality of the windows, and the efficiency of the furnace. If you consider each of these alternatives separately, you will likely follow the curve of diminishing returns. Each improvement reduces the energy consumption but eventually you get to the point of diminishing returns when the next improvement provides little if any energy savings.

However, if you look at the whole system the addition of better insulation and better windows will reduce the energy requirement. Now the builder can install a much smaller furnace and a smaller air conditioner. The additional cost of the insulation and windows can often result in overall savings because of the smaller furnace and A/C. Spending a little more money on compact fluorescent light bulbs for the house will not only save on the energy cost but also emit less heat. This will result in energy savings from the air conditioning unit.

INTERFACE PIPE DESIGN

Another example of “tunneling through the cost barrier” can be seen at the industrial level. Several years ago, the largest American commercial carpeting company, Interface, was building a new factory in Shanghai.

It hired an engineering company to design the industrial process which required 14 pumps. In optimizing the design, a top engineering firm sized those pumps to total 95 horsepower. The design included the smallest diameter pipes that could transfer the fluids effectively as well as the most efficient pumps. However, an Interface engineer decided that he would review the design and see if it could be made more efficient. He looked at the entire process as a system rather than optimize each component separately.

The new specifications required two changes in design. First, the Interface engineer chose to deploy big pipes and small pumps instead of the original design's small pipes and big pumps. Friction falls as nearly the fifth power of pipe diameter, so making the pipes 50% fatter reduces their friction by 86%. The system then needs less pumping energy—and smaller pumps and motors to push against the friction. If the solution is this easy, why weren't the pipes originally specified to be big enough? Because of a small but important blind spot: traditional optimization compares the cost of fatter pipe with only the value of the saved pumping energy. This comparison ignores the size, and hence the capital cost, of the equipment—pump, motor, motor-drive circuits, and electrical supply components—needed to combat the pipe friction.

The engineer did not have to calculate how quickly the savings could repay the extra up-front cost of the fatter pipe, because capital cost would fall more for the pumping and drive equipment than it would rise for the pipe, making the efficient system as a whole cheaper to construct.

Second, the engineer laid out the pipes first and then installed the equipment, in reverse order from how pumping systems are conventionally installed. Normally, equipment is put in some convenient and arbitrary spot, and the pipe fitter is then instructed to connect point A to point B. The pipe often has to go through all sorts of twists and turns to hook up equipment that's too far apart, turned the wrong way, mounted at the wrong height, and separated by other devices installed in between. The extra bends and the extra length make friction in the system about three to sixfold higher than it should be. The pipe fitters don't mind the extra work: they are paid by the hour, they mark up the pipe and fittings, and they won't have to pay the pumps' capital or operating costs.

By laying out the pipes before placing the equipment that the pipes connect, Interface was able to make the pipes short and straight rather than long and with many turns. That enabled them to exploit their lower friction by making the pumps, motors, inverters, and other electrical components even smaller and cheaper. The fatter pipes and cleaner layout yielded not only 92% lower pumping energy at a lower total capital cost but also simpler and

faster construction, less use of floor space, more reliable operation, easier maintenance, and better performance. As an added bonus, easier thermal insulation of the straighter pipes saved an additional 70 kW of heat loss, enough to avoid burning about a pound of coal every two minutes, with a three-month payback.

The Interface engineering department marveled at how they could have overlooked such simple opportunities for decades. The redesign required, as inventor Edwin Land used to say, “not so much having a new idea as stopping having an old idea.” The old idea was to “optimize” only part of the system—the pipes—against only one parameter—pumping energy. The engineer, in contrast, optimized the whole system for multiple benefits—pumping energy expended plus capital cost saved. Such whole-system life-cycle costing, in which all benefits are properly taken into account over the long run, is widely accepted in principle but almost always ignored in practice. Instead, single components are usually considered in isolation. You can actually make a system less efficient while making each of its parts more efficient, simply by not properly linking up those components. If they are not designed to work with one another, they will tend to work against one another.

REDUCING OIL IMPORTS

The United States has been importing oil for many years and may continue to do so for many more to come. While it has always been a major cost to US citizens, there have been some attempts to alleviate the situation.

In the early 1970s, there was a major shortage of oil production in the United States. Automobile owners were limited to the amount of gasoline that could be purchased at any one time. Consequently, there were long lines at gas stations, sometimes even reaching one-half block. This was primarily due to an oil embargo by the Arab members of the Organization of Petroleum Exporting Countries (OPEC). The embargo was instituted because of the US support for Israel. About the same time, US oil production peaked as the country had depleted 50% of its known (at that time) oil reserves [3]. This made the United States more concerned about oil imports as US production would start to decrease. The country was concerned about increasing oil imports while an oil embargo took effect.

Several years later as oil imports continued to increase, then President Jimmy Carter, who had the ability to think in systems, suggested that we impose a tax on gasoline that would be proportional to the fraction of US oil consumption that had to be imported. If imported oil consumption continued to rise, the tax on gasoline would rise in order to reduce the demand for

gasoline. The tax would be used to develop alternatives to the imported oil in order to reduce its consumption in the United States. Once the imported oil quantity fell to zero, the tax would fall to zero. Obviously, this tax never passed primarily because President Carter could not explain it to the press and public that didn't understand systems thinking.

REDUCING CARBON DIOXIDE EMISSIONS

If we think of climate change as a system consisting of all the sources of carbon dioxide emissions (elements), the impact of the emissions on our atmosphere (interconnections), and the warming of the planet (purpose), we can then suggest means of impacting the system in a beneficial way. In this particular example, the purpose or function is not necessarily a goal that we wish to achieve. Here we want to find a means to reduce the function of this system.

To achieve the goal of reducing the purpose, we must reduce the elements. However, the elements themselves can be considered the function of a system. A power plant can be considered a system with elements being the fuel, water/steam, generators, etc. The interconnections are all the controls that eventually lead to the production of electricity, which is the purpose or function. However, the production of carbon dioxide is also a "purpose," albeit a negative one. So one way of reducing the carbon dioxide is to find a system that allows for maximizing one purpose (electricity) while minimizing another purpose (carbon dioxide). This requirement can be achieved with renewable energy like solar, wind, or geothermal.

Another approach is to reduce the generation of electricity in order to achieve a corresponding reduction in carbon dioxide emissions. This can be done by imposing a tax on the carbon dioxide, which should reduce the demand for electricity. This would also apply to the carbon dioxide emitted from internal combustion vehicles that combust gasoline. This approach is not dissimilar to imposing a tax on imported oil as described above.

ILLEGAL IMMIGRATION

During his term of office, President Carter was also concerned with the large number of illegal immigrants from Mexico. While this problem seems to be of major concern today, it even existed in the late 1970s. Carter believed that nothing could be done about that immigration as long as there was a large gap in the living standards and working opportunities between the United States and Mexico. Keep in mind that the United States–Mexican

border is the only border in the world between a first- and third-world country. Carter went on to suggest that rather than spending money on border guards, barriers, and/or fences, we should spend the money helping to build the Mexican economy. In addition, he said that this should be done until the illegal immigration stopped—when there no longer would be this great desire for Mexicans to come to the United States. This didn't happen either.

Again President Carter was thinking of this problem as a system. It consists of Mexican people (elements), means, and barriers to immigrate to the United States (interconnections), and the need to have a better life (purpose). Instead of looking at the interconnections, we should be looking at the purpose. By improving the purpose—a better Mexican life—the rest of the system can function without a problem.

A good example where this system worked was in Greece. After World War II, Greece was an impoverished nation and for next few decades, the Greek people were coming to the United States any way possible—legally, illegally, through Canada, visiting and not returning, etc. But after Greece's economy improved and the country eventually became a member of the European Union and later joining the Euro Zone, immigration to the United States from Greece has slowed down considerably. The recent financial crisis there, however, may renew the emigration of the Greek people.

SAILBOAT DESIGN

There are situations where systems thinking is applied to a project that results in a minimum application. One such example is provided by Donella Meadows. Once upon a time, people raced sailboats not for millions of dollars or for national glory, but just for the fun of it.

They raced the boats they already had for normal purposes, boats that were designed for fishing, or transporting goods, or sailing around on weekends.

It quickly was observed that races are more interesting if the competitors are roughly equal in speed and maneuverability. So rules evolved that defined various classes of boat by length and sail area and other parameters, and that restricted races to competitors of the same class.

Soon boats were being designed not for normal sailing, but for winning races within the categories defined by the rules. They squeezed the last possible burst of speed out of a square inch of sail, or the lightest possible load out of a standard-sized rudder. These boats were strange-looking and strange-handling, not at all the sort of boat you would want to take out fishing or for a Sunday sail. As the races became more serious, the rules became stricter and the boat designs more bizarre.

Now racing sailboats are extremely fast, highly responsive, and nearly unseaworthy. They need athletic and expert crews to manage them. No one would think of using an America's Cup yacht for any purpose other than racing within the rules. The boats are so optimized around the present rules that they have lost all resilience. Any change in the rules would render them useless.

In this case, systems thinking was applied to such an extent that it optimized the project extremely well. However, in application the end result was a system with limited use.

Additional examples of systems thinking and tunneling through the cost barrier can be found in "Thinking in Systems" by Donella Meadows or "Natural Capitalism" by Amory Lovins, L. Hunter Lovins and Paul Hawken.

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Base of the Pyramid

In 1932, the United States president, Franklin Delano Roosevelt, made a radio address titled “The Forgotten Man.” As part of his address he said, “These unhappy times call for the building of plans that rest upon the forgotten, the unorganized but the indispensable units of economic power ... that build from the bottom up and not from the top down, which put their faith once more in the forgotten man at the bottom of the economic pyramid.” This is the first known use of the term “bottom of the pyramid,” which has also been modified slightly to “base of the pyramid.”

That was close to 90 years ago, but today this term refers to the billions of people in the world living on less than US\$2 per day. This was defined by the University of Michigan Professors C. K. Prahalad and Stuart L. Hart in 1998. They each wrote excellent books that thoroughly described the business model and its benefits. Prahalad published “*The Fortune at the Bottom of the Pyramid*” [1] in 2004, while Hart published “*Capitalism at the Crossroads*” [2] in 2005. Hart subsequently went to the University of North Carolina, then to Cornell University, and is currently at the University of Vermont.

Prahalad proposes that businesses, governments, and donor agencies stop thinking of the poor as victims and instead start seeing them as resilient and creative entrepreneurs as well as value-demanding consumers. He proposes that there are tremendous benefits to multinational companies (MNCs) who choose to serve these markets in ways responsive to their needs. After all, the poor of today are the middle-class of tomorrow. There are also poverty reducing benefits if multinationals work with civil society organizations and local governments to create new local business models. How can this business model be employed?

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One of the major environmental issues today is climate change. The United Nations' Intergovernmental Panel on Climate Change (IPCC) in its 4th Assessment Report concluded "Most of the observed increase in global average temperature since the mid-twentieth century is very likely due to the observed increase in anthropogenic greenhouse gas concentrations [3]." This has been confirmed by the IPCC in subsequent reports with the most recent in 2018 [4]. The big question now is what can we do about it? The obvious answer is to reduce the quantity of fossil fuels for energy by employing more efficient systems. But what else can we do?

Scientists and engineers have developed processes and products that are very innovative and utilize much less energy. In some cases, however, these products are not accepted by the major markets because they would destroy an existing system or infrastructure—thus creative destruction. An example is the auto industry, which along with electric power generation is the largest contributor to global warming.

The energy required to power an automobile is used primarily to move the vehicle itself, which accounts for about 95% of the total weight. This means that only 5% of the fuel is used to move the driver from one location to another. The simple answer is to manufacture lighter automobiles. Several years ago, the Rocky Mountain Institute [5] developed an automobile made from high-tech ultra-light materials, which are stronger than steel, and the resulting car can achieve over 100 miles per gallon. This technology has been rejected by Detroit because it would eliminate the assembly plants as they exist today. The investment in the auto assembly plants is too great to change to a new technology. Is there any alternative?

THE GREAT LEAP DOWNWARD

Economists have described the global economic pyramid as consisting of three layers as depicted in Figure 8.1. At the bottom, or base of the pyramid (BOP), are about 4.7 billion people earning less than US\$1500 per year. In the center are the 2 billion people in the emerging middle class earning between US\$1500 and US\$15,000 per year. At the top of the pyramid are the one billion "wealthy" people earning over US\$15,000 per year. Most manufacturing companies market their products to the top of the pyramid where the "wealthy" consumers exist. But should we be looking at the BOP?

Most companies market their new technologies to at the bottom of the top of the pyramid as this is the "wealthy" part of the global market. These technologies usually have costs associated with the development, marketing, and sale of the products that can only be accepted by the top of the pyramid. Reducing the cost of the product to reach the bottom of the pyramid

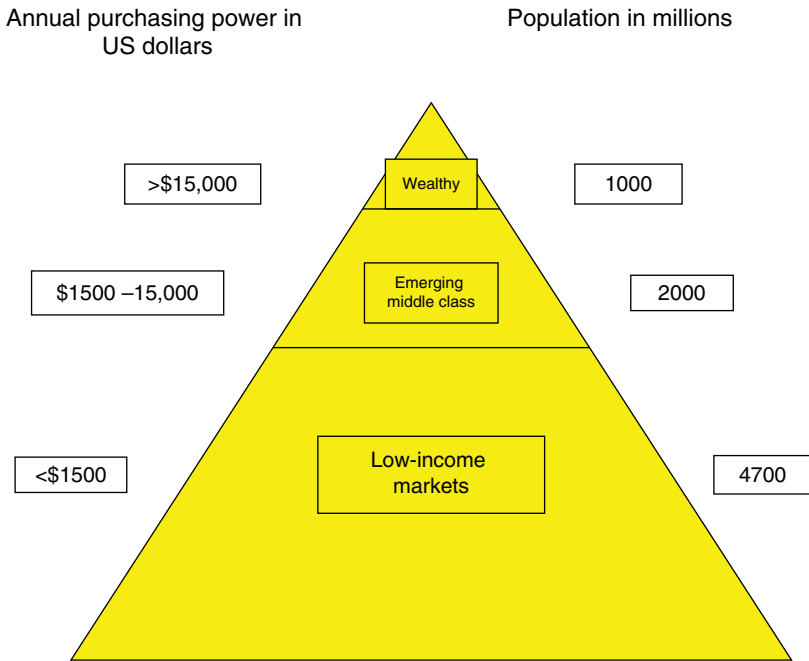


FIGURE 8.1 Base of economic pyramid. *Source:* Capitalism at the crossroads.

will usually reduce the quality and make the product of little interest. Clay Christensen [6] talks about disruptive innovations, which are products and services that are not as good as those in the mainstream markets, and therefore are of interest only to some niche non-traditional markets.

Hart and Christensen talk about the great leap downward [7] (Fig. 8.2) as the BOP is the ideal target for new disruptive technologies for at least two reasons. The opportunities for a new technology to penetrate a low-income market are much greater just based on the size of the market. In addition, by marketing to the BOP, a company can provide a market or service that would not otherwise be available.

An excellent example is that of the business started by Soichiro Honda of Japan [8]. In 1946, he started Honda Technical Research Laboratory and his first project was to make use of war surplus Tohatsu and Mikuni generator motors. With bicycles being the primary mode of transportation in Japan and other countries, he attached them to bicycles in order to provide basic transportation for this war-torn nation. Recognizing the diminishing supply of surplus motors, Honda formed Honda Motor Company, Ltd. in Hamamatsu in 1948. The company's first headquarters was a 12 ft × 18 ft shed that housed 13 employees. The company built the "A" model bicycle,

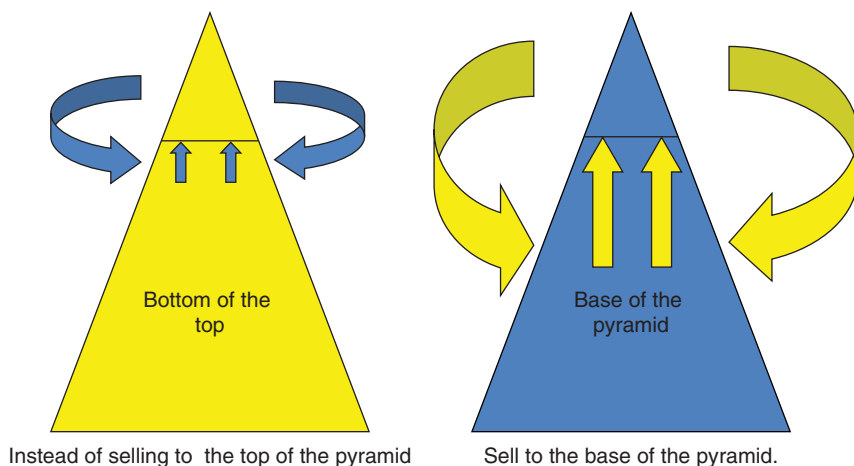


FIGURE 8.2 Great leap downward. (a) Instead of selling to the top of the pyramid. (b) Sell to the base of the pyramid. *Source:* Capitalism at the crossroads.

the “B” model tricycle, and then the “C” model motorcycle. The company then made larger motorcycle engines and by 1955 was the largest motorcycle manufacturer in Japan. Within four years, the company became the largest motorcycle manufacturer in the world.

Honda’s vision was to become an international company and that included selling its products in the United States. However, the US market for motorcycles was dominated by Harley Davidson, which made much larger motorcycles. In addition, the market was not that large compared to the US population. In 1965, Honda produced its first automobile and four years later imported its N600 sedan to the United States. Thus, Honda went from adding motors to bicycles, and then to manufacturing motorbikes for the BOP, to selling automobiles to the entire world population.

Another example of this great leap downward is from a Chinese company called Galanz. It was formed in 1978 to sell duck feathers and subsequently became a textile manufacturer. In 1993, it entered into an agreement to manufacture microwave ovens for Toshiba. The company started slowly and only manufactured 10,000 units per year. At that time, only 2% of the Chinese households owned a microwave oven. Using Chinese materials and Chinese labor, Galanz developed and manufactured an energy-efficient microwave that was affordable to the large middle and lower-middle class families in China. With the volume increasing, the average price of the unit started to decrease and became affordable to more families. As a result, Galanz’s domestic market share increased from 2% in 1993 to 76% by 2000 when the market was even much larger. Employing the “great leap downward” model,

it started selling its products worldwide, and by 2002 it enjoyed a 35% share of the global market. Today, it produces over 15 million microwaves per year and has a 40% share of the world market [9].

ELECTRIFY THE BOTTOM OF THE PYRAMID

There are about 1.1 billion people who live without electricity [10], and they are at the bottom of the pyramid. Extending the grid is not always practical, and in many cases it is almost impossible due to logistics and high costs. One answer may be distributive generation where there is local, self-contained power generation and storage systems.

The importance of providing electrical generation can be explained in Figure 8.3 [11]. It correlates per capita electricity consumption with the human development index, which is a very general measure of well-being by looking at life expectancy, literacy, education, and standard of living. Of course, as the electricity consumption of a country increases, the well-being of the people increases. Electricity provides lights, which, in turn, allow children to study in the evening, refrigeration, pumping of water into the home, power for computers and charging mobile phones, and many other benefits that enhance the people's well-being.

To provide this electricity to the bottom of the pyramid would normally require the development and installation of an electrical grid as well as the

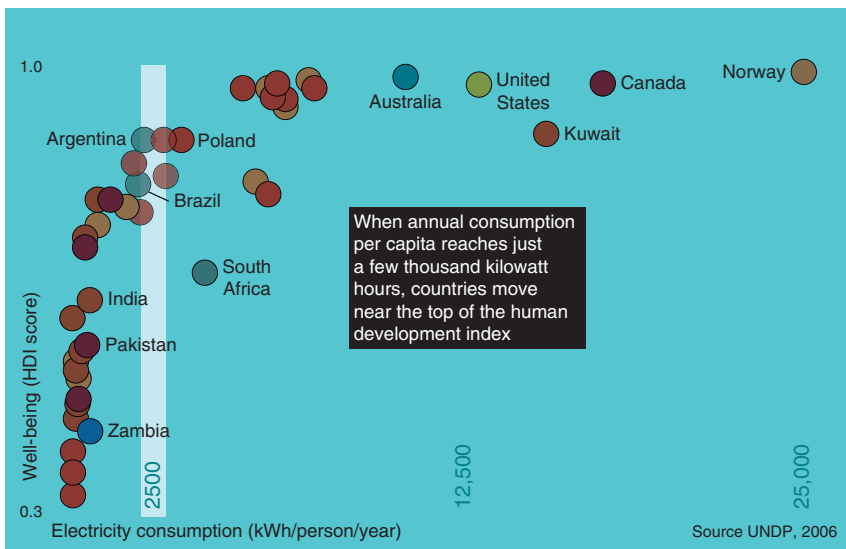


FIGURE 8.3 How electricity powers well-being.

generation of the electricity. The latter could be accomplished by a coal-fired power plant, a nuclear power plant, or diesel powered generators. For remote villages, this is not the most efficient means of delivering electricity even if there was no concern with carbon emissions from the coal power plant or any safety concerns with the nuclear power plant. Stand-alone power generation systems using renewable sources like solar, wind, biomass, geothermal, or hydroelectric would make more sense given the ability to store the energy to eliminate intermittent generation when the sun is not shining or the wind is not blowing. The critical aspect of this system is storage of the energy.

The US Department of Energy, through its Advanced Research Projects Agency, continues to invest in battery research in order to develop higher-efficiency batteries at significantly lower costs. Successful development of these batteries, while primarily targeted to increase the range of electric vehicles, will create a larger market at the bottom of the pyramid. This huge market will reduce the cost of the batteries and make them available to more people. Eventually, this technology could then leap up to the top of the pyramid and develop a distributive generation system for the developing countries as proposed by Jeremy Rifkin [12].

HINDUSTAN LEVER AND NIRMA

This is an interesting case study of two companies that focused on different levels of the economic pyramid. Hindustan Lever, Ltd. (HLL), a subsidiary of Unilever, the large international British company, was serving the people at the top of the pyramid [2]. This company noted that in the 1990s, a small Indian firm, Nirma, Ltd., was offering detergent products for the BOP. Nirma had developed a business model to serve the underserved by providing a new product formulation, a low-cost manufacturing process, a wide distribution network, special packaging for daily purchasing, and pricing for consumers with limited means. The Nirma price was Rs 3 per kilogram, while the HLL price for their detergent was Rs 13 per kilogram [13].

While HLL was content in serving the top of the pyramid, Nirma grew rapidly in a market completely ignored by HLL. As Nirma grew, it started to move up the pyramid as a result of its strong base at the bottom of the pyramid. HLL saw the growth of Nirma and in 1995 responded by developing its own business model for the BOP market, and diverging from its traditional business model.

HLL introduced a new detergent product called Wheel, which was reformulated for the poor people that wash their clothes in rivers and other public water systems. The company revised its production, marketing, and distribution to reach the large number of people living in the rural areas that

comprise the BOP. In addition to all of these marketing changes, it also reduced the price in order to be attractive to the BOP.

Today, Nirma and HLL are still competing as they are splitting evenly about 80% of the detergent market in India. It was a very wise move for HLL to make that leap downward as the BOP market accounts for over 50% of HLL's total revenues and profits. It is even a better market for Nirma.

BOP PROTOCOL

Stuart Hart started developing the BOP strategy while a professor at the University of Michigan. He then joined the faculty at the University of North Carolina Kenan-Flagler Business School where he and other faculty formed the BOP Learning Laboratory [14]. Working with Dan Vermeer, a member of Coca-Cola's strategic "think tank," they looked at the use of anthropological approaches to understanding the needs of people living at the BOP. They determined that a systematic approach, desired by companies such as Coca-Cola, could be used to understand the needs and opportunities at the BOP. In subsequent conversations, Hart and Vermeer fleshed out the broad contours of what would become the BOP protocol project that is now housed at Cornell University.

This project moved forward by engaging five different institutions:

- Johnson Graduate School of Management at Cornell University
- Stephen Ross School of Business at the University of Michigan
- World Resources Institute
- William Davidson Institute
- S. C. Johnson Foundation and the Wingspread Conference Facility

and four partner companies:

- DuPont
- Hewlett Packard
- S. C. Johnson, Inc.
- Tetra Pak

This BOP Protocol continues to move forward with projects in India with Solae Company and in Kenya with S. C. Johnson. Because the people at the bottom of pyramid can be found in almost any country, the BOP Protocol even developed a project in the United States. The project was initiated in 2008 by seeking to innovate and develop new grassroots enterprises

focused on improving the health and/or health care of Flint and Genesee County in the State of Michigan. By working with community resources, local partners, and poor and vulnerable families and individuals, this multi-year project would follow the BOP Protocol's participatory business innovation process to co-create and implement sustainable, community entrepreneur-led businesses.

INITIATIVES BY THE WORLD RESOURCES INSTITUTE

The World Resources Institute (WRI) is a global environmental think tank that goes beyond research to put ideas into action. It works with governments, companies, and civil society to build solutions to urgent environmental challenges. WRI's transformative ideas protect the earth and promote development because sustainability is essential to meeting human needs and fulfilling human aspirations in the future. The organization's efforts are directed to:

- Climate, energy, and transport
- Governance and access
- Markets and enterprise
- People and ecosystems

It applies various sustainable strategies to accomplish its goals in these various areas. One of the strategies that WRI applies is the BOP.

Jamshyd N. Godrej, a member of the WRI board of directors, was interested in bringing electrical energy to the village of Rupahi in Bihar, India, which is too isolated to be connected to the grid [15]. For energy, the residents rely on whatever is available to them—kerosene, firewood, dung, or diesel generators, all of which left their houses dim and full of smoke. Installed at the community level, small hydro and biomass gasification can supply electricity to an area not covered by the grid, and solar lanterns and energy-efficient stoves can replace “dirty” fuels like kerosene and wood. In Rupahi, villagers now get their power from Husk Power Systems (HPS), a company that converts rice husk into electricity via 35–100 kW mini power plants. HPS provides villages in the country's rice belt, like Rupahi, with cost-effective and environmentally friendly electricity. Providing electrical energy to the rural poor people of India has been estimated to be a US\$2.1 billion per year market.

WRI also believes that there are investment opportunities within the BOP. *Impact investing* [16] is gaining increasing traction among a wide range of

debt and equity investors, including pension funds, family offices, private wealth managers, foundations, individuals, commercial banks, and development finance institutions. Many impact investors choose to focus either in emerging markets or in developed markets. Part of the reason for this specialization is the significant regional differences that require local expertise. But another driver is investors' value sets: some prefer to help the world's poorest in emerging economies; others prioritize their local neighbors in need.

Within the developing world, the growing suite of impact investors is focusing on particular regions and sectors. Gatsby Charitable Trust and the Bill & Melinda Gates Foundation direct some of their investment capital to impact positively the lives of smallholder farmers in sub-Saharan Africa. Gray Ghost Ventures, Acumen Fund, and Omidyar Network all have programs that actively invest in alleviating poverty by financing innovations directed at India's low-income populations. In a report by The Rockefeller Foundation [17], it estimates the value of this emerging impact investing sector to be between US\$400 billion and US\$1 trillion, with profit potential between US\$183 billion and US\$667 billion over the next decade.

DEVELOPING THE BOTTOM OF THE PYRAMID

In *Capitalism at the Crossroads*, Hart describes the pathway to developing the BOP. Companies must move *beyond greening* and look at technological change and leapfrog to clean technologies that could result in disruptive business models. One of the challenges is to become indigenous and learn to co-develop technologies, products, and services with nature and local people. In this manner, the Multi-National Companies {MNCs} can become native to these regions.

The MNCs have been ingrained to develop strategies that take a top-down approach, and this must be altered if they wish to develop a strong market presence in the BOP. This was exemplified by HLL until it learned from Nirma. Nike also demonstrated its problems when it attempted to market the World Shoe. It was attempting to market the product to the BOP, but it was still employing top-down marketing.

In many cases with respect to marketing to the BOP, companies must think in terms of creative destruction rather than continuous improvement. Companies may have developed technologies that were never introduced because they would disrupt a current situation. However, at the base of pyramid, these same technologies could have an application as the potentially disruptive market doesn't exist. Introducing light emitting diode (LED) light bulbs to the developing countries had been very slow as the incandescent and compact fluorescent bulbs had dominated the market. However, combining

LED bulbs with inexpensive solar battery chargers can provide light to areas that are not on the electrical grid.

In Nike's World Shoe project, one of the failures was due to the pricing structure. The company failed to reduce the cost because it employed the same overhead as its standard products. China's Galanz was successful because it used local materials, local labor, and sold to local markets. Its cost structure was such that it was able to achieve a 75% market share in its own market.

Most MNCs think about large scale projects with large capital investments if they are going to make an impact. However, when looking at the base of pyramid, companies must consider starting at a smaller scale working with the smaller markets and growing organically as the markets respond. Disruptive technologies are typically smaller in scale and more distributive in scale.

Because the BOP presents many significant opportunities, there continues to be entrepreneurs with technological ideas for projects. CNBC has presented 10 technologies for the BOP [18]. One of the more interesting ones is from Embrace, which has developed an infant warmer resembling a mini-sleeping bag [19]. This portable incubator reduces the risk of hypothermia for newborns by providing a constant temperature for up to six hours without a continuous electricity source. The warmer contains phase-change material, a wax-like substance that can be heated using a hot water heating unit. Once heated, it slowly releases warmth over time. The cost is about 1% that of traditional incubators.

Another interesting technology was developed by Ann Makosinski. It is a Hollow Flashlight [20] powered solely by body heat based on the idea of Peltier tiles, where energy is created when one side is heated and the other side is cooled. Inside the flashlight air acts as a cooling agent; once the hand generates enough heat on the outside, the temperature achieves the necessary duality.

In summary, it is critical that the company must align the organization that will be pursuing the BOP project. Everyone involved in the project must buy into the elements of the organizational infrastructure. A misalignment can be fatal to a project as was demonstrated by Nike's World Shoe endeavor. The elements and alignment can be summarized as:

- *Vision/mission*—Setting the sustainability goal
- *Goals*—Establishing measuring targets
- *Strategy*—Identifying the sustainable value portfolio
- *Structure*—Creating separate experiments, ventures, and funding

- *Systems*—Designing new measurement, rewards, and project-evaluation tools
- *Processes*—Enabling new technology, product, market-development approaches
- *People*—Integrating sustainability into recruiting, leadership development, and performance evaluation

In order to assure success, aligning these elements of organizational infrastructure should not be underestimated.

IS THE BASE OF THE PYRAMID A MIRAGE?

Before a company decides to adopt the BOP strategy, it should carefully assess the market opportunity. Although this BOP strategy was initially developed at the University of Michigan, another professor at the same school, Aneel Karnani, questioned the validity of this strategy. He didn't believe that it could be adopted successfully and even harms the poor people in two ways. "First, it results in too little emphasis on legal, regulatory, and social mechanisms to protect the poor who are vulnerable consumers. Second, it results in overemphasis on microcredit and under-emphasis on fostering modern enterprises that would provide employment opportunities for the poor. More importantly, the libertarian proposition grossly under-emphasizes the critical role and responsibility of the state for poverty reduction [21]."

The developers of the BOP set the market size at US\$13–15 trillion, whereas Karnani believes it is really US\$360 billion. The major attraction of the BOP is this huge untapped purchasing power. The poor, however, may not really have the ability to save any money so their purchasing power may be limited. The BOP strategy relies on decent margins so the company can operate profitably. The company, on the other hand, may not be able to make a profit as the BOP market is very price sensitive and the cost of serving this market can be high. In addition, the poor spend 80% of their money on food, clothing, and fuel and really don't have much remaining to purchase a "technology" product that is planned for development through the BOP. Technology products may be the only sector that could be developed at the BOP because other products would result in reduced quality if the price is reduced. Karnani also believes that small to medium-sized companies are better suited to exploit the BOP rather than the large MNCs.

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Environmental Innovation Through Biomimicry

In the 1600s, Europe was hit with a crippling shortage. People had to deal with the fact that a valuable commodity was increasingly in short supply. What was it? Rags.

Rags were used to make paper, and paper was in great demand. Publishers of books, newspapers, and political pamphlets all clamored for more paper. But there just weren't enough rags. Advertisements appeared, asking women to "save their rags." In 1666, England banned the use of cotton and linen for the burial of the dead, decreeing they must be saved for making paper. One entrepreneur even suggested using the cloth from Egyptian mummies. The scarcity of rags led to fearful paper shortages in Europe and America.

Then a French scientist took a walk in the woods. Réne-Antoine Ferchault de Réaumur was an accomplished physicist and chemist. He was also a man who loved bugs. Walking in the woods one day, he came upon an abandoned wasp nest. Delighted, he began to examine it in detail, and an astounding fact dawned on him: the nest was made of paper, paper made by wasps, paper made without the use of rags. How? The wasps did it by chewing wood and plant fibers.

What wasps could do, he argued, man could find a way to do also. It took decades, but his discovery was the spark that inspired inventors to develop ways to make paper from wood pulp. Thanks to Réaumur's nature walk, we can now do what once would have been considered almost criminal: crumple up a piece of paper and throw it out [1].

Another interesting example is the process of photosynthesis, which means, "using light to put something together." Here plants, algae, and bacteria take carbon dioxide, water, and sunlight to produce energy-rich

Practical Sustainability Strategies: How to Gain a Competitive Advantage,
Second Edition. George P. Nassos and Nikos Avlonas.

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sugars while releasing oxygen. Animals, including humans, take that oxygen and the sugars and transform them back to carbon dioxide, water, and energy. Without the sunlight, this photosynthesis reaction could not take place.

A little over 100 years ago, an Italian chemistry professor, Giacomo Ciamician, wrote in *Science* [2] magazine that one day our landscape would change from power plants to a large array of glass that would mimic photosynthesis and produce another form of energy. Through his inspiration, we now have acres and acres of solar cells made of silicon, a material that allows for the production of energy, even though it is nowhere found in the plant structure.

This is just one of hundreds of examples of innovations inspired by nature. In the classic book, *Biomimicry* [3], Janine Benyus defines biomimicry in three different terms:

- (1) *Nature as model*—Biomimicry is a new science that studies nature's models and then imitates or takes inspiration from these designs and processes to solve human problems, for example, a solar cell inspired by a leaf.
- (2) *Nature as measure*—Biomimicry uses an ecological standard to judge the "rightness" of our innovations. After 3.8 billion years of evolution, nature has learned: What works. What is appropriate. What lasts.
- (3) *Nature as mentor*—Biomimicry is a new way of viewing and valuing nature. It introduces an era based not on what we can *extract* from the natural world, but on what we can *learn* from it.

What follows are examples of how nature has inspired or can inspire innovation that will lead to sustainable systems. Benyus now uses a better word than biomimicry for these new systems. She calls them "inspired by nature."

ABALONE

The abalone (Fig. 9.1) is a shellfish that is known for its delicious meat as well as its smooth inner coating that makes the shell as "hard as nails." A car could drive over an abalone shell and have no impact. It is stronger than any known ceramic, but why? It consists of an intricate crystal architecture that allows it to shrug off stress. If you can see the structure from the side, it consists of hexagonal disks of calcium carbonate stacked in a brick-wall formation. Between the "bricks" is a polymer that gives the formation some flexibility to ward off the head-on stress. The ceramics that are produced

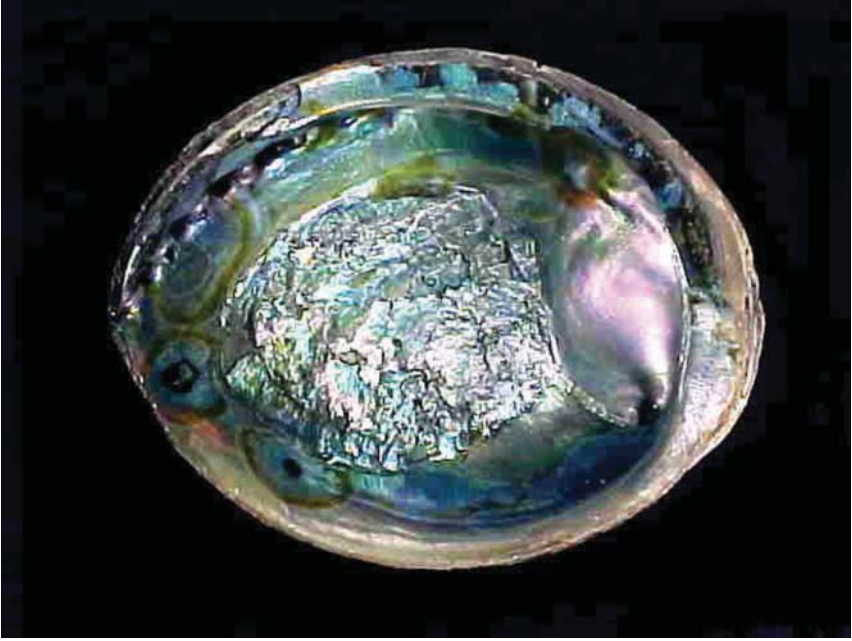


FIGURE 9.1 Abalone shell—stronger than ceramic.

commercially, such as glass, porcelain, and bricks, are manufactured by taking earthy inorganic particles and subjecting them to heat and pressure. The result is a very strong material, but brittle and subject to cracking. Scientists are now looking at natural designs like the abalone inner coating to determine how to reproduce that structure synthetically.

Just imagine trying to build a structure that could be as strong as the abalone shell. Suppose you wanted to build a very strong structure such as a house or small office, and after it is designed, structural forms are used connected to create the walls of the building. These forms are unique as they can contain liquids such as sea water. Suppose these forms are then filled with sea water and the secret chemical used by the abalone to create its shell is added to the sea water in the forms. After a short period of time, the water is drained and what remains are walls stronger than any ceramic or brick currently made. All that is needed to accomplish this feat is to learn how the abalone makes its shell.

SPIDER SILK

Another example is how spiders produce several different kinds of silk for various functions, such as forming a web or rappelling from drop-offs.

Each one is mixed in its own gland, extruded through its own spinneret, and endowed with its own chemical and physical properties. The properties of these silks are really astounding when compared with man-made materials. Compared with an equal weight basis, some silks are five times stronger than steel and five times tougher than Kevlar, the material used for bulletproof vests. At the same time, the silk can be very elastic and stretch up to 40% of its original length, something not possible by steel wire. Just imagine if someone could learn to do what the spider does, taking a renewable, soluble material and making an extremely strong water-insoluble fiber using very little energy and generating no toxic waste. By analyzing the spider's process and reproducing it, the entire fiber industry would change dramatically.

Today, research is being conducted to produce spider silk synthetically by creating the same proteins that spiders produce. While spider silk proteins are converted to threads in nature, these same proteins can be used to prepare other morphologies, such as films, gels, foams, capsules, non-wovens, spheres, and nanofibers using different processing methods [4].

Although spider silk is already tougher and lighter than steel, scientists have made it three times stronger by adding small amounts of metal. The technique may be useful for manufacturing super-tough textiles and high-tech medical materials, including artificial bones and tendons. Researchers at the Max Planck Institute of Microstructure Physics have found that adding zinc, titanium, or aluminum in a length of spider silk made it more resistant to breaking or deforming [5]. This idea was inspired by studies showing traces of metals in the toughest parts of some insect body parts. For example, the jaws of leafcutter ants and locusts both contain high levels of zinc, making them particularly stiff and hard.

BIVALVES

Bivalves is the name given to the seafood consisting of oysters, clams, mussels, and others. These creatures are usually found connected to some other object rather than just floating in the water. An amazing natural phenomenon is how these bivalves, or mussels in particular, can attach themselves to almost anything in water. If you look closely to a mussel, you can see hundreds of small translucent threads (about 2 cm long) extending like plastic tethers from the bivalve to the attached object. These tethers are called byssus, and they are truly amazing. When the bivalve wants to attach itself to an object, it will stick out a fleshy foot and creates a chemical complex at the end of the byssus where there is a small disk. Through the small disk, called a plaque, a natural adhesive is squirted between the plaque and the

attaching surface. Within three to four minutes, the entire process is complete including the curing of the adhesive.

At home or at the office, anytime we wish to use an adhesive to join two objects, both objects must be dry or the adhesive may not perform as expected. The adhesives industry has been trying for many years to find an adhesive that will function in moist conditions and stick to anything. The mussels can do just that.

In 2010, scientists at the University of Chicago have applied for a patent for a synthetic version of mussel glue, an adhesive, self-healing gel [6]. The potential remains exciting; biomedical transplants and repairs along with underwater mechanical engineering remain the chief areas where this material will have a significant impact. The innovation illustrates how important cross-collaboration over years of research can be.

Key to their breakthrough was using metal ions for molecular bonding rather than the typical permanent covalent bonding in most synthetic polymers. Other synthetic polymers had not performed well for both strength and ductility in the past. The tightly bonded material became too brittle as it was strengthened. As any structural engineer knows, a part of a bridge that is too strong can weaken the whole structure as much as a weak part. Changing the pH of the material also enabled them to change its properties. One of the bio-innovation is obtaining strength from shape even if it is at the molecular scale.

RHINOCEROS

The rhinoceros population in the world is dwindling very rapidly—like the black rhino population in Africa, which has gone from several hundred thousand around 1900 to 65,000 in 1970, and to around 2400 in 2004. As of November 2001, this species has been declared officially extinct.

The unfortunate reason for this decline is the illegal killing of these animals for the horns protruding from the rhino's head (Fig. 9.2). Because of their composition, these horns sold for about US\$65,000 kg⁻¹ in late 2012, or twice the value of gold [7]. The primary reason for this value is the booming demand in China and Vietnam for perceived medicinal and aphrodisiac qualities. They are also coveted in Yemen for making dagger handles.

One way to stop the slaughter of rhinos is to find a synthetic substitute for the horns. When analyzed, it was determined that these horns are made of keratin, the same tough, fibrous protein that is in our fingernails. However, grinding up our cut fingernails won't do any good because it is not the keratin itself that gives the horns its coveted strength and luster, but rather the structure. Upon careful examination of a horn cross-section, the



FIGURE 9.2 Rhinoceros horn—made of keratin. *Source:* Courtesy of Zimpapers Images.

pattern was recognized to be similar to the graphite-fiber-reinforced composite that have been recently engineered for race cars, tennis rackets, and the new lightweight Boeing 787 airplane. We just learned how to produce a very strong and lightweight material, whereas nature has been doing it for millions of years. However, there is still more to learn as most engineered composites do not have both compression and torsion strength like the rhino's horn—and they can't heal themselves.

FISH-INSPIRED TRAVEL

Nissan Motors was interested in developing an automobile that be driven in congested roads without the fear of a collision. So the company looked at the behavioral patterns of a school of fish, which avoid obstacles without colliding with each other (Fig. 9.3).

To develop this new automobile, Nissan noted that fish recognize the surroundings based on lateral-line sense and sense of sight, and form schools based on three behavior rules (Fig. 9.4): (i) change traveling direction without colliding with other fish, (ii) travel side-by-side with other fish while keeping a certain distance between each fish (to match the speed), and (iii) gain closer proximity to other fish that are at a distance from them.

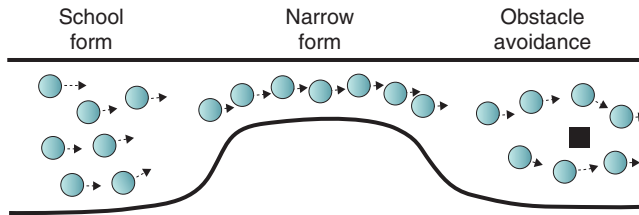
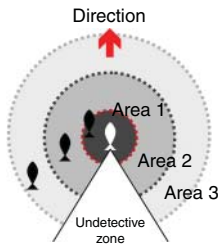


FIGURE 9.3 Fish behavior avoid collisions. *Source:* Courtesy of Nissan North America Inc.



Fish behavior rules

Area 1 – Collision avoidance

Change traveling direction without colliding with other fish.

Area 2 – Traveling side by side

Travel side by side with other fish while keeping a certain distance between each fish (to match the speed).

Area 3 – Approaching

Gain closer proximity to other fish that are at a distance from them.

FIGURE 9.4 Fish behavior rules. *Source:* Courtesy of Nissan North America Inc.

The company then applied laser rangefinder technology, a sensor that measures the distance to an obstacle with reflection of laser light, to allow the cars to travel side by side. It also applied ultra-wide band, short-distance radio communications technology that measures the position of the target and the distance to it through calculation of the time lag from transmission to reflection of the pulse signal. This technology assured the cars from colliding with an obstacle.

Through the bio-inspiration of a school of fish, Nissan has developed a robot car (Fig. 9.5) called the EPORO, an abbreviation of EPisode O (zero) RObot (episode aiming to be CO₂-free and accident-free).

CHEETAHS

Boston Dynamics, a private company with funding from the US Department of Defense DARPA Maximum Mobility and Manipulation (M3) program, has made and tested the world's fastest land robot, a four-legged machine based on the African cat (Fig. 9.6) [8]. Critical to the performance was the mimicking of the flexible spine of the cheetah, which allows for hyperextension, and the coordination of the striding legs.

The machine (Fig. 9.7) recently set the land speed record for robots at 28.3 mph, faster than the world's fastest human (Usain Bolt at 27.8 mph

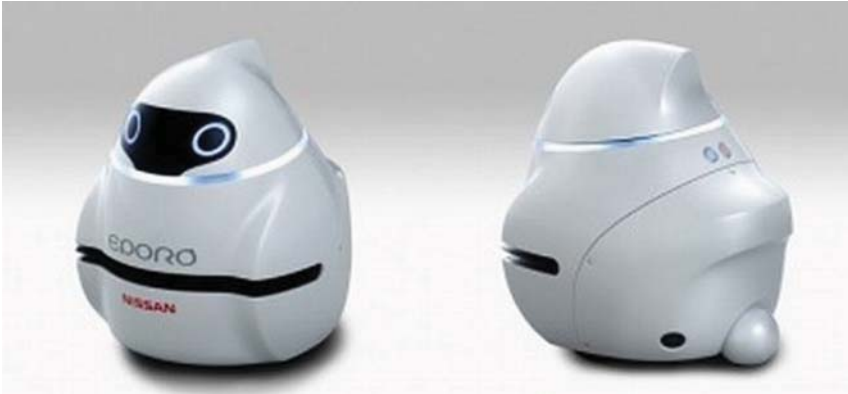


FIGURE 9.5 Nissan's robot car inspired by fish. *Source:* Courtesy of Nissan North America Inc.



FIGURE 9.6 Cheetah inspiring fast robots. *Source:* Courtesy of Boston Dynamics.

over 100 yd in 2009). The robot is not autonomous: an off-board hydraulic pump and a boom provide power and assist in making it steady as it runs on a treadmill. A free-running version, powered by a gasoline engine, is in the planning. When it does begin trials, it will still be far behind its natural mentor, which can reach speeds of up to 70 mph. All-terrain robots that are fast and reliable would be a distinct tactical advantage to military ground

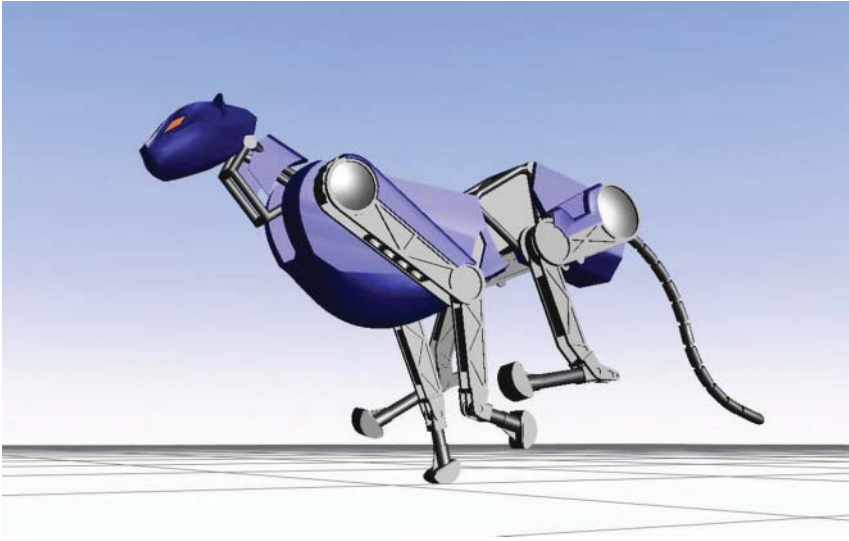


FIGURE 9.7 Fastest robot inspired by cheetah. *Source:* Courtesy of Boston Dynamics.

forces, and could be used for other hazardous work like disaster relief and medical rescue.

COMPACT AND EFFICIENT STRUCTURE

Many years ago, engineers were challenged to design a structural part that must be very rigid yet very light. The real design challenge was to employ the least amount of material to keep the weight very low. Perhaps the engineer was required to design, for example, an airplane wing. The goal is to develop an internal structure that could be covered with a skin which would make the wing very rigid yet very light.

If the engineer starts with a geometric design that utilizes the least amount of material for a given area, that geometry would be a circle. We know from geometry that the largest area for a given perimeter is in the form of a circle. However, if that design is a matrix of circles, there is a large amount of material where the circles are attached (Fig. 9.8). The engineer could then use the simplest geometric figure, which is the triangle, but even in this matrix there is a large amount of material at the vertex of each triangle (Fig. 9.9). Geometrically, the engineer would then go to the next design in the hierarchy, which would be a square. This design would show an improvement, but there is still too much material at the intersection of four lines (Fig. 9.10).

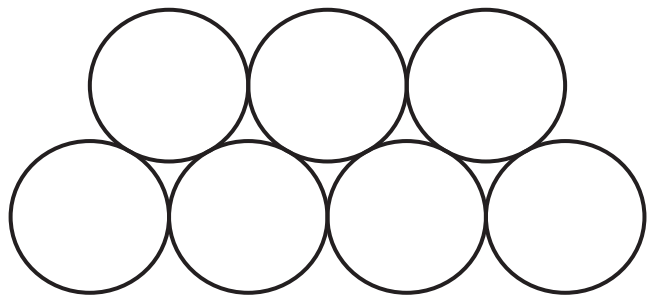


FIGURE 9.8 Structure with circles.

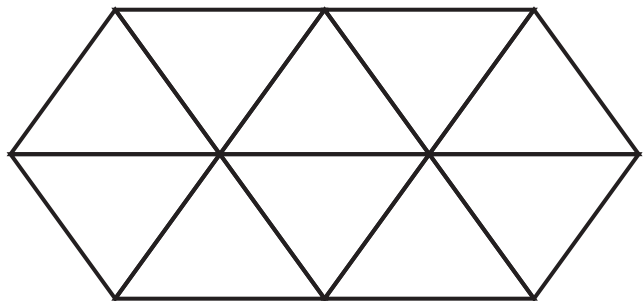


FIGURE 9.9 Structure with triangles.

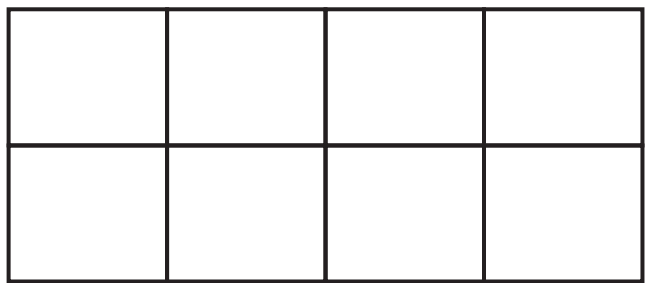


FIGURE 9.10 Structure with squares.

The next geometric design in the hierarchy is a pentagon, but the asymmetry makes it impossible to consider (Fig. 9.11).

Of course, the next step would be to employ a hexagonal type structure, which consists of the least number of lines at an intersection, and thus this would be the design of choice (Fig. 9.12). While this may be a design created by engineers, perhaps 100 years ago, bees have been using this same design for thousands of years in building their homes—the bee hive (Fig. 9.13).

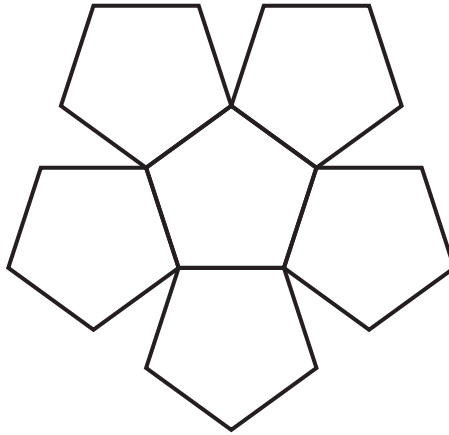


FIGURE 9.11 Structure with pentagons.

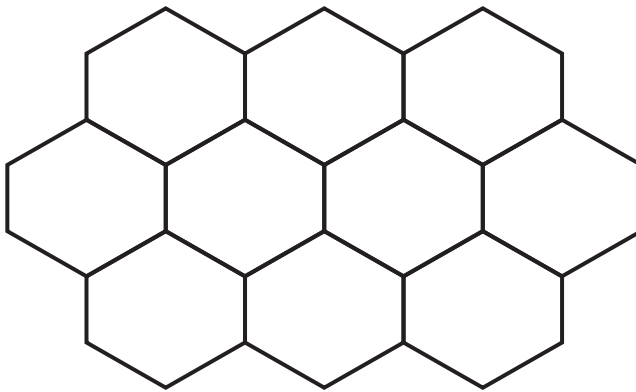


FIGURE 9.12 Structure with hexagons.

Bees learned how to construct the most rigid structure with the least quantity of material.

ETHANOL

During the past 15 years, the production of ethanol as a fuel has become a very important process. For many years, up to 10% ethanol has been added to gasoline in certain geographic areas to reduce emissions. Today, ethanol is the primary fuel for flex-fuel automobiles that can use up to 85% ethanol. In Brazil, ethanol is produced from sugarcane bagasse, but in the United States, it is produced from corn.



FIGURE 9.13 Beehive—most efficient structure. *Source:* Image used with permission from <https://www.symmetrymagazine.org/article/october-2012/beehives>.

The process to convert corn-to-ethanol consists of a chemical process whereby enzymes are added to convert the starch in the corn to a sugar. Yeast is then added and the sugar is converted to ethanol. The process may not be very efficient as some studies report that it takes as much as 1.29 units of energy during the process to produce 1.0 units of energy as ethanol [9]. On the other hand, other studies show just the reverse as the output-input energy ratio is 1.34 [10]. The difference between these two analyses has to do with the assumptions, primarily the allocation of the energy consumption to the by-products.

In any event, researchers have looked at ways to increase the efficiency of the conversion process. Currently, the corn-to-ethanol process utilizes only the corn kernel as this is the only part of the corn plant that can be converted to sugar and alcohol. Scientists at Michigan State University noted that the enzyme that allows a cow to digest grasses and other plant fibers can be used to turn other plant fibers into simple sugars. They then discovered a way to grow corn plants that contain this enzyme. They have inserted a gene from a bacterium that lives in a cow's stomach into a corn plant. Now, the sugars locked up in the plant's leaves and stalk can be converted into usable sugar without expensive synthetic chemicals [11]. This new approach should make the corn-to-ethanol process considerably more efficient, thanks to a cow's stomach.

COLOR FROM STRUCTURE RATHER THAN PIGMENTS

Like many other birds, a peacock's bright teal and brilliant blue feathers are not the result of pigments but rather nanoscale networks that reflect specific wavelengths of light. This is similar to the Mirasol technology developed by Qualcomm for its mobile phones and e-readers [12]. This so-called structural coloration has long interested researchers and engineers because of its durability and potential for application in solar arrays. But today's techniques to integrate structural color into materials are time-consuming and costly.

Now, researchers at the Harvard John A. Paulson School of Engineering and Applied Sciences (SEAS) [13], in collaboration with King Abdullah University of Science and Technology, have developed a new, more robust, and cost-effective system to build large-scale metamaterials with structural color.

A peacock's feather or butterfly's wing as replicated by Qualcomm for its Mirasol products rely on photonic crystals or highly ordered arrays of nanofibers to produce colors. Reproducing those structures in a lab requires precision and expensive fabrication. However, SEAS researchers were inspired by a very different kind of feather.

Contingas are one of the most flamboyant bird families on the planet. In a sea of Amazon green, their feathers pop with electric blues, bright oranges, and vibrant purples. Unlike a peacock's ordered array of nanostructures, contingas get their vibrant hues from a *disordered* and *porous* nano-network of keratin that looks like a sponge or piece of coral. When light strikes the feather, the porous keratin pattern causes red and yellow wavelengths to cancel each other out, while blue wavelengths of light amplify one another.

Inspired by the cotinga feather, the researchers used a simple etching process to create a complex but random porous nano-network in a metallic alloy. The structure was then coated with an ultra-thin transparent alumina layer. Depending on the metallic material and its thickness, different colors can be created. You may be thinking, what kinds of colors can a metallic alloy produce other than gray? As it turns out, lots. Ever since nineteenth century English scientist Michael Faraday, scientists have known that metals contain a plethora of colors but light doesn't penetrate deep enough to reveal them. The porous nanostructure creates localized hot spots of different colors in the alloy. The color that is reflected by the localized states depends on the thickness of the transparent coating. For example, with no alumina overlayer the material may look dark. With a coating 33 nm thick, the material reflects blue light. At 45 nm, the material turns red and with

a coating 53 nm thick, the material is yellow. By changing the thickness of the coating, the researchers could create a gradient of colors.

These researchers were able to show how to produce almost any color you want without using pigments. All they did was place a very thin alumina overlayer on the material that needed the color. This technology was inspired by the wing structure of peacocks and contingas.

CRICKET SOUNDS

If you think the sound of crickets is an essential part of any romantic summer evening, you have good reason [14]. Male crickets are trying to attract females by a species-specific chirp at a fixed frequency of about 4.5 kHz. But how do the females find them once they hear the call?

Female crickets are equipped with special receptors or tympani on their front legs that are used to compare the pressure fluctuations between right and left sides of the insect. The pressure is carried up into the cricket by means of auditory trachea. Pressure fluctuations reach each tympanum directly as well as indirectly, from the other side of the body. Phase shifts between these two sources strongly modulate the tympanum-oscillations and allow the female to determine the direction of the source of potential romance.

Crickets also have many tiny hairs or cerci along their abdomens that measure pressure fluctuations and scientists believe their purpose is to sense attacks. The sensitivity and precision of these hairs are remarkable, and inspired a research group from the University of Twente in the Netherlands to mimic their operation.

The team aimed at developing a bio-inspired perception system. They attached hundreds of 0.9 mm long plastic wires to sockets in silicon wafer sheets to form a receptor array. The wires rotate in their sockets as they are moved by minute air pressures and the smallest movements are registered by the flexibly suspended plate. The electrical capacity of the plate changes as a result, and this measure is fed into a central computer.

Further refinement allowed measurement at an even finer level of precision by adjusting the spring stiffness of the wires electronically. This was accomplished by investigating the alternating voltage needed to relax each wire at the required moment, enabling it to be extra sensitive to the related frequencies. The effect was significant and increased sensitivity at the adjusted frequency 10-fold.

This device will be a useful prototype for technologies like hearing aids and sensors as well as measuring devices for airflow in aeronautics.

The device already reaches levels of precision in measuring air pressure and particle velocity previously unmatched.

The innovation here is an example of successful scale change and use of modularity in order to improve an existing technology. The exploration of nature at the micro scale has informed the improvement, and it has the potential for creating a new paradigm for sensing in several different technologies.

This is another example of developing or extending a new technology by the inspiration from nature—in this case a cricket.

BIO-INSPIRED LEDS

Similar to inspiration provided by crickets, the night-time twinkling of fireflies has inspired scientists to modify an LED so as to make it over 50% as efficient as an original [15]. Researchers from Belgium, France, and Canada studied the internal structure of firefly lanterns, the organs on the bioluminescent insects' abdomens that flash to attract mates. The scientists identified an unexpected pattern of jagged scales that enhanced the lanterns' glow, and applied that knowledge to an LED design to create an LED overlay, which mimicked the natural structure.

The overlay, which increased LED light extraction by up to 55%, could be easily tailored to existing diode designs to help humans light up the night while using less energy.

Apparently, fireflies create light through a chemical reaction that takes place in specialized cells called photocytes. The light is emitted through a part of the insect's exoskeleton called the cuticle. Light travels through the cuticle more slowly than it travels through air, and the mismatch means a proportion of the light is reflected back into the lantern, dimming the glow. The unique surface geometry of some fireflies' cuticles can help minimize internal reflections, meaning more light escapes to reach the eyes of potential firefly suitors.

Using scanning electron microscopes, the researchers identified larger, misfit scales on the fireflies' cuticles. When the researchers used computer simulations to model how the structures affected light transmission, they found that the sharp edges of the jagged, misfit scales let out the most light. This finding was confirmed experimentally when the researchers observed the edges glowing the brightest when the cuticle was illuminated from below. Human-made light-emitting devices like LEDs face the same internal reflection problems as fireflies' lanterns, and researchers thought a factory roof-shaped coating could make LEDs brighter.

To create a jagged overlay on top of a standard gallium nitride LED, it is necessary to deposit a layer of light-sensitive material on top of the LEDs and then expose sections with a laser to create the triangular factory-roof profile. To produce a more efficient LED does not require creating a new LED, but rather just coat and laser patterns on an existing LED.

LESSONS FROM LAVASA

About 60 mi southeast of Mumbai, India, a new city, Lavasa, is under development [16]. It is a city that will consist of five small villages with populations of up to 50,000 people. The area under development will be a challenge as it is really a dense forest along steep slopes.

The real problem, however, is that this area is located in one of India's monsoon hot spots. It is subject to a very short but intense rainy season of about three months with total rainfall of up to 30 ft. On top of that, this rainy season is followed by months of drought. Even though this area with steep hills experiences extensive rain followed by drought, there are no signs of erosion and the water is used efficiently all year round.

A partnership between Janine Benyus and an international architectural firm, HOK, was retained to study the ecological system of this area and to integrate it into the design of the city. In this manner, whatever gets built will perform similarly to the natural environment. They will design the roofs of the buildings to simulate the environment and re-release the monsoonal water back into the air as water vapor. The pavements will be designed to allow water to permeate into the ground, and the building foundations will grip the hillsides like the roots of native trees.

To design the roads in the city, the team examined the behavior of anthills as they are able to remain structurally sound during the worst rains of the year. What was determined is the sinuosity (behaving like a sine wave) of the anthills allows for the water to slow down and channel over the anthills. The roads will be designed to mimic the characteristics of the anthills.

These designs will assure that the natural ecosystem of that region will not be disturbed, and at the same time the city will be enhanced by mimicking this natural environment. During the development of the city, some of the construction violated environmental regulations and the development was halted. Today, it is seeking funding to complete this project [17].

OWL WINGS INSPIRE WIND TURBINE BLADE DESIGN

Most birds make noise while flying, but not owls which allow them to surprise their prey. In fact, several species of owls have developed specialized plumage to minimize aerodynamic noise from their wings.

According to a study [18] published in *The Journal of Sound and Vibration*, owls' feathers move through the air at inaudible frequencies above 1.6 kHz. They have three plumage factors that make this possible. (i) A stiff comb on the leading edge of the wing breaks turbulence into small currents and reduces sound. (ii) An uneven fringe at the wing's trailing edge further muffles noise. (iii) The most notable mechanism, however, is the soft, velvety down that covers the owls' feathers. The down consists of hairs that grow perpendicular to the feather's surface but bend over in the flow direction to form a forest-like thatched canopy of interlocking fibers.

A team of researchers led by Justin W. Jaworski at Lehigh's P.C. Rossin College of Engineering and Applied Science are learning the effects of surface textures on a blade's acoustic signature. Through their efforts, they hope to improve the aerodynamic design of mechanical wings [19]. The team recently used the downy feathers as a model for a 3D printed wing attachment, which was able to reduce the wind turbine noise by 10 dB without impacting aerodynamics. "The canopy of the owl wing surface pushes off the noisy flow," Jaworski said. "Our design mimics that but without the cross fibers, creating a unidirectional fence—essentially going one better than the owl."

SHAVING RAZORS INSPIRED BY FROG AND CRICKET LEGS

Despite the fact that no animals (except for humans!) shave, it turns out the animal kingdom can teach us a thing or two about shaving [20]. Researchers at the Technion have found that by mimicking the texture of the legs of creatures that live in wet environments—they can create better razors.

"We use nature's solutions as inspiration for the development of new surface engineering technologies," explains Assistant Professor Michael Varenberg, the Head of the Tribology Laboratories at the Georgia Institute of Technology. The research group examined how the frog, salamander, and cricket, creatures that live in a wet environment, do not slip on wet surfaces.

The research was done using a one-of-a-kind device that was built by Varenberg's research group, which is capable of measuring the forces that develop between the contact surfaces, such as friction and adhesion, while working within a scanning electron microscope.

The scientists found that creatures living in wet environments improve their grip by draining the fluid out of contact using special channels that exist on the contact surfaces in their feet.

And it just so happens that similar problems exist in modern safety razors. The rubber strip ahead of the blades is used to improve shaving by increasing friction needed for stretching the skin before the blades slide over it.

The presence of the shaving lubricant on the skin reduces friction and therefore interferes with the skin stretching.

To examine real safety razors, the researchers removed the blades and replaced the original strip of rubber, which is covered today by a series of small fins, with a rubber strip bearing a biomimetic hexagonal surface texture that they built based on an extensive examination of frog, cricket, and salamander feet. “The surface we produced in the lab is twice more effective at increasing friction on the skin covered by shaving foam than the surfaces used in commercial razors,” explains Varenberg.

While the patented technology is not out on the market yet, it is not unlikely that it will be commercialized and that some of the world’s future razors will be inspired by frogs and crickets. Let’s just hope that they don’t leap off of the sink and onto the floor.

ADDITIONAL TECHNOLOGIES INSPIRED BY BIOMIMICRY

Here is a brief summary of some new technologies inspired by biomimicry [21].

- Hummingbird inspired wind turbines [22]

A North African company has developed what it contends is a paradigm-shifting wind turbine technology based on the flapping of hummingbird wings. Any potential paradigm shifts are down the road a bit, but based on some remarkable images and new demo videos, the clean energy machine may be the prettiest wind turbine ever invented. Developed by Tyer Wind, a startup based in Tunisia, the turbine uses biomimicry principles to replicate the mechanical action of hummingbird wings. The design is fundamentally different from standard rotor-based wind turbines because instead of converting linear motion—wind blowing across the land—into a circular motion, it converts it into a figure-eight pattern. Not only is that shape the same as the one hummingbird wings make while the birds hover, but it also generates energy on both the upstroke and the downstroke (see Fig. 9.14).

- Electric eels inspire a new kind of power source [23]

New power sources bear a shocking resemblance to the electricity-making organs inside electric eels. These artificial electric eel organs are made up of water-based polymer mixes called hydrogels. Such soft, flexible battery-like devices, described online October 13, 2017 in *Nature*, could power soft robots or next-gen wearable and implantable tech (see Fig. 9.15).



FIGURE 9.14 Hummingbird inspired wind turbines. *Source:* Reproduced with permission of Tyer Wind.

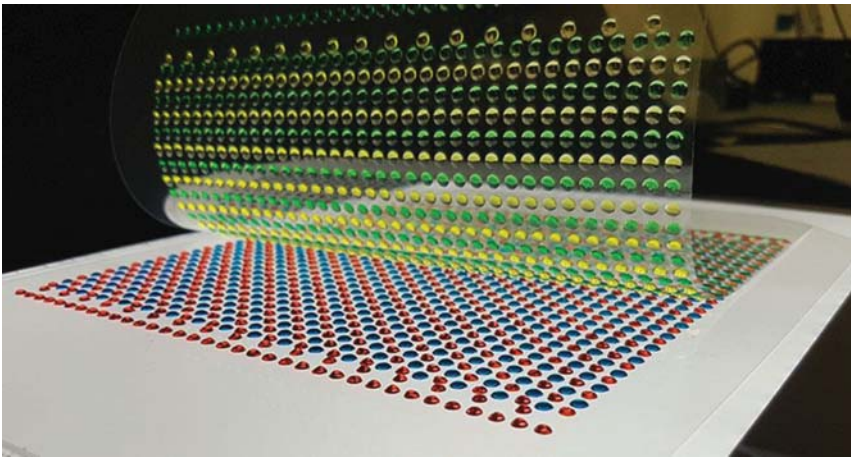


FIGURE 9.15 Eel inspired power. *Source:* Schroeder et al. (2017). Reprinted with permission of Springer Nature.

- Using shark scales to design better drones, planes and wind turbines [24]

To build more aerodynamic machines, researchers are drawing inspiration from an unlikely source: the ocean. A team of evolutionary biologists and engineers at Harvard University, in collaboration with

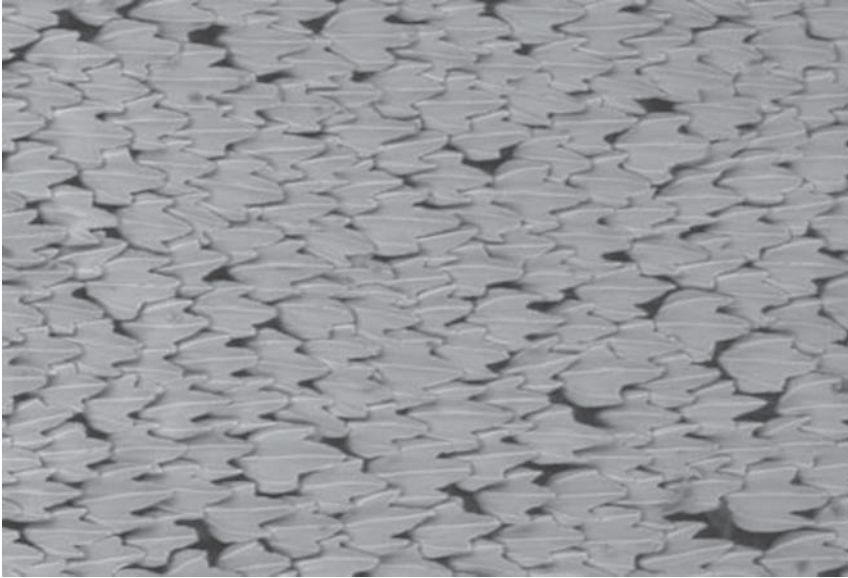


FIGURE 9.16 Shark scales. *Source:* Reproduced with permission of SEAS Harvard.

colleagues from the University of South Carolina, have shed light on a decades-old mystery about sharkskin and, in the process, demonstrated a new, bioinspired structure that could improve the aerodynamic performance of planes, wind turbines, drones, and cars (see Fig. 9.16).

- Pinecones could make buildings more energy efficient [25]
Cones produced by such trees as pines, spruce, hemlock, and fir respond naturally to different degrees of humidity by opening and closing, without consuming any electrical energy in doing so (Fig. 9.17). Designing window blinds based on their mechanical properties that could open and close in response to moisture—but use no energy in the process—could conserve a lot of energy.

MINDFUL MINING: A PROPOSAL

As a final project for a class, one student applied two bio-inspired technologies to make a mining operation considerably more efficient [26]. Below is a student's abstract of a proposal for a company like Caterpillar.

Introduction

Caterpillar is the world's leading manufacturer of construction and mining equipment, diesel and natural gas engines, industrial gas turbines, and

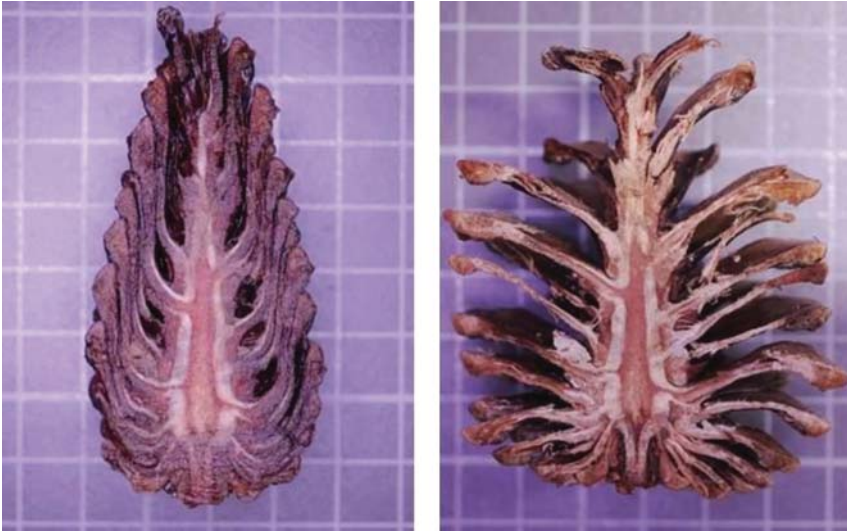


FIGURE 9.17 Pine cones for energy efficient buildings.

related services. Caterpillar is about a US\$50 billion company with profits of around US\$3 billion. Caterpillar provides production, support, and technology solutions for surface and underground hard-rock mining. Mining products include large mining trucks, wheel loaders, motor graders, scrapers, and track-type tractors.

Business as Usual

Mining is a very material intensive operation. Ore waste to metal ratio is often very high. In the case of gold it is as high as 10,000:1. This necessitates excessive amounts of earth to be moved to extract economically viable amounts of metal. The ecological footprint is very high. Caterpillar understands the importance of environment protection and sustainability. It has formed alliances with mining companies to reduce environmental impacts of mining. However, Caterpillar's present revenue stream is directly tied to the amount of earth that gets moved. Its product line is designed to move large amounts of earth. A larger ecological footprint translates to a higher sales and revenue.

Business Unusual: Proposed Business Model

The idea being proposed here is to use the science of biomimicry to perform precision metal and mineral extraction. Precision mining eliminates the need to remove vast amounts of earth. Using a combination of microbe-inspired

mining and ant-inspired micro swarm robotics, Caterpillar could possibly create a model wherein the revenue stream is dependent on the amount of pure metal and minerals extracted and not the amount of earth moved. This has the potential to radically transform the mining industry. In addition to the immense environmental benefits, this model could potentially help Caterpillar increase its revenue and market share by making other forms of mining equipment redundant.

CASE STUDY Watch a 25-minute movie with Janine M. Benyus produced by Leonardo DiCaprio. <https://nextnature.net/2016/03/biomimicry-the-documentary>.

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The Need and Growth of a Circular Economy

Since the beginning of production and consumption, the economy has been based on a linear business model. It consisted of the manufacturing of a product, the use of the product, and followed by the eventual disposal of the product. It was subsequently given the name *linear economy*.

It often leads to a system that is inefficient and costly and that harms the environment or depletes natural resources. For example, mining metals, from gold to coal, can spoil ecosystems and disrupt nearby communities. Making steel from ore requires a large amount of energy, which produces globe-warming carbon dioxide. A by-product of the linear model is material waste, which takes up space and may include contaminants. Trash ends up in undesirable places. The so-called Great Pacific Garbage Patch [1] is only the most well-known example of global-scale plastic pollution, yet products like steel and plastic can be reused, refurbished, and recycled to capture untapped value.

However, it is becoming worse. Due to the exponential growth of the global population, an increase of over five billion people since 1940, and the subsequent consumption of our natural resources, this linear economy can no longer be acceptable. As of 2018, the world is consuming the equivalent of 1.7 earths of its natural resources [2]. What is necessary today is a *circular economy*, which consists of take, make, use, reuse, and reuse again and again.

Practical Sustainability Strategies: How to Gain a Competitive Advantage,
Second Edition. George P. Nassos and Nikos Avlonas.

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Companion website: www.wiley.com/go/Nassos/PracticalSustainabilityStrategies_2e

KALUNDBORG SYMBIOSIS

One of the earliest examples of the circular economy was developed in Kalundborg, Denmark, in the early 1970s [3]. It was not originally created as a circular economy but has grown as reducing waste became more important, and has not been known as a circular economy until very recently when this sustainability model became necessary and was created.

There are currently over 30 exchanges of materials among the companies of Kalundborg [4]. Asnaes Power Station is at the heart of the network (Fig. 10.1). The power company gives its steam residuals to Statoil Refinery, meeting 40% of its steam requirements, in exchange for waste gas from the refinery. The power plant creates electricity and steam from this gas. These products are sent to a fish farm, to Novo Nordisk, which receives all of its required steam from Asnaes, and to a heating system that supplies 3500 homes. These homeowners pay for the underground piping that supplies their heat, but receive the heat reliably and at a low price. Fly ash from Asnaes is sent to a cement company, and gypsum from its desulfurization process is sent to Gyproc for use in gypsum board. Two-thirds of Gyproc’s gypsum needs are met by Asnaes. Statoil Refinery removes sulfur from its natural gas and sells it to a sulfuric acid manufacturer, Kemira. The fish farm

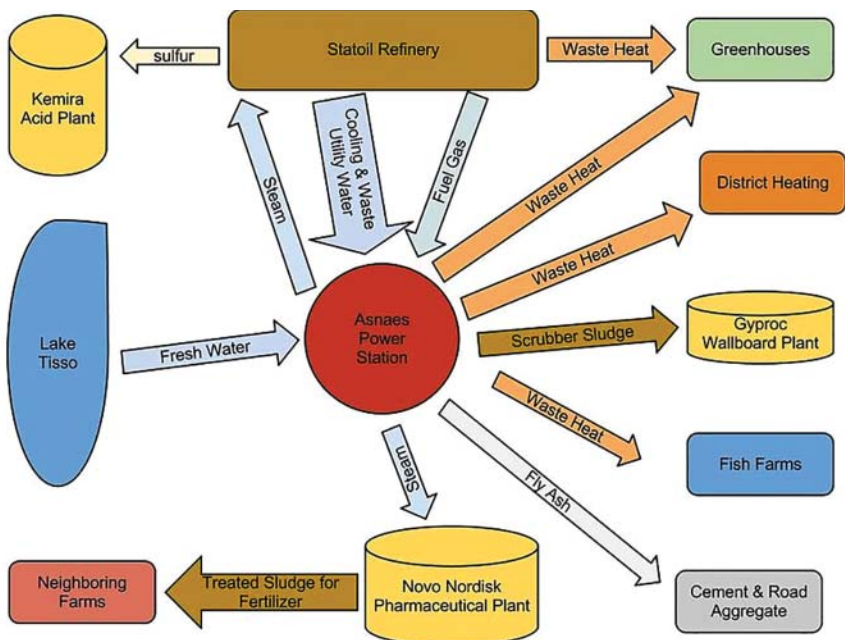


FIGURE 10.1 Flowchart of Kalundborg’s circular economy.

sells sludge from its ponds as fertilizer to nearby farms, and Novo Nordisk gives away its own sludge, of which it produces 3000 m³/day. The sludge is to be refined for biogas for the power plant.

Water reuse schemes have also been developed within Kalundborg. Statoil pipes 700,000 m³ of cooling water per year to Asnaes, which purifies it and uses it as “boiler feed-water.” Asnaes also uses approximately 200,000 m³ of Statoil’s treated wastewater per year for cleaning. The 90 °C residual heat from the refinery is not used for district heating due to taxes. Instead, heat pumps are used with the 24 °C wastewater as a heat reservoir.

Kalundborg was the first example of separate industries grouping together to gain competitive advantage by material exchange, energy exchange, information exchange, and/or product exchange. The very term, industrial symbiosis, was first defined by a station manager in Kalundborg as “a cooperation between different industries by which the presence of each increases the viability of the others, and by which the demands of society for resource savings and environmental protection are considered” [4].

Kalundborg’s success helped generate interest in industrial symbiosis. Other developed nations, including the United States, began to develop incentives for corporations to implement materials exchange with other corporations. This was the beginning of the implementation of more eco-industrial parks (EIPs). In 1996, the US President’s Council on Sustainable Development proposed 15 EIPs to pursue the idea of industrial symbiosis. These parks were created by grouping diverse stakeholders with common material flows together, with added governmental incentives to encourage materials exchange. The goal was to determine if the Kalundborg model could be replicated in the United States with the addition of five major characteristics: (i) some form of material exchange between multiple separate entities, (ii) industries within close proximity to each other, (iii) cooperation between plant managements of the different corporations, (iv) an existing infrastructure for material sharing that does not require much retooling, and (v) “anchor” tenants (large corporation with resources to support early implementation).

EARLIER MODEL

Although the Kalundborg symbiosis was probably the first industrial model of the circular economy, examples really existed long before. At least as early as the 1940s, the circular economy was employed by the dairy industry although it was not known by that name. Milk was delivered by truck in a quart or gallon glass bottle to the home of the dairy’s customers. It was placed outside the front door of the home. Upon consuming the contents

of the bottle, the customer would place the empty bottle outside the front door for the next milk delivery. The milkman would then pick up the empty bottle and replace it with a full bottle. The empty glass bottle would be taken back to the dairy, thoroughly cleaned, and refilled for the next delivery. This system created almost no waste at all except for possibly the bottle cap. (There is an even earlier model that was learned personally when I was very young. See “Personal Note”).

TERRACYCLE INTRODUCES “LOOP”

Following the milkman model of some 80 years ago, TerraCycle introduced its circular economy model in early 2019 [5]. It joined forces with some of the top consumer brand companies such as Häagen-Dazs, Procter & Gamble, Nestlé, PepsiCo, Unilever, Mars, Clorox, Coca-Cola, Mondelēz, and Danone. Another, and most critical, partner was United Postal Service (UPS).

Loop was created as a circular shopping platform that transforms the packaging of everyday essentials from single-use disposable to durable, feature-packed designs. They took some of the most popular brands and introduced them in more functional packaging. After the contents have been consumed, the empty container is placed back in the Loop Tote and scheduled for a free pick-up from the customer’s home by UPS. The containers are designed for a minimum of 100 uses.

Loop would determine the customer’s consumption of the various products based on the empty bottles that are returned. In turn, Loop would then replenish the products consumed by the customer. This would be an ongoing process that will not require the customer to continue to reorder. Loop would hygienically clean and sanitize the empty packaging and send back so they are ready for reuse, instead of ending up as waste after a single use.

The Loop tote became a breakthrough zero-waste delivery system that eliminates wasteful single use shipping materials. Durable packaging enables innovative features and unparalleled design, so that the customer can experience everyday products in a brand new way.

The circular economy is basically an industrial system that is restorative by intention and design for environmental reasons, not for economic reasons, although the circular economy leads to both cost reduction and material reduction. The idea is that rather than discarding products before their value are fully utilized, we should use and reuse them as much as possible. Presently only a few percentage points of the original product value is recovered after use. A good example is the mobile phone. With most people today

using a smart phone, we are encouraged to replace it at least every two years. The manufacturers offer improved phones in order to continue increasing sales. However, the old phones, which are still functional, become an unnecessary waste. Another example is the personal computer (PC). According to Moore's law [6], the efficiency of a PC will double every 18 months because that has been the time for Intel to develop a new computer processor with double the speed of the previous one. But instead of just replacing the processor, we were "trained" to replace the entire PC even though the case, the keypad, and most other parts were the same as the previous model.

DESIGNING FOR THE CIRCULAR ECONOMY

Products can be designed from the outset so that, after their useful lives, they will provide nourishment of something new. For example, they can be conceived as "biological nutrients" that will easily reenter the water or soil without depositing synthetic materials and toxins. Or they can be "technical nutrients" that will continually circulate as pure and valuable materials within closed-loop industrial cycles, rather than being "recycled"—really, down cycled—into low-grade materials and uses, making this process truly eco-effective. This is another example of learning from nature [7].

The products can also be designed for disassembly or deconstruction. It can make it easier for the product to be repaired or upgraded, prolonging its useful life. It can also help ensure that the product is recycled and enables whole components to be reused. The degree to which the product can be disassembled easily often determines how the product will end its life.

Another design feature is to consider flexibility whereby a product can have multiple applications. In this manner, it will be more efficient to fulfill the applications with the greatest demand. An example of this feature is the product line of Do It Right This Time (DIRTT) [8], which designs and manufactures interiors for office space, schools, and healthcare facilities. Given the allocated space, DIRTT will design and provide the components for space as initially required. However, the components can be disassembled and repurposed for a different design.

THE ELLEN MACARTHUR FOUNDATION

The Ellen MacArthur Foundation was launched in 2010 to accelerate the transition to a circular economy [9]. Since its creation the charity has emerged as a global thought leader, establishing the circular economy on

the agenda of decision makers across business, government, and academia. The Foundation's work focuses on six interlinking areas:

- (1) *Learning*—developing the vision, skills, and mindsets needed to transition to a circular economy.
- (2) *Business*—catalyzing circular innovation and creating the conditions to reach its scale.
- (3) *Institutions, governments, and cities*—again, catalyzing circular innovation and creating the conditions to reach its scale.
- (4) *Insight and analysis*—providing robust evidence about the benefits and implications of the transition.
- (5) *Systemic initiatives*—transforming key material flows to scale the circular economy globally.
- (6) *Communications*—engaging a global audience around the circular economy.

Through this effort, the Foundation developed the ReSOLVE Framework, which consists of:

- *Regenerate*—return recovered biological resources to the biosphere
- *Share*—underutilized assets
- *Optimize*—increase efficiency of product
- *Loop*—remanufacture products
- *Virtualize*—dematerialize books, CDs, travel
- *Explore*—apply new technologies

An infographic demonstrating the circular economy principle is shown in Figure 10.2. It is based on three principles:

- (1) Preserve and enhance natural capital by controlling finite stocks and balancing renewable resource flows. ReSOLVE levers: regenerate, virtualize, and explore.
- (2) Optimize resource yields by circulating products, components, and materials in use at the highest utility at all times in both technical and biological cycles. ReSOLVE levers: regenerate, share, optimize, and loop.
- (3) Foster system effectiveness by revealing and designing out negative externalities. All ReSOLVE levers.

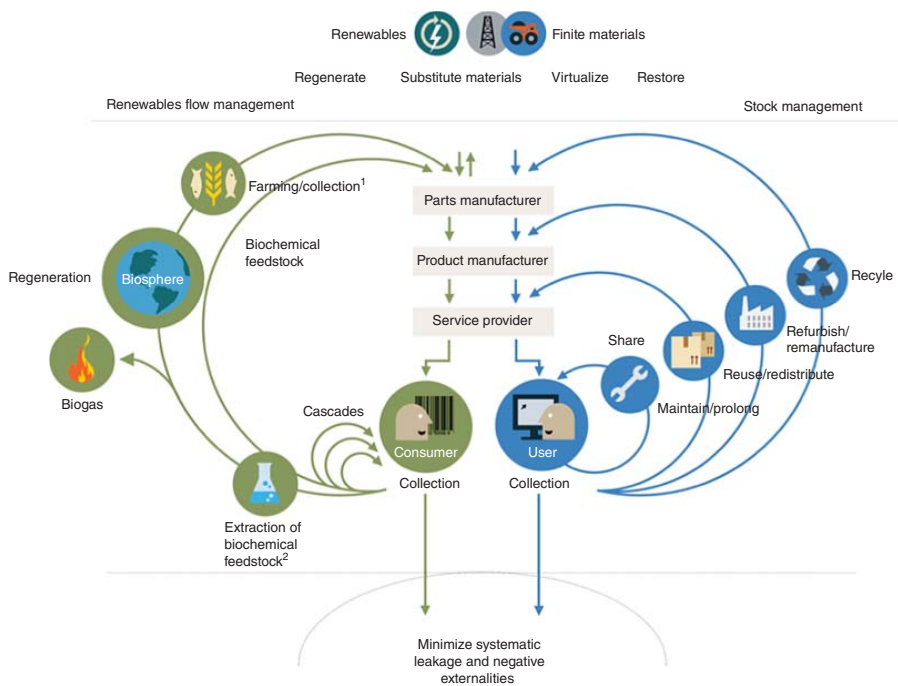


FIGURE 10.2 Infographic of circular economy.

The foundation has been responsible for many organizations adopting the circular economy principles with great success. Numerous case studies of these successes have been prepared [10].

QUANTITY OF WASTE FOR THE CIRCULAR ECONOMY

When talking about reducing the waste generated via the circular economy, most people normally refer to municipal solid waste (MSW), which is the waste generated by people and businesses. In 2015, about 260 million tons of MSW were generated in the United States. Basically, this is the quantity of material that is available for recovery and reuse. However, this is not the total waste generated in the United States.

A study was made in 2009 to determine the total amount of waste generated in the United States not only from the public and businesses but also from industry. This would include oil and gas waste, mining waste, and hazardous and non-hazardous industrial waste. It was difficult to determine this quantity of waste, given the name Gross National Trash (GNT), because these data are not provided by industry. But an earlier estimate was made in 1992 by Joel Makower that the GNT was about 13 billion tons with the MSW representing less than 2% [11] (see Fig. 10.3). This indicates that there is a massive amount of waste that can be considered for the circular economy. How to manage this waste is discussed in detail in Chapter 14.

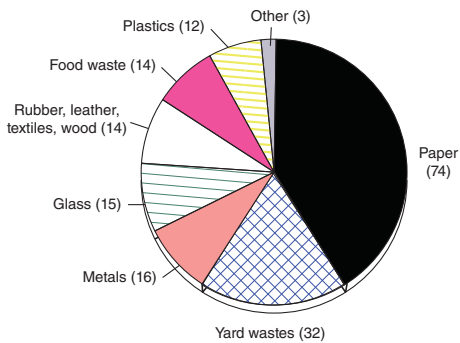
PERSONAL NOTE

While the term circular economy is rather new, the concept is definitely not. When I was a young boy over 70 years ago, I would spend time hanging out at my father's restaurant—eating and looking around. I still remember an operation that took place about once a month in the kitchen. My uncle, the chef, would trim fat from whichever piece of meat he needed, beef, lamb, or pork. He didn't throw away the fat but rather saved it for my father. Once a month, my dad would place the small pieces of fat in a very large pot, add lye, and boil the mixture. He would then let it cool until it became a solid mass. Cutting up this solid white mass into small pieces, he produced soap, which was used by the (human) dishwasher. This was my first exposure to a circular economy, making a waste product a feedstock for another product, although it was obviously not known as the "circular economy." Additional case studies on the circular economy are provided [12–14].

A tale of two circles

Municipal solid waste

Total: 180 million tons/year



Gross national trash

Total: 12,915 million tons/year

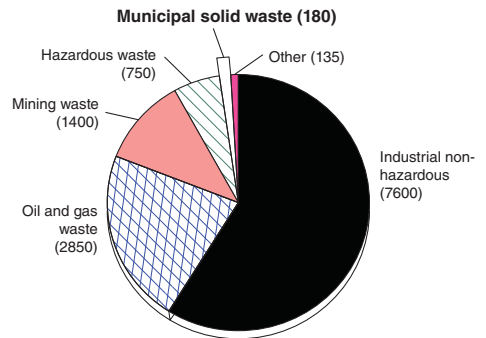


FIGURE 10.3 MSW versus GNT.

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Addressing Climate Change

In 1987, James Hansen,¹ director of the Goddard Space Institute, appeared before Congress to present the current state of the warming atmosphere. He indicated that the earth is warming due to an increase in carbon dioxide (CO₂) emissions. Historically, the concentration of CO₂ in the atmosphere has been 280 parts per million (ppm). However, primarily due to the industrial revolution, the concentration has been increasing. Hansen indicated that if the concentration exceeded 350 ppm, it could cause environmental problems like rising seas from melting glaciers and ice sheets, and greater and more frequent storms. While the earth was being subjected to *global warming*, the impact from this warming was addressed as *climate change*.

In 1992, the United Nations Intergovernmental Panel on Climate Change (IPCC) met in Rio de Janeiro, Brazil, to discuss how to address climate change. Five years later, the IPCC met again in Kyoto, Japan, and developed the Kyoto Principles, which established the reduction targets of CO₂ emissions such as 60–80% reduction from 1995 levels within 50 years. Target reductions were also set for shorter terms such as 15 years out. By 2012, which was 15 years after the Kyoto Protocol, the target was a reduction of 5% below the 1995 levels. Despite some efforts to reduce emissions of carbon dioxide and other greenhouse gases (GHGs), the emissions level were over 50% above the 1995 levels [1].

As mentioned in Chapter 1 (p. 16), it is imperative that everyone does what is necessary to reduce carbon emissions even if there were a slight possibility that climate change is not for real. We need a very rigorous plan to “draw down” carbon emissions. A team of 200 scientists, engineers, and scientists worldwide came together in Project Drawdown to develop 100

¹Currently member of Earth Institute at Columbia University.

TABLE 11.1 Drawdown: top solutions in each sector

Sector	Solution	Total atmospheric CO ₂ -Eq reduction (GT)	Net cost (billions US\$)	Net savings (billions US\$)
Energy	Wind turbines (onshore)	84.60	1225.37	7425.00
	Solar farms	36.90	−80.60	5023.84
Materials	Refrigeration	89.74	N/A	−902.77
	Alternative cement	6.69	−273.90	N/A
Food	Reduced food waste	70.53	N/A	N/A
	Plant-rich diet	66.11	N/A	N/A
Women and girls	Educating girls	59.60	N/A	N/A
	Family planning	59.60	N/A	N/A
Buildings and cities	District heating	9.36	457.07	3543.50
	Insulation	8.27	3655.92	2513.33
Land use	Tropical forests	61.23	N/A	N/A
	Temperate forests	22.61	N/A	N/A
Transport	Electric vehicles	10.80	14,148.03	9726.40
	Ships	7.87	915.93	424.35

creative ideas that will do exactly that, and it was compiled in “Drawdown,” a book edited by Paul Hawken [2].

These potential solutions to the climate change problem can be ranked as to how cost-effective they are; how quickly they can be implemented; or how beneficial they are to society. While these are all very interesting metrics for interpreting the result, the authors of *Drawdown* ranked the solutions based on the total amount of GHG they can potentially avoid or remove from the atmosphere. While the rankings are global, the relative importance of one solution over another may depend on geography, economic conditions, or which of the eight sectors is of most interest. Table 11.1 lists the seven different sectors, the top two solutions in each sector, and the different metrics: (i) total atmospheric CO₂ equivalent reduction in gigatons (GT), (ii) the net cost in billion US dollars, and (iii) net savings in billion US dollars.

The reduction in carbon dioxide equivalents is the quantity expected to be removed by a particular solution between 2020 and 2050. The total cost of each solution is the amount needed to purchase, install, and operate the system over the same 30 year period. The estimates tend to be conservative, but they still tend to offer an overwhelming net savings. For some of the

solutions, like a specific rainforest or support girls' education, the savings may not be calculable. A summary of these 14 solutions in the 7 sectors are shown in Table 11.1.

ENERGY: WIND TURBINES (ONSHORE)

Windmills have been around for over 1000 years, but it wasn't until the late 1800s that the kinetic energy was converted to electricity—thanks primarily to the Dutch. Basically, the blades on the turbine are shaped so the blowing wind causes them to rotate, which, in turn, rotates a generator that produces the electricity. This technology grew slowly but took off following the oil crisis in the early 1970s. In 2015, a record 63 GW of wind energy were installed around the world. Of this new capacity, almost half was brought on by China. Over 40% of the electricity demand in Denmark is supplied by wind power. The United States has a large quantity of wind energy potential down the center of the country where just three states—Kansas, North Dakota, and Texas—could meet the electricity demand of the entire country. The other big advantage of wind turbines is their small footprints, which are only about 1% of the land upon which they are installed. Farmers can continue to grow their crops or graze their cattle.

While wind turbines have great potential in providing carbon free energy, they do have some disadvantages. One potential problem is the variable nature of the wind when there are times that the turbines are not turning and thus not generating electricity. Other disadvantages include some noise and flickering of the turning blade shadows. There is also concern of being deadly to bats and migrating birds.

A monetary disadvantage may be the inequitable government subsidies. The International Monetary Fund believes that the fossil fuel industry receives much more funding than the wind energy industry. In 2015, the fossil fuel industry received about US\$5.3 trillion in direct and indirect subsidies. The indirect subsidies include health costs due to air pollution and global warming, which are not factors at all in the wind energy industry. In comparison, the wind energy industry has received only US\$12.3 billion in direct subsidies since 2000.

The cost of wind energy is very low and is expected to be the cheapest source of installed electricity capacity within the next 10 years. Currently, wind energy is about 2.9 cents/kWh, while the cost for natural gas combined-cycle plants is about 3.8 and 5.7 cents/kWh for large solar farms. Currently, wind energy benefits from tax credits, which will eventually disappear, but since the cost is expected to continue to decrease, it will make up for the lost tax credits.

Drawdown summarizes the impact of the onshore wind turbine industry as follows: *An increase in onshore wind from 2.9% of world electricity use to 21.6% by 2050 could reduce emissions by 84.6 gigatons of carbon dioxide. For offshore wind, growing from 0.1% to 4% could avoid 14.1 gigatons of emissions. At a combined cost of US\$1.8 trillion, wind turbines can deliver net savings of US\$7.7 trillion over three decades of operation. These are conservative estimates, however. Costs are falling annually and new technological improvements are already being installed, increasing capacity to generate more electricity at the same or lower cost [3].*

ENERGY: SOLAR FARMS

In the 1950s, while working on the invention of the silicon transistor, the silicon photovoltaic (PV) technology was invented by accident at Bell Labs. In 1954, this laboratory introduced the first solar battery to power a radio transmitter. The cost of this technology was so great at US\$1900 per watt that it could only be utilized on satellites. However, the cost has decreased so much that solar panels are being installed on rooftops around the world as well as developing “solar farms” consisting of thousands of panels. Solar farms enjoy much lower installation costs per watt than rooftop solar, and their efficacy in translating sunlight into electricity, known as the efficiency rating, is higher. Another advantage of solar farms is that the panels can rotate to follow the path of the sun, again increasing their efficiency.

One of the disadvantages of solar energy, like wind energy, is that it is available only when its fuel source, the sun, is shining. To some extent, the two technologies are complementary in that the sun is at its peak in the middle of the day while wind tends to pick up at night. A new energy project developed by Invenergy located near Lima, Ohio, is taking advantage of this complementary generation and is pairing the two technologies [4]. These two power generators hit their peaks at different times of day and night, allowing them to provide a steadier output together than if each were operating alone. There is also cost saving by sharing equipment and power lines. This project will consist of a 175 MW wind farm with a 150 MW solar farm built within the same property.

Solar energy has grown so fast; it is estimated that around 300 million tons/year of carbon dioxide have been eliminated despite meeting only 2% of the global energy demand. Some countries like Italy, Germany, and Greece already depend on solar energy to meet as much as 8% of their electrical demand. The development of battery storage technology will have a great impact on the growth of this electrical source of energy.

Drawdown summarizes the impact of global solar farms as follows: *Currently, 4% of global electricity generation, utility-scale solar PV grows to*

10% in our analysis. We assume an implementation cost of US\$1,445 per kilowatt and a learning rate of 19.2%, resulting in implementation savings of US\$81 billion when compared to fossil fuel plants. That increase could avoid 36.9 gigatons of carbon dioxide emission, while saving US\$5 trillion in operations costs by 2050 – the financial impact of producing energy without fuel [5].

MATERIALS: REFRIGERATION

All devices designed for cooling such as air conditioners and refrigerators require a chemical refrigerant that absorbs heat and releases it to cool buildings and vehicles as well as to chill food. The original refrigerants were chlorofluorocarbons (CFCs) and hydrochlorofluorocarbons (HCFCs), both of which were the cause of the depleting stratospheric ozone layer. This layer is essential for absorbing the sun's ultraviolet radiation. In 1987 the Montreal Protocol was enacted to phase out the use of these chemicals, only two years after the discovery of the gaping hole over the Antarctic.

While huge volumes of CFCs and HCFCs remain in circulation, the impact on the ozone layer has decreased. The replacement chemicals like hydrofluorocarbons (HFCs) have minimal effect on the ozone layer but contribute extensively in warming the planet. Their capacity to warm the planet is 1000–9000 times greater than that of carbon dioxide. In October 2016, officials from 170 countries met in Kigali, Rwanda, and negotiated an amendment to the Montreal Protocol to phase out HFCs beginning in 2019 for high-income countries and continuing through 2028 for the low-income countries.

During the next 10 years, there will be two competing activities—the research to identify new refrigerant chemicals that will not impact the ozone layer nor warm up the atmosphere against the increase in air-conditioning units resulting from the growth of the rapidly developing economies. Refrigerants currently cause emissions throughout their lifecycle beginning with the production, their use, servicing, leaking, and worst of all upon their disposal. When these chemicals are no longer needed, they must be transformed into some other chemicals that do not cause warming. In other words, these chemicals must be destroyed in order to reduce emissions.

Drawdown summarizes the impact of refrigeration chemicals as follows: *Our analysis includes emissions reductions that will be achieved through the 2016 Kigali accord, as well as additional practices to manage refrigerants already in circulations. We model adoption of practices to (1) avoid leaks from refrigerants and (2) destroy refrigerants at the end of life. Over thirty years, 87% of refrigerants that may be released can be continued, avoiding emissions equivalent to 89.7 gigatons of carbon dioxide. Although some*

revenue can be generated from resale of recovered refrigerant gases, the costs to establish and operate recovery, destruction, and leak avoidance outweigh the financial benefit – meaning that refrigeration, as modeled, could incur a net cost of US\$903 billion by 2050 [6].

MATERIALS: ALTERNATIVE CEMENT

The dominant material in the world's construction industry is concrete, which has a simple recipe of sand, crushed rock, water, and cement, all combined and hardened. Cement is a gray powder of lime, silica, aluminum, and iron and acts as the binder. It coats and glues the sand and rock together, thus producing the stone-like material that results after curing. To produce the cement, it is necessary to roast a mixture of crushed limestone and aluminosilicate clay in a giant kiln. Since the limestone is primarily calcium carbonate, the roasting process splits the material into calcium oxide, which is the desired lime content in cement, and carbon dioxide, which is the waste material. Heating the kiln to about 2640 °F requires large quantities of coal. Producing one ton of cement requires the burning of 400 pounds of coal. Between the decarbonization of limestone and the burning of the coal, each ton of cement produced will create about one ton of carbon dioxide.

To reduce the emissions caused by the fuel for the kiln will require alternative kiln fuels such as biomass. To reduce the emissions from the decarbonization of the limestone will require changing the formulation of the cement. Some possibilities are volcanic ash, certain type of clays, finely ground limestone, or blast furnace slag. As long as coal is being burned, fly ash can also be used in the formulation.

Drawdown summarizes the impact of alternative cement as follows: Because fly ash is a by-product of burning coal, each ton created is accompanied by 15 tons of carbon dioxide emissions. Using fly ash in cement can offset only 5% of those emissions. Even so, if 9% of cement produced between 2020 and 2050 is a blended mix of conventional Portland cement and 45% fly ash, 6.7 gigatons of carbon dioxide emissions could be avoided by 2050. The production saving of US\$274 billion are largely a result of longer cement life span [7].

FOOD: REDUCED FOOD WASTE

The production of food is one of largest use of manual labor as it requires about one-third of the world's labor force. Of all this food produced, about one-third does not make it from farm or factory to the dinner table. This is a sad fact given that about 800 million people worldwide are continuously

hungry from the lack of food. In addition, the food that is wasted contributes 4.4 GT of carbon dioxide equivalent into the atmosphere each year—about 8% of the total anthropogenic GHG emissions. If you were to rank food waste with the world's countries, it would rank the third largest emitter of GHGs globally behind the United States and China.

While food waste is a problem worldwide, the cause of it differs between low-income and high-income countries. In low-income areas, the waste is usually at the beginning of the supply chain and the loss of food is unintended. It could be the result of bad roads, lack of refrigeration or storage facilities, poor equipment or packaging, or high temperatures and high humidity. So the food may be rotting on the farm or spoiling during storage or distribution.

In high-income countries, the food waste is usually further along the supply chain and is more willful rather than unintended. The retailers waste the food primarily for esthetic reasons to satisfy their customers. Other waste may be from ordering too much food to avoid the risk of running short. Consumers also are involved in the waste process by making more food dishes than are eaten, food left in the refrigerator for a long period and thus thrown out, or being forced to throw out food because of confused labeling. Standardizing and regulating labeling would help alleviate this problem.

Adopting some national goals is critical in solving this problem. In 2015, the United States set a food-waste target based on the UN Sustainable Development Goals (see Chapter 3). France did something similar by passing a law that prevents supermarkets from disposing unsold food but rather requiring them to send this unsold food to charities, to be used as animal feed, or even to be used as compost. In terms of minimizing carbon emissions, the most effective policy is to minimize the waste.

Drawdown summarizes the impact of reduced food waste as follows: *After taking into account the adoption of plant-rich diets, if 50% of food waste is reduced by 2050, avoided emissions could be equal to 26.2 gigatons of carbon dioxide. Reducing waste also avoids the deforestation for additional farmland, preventing 44.4 gigatons of additional emissions. We used forecasts of regional waste estimates from farm to household. These data show that up to 35% of food in high-income economies is thrown out by consumers; in low-income economies, however, relatively little is wasted at the household level [8].*

FOOD: PLANT-RICH DIET

The Western diet, which consists of a large portion of animal protein, is also responsible for a large portion of the global carbon emissions. Depending on

how the emissions are measured, raising livestock may account for 15% or 50% of the global GHGs emitted each year. Regardless of how the emissions are measured, raising cattle contributes considerably more carbon emissions than growing various plants—vegetables, fruit, grains, and legumes. If cattle were their own nation, they would be the world’s third largest emitter of GHGs. One of the reasons for this large amount of emissions is the overconsumption of animal protein. On the average, world consumption of animal protein is 36% greater than necessary, while in the United States and Canada, it is 80% greater than necessary.

Switching to a plant-rich diet, while very important, will be difficult to achieve; eating meat is laden with meaning, blended into customs, and very appealing to the taste buds. It will be difficult to accomplish because a large percentage of the population has been eating meat as a personal requirement. Fortunately, there are companies like Beyond Meat and Impossible Foods that are developing meat substitutes that look and taste just like the real meat. Burger King has just announced the Impossible Burger, which will be available at a slight premium over the regular burger, but the company expects that this meat imitation will soon be cheaper than a regular burger.

Perhaps another strategy would be to increase the price of meat products in order to give them a “delicacy” classification rather than a staple. This could be done by eliminating the government subsidies benefiting the US livestock industry or by adding a tax on meat similar to taxes on cigarettes. The tax on cigarettes reflects the social and environmental negatives of smoking so a tax on meat could similarly reflect an environmental negative.

Drawdown summarizes the impact of a plant-rich diet as follows: *Using country-level data from the Food and Agriculture Organization of the United Nations, we estimate the growth in global food consumption by 2050, assuming that lower-income countries will consume more food overall and higher quantities of meat as economies grow. If 50% of the world’s population restricts their diet to a healthy 2,500 calories per day and reduces meat consumption overall, we estimate at least 26.7 gigatons of emissions could be avoided from dietary change alone. If avoided deforestation from land use change is included, an additional 39.3 gigatons of emissions could be avoided, making healthy, plant-rich diets one of the most impactful solutions at a total of 66 gigatons reduced [9].*

WOMEN AND GIRLS: EDUCATING GIRLS AND FAMILY PLANNING

These two solutions in the Women and Girls section are interrelated in this strategy for carbon emission reduction. The end goal in reducing emissions

is through the reduction in population growth. Women with more years of education have fewer, healthier children, and actively manage their reproductive health. An example of the impact of education on population growth is South Korea when it went from one of the least to most educated countries in the world. If all countries adopted a similar requirement to provide all girls with a minimum of 12 years of education, there would be 843 million fewer people worldwide than if current education enrollment rates remain as today.

Another example is a voluntary program adopted by Iran in the early 1990s where education was improved and provided free access to contraception. The result was a reduction in fertility rate by 50% within one decade. A similar program initiated in Bangladesh reduced the average birth rates from six children in the 1980s to two today. While the IPCC has been active since the early 1990s, it has been silent until very recently when in 2014 pointed to population growth as an important factor in GHG concentrations and included access to reproductive health services for family planning.

Drawdown summarizes the impact of educating girls and family planning as follows: Increased adoption of reproductive healthcare and family planning is an essential component to achieve the United Nations' 2015 medium global population projection of 9.7 billion people by 2050. If investment in family planning, particularly in low-income countries, does not materialize, the world's population could come closer to the high projection, adding another one billion people to the planet. We model the impact of this solution based on the difference in how much energy, building space, food, waste, and transportation would be used in a world with little to no investment in family planning, compared to one in which the projection of 9.7 billion is realized. The resulting emissions reductions could be 123.0 gigatons of carbon dioxide, at an average annual cost of US\$10.77 per user in low-income countries. Because educating girls has an important impact on the use of family planning, we allocate 50% of the total potential emissions reductions to each solution – 59.6 gigatons each [10].

BUILDINGS AND CITIES: DISTRICT HEATING

The center of cities tends to be very dense in terms of the close proximity of buildings. Instead of each building having its own boiler and air conditioning system, it is more efficient to have a central plant for heating and cooling, something called district heating and cooling (DHC). This central plant can provide hot and/or cool water to many buildings via a network of underground pipes. This idea was implemented in New York back in 1882;

steam was being pumped under Manhattan's busy streets to serve many customers with district heating. The ultimate use of this system can be found in Copenhagen, Denmark, where 98% of the heating demand is provided through DHC with the source of heat from waste heat in nearby coal-fired power plants and waste-to-energy plants. Since 2010, Copenhagen has also provided cooling by tapping into the cold waters from the Øresund Strait.

Once the distribution network is in place, the source of heating or cooling can be changed to more efficient systems. Coal-fired power plants can give way to wood pellets and even geothermal can be implemented. In Tokyo, its DHC system was able to reduce the energy use and subsequent carbon emissions by 50%. This technology is very common in Northern Europe but is still new and unfamiliar in other parts of the world. Designs of new cities will be incorporating DHCs as people move from rural to urban communities, particularly in China.

Drawdown summarizes the impact of district heating as follows: *By replacing existing stand-alone water- and space-heating systems, district heating can reduce carbon dioxide emissions by 9.4 gigatons by 2050 and save US\$3.5 trillion in energy costs. Our analysis estimates current adoption at 0.01% of heating demand, growing to 10% over the next thirty years. While natural gas is currently the most prevalent fuel source for district heating facilities, we model the impact only of alternative sources such as geothermal and solar thermal energy that will become more prevalent over time* [11].

BUILDINGS AND CITIES: INSULATION

In thermodynamics it is known that heat flows from warmer areas to cooler areas. One way to prevent this movement of thermal flow is to install a barrier—insulation. During the summer, hot air finds its way to infiltrate the indoor spaces that are being cooled. In the winter, the warm indoor air finds its way to cooler spaces through gaps around doors and windows or through chimneys in a house. When this occurs, more thermal energy is consumed during the winter months and more cooling energy is consumed during the summer months in order to maintain the desired comfort level.

The answer to maintaining the desired temperature without consuming additional energy is to create a building envelope with insulation. Insulation effectiveness is measured for its thermal resistance and given an R -value. The higher the R -value, the more effective the insulation, which may vary by type, thickness, density, and where it is installed. There are many types of insulation with the most popular being fiberglass, while others include

mineral wool (not really wool), polystyrene, and even recycled paper as a cellulose insulation.

When building a new house, adding the insulation with the highest *R*-value and adding highly insulated windows will reduce the energy costs. While this will add to the cost of the house, a smaller furnace and air-conditioner will offset the extra capital. But then, the operating cost will be reduced as well as the emissions making it very economic.

The extreme in insulation is a new building standard in Germany called Passive House, which was created in the early 1990s. This approach focuses on creating an airtight envelope for the building in order to separate it from the outside on all four sides plus top and bottom. This building is so hermetically sealed that air cannot escape to the outside. Some of these homes are so well insulated that they can be heated with a hair dryer. Basically, Passive House sets a high bar for insulation.

Drawdown summarizes the impact of insulation as follows: *Retrofitting buildings with insulation is a cost-effective solution for reducing energy required for heating and cooling. If 54% of existing residential and commercial buildings install insulation, 8.3 gigatons of emissions can be avoided at an implementation cost of US\$3.7 trillion. Over thirty years, net savings could be US\$2.5 trillion. However, insulation measures can last one hundred years or more, realizing lifetime savings in excess of US\$4.2 trillion [12].*

LAND USE: TROPICAL FORESTS

Tropical forests are defined as those that lie between 23.5° north or south of the equator. At one time, these forests covered 12% of the world's land-masses but now cover only 5%, primarily due to clearing, fragmentation, degradation, and depletion of plants and animals of that region. Fortunately, however, restoration is taking place both passively and intentional. This regrowth is sequestering about 6 GT of carbon dioxide per year.

According to the World Resources Institute (WRI), 30% of the world's forestland has been destroyed completely, while another 20% has just been degraded. More than 4.9 billion acres worldwide offer opportunities for restoration—an area larger than South America [13]. Restoration means bringing the damaged forests and ecosystem back to its original form and function. As the population continues to grow, these forests must also provide a social and economic function for the nearby communities. Forests are more than trees and fundamental for food security and improved livelihoods. They contribute to resilience of communities by regulating water flows, providing food, wood energy, shelter, fodder, and fiber, and

generating income and employment as well as harboring biodiversity. Furthermore, forests support sustainable agriculture and human well-being by stabilizing soils and climate [14].

Fortunately there is a global movement to restore the forests. In 2011, the Bonn Challenge set an ambitious goal to restore 370 million acres worldwide by 2020 [15]. In 2014, the New York Declaration agreed with the Bonn Challenge and added a cumulative target of 865 million acres to be restored globally by 2030. Under this scenario, a total of 12–33 GT of carbon dioxide would be removed from the atmosphere.

The cost to restore these forestlands can vary from US\$400 to US\$1200 per acre. These costs do not include the land cost and vary depending on the starting point, what will be grown, the size of the project, and the climate conditions. To restore 865 million acres would therefore cost between US\$350 billion and US\$1 trillion. However, according to estimates by the International Union for Conservation of Nature, this land could generate US\$170 billion/year in net benefits of improved yields and sequestering 1.7 GT of carbon dioxide [16]. In addition, the African Forest Landscape Restoration Initiative is committed to doing something similar for 247 million acres [17], while Brazil is committed to restoring 29 million acres of forests.

Drawdown summarizes the impact of tropical forests as follows: *In theory, 751 million acres of degraded land in the tropics could be restored to continuous, intact forest. Using current and estimated commitments from the Bonn Challenge and New York Declaration on Forests, our model assumes that restoration could occur on 435 million acres. Through natural regrowth, committed land could sequester 1.4 tons of carbon dioxide per acre annually, for a total of 61.2 gigatons of carbon dioxide by 2050* [18].

LAND USE: TEMPERATE FORESTS

Temperate forests, representing almost 25% of the world's forests, are defined as those that lie between 30° and 50°–55° latitude, mostly in the Northern Hemisphere. Almost all of these forests have been subject to deforestation since the beginning of history. These forests of about 1.9 billion acres are very resilient, and as a result of improved forest management and conservation, they are on the rise and are actually net carbon sinks of about 0.8 GT/year. According to WRI, there may be another 1.4 billion acres that can be restored.

There are many more opportunities for restoration of temperate forests as noted by the global Atlas of Forest and Landscape Restoration Opportunities

[13], a collaboration between WRI, the International Union for Conservation of Nature, and South Dakota State University. Some opportunities noted are in the eastern half of the United States and Continental Europe. There is already a start in the restoration of these forests in the Eastern United States just in the last 20 years, and it has resulted in a 33% increase in the carbon sink.

Just like the food sector where reducing waste is much more productive than generating switching to a plant-rich diet, preventing loss of forest is always better than trying to restore it. A restored forest, while being very beneficial, will not recover to its original state in terms of biodiversity and structure.

Drawdown summarizes the impact of temperate forests as follows: *We project that temperate forest restoration will expand to an additional 235 million acres through natural regeneration. Though this is much lower than the available area for tropical forest restoration, it still sequesters 22.6 gigatons of carbon dioxide by 2050* [19].

TRANSPORT: ELECTRIC VEHICLES

A prototype for the first electric car was built in 1828, but the first electric car in the United States was probably not built until 1891 [20]. Later on, Thomas Edison hired Henry Ford, and while Edison was working on building an electric car, he convinced Ford on pursuing an internal combustion gasoline vehicle. With improved roads, Americans started traveling further and an electric car did not make much sense because of its limited range.

Today, two-thirds of the world's oil is consumed by trucks and cars and account for 23% of all carbon emissions, second only to electricity generation. This is the result of about one billion cars on the road of which one million are electric vehicles (EVs). But as the developing nations become more industrialized, the number of vehicles in the world could reach two billion by 2035. And there is no question that the percentage of EVs will increase significantly. In fact, today there are close to 400,000 electric busses operating in the world and 99% of them are in China [21].

The growth of the EV market will continue rapidly because EVs are simpler to make, require less maintenance, have less moving parts, and do not require gasoline. The major resistance to growth is the question about range and the availability of battery charging. A typical hybrid can go 40 miles before switching to the gasoline engine, while the better EVs may have a range of 200 miles. In terms of carbon dioxide, EVs emitted 50% as much as the gasoline vehicles if the charging of the battery is off the grid. If the battery is charged from solar energy, the carbon emissions are reduced by

95%. The major question as to what impact the EVs will have depends on the growth projections.

Drawdown summarizes the impact of EVs as follows: *In 2014, 305,000 EVs were sold. If EV usage rises to 16% of total passenger miles by 2050, 10.8 gigatons of carbon dioxide from fuel combustion could be avoided. Our analysis accounts for emissions from electricity generation and higher emissions of producing EVs compared to internal-combustion cars. We include slightly declining EV prices, expected due to declining battery costs* [22].

TRANSPORT: SHIPS

Most of the global trade takes place via shipping, which accounts for 80% of the total. The only other alternative is by plane, which emits close to 50 times more carbon dioxide to transport the same quantity of goods for the same distance. Because the “world is flat” as described by Thomas Friedman [23], increase in world trade could be as much as 250% by 2050. Even though carbon emissions from shipping has not been as high priority as from vehicles, its market size plus its potential growth is drawing more attention.

Technical innovations are being recommended to make shipping more efficient. In 2011, the International Maritime Organization, a UN agency, established the Energy Efficiency Design Index (EEDI) [24] in order to make shipping safer and more efficient for new ships. It is similar to fuel-economy standards for cars, and the standards increase over time. Similar standards are being applied to the old shipping vessels.

One method for reducing fuel consumption and thus carbon emissions is by slowing down the ship, something that has become so common that it has been given a name—“slow steaming.” Adding this to improved design, new technology, improved maintenance, and operations, the efficiency of ships could reduce emissions by 20–40% by 2020 and 30–55% by 2030. However, emissions from marine vessels are not included in the global climate change agreements, and therefore no emission targets have been established. Also, currently there is no cap for carbon emissions even though this industry is expected to generate 17% of global carbon emissions in 2050.

Drawdown summarizes the impact of ships as follows: *With an efficiency gain of 50% across the international shipping industry, 7.9 gigatons of carbon dioxide emissions can be avoided by 2050. That could save US\$44 billion in fuel costs over thirty years and US\$1 trillion over the life of the ships* [25].

TOP 20 SOLUTIONS

What are presented in the previous sections are the top two solutions in each of the seven sectors. Table 11.2 shows a list of the top 20 solutions from all the sectors. You will note that 13 of the top 20 are from the food sector (8) and the energy sector (5).

TABLE 11.2 Drawdown: top 20 solutions

	Solution	Sector	Total atmospheric CO ₂ -Eq reduction (GT)	Net cost (billions US\$)	Net savings (billions US\$)
1	Refrigeration	Materials	89.74	N/A	–US\$902.77
2	Wind turbines (onshore)	Energy	84.60	US\$1225.37	US\$7425.00
3	Reduced food waste	Food	70.53	N/A	N/A
4	Plant-rich diet	Food	66.11	N/A	N/A
5	Tropical forests	Land use	61.23	N/A	N/A
6	Educating girls	Women and girls	59.60	N/A	N/A
7	Family planning	Women and girls	59.60	N/A	N/A
8	Solar farms	Energy	36.90	–US\$80.60	US\$5023.84
9	Silvopasture	Food	31.19	US\$41.59	US\$699.37
10	Rooftop solar	Energy	24.60	US\$453.14	US\$3457.63
11	Regenerative agriculture	Food	23.15	US\$57.22	US\$1928.10
12	Temperate forests	Land use	22.61	N/A	N/A
13	Peatlands	Land use	21.57	N/A	N/A
14	Tropical staple trees	Food	20.19	US\$120.07	US\$626.97
15	Afforestation	Land use	18.06	US\$29.44	US\$392.33
16	Conservation agriculture	Food	17.35	US\$37.53	US\$2119.07
17	Tree intercrop- ping	Food	17.20	US\$146.99	US\$22.10
18	Geothermal	Energy	16.60	–US\$155.48	US\$1024.34
19	Managing grazing	Food	16.34	US\$50.48	US\$735.27
20	Nuclear	Energy	16.09	US\$0.88	US\$1713.40

COMING ATTRACTIONS

In addition to the 80 top solutions for reducing carbon emissions by 2050, the authors of *Drawdown* present about 20 solutions that are in their early stage of development. These solutions may someday be very relevant while, on the other hand, may never come to be.

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Adapting and Building Resilience to Climate Change

The earth's atmosphere contains greenhouse gases (GHGs), and their ability to trap heat within the atmosphere is responsible for the ambient temperatures under which communities and economies have developed the last few thousands of years. Prior to that there was an ice age. And before that there was again a different climate to the one today. The earth's climate has always undergone some level of change throughout the geological ages. Large-scale climatic changes in the past have been caused by various factors including solar activity (radiation), variations in earth's orbit, volcanic eruptions, and even meteorites that may have caused the extinction of dinosaurs. However, the observed changes happening in our planet the last few decades have been overwhelmingly due to anthropogenic global warming associated with increased concentration of GHG emissions in the atmosphere after the industrial revolution. GHGs such as carbon dioxide and methane cause extra heat to be trapped and retained on earth (a greenhouse effect) that would have been otherwise deflected back into space. Human activities are estimated to have caused approximately 1.0 °C of global warming above pre-industrial levels, and global warming is likely to exceed 1.5 °C between 2030 and 2052 if it continues to increase at the current rate [1].

Even an apparently small increase of 1.5 °C can affect atmospheric circulation, and small variations in average conditions can have a big influence on weather extremes such as droughts, floods, and hurricanes, as have been witnessed around the world over the last decade and also cause dramatic large-scale climate shifts in the long term. Climate change poses significant risks to health, livelihood, food security, water supply, human security, and economic growth.

Practical Sustainability Strategies: How to Gain a Competitive Advantage,
Second Edition. George P. Nassos and Nikos Avlonas.

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Furthermore, it poses complex challenges because of the uncertainty associated with the exact timing, magnitude of projected changes, and the interconnectedness between risks and impacts in the modern globalized economy.

INTERCONNECTEDNESS OF CLIMATE RISKS IN A GLOBALIZED WORLD

In a globalized world, everything is becoming more connected including risks from climate change. Take, for example, the agriculture sector that is being impacted by persistent extreme weather such as flooding and droughts. Failure to adapt to such events can cause food shortage and water supply crises and subsequently increase the risk for other events such as conflict and unmanaged migration. Researchers have shown that an extreme drought in Syria between 2006 and 2009, the worst in the country in modern times, was most likely due to climate change, and the drought was a contributing factor in the violent uprising that began there in 2011 with all the domino effects that have caused in the Middle East region and even to Europe in the large-scale migration of Syrians [2]. When risk interconnectedness is “unpacked,” it becomes evident that there is a lot more at stake than just what is obvious.

Furthermore, as business becomes more globalized, interconnected, and interdependent, both vertically (throughout the value chain) and horizontally (companies among same sector), this can further exacerbate business risks. One of the best examples to showcase this domino effect was the 2011 floods in Thailand, which caused a number of computer hard disk (HD) drive manufacturers to shut down. Aside from businesses like Western Digital and other HD manufacturers who were directly impacted by flooding, the domino effects of this incident resulted in a global shortage of HD drives impacting the sector’s customers (the computer manufacturer Hewlett Packard lost approximately US\$2 billion) and employees (NEC slashed 10,000 jobs worldwide amid performance fears of its platform business that was hit due to flooding), all the way down to consumer and business level (adding US\$5–10 to the cost of each hard drive and ultimately computers) [3].

One way to analyze these interconnected risks and opportunities is by taking systems thinking approach (see Chapter 7). Systems thinking is a “dynamic thinking” approach, which promotes the understanding of risks and their boundaries within a system, how they influence one another over time and space, and what patterns emerge and manifest within a value-generating system. It encourages one to “push back” from events and

points in time to see the pattern of which they are a part. The assumption is that one will be capable of dealing with a dynamic, rather than only a static, view of reality and thus manage risks better.

Interconnected risks and systems are usually beyond one company's capacity to influence or control, but systems thinking combined with approaches like scenario planning offers management a means to entertain possibilities and attempt to both strengthen resilience and reduce the impact cost where this is needed.

LARGE-SCALE CLIMATIC SHIFTS

The risks from long-term climate change are not as easy to identify as those from extreme weather events because they are not one of big memorable events but can cause whole system shifts, that is, disrupt the sensitive equilibrium of the earth's biosphere with incalculable consequences. There is an anecdote about boiling a frog, which is useful in explaining our collective fundamental difficulty in reacting to significant changes that occur gradually. As the explanation goes, when a frog is placed in a pot of boiling water, it quickly jumps out of harm's way. However, when the frog is placed in a pot of water at room temperature and the temperature is slowly turned up over time, the frog adjusts its body temperature and by the time the temperature becomes deadly, it is too late for the frog to escape. Jumping into boiling water can be compared with experiencing extreme weather events—generally easy to detect and relatively easy to enhance resilience. On the other hand, the impact of incremental changes in temperature is not necessarily easy to identify, and it is important to think about how long-term climate change might affect us before getting too comfortable in the water and becoming less attuned to the changes taking place gradually.

In this context, forward-thinking businesses are beginning to identify their exposure not only to extreme events but also to gradual climate change. They are looking to understand the financial implications, enhance the resilience of their operations, and readjust their core business strategy going forward. Such companies already identify innovative advantage and competitive edge through adopting strategies, which provide solutions to real problems posed by climate change, in the same way that digital technology has been the driver for consumer growth in the last decade. For example, companies are already moving to take advantage of significant market opportunities that exist in providing solutions to water scarcity (e.g., desalination, water efficiency, water distribution, manufacture of pipes/pumps, wastewater treatment projects, and irrigation projects) to key industries such as power, steel, pulp and paper, and cement. Traditional risk management techniques can

help manage and adapt to the operational risks amplified by the changes in the frequency and severity of extreme weather events.

Long-term climate changes tend to be systemic, that is, influence many parts of the environmental, business, and societal systems with a wide web of impact connections and can bring risks and opportunities of a different scale. Resilience to risks or positioning to enhance opportunities can be built by taking a more holistic systemic and scenario planning approach to managing climate change and enhancing traditional risk management approaches.

BUILDING RESILIENCE

In order to understand the impacts of climate change and build resilience, it is first necessary to appreciate both the biophysical impacts of altered weather patterns and the operational, social, and institutional implications of these changes. Climate change impacts are seldom discrete as described in the case study below.

CASE STUDY: MINING COMPANY

A mining company operating in a water-stressed South American Country pumps water from a desalination plant at the coast to the mine in the mountains. Changes in precipitation could cause landslides and compromise the integrity of the water pipeline. To secure water supply, the site applies for and secures a temporary groundwater abstraction license. This source is already under pressure due to historical water stress conditions, and this is exacerbated by reduced groundwater recharge from rainwater due to changing precipitation patterns—the cause of the problem from the start. Abstraction of groundwater for industrial use at the mine further exacerbates drought in the area and has unintended and unforeseen impacts on local ecosystems and farming activities, erodes local community support, and seriously jeopardizes the project's license to operate.

Business leaders are becoming increasingly more adept at thinking commercially about the system within which they are operating. What this means is framing the systems boundary, identifying the critical interacting elements in the “business system” and how they relate with and affect each other, and keeping the big picture in mind without reducing it down to its component parts in such a way that ignores the interconnectedness. A key aspect to all this is the ability to see relationships between components where once they might have only seen individual parts and to be able to identify the critical relationships that cause the significant positive or negative impacts. It also

means being able to look at problem situations and knowing how to resolve them from a variety of points of view and using different system approaches in combination. This approach encourages a kind of creativity to managing a problem. A certain tolerance to messiness, uncertainty, and ambiguity is needed in order to avoid jumping to solutions too quickly. Spending time in ambiguity and confusion tends to lead to more elegant, resourceful, and effective responses. Mapping systems and identifying, understanding, and mitigating potential risks early give business leaders the opportunity to refashion their operations, reposition their products and services, and ensure an increasingly sustainable market position.

CASE STUDY: EXTRACTIVES SECTOR PARTNERS WITH GOVERNMENT AND COMMUNITIES TO FIGHT MALARIA IN MOZAMBIQUE

Changing temperatures and precipitation patterns can present increased health risks with knock-on effects on the extractives sector. Climate change could cause shifts in disease distribution, for example, in areas at risk of malaria [4]. BHP Billiton's Mozal aluminum smelter operation in Southern Mozambique is increasingly threatened by endemic malaria. The disease affected the facility's performance through absenteeism and low staff morale. Moreover, Mozal became an unattractive destination for skilled employees. The company realized that limiting malaria programs to the Mozal site itself by spraying its facilities and controlling nearby breeding areas would have little effect. Hence, it became a partner in the Lubombo Spatial Development Initiative (LSDI), a tripartite governmental program to develop the Lubombo region and decrease malaria prevalence. Within the LSDI, BHP Billiton was more capable of implementing an effective malaria control program in the area around its Mozal operations. BHP Billiton's partnership with the LSDI and actions taken to decrease the prevalence of malaria among employees have improved the company's productivity and supported the health of the community.

Climate risk management can be embedded within a systems approach to scenario planning and organizational management aimed at enhancing the resilience and, therefore, profitability of companies in the face of climate change. Scenario planning recognizes that many factors may combine in complex ways to create future scenario stories both plausible and surprising. Scenarios that take into account climate change and how it might interact with other risk drivers that are not necessarily solely climate-related can be extremely useful, for example, scenarios that take into account for

social, political, economic, and regulatory issues relating to water scarcity, temperature rise, etc.

THE STRATEGY DEVELOPMENT PROCESS

The aim of an adaptation strategy is to move toward embedding a systematic approach to climate risk and adaptation into business practices. How an organization chooses to do this depends on the nature of the risks identified, the business case for adaptation, and the company values/ambitions. An adaptation strategy will provide clarity and direction for the steps to address climate risks and opportunities.

The strategy development, as it relates to climate change, involves the exploration of answers to questions such as the following:

- Where are we as a business today?
- Where do we want to be in X years?
- What are the key climate change issues that our business (or project) strategy needs to address now and in the future?
- What capacities do we need to build and how can we best position ourselves to take effective action?

When developing a strategy, it is critical to “get the whole system in the room” in order to understand it, engage with it, and see how and where it is relevant for the organization. Rather than seeing strategic planning as the preserve of the “numbers people,” collaborative tools for strategy making become essential here in order to bring together all those affected within the system and on the systems boundaries, and to engage them in understanding the nature of the situation and its implications.

Data and analysis come to the process as inputs, but the key is bringing together different functional disciplines, stakeholders, and generations in a collaborative process to interpret the issues through multiple lenses and to obtain the deepest insights and uncover solutions to which all players can commit.

The strategy that is developed then not only is a strategy for the company adapting to climate change but also impacts the broader business environment and system in which the company exists. It recognizes interdependencies between players, climate, and time and puts a value to these.

Finally, the strategy looks to ensure that the organization maximizes opportunities and enhances resilience in the short, medium, and long-term depending on the nature of the impacts with which it is dealing. This

will help ensure that future expenditure and planning take place with an “adaptation lens” and that consideration of climate risk and adaptation is fully integrated within business processes.

Developing an adaptation strategy this way requires leaders and managers to have a level of fluency in systems thinking or the willingness to acquire this mindset and associated skills. The mindset and skills can be seen as a distinct portfolio of:

- Recognizing systems boundaries, perceiving interdependencies in and between actors, and going deep, not working with mere effects but with systemic root causes
- Appreciating the dynamics that underpin complex systems and accepting that building resilience to climate change will involve a large amount of complexity, ambiguity, and uncertainty that we need to learn how to be comfortable
- Seeking out the points of greatest influence and leverage within a system and designing and implementing interventions at these points
- Staying alert to the unintended, as well as the intended consequences, and responding to these in a way that optimizes outcomes.

This portfolio can enlarge the traditional expectations of business leaders around vision and strategy, planning, and execution. It can enable leaders to navigate more effectively through increasingly volatile, uncertain, complex, and ambiguous times and find the right path to address the significant problems that face, not only their business but also their broader communities and the earth. Using the systems approach as a leadership practice as well as a tool to understand how climate change might influence the relationships within the business system is the key element of the strategy making process and will be what ensure its implementation and sustainability. So, a company should seek out that its future managers and leaders participate in the strategy development process, because it is these individuals who need to:

- Own it, be committed to it, and take it forward
- Understand the climate-related risks and opportunities from a systems perspective, such that this becomes integrated into their way of understanding the business context in which they are operating and that they act from this standpoint
- Develop and deepen relationships with other stakeholders and build the reservoirs of trust that ensure effectiveness. Trust is the glue

that ensures that when unforeseen things happen, stakeholders draw together, rather than pull apart, and find the right path forward together.

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Emergent Technologies for Adaptation

There has been an increased attention toward sustainability-oriented innovation and research studies the past 10–15 years. The sustainability challenges from energy supply to climate change and huge financial needs in terms of infrastructure renewal and expansion are coupled with strong path dependencies [1, 2].

Already established technologies are connected with rigid user practices, lifestyles, and business models that make it difficult for radical changes in those areas, as they do not cope with the prevailing sustainability challenges. Emerging technologies, which are based on biotechnology, nanotechnology, and information and communication technology (ICT) systems can shape the future of a variety of industries and disrupt entirely others. In many cases, these technologies will define and restructure these sectors. In return these technologies will spark debates and changes in regulatory frameworks and corporate models.

According to the McKinsey report on sustainability [3], deploying current innovative technologies could improve resource productivity in energy, land, water, and materials up to 30% of total 2030 demand. The majority of these opportunities occur in developing countries and could save US\$2.9 trillion in 2030. Emergent technologies such as Artificial Intelligence (AI), Robotics, Blockchain, and Augmented Reality (AR) provide certain operational efficiencies that can transform globally the manufacturing, industrial processes, and labor. Since businesses recognize that it is impossible to meet the world demands in products and series in a linear increase in production and consumption, new innovation business models should be adopted leveraging emerging technologies. New digital technologies play an important role in

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greening the supply chain and building a circular economy by focusing on eliminating waste and enabling the use of natural resources. In practical terms, a sustainability strategy combined with the use of emergent technologies can drive solutions that would benefit both the environment but also business growth. Visualization technologies for transparent supply chain and sustainable source of raw material, capturing information of the produce life cycle to reduce waste, better data capture of assets for manufacturers, and using such technologies to visualize the carbon footprint of the operational processes to the business employs are only a few examples where innovative technologies can provide solution and be integrated to the strategy development of any business. Despite these examples that would transform the business processes, there are also simple examples that could benefit other sectors as well. For example, technologies might help to decrease inequality between developing countries by giving the opportunity to access systems such as e-health or online education. In a similar manner, Blockchain has the capability to improve sectors that range from education and energy to trade and food safety.

Even though the relation between digital technologies and individual Sustainable Development Goals (SDGs) has been discussed through the recent years, there has been a growing sense that the influence of emergent technologies can be more profound as the tremendous speed of innovation seems to cause social and economic disruptions that are not yet clear. As said earlier, new policies and measures would need to be employed to see this change into a direction that will not cause counterproductive impacts on society. For instance, rapid improvements in the wealthy countries and further isolation of the most vulnerable in society.

In an attempt to consider the challenges and the ways that innovation can affect SDGs, this chapter discusses the deployment of emergent technologies in those areas. It will present case studies and opportunities where the adoption of emergent technologies can drive sustainability.

NO POVERTY AND INNOVATION

The advancements in electronics and nanotechnology could provide systems that if they are implemented appropriately can reduce poverty on many fronts. According to Geldof et al. [4], there are three main areas where research–innovation technologies can help mitigating poverty. These are the following: energy generation, agricultural productivity, and provision for clean water. For example, in countries where pollution is a major problem and the quality of drinking water very low, nanotechnology can provide a solution. Currently, at Rice University in the United States, they are

developing a system solution for polluted ground water that exists in areas such as Bangladesh, China, India, and Nepal where arsenic is one of the core pollution causes. The system employs nanoparticles that can be bind to arsenic in water and being removed using magnet. This water-purification agent is very easy to prepare, as the only ingredients it needs is iron oxide particles and olive or coconut oil.

There is a range of research projects that are being funded by the Engineering and Physical Sciences Research Council (EPSRC) in the United Kingdom that aim to mitigate poverty in developing countries. University of Leeds has developed a sensor, which can gather data on environmental conditions, temperature, humidity, and light in agricultural areas. The purpose of this sensor is to inform the farmers in developing countries with necessary information to maximize their crop production. To reduce the harmful bacteria in the water, a blue light-emitting diode based on gallium nitride is being developed by the University of Cambridge to improve the water quality. Another initiative led by the University of Nottingham designed a smokeless cooking stove that can be fueled by burning wood or animal dung to tackle the problem of smoke produced by wood-burning stoves that cause many deaths due to the smoke produced in Asia and Africa. Employing simple physics, this system converts the heat that is being produced to acoustic energy and then into electricity.

While digital innovation and emergent technologies can contribute to reducing poverty in rural areas, it is not the magic wand either. These technologies cannot help with issues such as access to and ability to manage natural resources, for example, access to lands and other assets, labor. Further institutional and infrastructure development can be combined with policies and strategies for agriculture and rural development. The ultimate goal is to create mass-producing stoves with the use of low cost local materials to reduce the nation's carbon emissions.

ZERO HUNGER AND INFORMATION AND COMMUNICATION TECHNOLOGY (ICT) PLATFORMS

The 2018 Global Report on Food Crises argues that 124 million people in 51 nations experience face-crisis level hunger. For instance, during 2017 northeastern Nigeria, Somalia, Yemen, and South Sudan faced one of the worst food crises with more than 30 million people needing emergency aid. As the global population increases, the food shortage is likely to be bigger. To keep up with the need for food supply, emergent technologies need to be harnessed.

The new platform technologies including ICT for the agricultural sector, biotechnology, and nanotechnology have the potential to provide better market access as well as helping local farmers by empowering them. As there are many risks that smallholder farmers have to face before, during, and after production, solutions where information, which boosts their product, is delivered through smartphones can enhance the crop production. The mobile phone services are on several fronts, ranging from providing market and price information to knowledge sharing, insuring crop production to monitoring children's nutrition status. In the application of biotechnology, biofortification is among the most cost-effective ways of improving nutritional outcomes.

Project Healthy Children partnered with Telecommunications Company Vodafone aims to combine state-of-the-art dosifier technology and internet of things (IoT) systems at small mills throughout Africa to supply flour to millions of African people [5]. They have already rolled this technology to mills in Tanzania, Rwanda, Kenya, Malawi, and Mozambique with potential to extend beyond that.

Even though innovations have contributed to reducing the challenges of hunger and poverty, there is the problem such as "hidden hunger," which refers to the chronic deficiency of essential micronutrients [6]. To address those challenges innovation technology on its own will not achieve much. There should be a diffusion of technologies, policies, knowledge, and institutional arrangements to direct these complex challenges.

GOOD HEALTH, WELLBEING, AND VIRTUAL REALITY

Virtual reality (VR), even though is not considered to be an emergent technology since it exists many years now, has given the option to develop innovative design paradigms that have an impact on health and improve the quality of life for many individuals. It has been used quite a lot for training surgeons and medical specialists on specific processes much quicker and more efficiently. Field studies have shown the great effect that VR applications can have on well-being for older people and make them feel less isolated from social life [7]. In another recent study investigating whether VR can prompt greater empathy, an experiment with randomly assigning participants viewing a documentary of a young girl in a refugee camp either in a VR or a control two-dimensional format, showed that VR has greater potential to influential empathy [8].

There are numerous examples showing how emergent technologies have helped to improve our health and well-being; however the digitization era has also caused some issues. The excessive use of the Internet, the

interconnection of social media, and the vast gathering of information have caused mental health issues that did not exist before. Awareness, values, and knowledge should be transferred through societies to help preserve our physical, mental, and emotional state not to be transformed by the advancements of technologies.

QUALITY EDUCATION, MIXED REALITIES, AND 5G

New generation of students expect to be engaged in education with participatory, interactive, and sensory-rich activities. Enhancing student's skills, application, conceptualization, and visualization has become a standard practice in education. Technologies such as 3D printing (3DP), AR, VR, and mobile bring your own devices (BYODs) are some examples of disruptive technologies that have started to enter the classroom to shift the traditional face-to-face didactic lectures to a learning mode through different modes of delivery.

For instance, as Keefe and Laidlaw [9] suggest, VR makes it possible to take students outside the classroom into a virtual environment where science, technology, engineering, and math (STEM) concepts such as construction and painting could be much more realistically be presented. Toward a step further, Oner et al. [10] present the use of 3DP with secondary students, incentivizing students to model their designs and print them. The latest technology in AR/VR enables students to interact with visualization by navigating in comparative mixed reality systems, which in turn helps students to translate the 3D spatial objects in their head [11].

Although visualization systems are not one size fits all tool, the first results from research studies show promising results, and there is a potential of enhancing classroom practices.

Apart from the visualization tools, another cutting edge technology can help creating powerful education platforms to engage students. The fifth generation of wireless technology, 5G, can provide high speed and low latency wireless technology. NYC media lab [12] and universities all over the world are looking into ways of taking advantage of this technology to improve education by teacher training and student engagement with diverse learning needs.

Several attempts with AI software, mixed reality systems and IoT platforms have occurred over the recent years that can help students better inform their research and learn in a more engaging and fun way. We are not far away from embracing these technologies to build a sustainable education system. Unfortunately though, developing countries are excluded from this growth of technology, which can reshape dramatically the

education. Despite this, the usage of ICT in developing countries is not restricted and some countries have already adopted it [13]. For instance, the Minister of Education in Lebanon has already distributed 400 computers for public schools that are connected to the web through broadband [14]. There are many barriers that affect the implementation of cutting edge technologies in developing countries. As Khan et al. [15] suggests drawing from the example of Bangladesh, some examples of barriers are the lack of resources, the lack of necessary steps to increase enrolment of girls at schools and limited commitment and funds to implement those systems in schools.

GENDER EQUALITY AND MOBILE TECHNOLOGIES

Gender is one of the most challenging areas for generalization as it varies across countries and specific ICT technologies. There has been evidence showing that access to technologies such as mobiles displayed less of gender bias. Wherever there is available access to the Internet, it has an influence on the autonomy and social connection of (young) women. In some cases, mobiles are becoming vital for women's livelihood strategies. Two examples are in Ghana where poor women are engaged in trade and in Jamaica where a single parent depends on social networking of the resources [16].

There is a growing trend of organizations to build the IT skills of young girls and to make new ICT systems accessible to them. In Afghanistan, the United Nations Development Programme (UNDP) in relation to Women's Affairs Ministry has initiated computer training that targets women. Another program called Women's Information Resource Electronic Service (WIRES) aims to build the skills of women entrepreneurs in small-scale businesses in Uganda. This program allows women to access information on markets, prices, and trade support services [17].

Nonetheless ICT technologies have the ability to amplify women's voices and help publicize their experiences and perspectives. However, even though steps have been taken forward, even in some developing countries to empower women, we still have a long way ahead of us but cutting-edge technologies have given the power to build movements, allow organization networks, and create technology-based solutions to fundamental issues such as violence and political empowerment.

CLEAN WATER NANOTECHNOLOGY—SANITATION AND SMART CITIES

There are 600 million people that do not have access to safe drinking water. For that reason, there is a long list of innovations to tackle this challenging

issue. Researchers have found a solution for water purification using nanotechnology. This technology removes microbial and bacterial using composite nanoparticles [18]. Even though many of nanotechnologies are still in the research stage, there are two that have made their way into pilot testing and commercialization. For example, nanotechnology enabled membranes, through which the water passes is being filtered and purified. Despite the challenges with these technologies such as high cost and technical hurdles, these remain to be temporary, and with the collaboration between government and research, these barriers will be controlled.

Other technologies that combine IoT and AI can transform the provision and maintenance of sanitation services in cities. For instance, the Toilet Board Coalition [19] is pioneering work around the “digitization of sanitation” in cities such as Pune, India. Organizations and researchers try to integrate sensors and Wi-Fi into toilet networks in order to produce data to public health as well as the quality of the maintenance systems. Such “smart” systems may help optimize the routes for waste collection. Again, however, technology on its own can change the situation, further governance frameworks, social acceptance, and investment funds are important to be established for such initiatives.

AFFORDABLE CLEAN ENERGY AND BIOFUELS

Affordable, reliable, and clean global energy system requires a diverse portfolio of low-carbon technologies superior to existing options. Unfortunately, the advancement in technologies is possible, but they need either a fundamental breakthrough in the lab or a demonstration project in the field. For example, the innovation with the catalyst to use sunlight to split water has not yet produced a chemical and efficient solar power for widespread use. Nanoengineered membranes could block the carbon emission in fossil fueled power plants, and solar coatings can enable buildings to produce more energy than they consume [20]. However, the problem lies on the high prices of alternative fuels. A plunge in oil prices makes it very difficult for biofuels companies to compete, and it drives also consumers away from electric cars. However, positive examples such as Iceland that already get a quarter of its electricity from geothermal energy can provide a tremendous amount of data that can change the industry drastically and highlight an alternative way of energy.

DECENT WORK, ECONOMIC GROWTH, AND FUTURE TECHNOLOGIES

It has been suggested that in the near future the job landscape will change dramatically with new jobs that do not exist at the moment. Even though

cutting edge technologies enhance the productivity and pave the way for more opportunities, they also create the challenges that may diminish some areas of work. Indeed, in the past decade the computerization has led to the extinction of some jobs such as bookkeepers and cashiers [3]. For instance, autonomous vehicles are an example of how manual tasks in transport can be automated [21]. A study examined how susceptible jobs are to computerization showed that 47% of total US employment is at risk among 702 occupations.

On the contrary, re-organizing the economy around technologies has huge benefits of global economy. To minimize the disruptions and maximize the benefits, policies on digital data, labor policies, and competition between emerging realities should be incorporated. Several jobs such as data security and transactional services would be very popular and will open new avenues for employment.

INDUSTRY, INNOVATION, INFRASTRUCTURE, DRONES, AUGMENTED REALITY, AND SMART CITIES

Traditionally, infrastructure projects were considered labor intensive and not embracing the new cutting edge technologies. Recently though the reality has been changing, a wide range of emergent technologies are transforming the way infrastructure is built and how the industry generally operates. For instance, the use of drone technology enables easy supervision of investment programs, overlooking the progress as well as managing maintenance of the existing infrastructure systems.

AR provides the option of overlaying information and visuals on someone's physical space, which can serve both the workers and the customers. The use of smart glasses can accelerate and improve the quality of work for any employee. Construction companies have already embraced this technology to view plans and development strategies. An AR technology can also change the infrastructure. For example, use AR to help users navigate their way in an airport to avoid congestion in a city.

Smart cities that monitor and integrate conditions for all its infrastructures such as bridges and tunnels and will also monitor the security aspects are some of the goals to mitigate the issues that have been created by the urban population growth [22]. In combination with smart cities, both IoT and industrial IoT (IIoT), which is a subset used in the manufacturing and industrial sectors, open the way for a future in this industry where the availability of data from sensors can have a great impact. The opportunities that are created by IoT including remote monitoring, real-time tracking and smarter decision-making can deliver next level services in this industry.

REDUCING INEQUALITIES

With the appropriate policy context, there is a vast opportunity for cutting edge technologies to reduce inequality. A significant contributor to providing the poorest with access to basic services has been technological advancements. Simple examples would be the access to education and training, even including the world-leading universities (MOOCs) and solar homes have provided access to electricity in developing countries like Bangladesh. In the case of agriculture in many countries, there has been a technology extension to reduce informational inefficiencies and improve the finance programs that address the supply of credit and inform the farmers through smart applications [23].

Through public policies, governments have the power to impact how technology can be directed to address social challenges. For instance, there is a lack of access to financial services by the poor in several countries in the Asia-Pacific region. A digital platform based on biometric identification has been issued to Indian residents. It is linked to a mobile phone and a low cost bank account. The key technologies that it uses are AI, big data, cloud computing, machine learning, IoT, payment infrastructure, and mobile Internet. To motivate citizens to use this platform, the federal government has routed salary payments through this system. Despite its potential, systems like these cause the rise of challenges of privacy, user content, and biometrics, which are still open for debate.

SUSTAINABLE CITIES AND COMMUNITIES, IoT, AND SMART CITIES

As we mentioned earlier, cities are growing exponentially and the challenges remain constant: tackle air pollution, manage carbon emissions, recycle water, provide clean water, and ultimately improve the quality of life. Technological innovation can unlock new potential to target those challenges. Smart cities and big data are in the heart of the systemic innovation and have become a necessity.

Melbourne, Australia, has been one of the first cities to incorporate the technology of IoT early on. As it is part of the C40 cities (climate leadership group), the City Council has conducted various studies over the recent years in the implementation of different IoT applications from microclimate monitoring to infrastructure to service operations in health services [24]. In a similar manner, China has set up an IoT center in Shanghai to study technologies and industry standards. A group of 60 telecom operators (key drivers of the technology) has initiated “Sensing China” project.

This interconnection with fast broadband has a certain benefit for communities as well, especially those that live in rural areas in relation to economic growth and access to public service. For example, the Hyper Village concept, which is part of FutureScapes, Sony Europe, and Forum for the Future's collaborative project [25] that is looking at ways to develop equity between rural and urban communities toward a more sustainable future, is about sharing not only materials but also experiences and knowledge. This will remove the feeling of excursion that a lot of communities feel in the rural areas.

Nonetheless, all these examples have demonstrated the challenges that come with smart cities and IoT, which comes down to data storage and the need for common communication. To further develop those systems though and to identify the solutions for data storage, a business model of smart city has equal importance as innovative technologies.

RESPONSIBLE CONSUMPTION AND PRODUCTION

People need to fundamentally change the way they consume but also produce goods and services to drive economic growth, improved quality of life, and have as much as little environmental impact across the life cycle as possible.

Technology plays a vital role in addressing consumption and production. Smart refrigerators that have incorporated cameras that can be accessed through smartphone apps allow consumers to check their fridge contents before they make a purchase, which can reduce the food waste [26]. Similarly, smart meters and smart buildings help users to consume less energy and enable them to understand the impact of their consumption through graphs and feedback. This technology may not be available to everyone at the moment due to cost; however, in future years the prices of such technology systems will decline and it will be available to low-income individuals. Other emergent technologies such as blockchain along with data analytics will guide the development with tracking systems that can process user data allowing efficiency at the stages of product development [27]. AI enabled robots will be used to recover, reuse, and recycle products. Robotic capabilities include weight measurement and 3D scanning, which helps in sorting out waste more efficiently.

A significant challenge is that the new innovative products are becoming more and more complex, which makes the recovery and reuse even more challenging. For example, electronics are small and sophisticated. Product designers need to consider these challenges but an appropriate infrastructure to support the repair and reuse of them needs to take place.

CLIMATE ACTION—LIFE BELOW WATER—LIFE ON LAND

Technological options for zero or low emission energy technologies are needed to solve the climate problem and ensure that the life on water and on land will continue to exist in a balanced way.

Apart from the policies that have taken place, technologies have been developed technology to tackle issues such as global emissions. For instance, using nuclear power to produce electricity free of carbon emissions is a technology that has been developed but not in stage where it is safe and cost effective. Powering our vehicles using electricity has already been in place. However, more efficient batteries need to be developed with a more efficient charging technology. Regarding the mass production of meat that emits enormous amounts of carbon dioxide, there are alternatives to produce lab-grown meat and meat substitutes. Of course, smart buildings play a fundamental role in enabling users to reduce their energy consumption.

Countries such as Argentina, China, India, and Tanzania among others have started to incorporate drones to monitor sea level rise and off grid solar to deliver lighting. Automated technology that collects data from the oceans can provide information about ocean health and manage ecosystems better. Satellites, sensors, and other similar technology can help to alert of fish stocks and pollution.

PEACE, JUSTICE, STRONG INSTITUTIONS, AND NEW TECHNOLOGICAL CHALLENGES

Transparency of governments as well as of supply chains are a couple of steps that need to be taken to achieve a more stable and strong society. Even though AI and fast Internet connections can help informed decision making and lead to transparency through the use of applications that allow to track processes or be aware of origins within the supply chain, new technologies have also created a new set of concerns. Issues such as cybercrime or autonomy weapons can have a significant impact on the stabilization of peace in societies. This particular sustainable goal reminds us that technology does not only offer solutions but also create challenges that need to be addressed with the appropriate policies and government actions.

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The Circular Economy Through Energy Recovery

The circular economy has been defined (Chapter 10) as take, make, use, and reuse and reuse again and again. This basically refers to products that can be reused or reprocessed into another product. However, a last case alternative is to convert the product to some form of energy. This may be considered as a last resort in avoiding landfilling. The ultimate goal is to move away from waste disposal to the more efficient management of waste. It is important to convert all of the waste to resources—moving from that linear economy to a circular one and that is where waste-to-energy (WtE) initiatives make a difference as WtE is a sustainable, clean energy option for the future.

UNDERSTANDING WASTE MANAGEMENT

There are five options available within the waste management hierarchy: reduce, reuse, recycle, recover, and reject. Of course, the two most ideal scenarios are to reduce and reuse. Both prevent the generation of waste and ensure the direct reuse of waste materials without additional processing. This is closely followed by recycle and recover, the stages of waste management where the value of waste starts to be realized in the form of WtE initiatives. An example of recovery includes the incineration of waste with a high level of energy recovery and the reprocessing of waste into materials that can be used as a solid, liquid, or gaseous fuel, for example, the anaerobic digestion of organic waste to produce fertilizer and biogas, which can be burnt to release the energy. This is the literal example of turning our trash into treasure. WtE initiatives are a sustainable way to address the global waste stream issue, and an excellent example of the circular economy.

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There is also the question as to how much waste is generated and therefore how much is available for management. Most of the waste that is considered available for reuse, recycle, or recovery is the everyday waste (municipal solid waste (MSW)) generated by households and businesses and collected usually on a weekly basis by waste management company. However, a closer look at the total waste generated in the United States shows that the MSW is a very small fraction of the total waste generated, perhaps only 2%. This total waste has been called the gross national trash (GNT) [1] and is depicted in Figure 14.1. From this figure, you will note that the GNT in 1992 was about 12.9 billion tons, while that of MSW was only 180 million tons, or 1.4% of the GNT. So the entire circle on the left represents a mere sliver of the circle on the right.

The entire circle on the left represents a mere sliver of the circle on the right, which consists mostly of manufacturing waste from oil and gas fields, mining waste, and other hazardous and non-hazardous industrial waste.

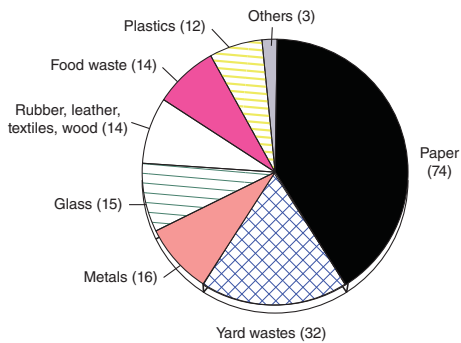
To understand the problem, it helps to understand the bigger picture. The GNT pie includes five major slices as follows:

- The biggest slice, about 59% of the total, consists of industrial wastes from industries involved with pulp and paper, iron and steel, stone, clay, glass, concrete, food, textiles, plastics and resins, chemicals, water treatment, and other processes. All of it results from fabricating, synthesizing, modeling, molding, extruding, welding, forging, distilling, purifying, refining, and otherwise concocting what are collectively referred to as the finished and semi-finished materials of our manufactured world.
- A slightly smaller slice, about 22% of the total GNT, is something called RCRA special waste, referring to a category of wastes defined under the US Resource Conservation and Recovery Act of 1976. This includes medical waste, septic tank pumpings, industrial process waste, slaughterhouse waste, pesticide containers, and incinerator ash.
- A third slice, about 11% of the total, consists of all the different forms of mining waste.
- A fourth slice, about 6% of the total, consists of hazardous waste as defined by RCRA.
- A fifth slice, about 1% of the total, is all other non-MSW waste.
- The final slice of the pie, about 1.4% of the total, is MSW and is represented by the entire circle on the left in Figure 14.1.

A tale of two circles

Municipal solid waste

Total: 180 million tons/year



Gross national trash

Total: 12,915 million tons/year

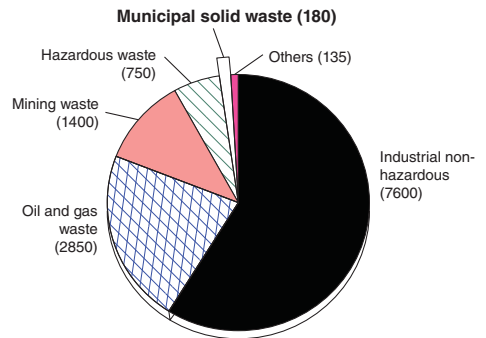


FIGURE 14.1 Gross national trash.

WASTE-TO-ENERGY SYSTEMS

There are a number of different WtE systems as shown in the following text:

- Incineration
- Combined heat and power (cogeneration)
- Combined heat, power, and cooling (trigeneration)
- Pyrolysis
- Gasification
- Anaerobic digestion

Incineration

Incineration is a waste treatment process that involves the combustion of organic substances contained in waste materials. Incineration and other high-temperature waste treatment systems are described as “thermal treatment.” Incineration of waste materials converts the waste into ash, flue gas, and heat with the heat being the most important component. The non-organic components of the waste stream form an ash consisting of the inorganic metals. The ash is deposited at the bottom of the combustion unit, or some of it may be dispersed in the atmosphere as fine particulates in the flue gas. In some cases, the heat generated by incineration can be used to generate electric power.

As seen in Figure 14.2, the waste collected from the source is deposited in a bin (1) from where it is picked up and fed (2) into a combustion

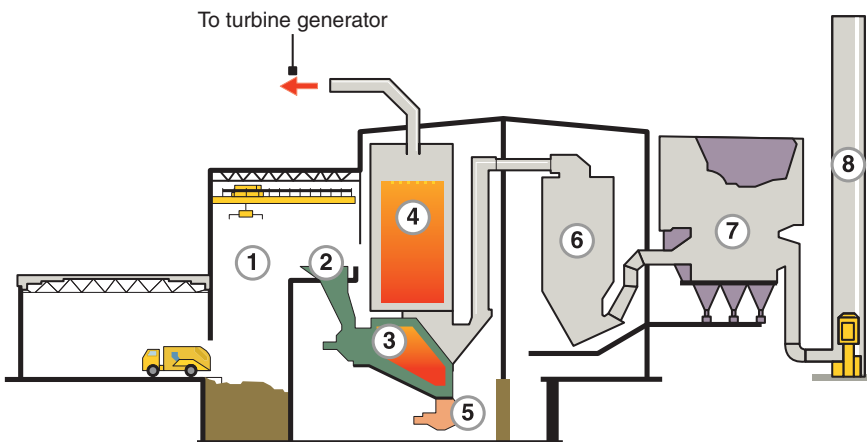


FIGURE 14.2 Waste incinerator.

chamber (3) where it is burned at a high temperature. The hot gases heat a steam chamber (4), which drives a turbine generator. Any inorganics in the combustion chamber are deposited as ash (5). The effluent gases are cooled and cleaned (6) before passing through a baghouse (7) to remove fine particulates. The clean gases are then directed through the emission tower (8), which is monitored to meet the emission regulations.

Combined Heat and Power (Cogeneration)

Incineration with energy recovery is one of several other WtE technologies such as gasification, pyrolysis, and anaerobic digestion. While incineration and gasification technologies are similar in principle, the energy produced from incineration is high-temperature heat, whereas combustible gas is often the main energy product from gasification. Incineration and gasification may also be implemented without energy and material recovery.

Combined heat and power (CHP) is not a technology, but an approach to applying technologies. Heat that is normally wasted in conventional power generation is recovered as useful energy, which avoids the losses that would otherwise be incurred from separate generation of heat and power. By combining heat and power in a single system, the overall efficiency of converting the feedstock to energy can be as high as 80% rather than much lower rates of the individual systems.

Conventional generation is typically inefficient when only about a third of the input fuel's potential energy is converted to usable energy. These systems were first used when a boiler was installed in a building or factory to generate heat, but engineers noticed that much of the generated heat was wasted. By using the waste heat to generate steam and drive a turbine generator, the process became much more efficient and thus the word "cogeneration" was first used in the 1970s. A typical system is shown in Figure 14.3.

The fuel is combusted in the boiler, which produces steam to drive the turbine which, in turn, drives the generator to produce electricity. The excess steam from the turbine is used to produce hot water or air for heating. It is also possible for the fuel to be used as the feed to a turbine and the excess heat would be fed to the heat recovery unit to produce steam or hot water.

The Environmental Protection Agency (EPA) provides a catalog of various CHP technologies that also includes reciprocating engines, microturbines, and fuel cells [3].

A typical commercial CHP system is produced by a company called AgriPower [4], founded in 2004. This company was founded by an engineer who wanted to provide electricity for villages in developing countries that relied on farming for their livelihood. The system was designed to convert the agricultural waste to electrical power, thus "AgriPower." The company

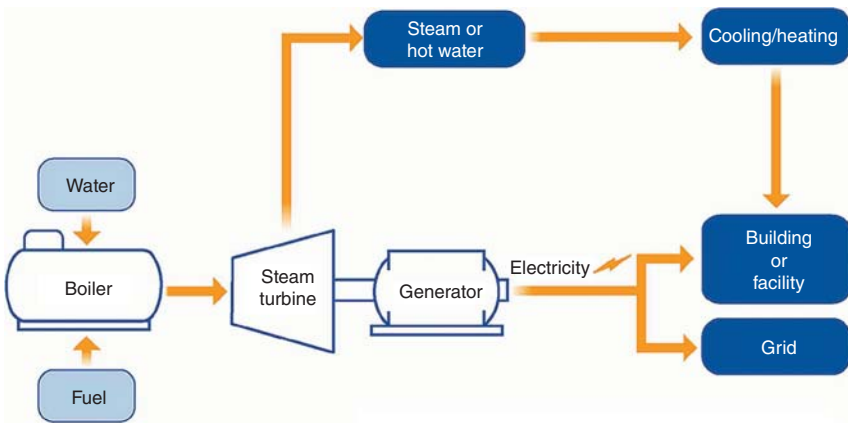


FIGURE 14.3 Steamboiler with steam turbine [2].

manufactures two WtE systems, one called “Heat Only” and the other is a “CHP.” The primary difference between the two systems is that the CHP system utilizes the waste heat from the Heat Only system to drive a turbine generator to produce electricity. The feedstock for either system consists of wet and dry (or mixed) customer waste materials including biomass such as all types of wood, wood debris, cardboard, paper, agricultural by-products, forest residuals, and animal manure, with up to a 50% moisture content. These systems can also accept many types of plastic.

Combined Heat, Power, and Cooling (Trigeneration)

By combining a CHP system with adsorption chilling, the resulting system will result in combined cooling, heat, and power (CCHP). Thus, a plant producing electricity, heat, and cold is called a trigeneration or polygeneration plant. Absorption cooling works on the principal that evaporation requires an increase in temperature. If a fluid evaporates due to a change in pressure, it must remove heat and thus causes a cooling effect. By reducing the air pressure so the liquid evaporates at a lower temperature, cooling is achieved. If this effect is combined with a heat and power system, the result will be a CCHP.

Some of the benefits of trigeneration are the following:

- The cost of fuel and energy can be reduced.
- There is a high efficiency in the production of electricity and heat.
- Lower electrical usage can be achieved during the summer when demand is very high.

- Steam or hot water for onsite use can be generated by the hot engine.
- The carbon equivalent emissions are reduced.
- With water being the refrigerant, there would be no harmful chemical pollutants.
- This form of energy production can improve the building's Leadership in Energy and Environmental Design (LEED) rating.

Pyrolysis

Pyrolysis is the heating of an organic material, such as biomass, in the absence of oxygen [5]. Because no oxygen is present, the material does not combust but the chemical compounds, such as cellulose, hemicellulose, and lignin, which make up that material, thermally decompose into combustible gases and charcoal. Most of these combustible gases can be condensed into a combustible liquid, called pyrolysis oil or bio-oil. The pyrolysis of biomass produces three products: one liquid (bio-oil), one solid (bio-char), and one gaseous (syngas). The proportion of these products depends on several factors including the composition of the feedstock and process parameters. However, all things being equal, the yield of bio-oil is optimized when the pyrolysis temperature is around 500 °C and the heating rate is high.

Bio-oil is a dense complex mixture of oxygenated organic compounds. It has a fuel value that is generally 50–70% that of petroleum base fuels and can be used as boiler fuel or upgraded to renewable transportation fuels. The bio-char produced can be used on a farm as an excellent soil amender that can sequester carbon. Bio-char is highly absorbent and therefore increases the soil's ability to retain water, nutrients, and agricultural chemicals, preventing water contamination and soil erosion. Soil application of bio-char may enhance both soil quality and be an effective means of sequestering large amounts of carbon, thereby helping to mitigate global climate change through carbon sequestration. Syngas, or synthesis gas, is a fuel gas mixture consisting primarily of hydrogen, carbon monoxide, and very often some carbon dioxide. The name comes from its use as intermediates in creating synthetic natural gas (SNG) and for producing ammonia or methanol.

Gasification

Gasification is a process that converts organic or fossil fuel-based carbonaceous materials into carbon monoxide, hydrogen, and carbon dioxide [6]. This is achieved by reacting the material at high temperatures (>700 °C), without combustion but with a controlled amount of oxygen and/or steam.

The resulting gas mixture is called syngas or producer gas and is itself a fuel. The power derived from gasification and combustion of the resultant gas is considered to be a source of renewable energy if the gasified compounds were obtained from biomass.

The advantage of gasification is that using the syngas (synthesis gas H_2/CO) is potentially more efficient than direct combustion of the original fuel because it can be combusted at higher temperatures. Syngas may be burned directly in gas engines, used to produce methanol and hydrogen, or converted via the Fischer–Tropsch process into synthetic fuel. Gasification can also begin with material, which would otherwise have been disposed of such as biodegradable waste. In addition, the high-temperature process refines out corrosive ash elements such as chloride and potassium, allowing clean gas production from otherwise problematic fuels. Gasification of fossil fuels is currently widely used on industrial scales to generate electricity [5].

Anaerobic Digestion

Anaerobic digestion is a series of biological processes in which microorganisms break down biodegradable material in the absence of oxygen [7]. One of the end products is biogas, which can be combusted to generate electricity and heat, or can be processed into renewable natural gas and transportation fuels.

A range of anaerobic digestion technologies are converting livestock manure, municipal wastewater solids, food waste, high strength industrial wastewater and residuals, fats, oils and grease, and various other organic waste streams into biogas on a continuous basis. Separated digested solids can be composted, utilized for dairy bedding, directly applied to cropland, or converted into other products. Nutrients in the liquid stream, the digestate, can be used in agriculture as fertilizer.

The digestion process begins with bacterial hydrolysis of the input materials in order to break down insoluble organic polymers such as carbohydrates and make them available for other bacteria. Acidogenic bacteria then convert the sugars and amino acids into carbon dioxide, hydrogen, ammonia, and organic acids. Acetogenic bacteria then convert these resulting organic acids into acetic acid, along with additional ammonia, hydrogen, and carbon dioxide. Finally, methanogens convert these products to methane and carbon dioxide.

There are many different anaerobic digester systems commercially available. Some of them are the following:

- *Covered anaerobic lagoon digester*—Sealed with flexible cover, with methane recovered and piped to the combustion device.
- *Plug flow digester*—Long, narrow concrete tank with a rigid or flexible cover. The tank is built partially or fully below grade to limit the demand for supplemental heat.
- *Complete mix digester*—Enclosed, heated tank with a mechanical, hydraulic, or gas mixing system.
- *Dry digester*—Upright, silo-style digesters made of concrete and steel with rigid cover.

THE CHALLENGES OF WASTE-TO-ENERGY INITIATIVES

There are a number of challenges attached to renewable energy options: with wind power, it is reliant on the wind; with solar, reliant on the sun—both of which are plentiful and provided without contracts. WtE initiatives are reliant on a steady stream of waste provided by a third party.

Customers need to secure a waste contract to guarantee the operation of the plant—as a tipping fee is paid by companies which deliver their waste. As an example: one plant built in the United Kingdom has a throughput of 150,000 tons of food waste per year, which equates to about 20 trucks/day. The revenue on this, plus financial incentives from generating low carbon electricity, heat, and biogas, makes WtE initiatives a viable option for the customer. Following on from this, there are other by-products in the process such as compost, recoveries, and aggregates that can also be sold into the market.

Another important aspect is the planning permission. The large plants needed to maintain WtE initiatives tend to be close to cities—sometimes nearby landfill sites—because that’s where the large volumes of waste come from and you need to make sure the plants are compliant with emission and odor targets.

THE FUTURE OF WASTE-TO-ENERGY INITIATIVES

The future of this emerging form of new energy can be described as innovative, transitional, and sustainable. All these things are important when meeting the world’s changing resources and energy needs. Waste is a global challenge and one that will not be rectified in a short time scale, perhaps not even in this century. Therefore, solutions such as WtE initiatives are essential to helping stem the tide of plastics and waste into the oceans and landfill sites.

It is important to prioritize prevention, reuse, and recycling of waste, followed by recovery and disposal: WtE is categorized as recovery, the preferred option for materials, which cannot be reused or recycled.

WtE plants burn household and household-adjacent waste that cannot be recycled, turning rubbish into energy in the form of electricity or heat. The volume of waste in the European Union is reduced by 90%, in keeping with both the EU Industrial Emissions Directive—which limits the emissions of energy plants—and the EU Landfill Directive, which mandates no more than 10% of a Member State's waste can end up in landfills by 2030. This should become a requirement worldwide.

Landfilling, the burial of primarily non-recyclable waste on designated tracts of land, is considered less than ideal from a sustainability perspective. Using an area as a landfill results in potential groundwater pollution, high-methane discharge, and explosion hazards from trapped gas emissions, and as much of the material deposited in landfills is not biodegradable it can take decades to break down. Where waste can be safely used as fuel, therefore, it is recommended to do so rather than depositing it in landfills.

WtE provides the dual service of disposing of non-recyclable waste and replacing fossil fuels in energy production—in reducing fossil fuel use, the process decreases the impact of energy production plants on climate change and maintains security of energy supply. Efficient WtE plants thermally treat waste materials to minimize carbon dioxide and methane emissions produced by burning them. The ash produced by burning treated waste is collected, stored, and used in road construction or covering landfill sites.

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Environmentally Effective Buildings

In 1981, Bill McDonough¹ started his architectural practice, and his first major commission was the design of the Environmental Defense Fund (EDF) headquarters in New York City in 1984 [1]. Being very concerned with the environment, EDF made a stipulation that the design of the building should provide the highest possible air quality for the benefit of the EDF employees. He was able to accomplish his goal while working with David Gottfried, a construction manager, and Michael Italiano, an environmental attorney.

During the next few years, David Gottfried and Michael Italiano continued their interest in environmentally efficient and effective buildings. They conducted many informal meetings with Rocky Mountain Institute, Green America, Carrier Corporation, Herman Miller, and Interface. In 1993, Gottfried, Italiano, and S. Richard “Rick” Fedrizzi, the marketing manager for Carrier’s parent company, United Technologies Corporation, founded the US Green Building Council (USGBC) with Fedrizzi as the chair of the USGBC.

As the organization was created for the development of environmentally effective, as opposed to efficient per *Cradle to Cradle* [2], buildings, the first task for the USGBC was to define such a “green” building. In simple terms, a green building was defined by five factors:

- *Energy efficiency*—The consumption of energy for the construction of the building and the operation of the building must be at a minimum. Use of renewable energy for any portion of the building’s operation enhances its definition.

¹ Author of “Cradle to Cradle” with Michael Braungart.

- *Water efficiency*—The consumption of water for the construction of the building, but more important for its operation should be at a minimum.
- *Efficiency in materials*—There should be efficiency in the use of materials in the construction of the building. This includes the use of recycled materials, sourcing materials from nearby locations, and the use of renewable materials.
- *Reducing health impacts*—The design and construction of the building should minimize the impact on human health by incorporating the highest quality of air, and minimizing the emission of toxic vapors from the materials or coatings.
- *Reducing building impact on the environment*—Considerations in the design of the building should include minimizing the impact on the environment in terms of air, water, soil, and vegetation.

Once the organization defined a green building, its next initiative was the creation of the Leadership in Energy and Environmental Design (LEED), a certification program for new buildings that would include a more specific definition of a green building. USGBC had six primary goals [3] in mind when it developed LEED:

- (1) Define “green building” by establishing a common standard of measurement.
- (2) Promote integrated, whole-building design practices.
- (3) Recognize environmental leadership in the building industry.
- (4) Stimulate green competition.
- (5) Raise consumer awareness of green building benefits.
- (6) Transform the building market.

The first version, called LEED 1.0, was launched in 1998 and provided new buildings for earning up to 40 points. Depending on the number of points awarded to a new building, it could earn a bronze, silver, or gold award. However, there were many shortcomings and during the next two years this initial version was subject to testing and review, and it led to the development of LEED 2.0, which expanded to 69 points. Although these standards were designed primarily for new construction, they were also applied to existing construction in the event there was a movement to convert an existing building. More testing and reviews resulted in LEED 2.1 in early 2003.

It wasn't until November 2005 when USGBC introduced LEED-NC 2.2, which was specific for new construction. Modifying an existing building to achieve LEED status is considerably different than constructing a new building so a new set of standards was developed. For instance, points might be awarded for the material of construction of a new building, but this would not be possible for an existing one. In addition to LEED for new construction, USGBC also developed the following standards:

- LEED-EB for existing buildings
- LEED-CI for commercial interiors
- LEED-CS for core and shell
- LEED-H for houses
- LEED-ND for neighborhood development

USGBC also developed a series of application guides for the different type of buildings and building complexes. Some of the application guides are for:

- Retail
- Multiple buildings
- Campuses
- Lodging
- Healthcare
- Laboratories
- Schools

The maximum number of points that can be earned by new construction through LEED-NC 2.2 was set as follows:

• Sustainable sites	14
• Water efficiency	5
• Energy and atmosphere	17
• Materials and resources	13
• Indoor environmental quality	15
• Innovation and design process	5
• Total	69

Some of the approaches that can lead to the points are as follows:

- *Sustainable sites*—Not on prime farmland; not lower than 5 ft above the 100-year flood plain; not within 100 ft of wetlands or brownfield; bicycle storage and shower for riders; preferred parking for carpool vans, hybrid or electric automobiles; green roof.
- *Water efficiency*—Use captured rainwater; no irrigation; treat water on-site.
- *Energy and atmosphere*—No use of chlorinated fluorocarbons; use renewable energy.
- *Materials and resources*—Recycle waste materials; divert demolition waste from landfill; source materials from within 500 miles of site.
- *Indoor environmental quality*—Use low volatile organic compound (VOC) coatings; recycled carpeting; operable windows; daylight use.
- *Innovation and design process*—Retain an LEED accredited professional (AP); innovations are not included in LEED.

By earning as many points as possible, a newly constructed building can receive an award as follows:

• Certified	28–32 (under LEED 2.0, this level was called bronze)
• Silver	33–38
• Gold	39–51
• Platinum	52–69

For an existing building, the total number of points available for earning an LEED-EB award is significantly more than that for new construction:

• Sustainable sites	14
• Water efficiency	5
• Energy and atmosphere	23
• Materials and resources	16
• Indoor environmental quality	22
• Innovation and design process	5
• Total	85

You will note that the larger number of points for the LEED-EB is confined to just three of the six categories, energy and atmosphere, materials

and resources, and indoor environmental quality. It is in these categories that exhibit challenges to convert an existing building to a LEED certified building.

Despite several upgrades of the LEED system over a seven years period, there were still several shortcomings:

- *Go after easy points*—Since there are alternative means to earn points, consultants might recommend, for example, installing a bicycle rack and a shower rather than installing a green roof. Either option is worth one point, but the costs are much different.
- *Regional differences*—There is a big difference between, say, Arizona and Illinois in terms of solar energy or water availability.
- *Need weighting factors*—While some enhancements earn the same number of points, the benefits may differ. Bicycle racks and green roofs earn the same points, but green roofs are more beneficial in terms of water retention and building cooling.
- *LCA is missing*—There is no life cycle assessment made on the various alternatives.
- *Definition of green material or product*—Without standards for green products, there is not consistency for awarding points for the products used.
- *Analysis of cost versus benefit*—It would be helpful to determine the environmental, social, and/or economic benefit as a function of the cost.
- *Point totals confusing*—There is no consistency in the total points as a function of the type of building.

Many of the shortcomings and other improvements were incorporated in LEED 3.0, which expanded the certification requirements to seven categories allowing for a total of 110 points.

• Sustainable sites	26
• Water efficiency	10
• Energy and atmosphere	35
• Materials and resources	14
• Indoor environmental quality	15
• Innovation in design	6
• Regional priority	4
• Total	110

Regional priority is a new category that was added in this version. This new category took into consideration the environmental priorities of different regions within the United States.

Within each of the seven categories were two sublevels of requirements. The first sublevel of requirement was the “prerequisite,” which consisted of one to three prerequisites for each of the first five categories with total of eight in all. The second level of requirements was the credits, which could be earned only if the prerequisites were achieved.

In addition to increasing the total number of points that could be earned to 110, LEED 3.0 was also modified to provide for a consistent number of points for each certification level regardless of the building category. Based on LEED 3.0, the required points became:

• Certified	40–49
• Silver	50–59
• Gold	60–79
• Platinum	80+

The next version of this certification process, LEED 4.0, was released in November 2013 [4]. It features new market sectors that address the unique needs of specific building types from data centers to existing school buildings and changed the credit rating system to include points for resource savings and emissions abatement. LEED 4.0 addressed 21 different market sector adaptations to identify and focus on the unique needs of each market. In particular, it provided new solutions for the following sectors:

- Existing schools
- Existing retail
- Data centers (new and existing)
- Warehouses and distribution centers (new and existing)
- Hospitality
- Mid-rise residential

Earning any level of accreditation is an excellent achievement of creating an efficient and healthy building. Many of the building owners seek LEED certification not only for the benefits but also for marketing purposes. If it is a rental building, the owner can attract quality tenants and can charge higher

rates. Other owners seek the accreditation for the human and social benefits. Schools with day lighting showed improved test scores, while stores with sky lights can achieve greater sales. Productivity in business settings is also increased with improved lighting and clean air circulation. Reduced absenteeism due to illnesses not only improves productivity but it can also reduce health insurance costs.

An example of an LEED Platinum building is the Bank of America Tower (Fig. 15.1) with the following features:

- (1) *Water from sky and earth*—Rain water is collected and used for flushing toilets, irrigating the green roof, and to run the air conditioning.
- (2) *Daylight savings*—Floor to ceiling windows reduces the lighting costs and improves productivity.
- (3) *Chill factor*—Chilled air is pumped into a space below raised floors, and as it warms up it rises and pulls more cool air with it.
- (4) *Aired out*—Air is brought into the building from above the 10th floor as it is free of tailpipe emissions from vehicles. Despite the use of cleaner air it is still filtered to remove particulate matter, allergens, ozone, and other compounds.
- (5) *Homemade juice*—The Bank of America Tower contains its own power plant running on natural gas. The efficiency is relatively high as there are no losses from shipping electricity over power lines, and the waste heat is reused.
- (6) *No parking*—There is very limited parking for automobiles, but rather the lower level is accessible to a network of public transportation. There is also secured bicycle parking.
- (7) *Ice storage*—During the night when electricity costs are much less, ice is produced to provide cooling for the air conditioning system.

An improvement of LEED 4.0 was released in December 2018 as LEED 4.1. The changes were actually a series of stepping stones to improve credit utilization. For example, carbon becomes a metric in order to minimize or optimize energy performance. Further evaluation of LEED 4.1 took place in early 2019 when a beta version was released for testing. This was followed by beta versions for building design and construction (BD + C), interiors design and construction (ID + C), residential, and for cities and communities. The new version will continue to employ Arc, a digital platform for tracking and measuring building performance [5].



FIGURE 15.1 Bank of America Tower.

NET-ZERO ENERGY BUILDINGS

As more buildings are being constructed worldwide, it is imperative that the energy and environmental impacts are minimized. One solution is to design buildings that produce as much energy as they consume. This refers to net-zero energy (NZE) buildings.

In 2007–2008, Congress expressed its support by establishing the Zero Net Energy Commercial Initiative, which requires all new commercial buildings in the United States achieve NZE by 2030 [6].

As a result of this initiative, the USGBC unveiled its LEED Zero certification in November 2018. This certification can be achieved when projects demonstrate one of the following: net zero carbon emissions, NZE use, net zero water use, or net zero waste. The USGBC has awarded the first LEED Zero to the corporate headquarters of the engineering firm Petinelli in Curitiba, Brazil [7]. While this building had already achieved Platinum LEED v4 for Existing Buildings: Operations and Maintenance, it was also able to earn the LEED Zero for energy but will be seeking LEED Zero for water.

LEED PROJECT CERTIFICATION PROCESS

The owner of a building or project must determine which LEED rating system will be the most appropriate for the proposed project.² Once the rating system has been selected, there is a five-step process required to achieve certification.

- (1) *Project registration*—This is the first step and it must be done through the Green Building Certification Institute (GBCI), which has assumed the administrative responsibility for all LEED project certifications. The registration process itself, which is managed online, consists of five steps:
 - a. *Eligibility*—Verifying that the project meets all the requirements.
 - b. *Rating system selection*—Actual selection of the rating system to be used.
 - c. *Rating system results*—Project administrator receives an online LEED scorecard that identifies all the prerequisites and possible credits.
 - d. *Project information*—Project administrator inputs information on the project.
 - e. *Review*—Computer reviews the project information and registration information for completeness.
 - f. *Payment*—Administrator pays necessary fees.
 - g. *Confirmation*—Computer issues a final confirmation that the project is registered.
- (2) *Application preparation*—This step is conducted online over a period of time. As each activity for a prerequisite or credit is completed, the information requested is entered online.

²Farmer, p. 24.

- (3) *Application submittal*—The application submittal step requires two types of information for review by the GBCI. One requires information on the design of the building including site location and building design information. The second type of information shows how the project meets with compliance during the actual construction process.
- (4) *Application review*—This is the review by the GBCI to determine whether the project actually meets the requirements. The GBCI will reward the points or deny them. If there are any points denied, the project administrator can appeal the decision, accept the decision, or add additional points in other categories to make up the deficit.
- (5) *Certification award*—The final step is the actual award of certification by the GBCI.

LEED ACCREDITED PROFESSIONAL

An LEED accredited professional, LEED AP, is highly recommended for anyone pursuing a project seeking LEED certification. This professional has been trained to understand fully the certification process and what is needed to achieve certification. From 2001 to 2009, exams were given by the USGBC to those interested in becoming an LEED AP, and each year the exam was more intensive as the project requirements became more specific as the process went from LEED 1.0 to LEED 2.2.

In 2009, the GBCI took over the administration of this process and developed a new pathway for becoming an AP [8]. The first step is to become a Green Associate, which is intended for those professionals who want to demonstrate green building expertise in non-technical fields of practice. There is no need, however, to start with the Green Associate credentials as one can pursue the LEED AP accreditation directly. While LEED AP was the only position possible through 2009, now it is possible to earn LEED AP for specific types of projects. Available today is the following:

- *LEED AP BD + C*—For building design and construction and includes new construction, schools, and core and shell.
- *LEED AP ID + C*—For interior design and construction and includes commercial interiors.
- *LEED AP Homes*—For the design and construction of high performance green homes.
- *LEED AP O + M*—For the operation and maintenance of existing buildings.

- *LEED AP ND*—For the design and development of neighborhoods that meet the high levels of environmentally responsible and sustainable development.

Additional AP positions will, no doubt, be developed as the LEED process continues to evolve.

LIVING BUILDING CHALLENGE

In 2007, a Seattle architect, Jason McLennan, introduced the Living Building Challenge (LBC) whereby new structures would produce all of their own energy and use only water that falls on-site. This new building standard has been adopted by the Cascadia region green building council for Oregon, Washington, and British Columbia. Builders would be required to use sustainably sourced materials and avoid toxic materials like asbestos, mercury, and polyvinyl chloride (PVC). They must build on previously developed sites and meet measurements of livability, social equity and beauty.

When Cascadia released the LBC, its objective was not to compete with LEED. In fact, at that time the thought was to provide some ideas for the development of LEED 3.0. The plan was to make the LBC requirements more sustainable than that for LEED Platinum. The LBC is not a rating system so there are no points to be earned. The basis of LBC was a series of 16 prerequisites of which all must be met. It has since been increased to 20 prerequisites through version LBC 2.1 [9]:

- (1) *Limits to growth*—Projects may only be built on previously developed sites, either grayfield or brownfield and may not build within 50 ft of wetlands or adjacent to sensitive ecological habitats.
- (2) *Urban agriculture*—The project must integrate opportunities for agriculture appropriate to its scale and density.
- (3) *Habitat exchange*—For each acre of development, an equal amount of land must be set aside as part of a habitat exchange.
- (4) *Care free living*—The project must contribute toward the creation of walkable, pedestrian-oriented communities.
- (5) *Net zero water*—100% of the occupants' water use must come from captured precipitation or reused water that is appropriately purified without the use of chemicals.
- (6) *Ecological water flow*—100% of storm water and building water discharge must be handled on-site.
- (7) *Net zero energy*—100% of the building's energy needs supplied by on-site renewable energy on a net annual basis.

- (8) *Civilized work environment*—Every occupiable space must have operable windows that provide access to fresh air and daylight.
- (9) *Healthy air*—All buildings must meet specific criteria for air sources and control depending on the space itself.
- (10) *Biophilia*—The project must be designed to include elements that nurture the innate human attraction to natural systems and processes.
- (11) *Materials red list*—The project cannot contain certain materials or chemicals such as lead, cadmium, mercury, PVC, or CFCs (chlorofluorocarbons).
- (12) *Embodied carbon footprint*—The embodied carbon footprint of its construction must be offset through carbon credits.
- (13) *Responsible industry*—All wood must be FSC (Forest Stewardship Council) certified or from salvaged sources.
- (14) *Appropriate sourcing*—Depending on the material or service, it must be obtained within a certain radius that can vary from 250 mi (heavy materials) to 7000 mi (renewable energy technologies).
- (15) *Conservation and reuse*—Construction waste must be diverted from landfills that can vary, depending on the waste, from a minimum of 95–80%.
- (16) *Human scale and humane places*—The project must be designed to create human-scaled rather than automobile-scaled places, so that the experience brings out the best in humanity and promotes culture and interaction.
- (17) *Democracy and social justice*—All primary transportation, roads, and non-building infrastructure that are considered externally focused must be equally accessible to all members of the public.
- (18) *Rights to nature*—The project may not block access to, nor diminish the quality of fresh air, sunlight, and natural waterways for any member of society or adjacent developments.
- (19) *Beauty and spirit*—The project must contain design features intended solely for human delight and the celebration of culture, spirit, and place appropriate to the function of the building.
- (20) *Inspiration and education*—Educational materials about the performance and operation of the building must be made available to the public.

The philosophy of the LBC is for every single act of design and construction to make the world a better place. This should be the philosophy of all designers and builders.

WORLD'S GREENEST BUILDING

On Earth Day in April 22, 2013, the Bullitt Foundation opened its new building, the Bullitt Center (Fig. 15.2), in Seattle, Washington [10]. The 50,000 ft² building was considered by the builders to be the greenest building in the world as they by-passed the USGBC's LEED certification in favor of the more stringent 20 prerequisites of the LBC.

The Bullitt Center's approach toward environmental sustainability started with the design of the site. Cisterns store rainwater and "gray water" from sinks and showers funnel through the building's green roof. The Bullitt Center is flanked by a planting strip that makes the approaching sidewalk more pleasant for local workers and residents. Solar arrays provide as much electricity as the building requires. Medium-height sidewalk plantings also create a physical separation between pedestrians and vehicle traffic. The building's planners chose a thriving residential neighborhood intermixed with shops, restaurants, and bars for the opportunity to add more commercial space. This approach enhances the vision of people living closer to their place of work so that commutes are very short or, better yet, sufficiently close to walk to the office.

In tune with the ideals behind the LBC, the Bullitt Center took inspiration from nature and created a work environment that is practical, yet also healthy for its inhabitants. Architectural details that are aesthetically pleasing yet practical include higher ceilings (eliminating an additional floor possible under local building codes) and a central glass-enclosed staircase that encourages tenants to use the stairs instead of the elevator. Exposed



FIGURE 15.2 The Bullitt Center. *Source:* Reproduced with permission of the Bullitt Center.

wood, FSC certified, is a reflection of the local Pacific Northwest natural environment.

In order to meet the water requirements of the LBC, the building must be designed to provide its own supply of water through rainwater capture. Until that really happens, water will be supplied by the city of Seattle. It is not only a question of quantity, but the building owners must also prove that the water is safe for health reasons.

The Bullitt Center also needs to generate as much energy as it uses to comply with the LBC, so it is topped off with a giant solar array that is bigger than the building's footprint (Fig. 15.3). It is tied into the grid but is projected to be electricity neutral, using a third of the energy of normal buildings. The building is heated using ground source heat pumps. As a result of these energy efficiency features, the Bullitt Center consumes 75% less energy than the average commercial building as shown in Figure 15.4.

Some of the other features include great windows that open, high ceilings for lots of natural light, greener concrete, careful choice of materials, and bike racks instead of parking spaces. One of the strictest LBC requirements is that of durability. This building was designed to last 250 years. No one today can be sure that it will meet that requirement. In fact, the building cannot be LBC certified until it has been occupied and operating for a minimum of one year.



2012-12.31_e5680

Bullitt center under construction

Photo: John Stamets

FIGURE 15.3 Bullitt Center solar array. *Source:* Reproduced with permission of the Bullitt Center.

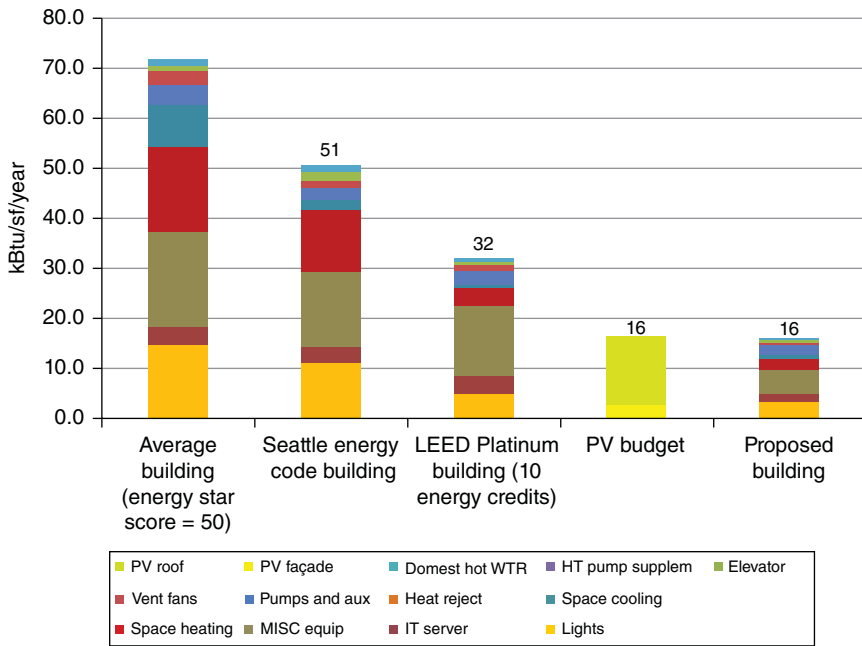


FIGURE 15.4 Bullitt Center energy consumption.

The vision of the Bullitt Foundation was to develop a model to meet the goal of the LBC where every feature and benefit of the building design will make the world a better place. If all new buildings in the world met this goal, the earth would certainly be a better place.

THE WELL BUILDING STANDARD

The WELL Building Standard is the world's first architectural benchmark focused exclusively on human health and wellness to improve sustainability [11]. It presents a performance-based system for certifying features of the built environment that impact health and well-being. Its focus is to integrate sustainability in architectural design so that occupants are healthier.

It was initially launched to improve the way people live by developing spaces that enhance workers health and quality of life. The standard sets performance requirements in seven areas—air, water, nourishment, light, fitness, comfort, and mind. All these concepts are based on medical research concerning how environments affect human health.

The WELL Building Standard was launched in 2013 by design firm *Delos* to improve peoples' health in the work environment. The standards

are administered by the International WELL Building Institute (IWBI) and third-party certified by Green Business Certification, the same organization that certifies LEED.

BREEAM

The Building Research Establishment Environmental Assessment Method (BREEAM) was the first established method for assessing, rating, and certifying sustainable buildings have been launched in 1990 [12]. It uses scientifically based metrics and indices in the areas of energy and water use, health and wellbeing, pollution, transport, materials, waste, ecology, and management processes. As opposed to the LEED ratings of Platinum, Gold, Silver, etc., BREEAM rates its buildings as Pass, Good, Very Good, Excellent, and Outstanding.

Although BREEAM is used in more than 70 countries, several European countries have developed their own country-specific criteria. This has been done in The Netherlands, Spain, Norway, Sweden, and Germany. Each is operated and administered by the country's green building council.

OTHER BUILDING CERTIFICATIONS

The Building Owners and Managers Association's 360 Performance Program [13] evaluates six areas of operations and management practices, including Building Operations and Maintenance, Life Safety/Security/Risk Management, Training and Education, Energy, Environment/Sustainability, and Tenant Relations/Community Involvement. As you can tell, it is a holistic building evaluation that moves beyond just green criteria. The certification only applies to occupied office building and industrial space.

The green certification created by the Society for Environmentally Responsible Facilities (SERF) [14] is one of the newer certifications. It was founded in mid-2011, and according to its founder, Joe Maguire, SERF certification aims to streamline the application and submission process, and reduce the time and money it takes for a building to be certified green. The process should take between four and six weeks. SERF certification can apply to a variety of building types, including office, multi-family residential and institutional.

Another certification system is Green Globes [15], a rating system offered in the US by an organization called the Green Building Initiative [16]. Using an online questionnaire and evaluation tool, the rating system scores facilities on a 1000-point scale in several categories. Green Globes can be used as

a self-assessment tool, or, if owners prefer, they can hire third-party assessors to certify facilities.

In the United States, California has established its own building certification systems called CAL Green [17], which became effective in 2009. Buildings in California that seek a sustainable building certification will more often seek LEED as it is a worldwide certification. Some building developers in California will seek the CAL Green certification in addition to LEED primarily for marketing purposes.

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Green Chemistry, Nanotechnology, and “Big Hairy Audacious Goal”

In the preceding chapters, we have provided a summary of various strategies that can be employed by corporations, government agencies, and non-profits to offer products and services of the highest quality without having a negative impact on the environment. In the next decades, it will be extremely critical to be aware of the declining health of the environment and that everything we do should improve its health before it is too late. In addition to employing strategies like those described in the earlier chapters, there may be many others that can be integrated into an organization's operation to accomplish the same goal. Here we summarize three additional strategies.

GREEN CHEMISTRY

Green chemistry is a revolutionary approach to the way that products are made; it is a science that aims to reduce or eliminate the use and/or generation of hazardous substances in the design phase of materials development. It requires an inventive and interdisciplinary view of material and product design. Green chemistry follows the principle that it is better to consider waste prevention options during the design and development phase than to dispose or treat waste after a process or material has been developed. This not only reduces the amount of waste material generated, but also reduces the quantity of feedstock required to manufacture the product.

Practical Sustainability Strategies: How to Gain a Competitive Advantage,
Second Edition. George P. Nassos and Nikos Avlonas.

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Companion website: www.wiley.com/go/Nassos/PracticalSustainabilityStrategies_2e

For a technology to be considered green chemistry, it must accomplish three things:

- It must be more environmentally benign than existing alternatives.
- It must be more economically viable than existing alternatives.
- It must be functionally equivalent to or outperform existing alternatives.

Green chemistry presents industries with incredible opportunity for growth and competitive advantage. This is because there is currently a significant shortage of green technologies: it is estimated that only 10% of current technologies are environmentally benign; another 25% could be made benign relatively easily [1]. The remaining 65% have yet to be invented! Green chemistry also creates cost savings: when hazardous materials are removed from materials and processes, all hazard-related costs are also removed, such as those associated with handling, transportation, disposal, and compliance.

Through green chemistry, environmentally benign alternatives to current materials and technologies can be systematically introduced across all types of manufacturing to promote a more environmentally and economically sustainable future.

In 2007, John Warner and Jim Babcock partnered to found the first company completely dedicated to developing green chemistry technologies, the Warner Babcock Institute for Green Chemistry. The Institute was created with the mission to develop nontoxic, environmentally benign, and sustainable technological solutions for society.

The institute works with corporations to find a more sustainable process for the manufacture of chemical products. The methodology by which the institute develops these processes is based on 12 principles:

1. *Pollution prevention*—It is better to prevent waste than to treat and clean up waste after it is formed.
2. *Atom economy*—Synthetic methods should be designed to maximize the incorporation of all materials used in the process into the final product.
3. *Less hazardous synthesis*—Whenever practicable, synthetic methodologies should be designed to use and generate substances that possess little or no toxicity to human health and the environment.
4. *Design safer chemicals*—Chemical products should be designed to preserve efficacy of the function while reducing toxicity.

5. *Safer solvents and auxiliaries*—The use of auxiliary substances (solvents, separations agents, etc.) should be made unnecessary whenever possible and, when used, innocuous.
6. *Design for energy efficiency*—Energy requirements should be recognized for their environmental and economic impacts and should be minimized. Synthetic methods should be conducted to ambient temperature and pressure.
7. *Use of renewable feedstocks*—A raw material or feedstock should be renewable rather than depleting whenever technically and economically practical.
8. *Reduce derivatives*—Unnecessary production of derivatives (blocking group, protection/deprotection, temporary modification of physical/chemical processes) should be avoided whenever possible.
9. *Catalysis*—Catalytic reagents (as selective as possible) are superior to stoichiometric reagents.
10. *Design for degradation*—Chemical products should be designed so that at the end of their function they do not persist in the environment and instead breakdown into innocuous degradation products.
11. *Real-time analysis for pollution prevention*—Analytical methodologies need to be further developed to allow for real-time in-process monitoring and control prior to the formation of hazardous substances.
12. *Inherently safer chemistry for accident prevention*—Substance and the form of a substance used in a chemical process should be chosen so as to minimize the potential for chemical accidents, including releases, explosions, and fires.

When working in the various industry sectors, the approach to green chemistry may vary slightly. For example, here are some of the approaches taken for the following industry sectors:

- *Energy, natural resources, and environment*—Novel photon to chemical energy mechanisms may be used for the generation or storage of energy. Alternative fuels may be developed from renewable feedstocks and intermediates. Diverse approaches may be used for the sustainable reclamation and/or purification of water.
- *Industrial chemicals and materials*—When attempting to manufacture a specific product, there may be different approaches to achieve the ultimate chemical product. Here is where one must look at alternative feedstocks that are more sustainable and economic.

- *Industrial products*—It is important for the products to be void of volatile organic compounds (VOCs) as well as other chemical coatings that are prohibited by new regulations.
- *Pharmaceuticals and biotechnology*—New methods are sought in order to reduce the non-active compounds in a particular product while developing new synthetic methods for the active ingredient.
- *Personal care and cosmetics*—Must develop new materials and application mechanisms that avoid VOCs and other potentially harmful materials to the skin, hair, and nail products.
- *Retail, consumer, supply chain*—Must assess the entire supply chain to be sure the materials used in the manufacture of products are sustainable and meet current and potentially future regulations.

An example of applying green chemistry to a pharmaceutical product is the synthesis of ibuprofen [2]. Ibuprofen is the active ingredient in many analgesic and inflammatory drugs such as Advil, Motrin, and Medipren. Beginning in the 1960s, ibuprofen was produced by a six-step synthesis with an atom economy of only 40%. This meant that less than half (40%) of the weight of all the atoms of the reactants were incorporated in the ibuprofen, and 60% were wasted in the formation of unwanted by-products. The annual production of approximately 30 million pounds of ibuprofen by this method resulted in over 40 million pounds of waste. But during the 1990s, the BHC Company developed a new synthesis of ibuprofen with an atom economy of 77–99%. This synthesis not only produces much less waste, it is also only a three-step process. A pharmaceutical company can thus produce more ibuprofen in less time and with less energy, which results in increased profits.

In another example, the shoe company Nike identified five chemicals in its original shoe rubber that were hazardous and worked diligently to eliminate these chemicals [3]. In the original rubber, those five toxic chemicals made up 12% of the product by weight. The “green rubber” that Nike eventually created by applying green chemistry has only one of the five chemicals in it, and that chemical makes up only 1% of the product by weight. That reduced the toxic composition by 96%, or 3000 metric tons/year.

The company was also successful in reducing zinc in its shoes. Using zinc meant emitting 340 g of VOCs for every pair of shoes during the manufacturing process. But Nike engineers discovered the zinc wasn't really that essential to the shoes. They were able to remove 80–90% of the zinc in the shoe manufacturing process—reducing toxic emissions from 340 g per pair in 1995 to 15 g in 2006. This is a huge difference that would never be noticed by the consumer, but would be noticed by the workers in the factories.

NANOTECHNOLOGY

When K. Eric Drexler popularized the word “nanotechnology” in the 1980s [4], he was talking about building machines on the scale of molecules, a few nanometers wide—motors, robot arms, and even whole computers, far smaller than a cell. Drexler spent the next 10 years describing and analyzing these incredible devices, and responding to accusations of science fiction. Meanwhile, mundane technology was developing the ability to build simple structures on a molecular scale. As nanotechnology became an accepted concept, the meaning of the word shifted to encompass the simpler kinds of nanometer-scale technology. The US National Nanotechnology Initiative was created to fund this kind of nanotech: their definition includes anything smaller than 100 nm with novel properties, where a nanometer is one-billionth of a meter.

To put in perspective of just how small is a nanometer, a visual comparison is helpful [5]. If a 20-nm object were blown up to the size of a soccer ball, a virus would be the size of a person, a red blood cell would be the size of a soccer field, a doughnut would be the size of the United Kingdom, and a chicken would be the size of earth. In other words, a nanoparticle is really small.

Much of the work being done today that carries the name “nanotechnology” is not nanotechnology in the original meaning of the word. Nanotechnology, in its traditional sense, means building things from the bottom-up, with atomic precision. This theoretical capability was envisioned as early as 1959 by the renowned physicist Richard Feynman [6] when he said;

I want to build a billion tiny factories, models of each other, which are manufacturing simultaneously ... The principles of physics, as far as I can see, do not speak against the possibility of maneuvering things atom by atom. It is not an attempt to violate any laws; it is something, in principle, that can be done; but in practice, it has not been done because we are too big.

Based on Feynman’s vision of miniature factories using nanomachines to build complex products, advanced nanotechnology (sometimes referred to as molecular manufacturing) will make use of positionally controlled mechanochemistry guided by molecular machine systems. Formulating a roadmap for development of this kind of nanotechnology is now an objective of a broadly based technology roadmap project led by Battelle (the manager of several US National Laboratories) and the Foresight Nanotech Institute.

Shortly after this envisioned molecular machinery is created, it will result in a manufacturing revolution, probably causing severe disruption. It also has serious economic, social, environmental, and military implications. Just how can this concept be applied?

One of the greatest environmental concerns on a worldwide basis is global warming. Scientists have positively concluded that this event has and continues to occur, and that it is the result of human activity. Carbon dioxide emitted from numerous sources, primarily electrical power plants and transport vehicles, has increased the atmosphere's retention of heat generated by the sun's rays. This is the result of a thick carbon dioxide layer in the atmosphere that is expected to remain, possibly forever. However, there may also be another phenomenon caused by pollutant particles in the atmosphere that produces an opposite cooling effect called global dimming. Separate studies measure the amount of sunlight passing through the atmosphere and also the rate of evaporation of water. Each of these two studies indicates that there has been a change in the measurements over a period of time causing global dimming. If this is truly the case, the warming effect from carbon dioxide must be greater than we thought. And, if these particles are the result of power plant emissions, and if some of these power plants are replaced by renewable energy sources, the resulting reduction in particles could subsequently increase global warming. So, perhaps, the real answer to these opposing phenomena can be found if we examine them at the nanoparticle level.

Another critical environmental issue is the depletion of our natural resources. In the book entitled *Biomimicry* (see Chapter 9), Janine Benyus suggests that we take a more careful look at nature's amazing processes to become more efficient in the manufacturing of our products. An example of biomimicry is how spiders produce several different kinds of silk for various functions, such as forming webs or rappelling from drop-offs. The properties of these spider-produced silks are astounding when compared with man-made materials. Compared with an equal weight basis, some of these spider-produced silks are five times stronger than steel and five times tougher than Kevlar, the material used in bulletproof vests. At the same time, the silk can be very elastic and stretch up to 40% of its original length, something that steel wire is incapable of doing. Just imagine if someone could learn to do what the spider does, taking a renewable, soluble material and making an extremely strong water-insoluble fiber using very little energy and generating no toxic waste. By analyzing the spider's process and reproducing it, the entire fiber industry would change dramatically. This may be another role for nanotechnology.

Ms. Benyus provides other examples of natural phenomena that are extremely interesting. These include the abalone shell that is stronger than

any known ceramic. If we look at the natural design of the inner coating of these shells, we may learn how to manufacture stronger materials and more sustainably. Still another example is the adhesive created by mussels and other bivalves, allowing them to attach to almost anything, and these adhesives are waterproof. By examining these natural phenomena at the nanoscale, new doors may be opened in our continuous search for a more sustainable environment.

Another example of the application of nanotechnology is the search for the perfect sunscreen. Zinc oxide would be the perfect sunscreen ingredient if the resulting product didn't look quite so silly. Thick, white, and pasty, it once was seen mostly on lifeguards, surfers, and others who needed serious sun protection. But when the sunscreens are made with nanoparticles, they turn clear—which makes them more user-friendly.

Improved sunscreens are just one of the many innovative uses of nanotechnology, which involves drastically shrinking and fundamentally changing the structure of chemical compounds. But products made with nanomaterials also raise largely unanswered safety questions—such as whether the particles that make them effective can be absorbed into the bloodstream and are toxic to living cells [7].

While the development of nanotechnology is over 30 years old, the booming nanotech industry is less than two decades old. Nanoparticles are already found in thousands of consumer products, including cosmetics, pharmaceuticals, anti-microbial infant toys, sports equipment, food packaging, and electronics. In addition to producing transparent sunscreens, nanomaterials help make light and sturdy tennis rackets, clothes that don't stain and stink-free socks.

The particles can alter how products look or function because matter behaves differently at the nanoscale, taking on unique and mysterious chemical and physical properties. Materials made of nanoparticles may be more conductive, stronger, or more chemically reactive than those containing larger particles of the same compound.

One of the problems is that the development of applications for nanotechnology is rapidly outpacing what scientists know about safe use. The unusual properties that make nanoscale materials attractive may also pose unexpected risks to human health and the environment. "We haven't characterized these materials very well yet in terms of what the potential impacts on living organisms could be," said Kathleen Eggleson, a research scientist at the Center for Nano Science and Technology at the University of Notre Dame [8].

Scientists don't know how long nanoparticles remain in the human body or what they might do there. But research on animals has found that inhaled

nanoparticles can reach all areas of the respiratory tract; because of their small size and shape, they can migrate quickly into cells and organs. The smaller particles also might pose risks to the heart and blood vessels, the central nervous system, and the immune system, according to the US Food and Drug Administration (FDA).

Though nanomaterials have been used in consumer products for more than a decade, the FDA acknowledged for the first time in April 2012 that they differ from their bulk counterparts and have potential new risks that may require testing [7]. In draft guidelines on the safety of nanomaterials in cosmetic products, the agency advised companies to consult with the FDA to find out the best way to test their products. Rather than adopting a one-size-fits-all approach, the FDA plans to assess nano-enabled products on a case-by-case basis, according to the guidelines. "There is nothing inherently good or bad about a nanomaterial," said Chad Mirkin [8], Director of the International Institute for Nanotechnology at Northwestern University who nevertheless thinks each class of material should be considered a new form of matter and reviewed for safety.

Proponents of nanotechnology say the potential benefits reach far beyond sunscreen. Controlling matter at the atomic scale is being hailed as the next "industrial revolution" because it could help solve everything from climate change and world hunger to energy shortages and biodiversity loss. In medicine, scientists envision microscopic robots that could swim around in the bloodstream, repairing cells, and diagnosing diseases. Nanotechnology also might unleash powerful new therapeutic weapons for treating many of the worse forms of cancer, cardiovascular problems, and neurodegenerative disease.

Researchers at the University of Illinois at Urbana-Champaign College of Liberal Arts and Sciences are applying nanotechnology to a number of medical projects [5]. They are developing a sensor that will give surgeons a way to collect significant amounts of information about, say, a tumor while the patient is under the knife. They are also searching for a better way to screen for prostate cancer. They are working on a way to tag the actual cells that cause cancer and determine whether and to what extent they exist in the patient. By making it easier to study membrane proteins, researchers have developed nanodiscs that allow for probing the vast number of possible pharmaceutical targets and develop more efficient drugs.

If the new nanomaterial is proved to be safe, what are the steps to manufacture it? Manufacturing at the nanoscale is known as **nanomanufacturing**. Nanomanufacturing involves scaled-up, reliable, and cost-effective manufacturing of nanoscale materials, structures, devices, and systems. It also includes research, development, and integration of top-down processes and increasingly complex bottom-up or self-assembly processes.

In more simple terms, nanomanufacturing leads to the production of improved materials and new products. As mentioned earlier, there are two basic approaches to nanomanufacturing, either **top-down** or **bottom-up**. Top-down fabrication reduces large pieces of materials all the way down to the nanoscale, like someone carving a model airplane out of a block of wood. This approach requires larger amounts of materials and can lead to waste if excess material is discarded. The bottom-up approach to nanomanufacturing creates products by building them up from atomic- and molecular-scale components, which can be time-consuming. Scientists are exploring the concept of placing certain molecular-scale components together that will spontaneously "self-assemble," from the bottom-up into ordered structures.

Within the top-down and bottom-up categories of nanomanufacturing, there is a growing number of new processes that enable nanomanufacturing. Among these are the following:

- Chemical vapor deposition is a process in which chemicals react to produce very pure, high-performance films.
- Molecular beam epitaxy is one method for depositing highly controlled thin films.
- Atomic layer epitaxy is a process for depositing one-atom-thick layers on a surface.
- Dip pen lithography is a process in which the tip of an atomic force microscope is "dipped" into a chemical fluid and then used to "write" on a surface, like an old fashioned ink pen onto paper.
- Nanoimprint lithography is a process for creating nanoscale features by "stamping" or "printing" them onto a surface.
- Roll-to-roll processing is a high-volume process to produce nanoscale devices on a roll of ultrathin plastic or metal.
- Self-assembly describes the process in which a group of components come together to form an ordered structure without outside direction.

As new technologies allow for smaller and smaller materials, nanotechnology will become a major force in the future.

"BIG HAIRY AUDACIOUS GOAL"

In the earlier chapters, many different strategies have been discussed to provide a means for developing, extending, or maintaining a competitive advantage while being sustainable. There are, however, ideas for sustainability that

don't fit any prescribed strategy. These are adopted when "thinking out of the box" or, in other words, seeking a "big hairy audacious goal" (BHAG). Three different examples are provided to show how one might develop such a project.

Washing Machines

For many years, the washing of clothes has been accomplished by adding a detergent to hot or warm water and then rinse with cold water. This obviously is very energy intensive from the heating of the water as well as the requirement for operating the washing machine. Usually the hot water was obtained from a central hot water tank, but some machines were developed for adding cold water and heating it in the washing machine.

The soap companies eventually produced a detergent that is effective in cold water. This certainly reduces the energy requirement for washing clothes. However, the process still requires the use of a large quantity of water that is usually discharged to a sanitary sewer rather than being reused.

Applying the BHAG philosophy, how could the clothes be cleaned with minimal energy usage and no water? Perhaps the use of some non-toxic chemical in its vapor form could be used to remove dirt and grime from clothes. This chemical in pellet form could be added to, say, a container with the clothes, the pellet activated to release the vapor, and then the clothes tumbled so that the vapor makes contact with all the clothes surfaces. This, in theory, will make clothes cleaning more sustainable, and it is expected that soap manufacturers are conducting research in this area.

Toilets

In the United States almost all toilets are furnished from the same water source as our drinking water. Is it really necessary to flush liquid waste or solid waste with such clean water? If the BHAG philosophy is applied, perhaps another source of water such as gray water would be used. But where would this water be obtained for a home? Again, thinking BHAG will lead to sources of water like collected rainwater, washing machine waste water, dishwasher waste water, or even water going down the drain from a shower. A simple solution would be to collect the drain water from a bathroom sink and divert it to the toilet tank. This gray-water concept has been converted to a number of different designs, one of which is shown in Figure 16.1. Another example of applying the same thinking is in a public bathroom for men. Each of the urinals can have a small water dispenser and sink above



FIGURE 16.1 Toilet using gray water. Source: Reproduced with permission of Victorian Plumbing.

each urinal so the person can wash his hands without leaving the urinal as shown in Figure 16.2. The wash water from the sink would then drain down to flush the urinal. Thus the water would be used twice.

Urban Farming

Prior to the nineteenth century, a typical home was four walls and a roof—a very simple structure. As people started living closer together, anyone needing more space added a second floor. But then it seemed to be more cost-effective for more than one family to live in a building with four floors and a roof. So this became the advent for apartment buildings. As land became more valuable and people continued to live in cities, developers started constructing taller buildings. So basically, starting with living at ground level, we have gone to living in high-rise buildings.

The need for more food created a demand for more agricultural land. But because the farm land is usually far from the large cities where most of the population is located, food travels great distances from its source to



FIGURE 16.2 Urinals with sink for flushing. Source: Courtesy of Kaspar Jursons.

the dinner table, estimated to be about 1500 miles. Implementing the BHAG concept, why can't agricultural land go up just like homes?

A number of entrepreneurs have begun with this concept by developing fruit and vegetable growing in previously vacant warehouses. One such grow house is AeroFarms [9] in Newark, NJ, which consists of 70,000 ft² in a former indoor arena (Fig. 16.3). The vegetables and/or fruits are grown without sun or soil in a fully controlled indoor environment. The company uses aeroponics to mist the roots of the greens with nutrients, water, and oxygen. The aeroponic system is a closed loop system, using 95% less water than field farming and 40% less than hydroponics.



FIGURE 16.3 AeroFarms, Newark, NJ. Source: Reproduced with permission of AeroFarms.

Another example of vertical farming is a new facility that combines the vertical farming concept with solar energy to create a “net-zero vertical farm.” This new farm, a joint effort between University of Toronto—Scarborough and Centennial College—will provide all the electrical power needed for the building which includes research laboratories and a training facility [10]. It will provide all the power needed from the heating and cooling to the lighting of the vertical farm.

Besides producing more food for the growing population, there are numerous other advantages of vertical farming:

- Year-around crop production
- No weather problems
- All food grown organically
- Eliminates agricultural runoff
- Farming in urban cities
- Reduces fossil fuel use
- Converts black and gray water
- Provides jobs for local residents

CASE STUDY

Shell Oil Company: Eureka <https://www.youtube.com/watch?v=L7mfDEJRslY>.

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Sustainable Strategies and Beyond

There are many opportunities to implement sustainable strategies and reduce the impact on the environment, provide social benefits, and do it profitably. It is a matter of thinking on how to make a service or product more sustainable, or even develop a new product or service for an unmet need.

While teaching a course titled “Business Strategy: The Sustainable Enterprise” for over 12 years and another course on “Sustainability Management” for the past two years, the final project for each student was to take a strategic business unit (SBU) at their place of employment and propose how to make it sustainable. The proposal must show how the new business model would possess environmental integrity, social equity, and economic viability. For those students that were not employed, they had the option to propose a project for a publicly held company or to propose a new business. Some of the proposed projects were the following:

- *Medication delivery*—Proposing the reuse of bottles and tubing rather than disposal after one use.
- *Farm equipment*—A central logistics service providing tractor, huskers, and others to farms in a region rather than every farmer required to possess each piece.
- *Eco-toys*—Lease toys to families as they are used for a period of time considerably less than the life of the toy.
- *Endless Christmas*—Since both buying a plastic artificial tree from China and cutting down a real tree for two-weeks use are not good for the environment, why not lease a tree for the holidays; and allow someone else to lease it the following year.

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- *Ford Base of the Pyramid (BOP)*—Proposed a joint venture between Ford and Infosys, a company in India, to produce a low cost car employing the BOP strategy.
- *Communal refrigeration*—Large refrigeration and freezers to be shared by people living in small apartments in a large building.
- *Highway solar panels*—Install solar panels along or on highways where land is not used.
- *Home furnishings*—Expensive long lasting home furnishings can be exchanged for a different long lasting style.
- *eClub*—For those that want the latest electronic equipment, lease it and then allow someone who doesn't need the latest to buy it.
- *Moo-trient*—Collect cattle methane waste and convert it energy and cattle manure to fertilizer.
- *Horizontal refrigeration*—Migration of users to a horizontal orientation for refrigeration with present level of technology used by the chest freezers: thicker insulation and larger cooling plate.
- *Reuse furniture scraps*—Collect the textile and leather remnants, primarily from high end and luxurious furniture companies, and manufacture new smaller products like pillows, wallets, hats, gloves, handbags, and jewelry.
- *Home delivered meals*—Deliver the dinner that will come along with reusable and compostable options giving the customers the option to return the reusable containers or the compostable containers.
- *Refrigeration for sub-Saharan Africa*—A company will provide on-demand transportation services connecting the farmers with refrigeration facilities.
- *Eco-travel*—Provide tourism options that are all sustainable in order to reduce environmental issues at highly visited locations such as the Phi Phi Islands in Thailand, or pollution and erosion in Machu Picchu (Peru) due to massive tourism, and the ecological deterioration of Mount Everest.
- *Ice cream shop*—Collecting the plastic wrap and wooden sticks, after popsicles are consumed, to recycle.

Once the concept of sustainability is embraced, it will become automatic to think of ways to make a product or service more sustainable. Just thinking about everyday products and services, one might think of the following:

- A pizza container may be used for only five minutes, the time it takes from the restaurant to your home. Then it may not be recyclable if it is

greasy or even if it is recyclable, the process is costly for a product used for only five minutes. Why not provide returnable pizza containers?

- Most hotel rooms provide small bottles of shampoo, conditioner, and/or skin lotion. Each may be partially used and then disposed or sent to a community center. Why not provide a large refillable container in the bathroom so the occupant uses only what is needed and provides no waste.
- When flossing teeth, a piece of floss about 12 in. long is needed for wrapping around the fingers. However, only 0.5 in. of the floss is actually performing the work of cleaning the teeth. The other 11 ½ in. is actually waste. Why not provide a device containing the floss so that only a small piece is used and then advanced on the tool to expose a clean section.

Many more ideas can be developed as people accept sustainable strategies and the need for them becomes more critical. So start thinking.

PRACTICAL TOOLS AND GUIDELINES

Standards and Guidelines for Managing Sustainability (CSR)

NEED FOR A SUSTAINABLE STRATEGY

The need for managing corporate social responsibility (CSR) is a rising necessity nowadays, and the business world needs practical tools and guidelines to integrate sustainability into operations.

The reasons why sustainability is extremely urgent are numerous. Firstly, as pointed out by the Living Planet Report 2010 released by WWF, the Ecological Footprint (the measure that defines the extension of land needed to produce and dispose the goods produced) has doubled since 1966, leading humanity, in 2007, to consume the equivalent of 1.5 planets [1]. Furthermore, the gap between poor and rich and problems related to health, education, and social justice are increasing [2]. In addition to this the situation will probably get worse, if we do not take serious measures, since the planet will be inhabited by 9 billion people by 2050, according to the projection on current growth of the United Nations.

Sustainability management should go beyond the usual practices, and all the possible actors should take a more proactive and effective way of approaching it. Including sustainability management into the core business of companies, non-governmental organizations (NGOs) and governments are the only way to have a more sustainable and balanced planet. If all organizations integrate sustainability into their day to day business, there will be good chances of facing (and maybe solving) the challenges of scarce resources and of climate change [3].

In this long and hard path toward sustainability, organizations have a fundamental key role to play. But to do so, they have to implement factors

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that contribute to the success of a sustainable business model. The International Institute for Environment and Development has undertaken research to identify those factors that contribute to the success of business models for sustainable development. Their findings include the following:

- It is necessary that businesses start to approach different actors (companies, government agencies, and development practitioners) to build constructive and strategic alliances.
- Local communities need to be included as partners and co-designers of new models and to develop the local capacity of buying.
- The introduction of sustainable business models for development has to be self-sustaining in the long term; a significant investment of time and resources at the start is key for successful innovation and growth.
- The importance of trade-offs among different sustainable development targets (economic, social, environmental) needs to be recognized and addressed.
- The sustainable business model has to embed an ongoing monitoring and evaluation process.

“Sustainable development is good business in itself. It creates opportunities for suppliers of ‘green consumers’, developers of environmentally safer materials and processes, firms that invest in eco-efficiency, and those that engage themselves in social well-being. These enterprises will generally have a competitive advantage. They will earn their local community’s goodwill and see their efforts reflected in the bottom line.” Therefore, there is no reason why business (together with society) should not move toward a sustainable development.

MANAGING SUSTAINABILITY AND STANDARDS

In general sustainability management is described as the intersection/interaction of the environmental, the economic, and the social sphere. Thus, sustainability is met when all these three criteria are considered and met. It is interesting to report how, in June 2010, United Nation’s Global Compact (UNGC) and Accenture undertook a survey among numerous CEOs, on sustainability. This survey, among others things, has underlined how most of the CEOs interviewed see sustainability as a critical issue for the future of their companies [4].

In order to create the condition to allow sustainability to flourish, there is a strong need of collaboration between different social partners in order

to define a more sustainable world and lifestyle. The role of companies is indeed extremely important for this achievement; in fact companies can significantly contribute by reducing the ecological footprint of their products, as well as adopting smarter solutions for the engagement of their customers, employees, and society.

The Roadmap for Sustainability developed by Ceres presents the main issues that a company should face to embed sustainability within its business strategy. To do so, this document identifies four areas considered key for the successful development of a sustainable business strategy, and those areas are governance, stakeholder engagement, , and performance.

Stakeholder engagement is a key aspect to include on the path toward sustainability. In fact, as it is reported in Ceres' Roadmap for Sustainability, the stakeholders are important actors in helping to identify, and reduce, the environmental and social impacts of the company. Internally the company has to engage itself with the employees and communicate its strategy to them. The way to do it can be through the engagement of people from different business lines, areas, and regions, as well as through the creation of dedicated CSR teams. The external stakeholders are important actors, particularly for multinational companies, in order to understand the impact in other locations within the social and environmental sphere.

A comprehensive disclosure is a useful tool to disclose and manage the sustainable initiatives undertaken by the company; it is also an efficient way to include different stakeholders. In several countries there are requirements on corporate responsibility reporting. Most of the times, the disclosures include credible and standardized metrics, which deal with stakeholders concerns and express clear targets and goals.

Finally, one needs to examine the performance of the company and its relation to sustainability. According to Ceres' Report, undertaking sustainable initiatives helps the company to reduce its costs, transform wastes into resources, eliminate costs inefficiency, avoid conflict within the operations or the supply chain, and, last but not least, increase innovation [5].

There are different ways of managing sustainability, and probably the one on whose effectiveness there is more agreement is embedding sustainability to the core business of the company. Eric Lowitt wrote an interesting article in the Guardian, where he presented how companies such as Starbucks, UPS, Centrica, and Hitachi are adopting three common steps to embed sustainability to their competitive strategies: seek natural ways to tie sustainability and strategy together, connect sustainability to opportunity, and integrate materiality issues into competitive strategy.

The combination of sustainability and strategy design is successful when the sustainable issues faced are relevant for the company management

strategy. For instance, UPS communicated with its stakeholders using a strategic framework, which was already in used, and so familiar, that it was integrated into other new initiatives. Secondly, linking sustainable initiatives to how they can bring new business opportunity is an important and useful process, which helps to integrate them into the corporate strategy of the company. Finally, the material assessment allows companies to prioritize the issues that are more relevant for them and their stakeholders and find a different, and more sustainable, way of solving the issue [6].

To more effectively manage sustainable practices within business, there are also numerous standards that provide help on how to integrate sustainability in the daily strategy including the GRI, ISO 26000, BS 8900, and UNGC.

There are also other standards and guidelines, which, even if not directly, always ask to be strongly embedded to the governance of the organization in order to be more effective and efficient in their roles of supporter of a sustainable development of the organization.

CASE STUDY ON SUSTAINABLE STRATEGY

Wal-Mart is the world's biggest retailer. After years of questionable environmental initiatives, in 2005 Wal-Mart started to build employees, nonprofit, governance agencies, and develop a greener supply chain. Its goal is to decrease environmental footprint and, at the same time, increase profit. To do so Wal-Mart intends to be supplied 100% by renewable energy, to create zero waste, and to sell products, which sustain resources and the environment. Wal-Mart realized that addressing just its operational impact would have faced just 10% of the total impact; therefore it started to engage the external stakeholders, pushing them to join 14 "sustainable value network," within areas of impact identified by the company "to work toward business and environmental sustainability in each area," and to create in this way the Wal-Mart's Sustainable Value Networks.

At the core of the sustainable strategy of Wal-Mart, there is a shift toward the creation of value from long term relationships (with nonprofits, suppliers, and other external stakeholders), which finds a profitable way to approach environmental issues, since it also increase efficiency.

Another interesting and important initiative undertaken by Wal-Mart, in order to integrate sustainability in its strategic management, is that of giving responsibility on sustainability to current employees, in order to better embed the concept within the everyday strategy and management, rather than creating a new department of sustainability. So, with the few exceptions of the small dedicated team of five persons and few people within the

textile and global logistics, there is no employer of Wal-Mart who is working on sustainability full time. The reason for this is given by Tyler Elm, senior director of corporate strategy and business sustainability, which affirms that “Business sustainability is not something you are doing in addition to your job. It is a new way of approaching your job.”

“More than anything else, Wal-Mart’s network approach must remain profitable if it is to be sustainable in the long run and achieve Scott’s environmental goals”.

GLOBAL REPORTING INITIATIVE (GRI) AND STAKEHOLDERS

The Global Reporting Initiative (GRI) is one of the most widely adopted methods of sustainability reporting. GRI is a nonprofit organization that introduced a sustainability-reporting framework in 2000. Today, more than 3000 companies across the world use GRI’s reporting system. According to the organization’s website, “The GRI Reporting Framework is intended to serve as a generally accepted framework for reporting on an organization’s economic, environmental, and social performance. It is designed to be used by organizations of any size, sector, or location.”

Naturally, the stakeholder approach is an essential element of this framework. The GRI Sustainability Reporting Guidelines note, “[The] multi-stakeholder approach to learning has given the Reporting Framework the widespread credibility it enjoys with a range of stakeholder groups” [7]. In other terms, it is widely accepted that businesses must consider the interests and perspectives of stakeholders in order to make significant strides in sustainability.

The GRI framework identifies “stakeholder inclusiveness” as an essential measure of sustainability. GRI advises, “The reporting organization should identify its stakeholders and explain in the report how it has responded to their reasonable expectations and interests” [8]. In other terms, it is essential to measure a company’s stakeholder strategy and engagement in order to properly assess and disclose its sustainability structure.

It is important to understand that stakeholder theory does not stand-alone; rather it is inextricably linked to many aspects of sustainability. For example, the concept of *transparency* is primarily fostered through stakeholder engagement. GRI defines transparency as “the complete disclosure of information on the topics and indicators required to reflect impacts and enable stakeholders to make decisions, and the processes, procedures, and assumptions used to prepare those disclosures” [7]. In this sense, a company

must ensure transparency in order to properly engage stakeholders and further its sustainability goals. One cannot achieve transparency or proper stakeholder engagement without the other.

GRI INTERPRETATIONS OF STAKEHOLDER ENGAGEMENT

The term “stakeholder” conjures different meanings in different contexts. Given the ambiguity of the term itself, GRI explicitly offers its interpretation of “stakeholder inclusiveness.” According to GRI guidelines, “Stakeholders are defined as entities or individuals that can reasonably be expected to be significantly affected by the organization’s activities, products, and/or services; and whose actions can reasonably be expected to affect the ability of the organization to successfully implement its strategies and achieve its objectives” [7]. Compared with other interpretations, GRI’s definition is rather broad.

One notes that the definition itself can be interpreted in various ways, depending upon the reader’s interpretation of the term, “reasonably.” This realization underlines the elusiveness of key terms within sustainability and alludes to the difficulties associated with establishing overarching international standards. The ambiguity of key terms within sustainability can become problematic as different parties often subscribe to their own interpretations. In order for interested parties to collaborate and progress sustainability issues, they must agree on a single interpretation of the key terms.

THE STAKEHOLDER REPORTING PROCESS

GRI’s interpretation of “stakeholder inclusiveness” introduces an important challenge relating to reporting and stakeholder engagement. Considering that a sustainability report must speak to the needs and perspectives of those “reasonably” affected, how does the company determine which stakeholders it should address in its report? This question relates to the stakeholder identification.

A company must act in a systematic and organized fashion in order to respond to this challenge. GRI explains, “For a report to be reliable, the process of stakeholder engagement should be documented. When stakeholder engagement processes are used for reporting purposes, they should be based on systematic or generally accepted approaches, methodologies, or principles” [7]. A company should report its process of stakeholder identification in an open and honest manner.

Similarly, a report should succinctly document how and when the company interacted with those parties and identify measures to balance divergent stakeholder expectations. Finally, it should address how stakeholder engagement affects, influences, and relates to the content of the report and the sustainability initiatives of the company [7].

GRI TESTS FOR STAKEHOLDER INCLUSIVENESS

As a comprehensive sustainability framework, GRI explicitly identifies those aspects of stakeholder engagement a company should address. The following list is an extract from GRI's guidelines, which outline categories of information companies should disclose in order to foster proper stakeholder engagement [7]:

- List of stakeholder groups engaged by the organization.
- Basis for identification and selection of stakeholders with whom to engage.
 - This includes the organization's process for defining its stakeholder groups, and for determining the groups with which to engage and not to engage.
- Approaches to stakeholder engagement, inducing frequency of engagement by type and by stakeholder group.
 - This could include surveys, focus groups, community panels, corporate advisory panels, written communication, management/union structure, and other vehicles. The organization should indicate whether any of the engagement was undertaken specifically as part of the report preparation process
- Key topics and concerns that have been raised through stakeholder engagement and how the organization has responded to those key topics and concerns, including through its reporting.

PRESENTATION OF REPORTED STAKEHOLDER DISCUSSIONS

Stakeholder engagement is an essential component of sustainability reporting. Yet, there are a variety of ways companies can choose to report their engagement efforts. While some companies opt to construct a narrative of the ways they interact with stakeholders, such as the selected case studies in this chapter, others choose a more visual or graphic approach. For example,

BMO Financial Group's 2010 Sustainability Report presents its stakeholder engagement activities in outline format [8].

BMO engages regularly with a wide range of stakeholders as we work to identify key issues affecting sustainability in all of its dimensions—social, environmental, and economic. The valuable insights provided by these diverse groups guide us in better managing our business to meet their needs.

Our responsibilities to our stakeholders are as follows:

Customers:

- *Satisfaction*—Our vision, defining great customer experience, means we work to exceed customers' highest expectations.
- *Trust*—Build customer relationships based on reciprocal trust, honesty, integrity, open dialogue, and mutual respect.
- *Access*—Provide convenient access to banking services wherever our customers live and regardless of differences in abilities.
- *Products*—Provide financial products and services that address our customers' needs at every stage in their lives.

Communities:

- *Investment*—Through donations and sponsorships, seek opportunities to support initiatives that improve the quality of life in communities where we live, work and do business.
- *Support*—Maintain our commitment to supporting BMO employees' involvement as volunteers and donors in charitable campaigns and programs.

Employees:

- *Talent*—Cultivate an environment where all employees are encouraged to explore their potential and contribute to their fullest—and where strong performance is rewarded.
- *Leadership*—Develop current and future leaders' capabilities at key stages in their careers, using a consistent leadership framework across all lines of business.
- *Inclusion*—Promote an open and supportive workplace, and insist on diversity, equity and inclusion in every area of the organization.
- *Well-being*—Foster an environment that creates opportunities for employees, where change can be viewed positively and feedback is frequently sought. Advocate work-life effectiveness and other approaches that provide flexibility to employees, with managers committed to providing coaching and guidance.

Shareholders:

- *Performance*—Maximize returns to investors with sustainable improvements in performance achieved through effective growth strategies and sound fiscal management.
- *Governance*—Maintain responsible, rigorously ethical, and fully accountable corporate governance standards and practices.
- *Transparency*—Provide financial disclosure and environmental, social, and governance information that is clear, comprehensive, and meaningful to all shareholders.

Regulators:

- *Compliance*—Remain compliant with all relevant laws and government regulations in the countries where we operate.

Non-governmental organizations (NGOs):

- *Responsibility*—Address concerns raised by various NGOs by maintaining responsible lending practices and considering the environmental and social impacts of our business decisions.

Suppliers:

- *Fairness*—Strive to ensure equity in the selection and management of all vendors supplying goods and services to BMO.

ISO 26000 FRAMEWORK

Although GRI is among the most popular and highly regarded CSR frameworks, it is by no means the solitary golden standard of sustainability reporting. In 2010, the International Organization for Standardization (ISO) published ISO 26000, which introduced a new set of “Social Responsibility” guidelines. ISO is the world’s largest developer and publisher of International Standards, as it is a network of national standards institutes from 163 countries [9]. The members of this vast network, represented by industry, government, labor, consumers, NGOs, and service representatives, spent five years creating the ISO 26000 standard.

According to ISO, “The International Standard ISO 26000:2010, *Guidance on social responsibility*, provides harmonized, globally relevant guidance for private and public sector organizations of all types based on international consensus among expert representatives of the main stakeholder groups, and so encourage the implementation of best practice in social responsibility worldwide” [10]. In more succinct terms, this initiative

provides flexible guidance for organizations of all types on sustainability strategies.

ISO explicitly states that its 26000 should not be used for certification. According to the website, “Any offer to certify, or claims to be certified, to ISO 26000 would be a misrepresentation of the intent and purpose and a misuse of this International Standard” [10]. As ISO 26000 does not issue requirements, only detailed guidelines, the framework is not appropriate for certification.

The fifth clause of ISO 26000 is entitled, “Recognizing social responsibility and engaging stakeholders.” The ISO sustainability framework, like GRI, emphasizes the importance of stakeholder engagement. In fact, the ISO standard identifies stakeholder engagement as one of two “fundamental” practices of social responsibility. These fundamentals include “an organization’s recognition of its social responsibility, and its identification of and engagement with its stakeholders” [11].

In addition, the framework establishes seven core subjects, the first of which is “accountability.” In this context, accountability describes the company’s responsibility to society as well as those affected by its decisions and operations. Evidently, this term relates directly to the stakeholder approach. The fourth core area is “respect for stakeholder interest.” In conclusion, ISO “provides guidance on the relationship between an organization, its stakeholders and society, on recognizing the core subjects and issues of social responsibility and on an organization’s sphere of influence” [12].

UNITED NATIONS SUSTAINABLE DEVELOPMENT GOALS (UN SDGS)

At the UN Sustainable Development Summit (September 25, 2015), world leaders adopted the 2030 Agenda for Sustainable Development, which includes 17 Sustainable Development Goals (SDGs), also known as the Global Goals, to end poverty, fight inequality and injustice, and tackle climate change by 2030 [13]. “World leaders have an unprecedented opportunity this year to shift the world onto a path of inclusive, sustainable and resilient development,” Helen Clark, the UN Development Programme (UNDP) Administrator, indicated.

Through the MAPS approach, which is about Mainstreaming, Acceleration, and Policy Support [14], UNDP supports countries in three different ways—firstly, by providing support to governments to reflect the new global agenda in national development plans and policies, which is underway in many countries; secondly, by supporting countries to accelerate progress on

SDG targets; and thirdly, by making the UN's policy expertise on sustainable development and governance available to governments at all stages of implementation.

While the SDGs are not legally binding, governments are expected to take ownership and establish national frameworks for the achievement of the 17 goals. Countries have the primary responsibility for follow-up and review of the progress made in implementing the Goals, which will require quality, accessible, and timely data collection. However, the Goals are not intended only to governments and countries, but also businesses.

CASE STUDY: BMO FINANCIAL GROUP—2017 ENVIRONMENTAL, SOCIAL, AND GOVERNANCE REPORT AND PUBLIC ACCOUNTABILITY STATEMENT

Report Extract

Sustainable Development Goals The UN SDGs are a universal call to action to end poverty, protect the planet, and ensure that all people are able to enjoy peace and prosperity. The 17 goals have specific targets to be reached by 2030, but this can only be achieved if governments around the world work in concert with the private sector and civil society organizations [15].

The SDGs provide a common set of objectives that governments, companies, and investors can cite in order to consistently communicate how their activities are resulting in positive changes. Through BMO's participation in the Global Compact Network Canada (GCNC), we collaborate with other Canadian businesses on how best to report the progress being made toward achieving the SDGs.

Throughout this report, we indicate where our efforts are in alignment with specific SDGs. Here, we highlight the five SDGs for which we believe BMO can have the greatest impact, with relevant examples of our accomplishments in 2017 (Table 18.1).

United Nations Global Compact Initiative The UNGC in another important initiative serves to promote sustainability reporting. The compact is a multinational policy plan to encourage companies to act in a socially responsible manner. According to the official website, "The UN Global Compact is a strategic policy initiative for businesses that are committed to aligning their operations and strategies with 10 universally accepted principles in the areas of human rights, labor, environment and anti-corruption." [16] The Compact's framework boasts more than

TABLE 18.1 Sustainable Development Goals for which BMO can have the Greatest Impact

Social change	SDG 5: Gender equality	BMO has multi-year workforce representation goals that include gender equity in senior leader roles—Canada and United States (%). Our 2020 goal is a minimum of 40% women and men in each line of business.
	Achieve gender equality and empower all women and girls	
Community building	SDG 9: Industry, innovation, and infrastructure	BMO Harris Bank provided US\$552 million in community development loans in 2017 to support economic development, community services and affordable housing.
	Build resilient infrastructure, promote sustainable industrialization, and foster innovation	
Financial resilience	SDG 10: Reduced inequalities	In 2017, our Aboriginal Banking unit celebrated 25 years of working to meet the personal banking needs of Aboriginal peoples, provide financial advice to on reserve businesses, partner with communities on their economic development initiatives, and offer customized trust and investment management solutions.
	Reduce inequality within and among countries	
Environmental impact	SDG 12: Responsible consumption and production	BMO has been carbon neutral since 2010. Early in 2017, we joined the Carbon Pricing Leadership Coalition (CPLC), a voluntary initiative that supports the implementation of carbon pricing around the world.
	Ensure sustainable consumption and production patterns	
	SDG 13: Climate action	BMO Capital Markets participated in US\$6.4 billion of renewable energy equity and debt financing and provided US\$4.2 billion of loan commitments to renewable entities and renewable projects.
	Take urgent action to combat climate change and its impacts	

8700 corporate participants, 6000 businesses, and stakeholders from 135 countries making the UNGC the world's largest voluntary CSR reporting initiative. The 10 principles of UNGC are listed in detail as follows:

Human rights:

- *Principle 1*—Businesses should support and respect the protection of internationally proclaimed human rights.
- *Principle 2*—Make sure that they are not complicit in human rights abuses.

Labor:

- *Principle 3*—Businesses should uphold the freedom of association and the effective recognition of the right to collective bargaining.
- *Principle 4*—The elimination of all forms of forced and compulsory labor
- *Principle 5*—The effective abolition of child labor.
- *Principle 6*—The elimination of discrimination in respect of employment and occupation.

Environment:

- *Principle 7*—Businesses should support a precautionary approach to environmental challenges.
- *Principle 8*—Undertake initiatives to promote greater environmental responsibility.
- *Principle 9*—Encourage the development and diffusion of environmentally friendly technologies.

Anti-corruption:

- *Principle 10*—Businesses should work against corruption in all its forms, including extortion and bribery.

Overall, the Global Compact pursues the following two complementary objectives:

1. Mainstream the 10 principles in business activities around the world
2. Catalyze actions in support of broader UN goals, including the Millennium Development Goals (MDGs)

With these objectives in mind, the Global Compact has shaped an initiative that provides collaborative solutions to the most fundamental challenges facing both business and society. The initiative seeks to combine the best

properties of the UN, such as moral authority and convening power, with the private sector's solution-finding strengths, and the expertise and capacities of a range of key stakeholders. The Global Compact is global and local; private and public; and voluntary yet accountable.

SUSTAINABILITY ACCOUNTING STANDARDS BOARD (SASB)

While previous initiatives focus more on environmental, social, governance (ESG) and sustainability information, Sustainability Accounting Standards Board (SASB) standards are mainly targeted at financially material issues. The SASB [17], which was founded in 2011, aims to help companies around the world report on the sustainability topics that are reasonably likely to impact the financial condition and therefore matter most to their investors [18]. The set of 77 standards is ideal for company comparison and benchmarking because of its special industry-specific concept.

INTERNATIONAL INTEGRATED REPORTING COUNCIL (IIRC)

Another global coalition, which was convened in order to promote the reporting of financial stability and sustainable development, is the International Integrated Reporting Council (IIRC). More specifically, this allied group, which consists of “regulators, investors, companies, standard setters, the accounting profession and NGOs,” [19] aims to direct businesses and investors to adoption of integrated reporting (IR). This has to do with the integrated representation of a company's performance about value creation, under the concept of corporate communication.

CORPORATE SUSTAINABILITY TRENDS

In recent years, many studies have been conducted to provide a useful and convenient representation of the current conditions regarding Sustainability Reporting by companies and organizations. In this context, in 2017 and 2018 the Centre for Sustainability and Excellence (CSE) carried out annual Research of Sustainability (ESG) Reporting Trends in North America. The research concerns the analysis of the Sustainability Reports that are published by 1200 companies and organizations stationed in North America (United States and Canada), many of which are included in the Fortune 500 companies. More specifically, for the first time, the report included a more detailed survey of the correlation between Sustainability and Financial

results and provided insight into Artificial Intelligence (AI) and Blockchain applications that will transform the sustainability field over the next years.

The most important trends seem to be the following:

1. *Highest sustainability score may indicate better financial results:*

There is a strong correlation between the financial performance of the organization and the sustainability performance, as indicated by the analysis of 50 companies with the highest ESG score (CSRHub Rank from July 2018). In particular, 80% of these companies had an increase in their revenues between 2016 and 2017.

2. *Artificial intelligence (AI) and blockchain:*

AI is gaining more followers as it offers useful tools for almost every kind of industry. Technological advances and increasing demand for real-time updates are going to bring a number of transformations in the way corporate sustainability is monitored, implemented, and reported. Respectively, Blockchain can change the way business transactions take place. From a supply chain perspective, such visibility will help ensure efficient transactions, while promoting food safety, efficient recalls, the elimination of counterfeits, and the assurance of ethical trading partners.

3. *Sectors with high volume of sustainability reports:*

The sectors with the highest presence in the research sample, regardless of whether the organization has used any reporting guidelines for publishing the report, are Financial Services (10.0%), Energy and Energy utilities (9.5%), Food and Beverage (7.0%), Healthcare Products (5.3%) and Mining (5.1%). The top sectors in the United States are Energy and Energy Utilities (8.5%), Food and Beverage (8.5%), and Financial services (8.1%), while in Canada the top sectors are Mining (19.4%), Financial Services (15.7%), and Energy and Energy Utilities (13.4%).

4. *Most frequent terms used for sustainability reporting:*

The two most frequent terms used for reporting are “Sustainability Report” and “Corporate Responsibility Report” among a wide range of combinations of terms like “sustainability,” “corporate,” “citizenship,” “social,” “responsibility,” and “sustainable development.”

5. *Transparency of sustainability reports:*

The use of the GRI reporting guidelines results in larger, more comprehensive, sustainability reports, in the United States and Canada, while non-GRI reports tend to be smaller in size, mostly focusing

on the companies' sustainability programs and initiatives and not their sustainability performance.

6. Most frequent standards and guidelines for reporting:

58.7% of the North American companies and organizations that published sustainability reports for the 2016–2017 reporting period used some version of the Reporting Guidelines of the Global Reporting Initiative or used it as the reference point. 24% of the reports include the information about the company/organization's compliance with the Carbon Disclosure Project (CDP). 14% include the information about the company/organization's compliance with the 10 principles of the UNGC. 13.9% include the information about the company/organization's compliance with the 17 SDGs.

7. Use and compliance to GRI standards for sustainability reporting:

15.7% of the companies/organizations that published the sustainability reports for 2016–2017 period used the latest version of the GRI Standards and 32.4% the GRI's Reporting Guidelines (G4). In terms of the compliance levels ("in accordance-core" and "in accordance-comprehensive"), which indicate the number of indicators the company/organization has chosen to disclose in the report for each identified material issue, most companies/organizations (76.4%) have chosen the "core" option, only a small number of them (7.1%) have chosen "comprehensive" option, and a significant percentage (16.5%) have not declared a compliance level.

8. The decline of external assurance:

Only 25.7% of all GRI reports have been externally assured. The most commonly used, internationally recognized assurance standards are the Accountability 1000 Assurance Standard (AA1000AS) and the International Standard on Assurance Engagement (ISAE 3000). 28.4% of the external assurances have been conducted using the ISAE 3000, while 10.8% the AA1000AS. 3.9% of the external assurances were conducted using both assurance standards.

ISSUES FOR LEARNING AND DISCUSSION

1. Do product and service innovation belong to Growth, Productivity, or Risk management of a company's sustainable business strategy?
2. How many principles does the UNGC have?

3. Which are the top sustainability trends that were identified in CSE's North America Research 2017–2018?
4. Do the SDGs deal only with social issues, or do they cover environmental as well? Can companies adopt them and link them with their initiatives and sustainability efforts or are the SDGs used only by countries?

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The Corporation and its Stakeholders

EXAMINING THE STAKEHOLDER CONCEPT

The relationship between the organization and its stakeholders has evolved over time. During previous years, the only groups that seemed to attract an organization's attention were those related to the production and purchase of their products or services. These groups, also known as the three-legged stool, included the employees, investors, and customers. More specifically, during the pre-corporate period, investors, who supplied their money in order to support the business, were the only individuals to have stake in its operations and successes. These investors/owners, typically the managers of the firm, required assistance from employees to operate the business. All in all, the procedures were less complex, but so were, moreover, the expectations among the various parties involved with the organization.

As years went by, various forces triggered society toward transformation. A principal factor that greatly contributed to the gradual change of business conduct was the recognition by the public, or society, that any organization is not the sole property or interest of the founder or owner. This and many other driving forces have created the need for businesses today to be responsive to individuals and groups they formerly viewed as powerless and indifferent toward their conduct. Thus, the stakeholder concept emerged. Contemporary stakeholder focus includes an augmented number of groups, and it reveals a more sensitive approach toward them. Modern businesses, non-governmental organizations (NGOs), and governments demonstrate the importance of stakeholder relationships and seem to grasp the vital role they play in terms of the organization's success.

Concurrently, during the last decades, the business world appears preoccupied with an interesting debate. This debate namely concerns the issue

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of whether corporations should focus on shareholders or stakeholders. The adoption of either one of these two prominent positions actually dictates the organizational policy and strategy and moreover provides a theoretical framework upon which organizational culture is formatted. Are corporations exclusively responsible to their shareholders and therefore their only objective and responsibility is to make profit for them, or are corporations responsible for all stakeholders?

One of the most absolute advocates of the first position is undoubtedly Milton Friedman, who in 1970 wrote: *“the one and only social responsibility of business is to use its resources and engage in activities designed to increase its profit.”* According to this theory, shareholders provide capital to corporate managers in order to achieve certain final results, usually the maximization of share values. The shareholder theory contrasts with the stakeholder theory in the sense that it focuses particularly on value creation for shareholders. Furthermore, advocates of the shareholder theory argue that the stakeholder theory does not provide a clear idea of how the corporation should address conflicting stakeholder expectations. However, advocates suggest that the principle of value maximization, of the shareholder theory, gives a clear way to conceptualize and manage trade-offs between corporate stakeholders.

In spite of this, during the 1990s, the second position of stakeholder focus was highlighted by several important conferences on stakeholder theory. “The late Max Clarkson of the University of Toronto convened two conferences in 1993 and 1994 on this topic. In 1994, Juha Näsi convened a conference on stakeholder theory in Finland.” [1] Stakeholder focus is embedded in the idea of sustainability (corporate social responsibility (CSR)). One can argue that one of the most essential contributions of sustainability is the critical shift from shareholders to stakeholders. Indeed businesses that align sustainability with their operations are focusing on the entire spectrum of their stakeholders rather than shareholders alone.

STAKEHOLDERS: DEFINITION—PRIMARY AND SECONDARY STAKEHOLDERS

The stakeholder theory aims to systematically address the fundamental question of which groups are stakeholders and which are not. There is no actual disagreement among scholars involved in this issue, on what kind of entity is or may potentially be a stakeholder. In general, persons, groups, neighborhoods, organizations, societies, etc. are thought to be eligible stakeholders. In order to better appreciate the concept of stakeholders one needs to first examine the idea of a stake. A stake is “an interest or a share in an

undertaking” [1], as well as a claim to a title or a right to something. The idea of a stake, therefore, lies on a continuum where on one end there is the simple interest in an undertaking and on the other end a legal claim.

Proceeding to the definition of the concept, stakeholder theorists differ considerably on whether they take a broad or narrow view of the issue. For example, Freeman, adopting a broader perspective, defines a stakeholder as, “any group or individual who can affect or is affected by the achievement of the organization’s objectives” [2]. This definition is certainly among the broadest in the literature and lies in stark contrast to Clarkson’s narrower definitions of stakeholders. In fact, Clarkson creates a distinction between voluntary or involuntary risk-bearers. He writes, “Voluntary stakeholders bear some form of risk as a result of having invested some form of capital, human, or financial, something of value, in a firm. Involuntary stakeholders are placed at risk as a result of a firm’s activities. But without the element of risk there is no stake” [3].

Between the broad and narrow are many other efforts to define stakeholders. Nevertheless, we will address the issue with respect to Freeman’s definition since it includes a broader spectrum of actors. Therefore, stakeholders can be defined as categories of people that affect or can be affected by the company’s decisions, policies, and operations. These parties, as well as their interests and expectations, vary considerably, and can be contradicting. An organization’s main goal is to balance stakeholders’ needs in a general attempt to satisfy them. A company’s primary relation with society includes all the interactions with groups that result in its fulfilling its ultimate goal of producing goods and services for customers. Therefore, stakeholders can be segmented into two main groups. The first group is primary stakeholders, who are considered critical to the organization’s survival and actions, and include customers, communities, suppliers, employees, and investors. The second group is secondary stakeholders, groups that are interested in or affected by organizational conduct; these include the general public, media, NGOs, government, activists groups, and others. However, secondary stakeholders should not be considered in any case less important and in some cases could be primary stakeholders depending on the company sector.

CASE STUDY: CAMPBELL SOUP COMPANY—2018 CORPORATE RESPONSIBILITY REPORT EXTRACT

Stakeholder Engagement

Proactive and consistent engagement with both internal and external stakeholders is critical to effectively executing our corporate responsibility strategy [4]. Our approach to stakeholder engagement includes [4] the following:

- Identification of issues that are most important to stakeholders
- Establishment of processes for meaningful and effective engagement
- Benchmarking for leadership performance within and outside our sector

Table 19.1 summarizes our engagement with major stakeholder groups.

TABLE 19.1 Stakeholder engagement by Campbell Soup Company

Stakeholders	Types of engagement	Issues
Advocacy groups/non-governmental organizations (NGOs)	• Organizational memberships • Direct engagement • Strategic issue partnerships • Professional conferences • Roundtable events • Social media	• Health, wellness, and nutrition • Nutrition and labeling standards • Animal welfare • Deforestation • Climate change • Human rights • Sustainable agriculture
Communities	• Direct, on-the-ground relationships • Corporate and employee giving • Networking events • Surveys • Signature partnerships • Regional priority plans • Employee volunteerism	• Critical local needs (such as food access and nutrition education) • Disaster relief • Environmental initiatives • Health, wellness, and nutrition • Strategic partnership priorities (such as health and well-being)
Consumers	• Dedicated 24-h toll-free hotline • Consumer insights • Branded websites • Campbell's Kitchen® website • Campbell's Nutrition website • What's in my food website • Social media • Uncanned™ by Campbell's website	• Food safety and quality • Transparency • Packaging information • Health, wellness, and nutrition • Affordability • Food waste • GMOs • Environmental practices
Customers	• Dedicated customer teams • Top-to-top customer meetings • Customer surveys • Industry trade groups (such as FMI and consumer goods forum)	• Product innovation • Trade practices • Social accountability • Environmental practices • Transparency • Sales support and service • Health, wellness, and nutrition

TABLE 19.1 *(continued)*

Stakeholders	Types of engagement	Issues
Employees	• Quarterly employee forums • Town hall meetings • Campbell ID Networks • Interactive workplace by Facebook site • Anonymous hotline	• Diversity and inclusion • Occupational health and safety • Recognition • Equitable compensation and benefits • Talent management • Business trends • Health, wellness, and nutrition
Investors	• Meetings with institutional and socially responsible investors • Direct engagement on specific topics of interest • Ratings/rankings and indices • Annual report • Annual shareholder meetings • Investor calls and forums	• Business performance • Governance practices • Sustainability strategies • Health and well-being strategies • Ethical business practices • Human rights • Nutrition
Regulators and policymakers	• Campbell political action committee • Direct engagement on issues important to Campbell • Advocacy • Significant monitoring and communication of regulatory activities • Industry trade associations	• Food safety and quality • Product labeling • Trade policy implications • Health, wellness, and nutrition policy • Environmental policy • Sustainable agriculture
Suppliers	• Collaborative partnerships • Responsible sourcing supplier code • Strategic relationship management program • Face-to-face meetings • Supplier diversity program • Surveys • Assessments and audits	• Product safety and quality • Human rights • Ethical sourcing • Sustainable packaging • Ingredient traceability • Cost-savings opportunities • Supply chain risk mitigation • Sustainable agriculture

Stakeholder Relations and Attributes—Power, Legitimacy, Urgency

Evidently, firms and stakeholders have a relationship most of the time based on power dependence. Some definitions focus on the firm's dependency on its stakeholders in order to survive, while others focus on the stakeholders' dependency on the firm for upholding their rights, minimizing problems, or achieving their interests. In general, broader definitions emphasize stakeholders' power to influence the firm's conduct, irrespective of legitimate claims. In order to better illustrate the issue, let us define power. Power is "the probability that one actor within a social relationship would be in a position to carry out his own will despite resistance" [5]. Therefore, a stakeholder group has power as it has or can obtain access to coercive, utilitarian, or normative means in order to impose its will in the relationship with the organization. Coercive power is based on the physical resources of force; violence, or restraint, utilitarian power is based on material or financial resources, and normative power on symbolic resources.

Another stakeholder attribute is legitimacy, which refers to "the perceived validity or appropriateness of a stockholder's claim to a stake." [1] Consequently, owners, employees, and customers represent a high degree of legitimacy due to the nature of their relation with the organization, which can be characterized as explicit, formal, and direct. The last attribute is urgency, which refers to "the degree to which the stakeholder's claim calls for the business's immediate attention or response" [1]. Based on these three attributes, Mitchell et al. [6] created a typology of stakeholders, which depicts the degree toward which managers must attend stakeholders based on their assessment, and of competing stakeholder claims using the criteria of legitimacy, power, and urgency.

CASE STUDY: TD—2017 CORPORATE RESPONSIBILITY REPORT EXTRACT

Stakeholder Engagement

There are many environmental, social, and governance issues that demand our attention [7]. We give more importance to those risks or opportunities that directly relate to TD's core business, are stated priorities for our stakeholders, and are areas where we can effect change. "We understand the importance of not only listening to stakeholders, but using that feedback to take action and improve" (Table 19.2) [7].

TABLE 19.2 Stakeholder engagement by TD

Stakeholder group	Ways we interact	Key topics in 2017
Customers	• Solicit feedback by phone and online • Formal process for handling complaints • TD ombudsman • Consumer associations • Social media team	• Fee changes • Processing delays • Credit decisions • Access to banking • Financial education • Sales practices
Colleagues	• Employee surveys, focus groups, and HR meetings • Executive leadership visits • Intranet comment engine and online communities (connections) • Employee ombudsman (between us) • Employee assistance program • Whistleblower hotline	• Improving work processes • Career development • Diversity and inclusion in the workplace • Increased emphasis on employee wellness • Sales practices
Shareholders and investors	• Annual meeting and quarterly earnings calls • Shareholder proposals • Shareholder relations team • Regular meetings with investors • Investor relations website • Industry conferences	• Transition to a low-carbon economy • Pipeline expansion projects • Sales practices • Interest rates • Digital evolution in the banking sector and competing with non-traditional players • Proxy access policy
Government	• Government relations teams for Canada and the United States • Ongoing dialog with regulators and policy-makers	• Progress on implementing regulations • Evolving credit card and payments landscape • Working to create greater financial literacy • Sales practices
Suppliers	• Website for prospective suppliers • Email responses to supplier questions	• Third party risk management • Education for small and diverse-owned suppliers on doing business with large companies

TABLE 19.2 (continued)

Stakeholder group	Ways we interact	Key topics in 2017
Industry associations	• Industry association memberships • Memberships with various multi-stakeholder groups • Participation in financial center bodies	• Meeting needs of customers and evolving customer expectations • Coordination among authorities for more workable regulation • Oversight of unregulated and under-regulated market participants
Communities	• Citizenship team • Diversity team • Ongoing dialog with community organizations • Volunteer network • TD friends of the environment foundation local chapters	• Financial education • LGBTQ rights • Indigenous communities • Opportunities for young people • Disaster support • Building operational and measurement capacity
Non-governmental organizations	• Meetings, phone calls, face-to-face consultation • Funding research projects • Conferences and forums • Over 250 engagements on environmental topics	• Environment and climate change • Low-carbon economy • Financing of fossil fuel energy developments • Sustainable investing • Free prior and informed consent • Natural capital

Balancing Stakeholders’ Expectations

Once the organization successfully identifies and classifies its stakeholders, it must then identify the needs and expectations of the stakeholders. Those expectations and needs are often expressed in various forms, such as letters of complaint or direct dialog. However, sometimes there are needs and expectations that are more difficult to define, as they are not expressed in a direct way, or even not (“expressed”)/communicated at all. However, the major question is not how to identify those needs and expectations, but rather how to address their conflicting nature.

Conflicting needs and expectations exist among all stakeholder groups. For example, the desires of internal stakeholders may differ from those of external stakeholders. For example, customers want low prices, high quality, and excellent service. On the other hand, employees want high wages, high-quality working conditions, and fringe benefits, while suppliers want low risk and high rate of returns; Communities want companies to benefit

the community at large, increased local investment, and employment and the list goes on. Evidently, stakeholder relations are more multifaceted than simple PR work and must be tailored to meet individual stakeholder requirements. These procedures are as diverse as the list of key stakeholders, and should take into account each party's appetite for direct involvement, preferred communications mechanisms, and key motivators.

In modern dynamic environments, one should not underestimate the value of strong stakeholder relationships, given the power of external influences to either enhance or disrupt organizational stability and viability. The benefits of good stakeholder relations are usually intangible and therefore hard to quantify. However, they do offer some form of insurance in an uncertain world. Furthermore, opening new channels of communication can offer fresh insight and value, which would have been otherwise inaccessible.

CASE STUDY: HESS CORPORATION—2017 SUSTAINABILITY REPORT EXTRACT

Stakeholder Engagement

Effectively engaging with stakeholders in our host communities helps Hess to responsibly access the resources we require as an international energy company—from the earliest phases of our new country entry process through the decommissioning of an asset [8]. Proactive, two-way and ongoing stakeholder engagement helps to establish a mutual understanding of expectations between Hess and those who live and work in our areas of operation. We understand that our success is in part tied to our ability to mitigate the potential risks associated with our activities, which could impact stakeholder relationships and public perception. We aim to create value through mutually beneficial opportunities with our stakeholders, which include communities, employees, contractors, suppliers, customers, industry members, governments, and investors.

Stakeholder Engagement Process

Our stakeholder engagement process focuses on the proactive relationship—and trust—building opportunities created by meaningful engagement, as well as the business value that engagement brings when it is integrated as a part of project risk management. Consistent with our continuous improvement methodology, we follow a “Plan, Do, Check, Adjust” cycle to direct our stakeholder management efforts. As the expectations, priorities, and

concerns of our stakeholders evolve, we adjust our approach to align with their needs and our business activities.

In 2017 we reviewed our stakeholder engagement process and associated tools at our Bakken asset as part of the regular planning cycle. The result was a refreshed and robust engagement strategy, which will be integrated into the asset's 2018 business plan. We will also continue to integrate stakeholder issues and engagement into our enterprise risk workshops and value assurance reviews.

We are working with our other assets, as part of implementing our environment, health, safety (EHS) and social responsibility (SR) strategy, to develop individualized plans for engaging with their priority stakeholders. Our Gulf of Mexico, Denmark, and Ghana assets developed engagement plans in 2017, and our other assets are in various stages of developing their plans.

EXTERNAL STAKEHOLDERS

Collaborating with stakeholders helps us to identify opportunities for benefiting our host communities while improving our business and strengthening our license to operate. We engage with a wide range of external stakeholders—from local landowners and governments to community service agencies and indigenous groups. Recent examples of our stakeholder engagement activities are provided in Table 19.3 [8].

Materiality Assessment

In keeping with the philosophy of sustainability, every company is committed to objectively identifying its material issues and evaluating the level of their impact to its stakeholders. Accordingly, one of the key principles of the Global Reporting Initiative (GRI) and any sustainability report is the Materiality Principle. “Material issues are those that have direct or indirect impact on the organization’s ability to create, maintain or lose environmental and social value for itself and its stakeholders,” as defined by the GRI [9]. Therefore, an organization must report on topics that cause the most significant economic, environmental, and social impact, or those considered more important by an organization’s internal and external stakeholders. In order to determine these issues, an organization must actively involve its stakeholders and examine all the topics for the time period that the sustainability report will refer to.

The methodology most of organizations follow is described in the following text:

TABLE 19.3 Stakeholder engagement by Hess Corporation

Category	External stakeholder groups	Recent examples of engagement
Land users/landowners	Residents, landowners, commercial land interests, farmers, ranchers	Continued to work through the Hess Community Connection to increase communication with stakeholders. Conducted annual meetings with landowners. Developed a new process for monitoring pipelines after construction to prevent potential issues. Donated funds to landowners who are ranchers and farmers to help feed their livestock during a drought
Resources users/rights holders	Mineral rights owners, water rights owners and users, hunters/fishers/gatherers	Integrated our Stakeholder management system with Hess' Owner Relations function, enabling us to improve information sharing between systems and decrease the turnaround time for answering questions and resolving grievances. Also, improved the user interface and accessibility of the website for owners
Governments	Local, regional and national authorities, national militaries, international governing authorities	Hosted the governor of North Dakota and his staff on a field tour of Hess facilities in the Bakken, to increase awareness of our operations, technology, and safety. Hosted the director of the Bureau of Safety and Environmental Enforcement at Hess' operational headquarters in Houston, to tour Hess' Emergency Operations Center and brief him on safety measures and procedures in relation to severe weather and the impact on offshore operations. Also, a Hess representative serves on the North Dakota Oil and Gas Research Council, a public private partnership
Direct economic interests	Investment partners, vendors and suppliers, contractors, unions, shareholders	Implemented the Job Experience Training program in North Dakota, providing valuable work experience to students from Bismarck State College. Also, a Hess representative was appointed by the North Dakota governor to sit on the Innovative Education Task Force Advisory Committee, which is aimed at helping colleges meet the state's workforce needs. Through our joint venture in Guyana, hosted open houses for key business associations and launched an oil and gas literacy series through the newly opened Centre for Local Business Development. Also through the joint venture, opened an online supplier registration portal aimed at providing greater opportunities to Guyanese suppliers

TABLE 19.3 (continued)

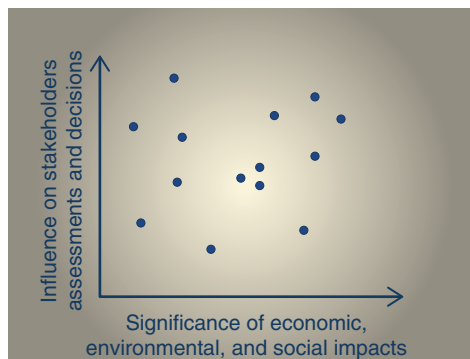
Category	External stakeholder groups	Recent examples of engagement
External business interests	Chambers of commerce, industry organizations, local businesses, sustainability initiatives	A Hess representative served on the executive committee of IPIECA. Leaders of Hess' EHS and SR functions actively participated in IPIECA and International Association of Oil and Gas Producers working groups and stakeholder engagement activities focused on topics such as social responsibility, water, climate change, environment, health, safety, and sustainability reporting. A Hess representative also served on the advisory board for the University of North Dakota's Petroleum Engineering Program, the board of directors for the Greater North Dakota Chamber of Commerce and the executive committee and board of directors for the North Dakota Petroleum Council
Special interest groups	Non-governmental organizations, religious groups, cause-oriented nonprofits, community groups	Participated in multi-stakeholder initiatives, including the Voluntary Principles on Security and Human Rights, the EITI, and the UN Global Compact, to advance responsible business practices globally. A Hess representative served on the board of the North Dakota Outdoor Heritage Fund, a governor-appointed position. Helped to fund the construction of a mountain bike trail in Theodore Roosevelt National Park in North Dakota
Community services	Local police/fire/emergency medical services, health care services, education, human service agencies	Continued to sponsor the Scholars Program in Ghana, which has provided secondary, vocational, technical, and health education to underprivileged students since 2012. Distributed food supplies, stoves, and electrical equipment to almost 500 families affected by monsoon flooding in Malaysia, where we work with nonprofit groups to identify and respond to community needs. Donated significant funds and volunteer time to Hurricane Harvey relief efforts in Houston
Indigenous groups	Formally recognized groups, tribal coalitions, government supporting agencies, indigenous advocacy groups	Continued engagement with local fishing villages in Ghana, through their Paramount Chiefs and community members, including providing updates on Hess business activities

Step 1—Determination and prioritization of the stakeholders, who are to be involved in the process.

Step 2—Identification and prioritization of all sustainability topics (economic, environmental, workplace related, human rights, social, responsible products, etc.) using the GRI Standards. During this step the material topics, as identified by other organizations in the same sector, are taken into account (sector benchmarking).

Step 3—A stakeholder materiality survey is conducted, using a qualitative questionnaire, in order to identify the material topics. For every question a qualitative scale (very high, high, medium, low) is used in order to evaluate the importance of each aspect. The same questionnaire is answered by members of the company's board of directors.

Step 4—A materiality matrix is created representing the topics of each aforementioned level of importance. A numerical value is assigned to each response in the questionnaire (very high = 4, high = 3, medium = 2, low = 1). For every topic an average value is calculated based on all the replies. The Y-axis represents the opinion of the stakeholder groups that are consulted during the materiality assessment process. The X-axis represents the impact of the organization on each topic, based on the opinion of the Board of Directors. Hence, and according to the GRI Standards, the factors of the highest importance to the stakeholders, and/or those where the organization has the highest impact must be included in the sustainability report. For the remaining topics is up to the organization to decide whether or not it wishes to include them in the sustainability report.



Materiality matrix

In defining material topics, there are some crucial factors that every reporting organization must take into account. Firstly, the economic,

environmental, and/or social impacts (such as climate change or poverty) assessed by experts. Secondly, the interests and anticipation of the people who are directly connected to the company, such as employees, as well as those belonging in a wider set, such as suppliers and local communities. Furthermore, the material topics are identified by other organizations in the same sector and the legislation considered to the organization and its stakeholders, too. Moreover, extremely significant are the company's strategies and goals and the way it manages sustainability issues. Finally, the appropriate prioritization is carried out in the report.

CASE STUDY: CAMPBELL SOUP COMPANY—2018 CORPORATE RESPONSIBILITY REPORT EXTRACT

Materiality Assessment

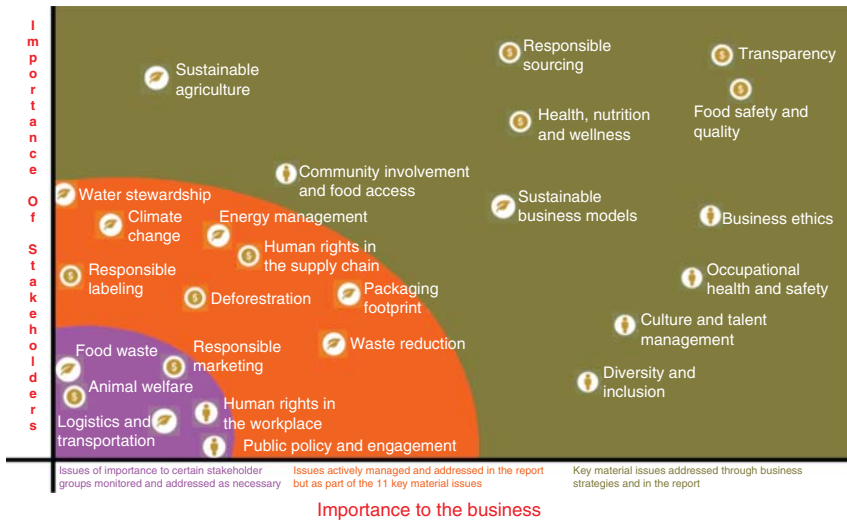
Material corporate responsibility issues are those topics that influence the judgment and decisions of our external and internal stakeholders, as well as having an impact on them and our business [10]. We identify and prioritize material corporate responsibility issues through the following four-step assessment process.

Step 1: Identify issues—We review the priorities publicly reported by our peers, conduct a media audit of trending issues and discuss issues with NGOs and investors.

Step 2: Survey stakeholders—We send the survey to employees, customers, suppliers, NGOs, community partners, government or regulatory agencies, investors, trade organizations, and consumers, and post it publicly on social media. More than 1250 participants responded to our most recent survey.

Step 3: Interview stakeholders—We conduct interviews internally with global employees and external stakeholders. In addition, as part of our most recent assessment, sustainable brands interviewed five internal stakeholders, and Business for Social Responsibility interviewed five external stakeholders.

Step 4: Validate—Campbell's Sustainability Steering Committee and various leaders around the organization provide feedback on the issues and validate the results of the research, survey, and interview process. The results are shared in the following text, and are reevaluated and revalidated throughout the year to allow for the inclusion of major or minor shifts in issues of priority [10].



Benefits from the Corporation's Responsible Behavior: The Sustainability (CSR) Debate

Sustainability (CSR) performance can be extremely beneficial for enterprises in many different ways. Although some companies may be forced to join the new movement of social and ethical activities, the long-term multi-effectiveness of sustainability seems apparent.

Some of the primary benefits are as follows:

1. *Improved financial performance*—There was a long debate among investment and business communities on whether there is a real connection between socially responsible business practices and positive financial performance. The results of several studies show that improved financial performance seems to be undisputable. Companies have a sustainability strategy if someone considers the fact of 100% transparency of financial records and awareness for each financial change on behalf of all parties. This creates a more precise economical management as well as more trust based financial exchanges, both factors that lead to improved economic results.
2. *Reduced operating costs*—Many of the sustainability practices and initiatives, such as those that address the environment or workplace, may lead to significant cost savings. For example, efforts that aim to reduce climate changing greenhouse gases also increase energy efficiency and reduce utility bills. In addition, many recycling initiatives

cut waste disposal costs and generate income by selling recycled materials. Although, these examples cover only 30% of the possible sustainability initiatives, they account for 70% of cost reductions.

3. *Added value for the product leading to customer loyalty*—In a global market where competitive products and services are becoming more homogeneous on price and quality, a socially responsible profile can serve as the decisive element of customer choice. The initiative for more socially and environmentally oriented production results in products and services that go beyond the customer's perceived value.
4. *Enhanced brand value and reputation*—At times, sustainability by businesses positively influences the public. Even though a corporate donation or an environmentally driven action might be expensive for the business, it seems to establish long-run brand image. Brands are the expression of the corporate value system, and as such the company's reputation. A brand no longer represents simply the finished product or service; rather it is the culmination of all aspects of business, including how and with whom it does business and the influence it can have on external and internal parties. Thus, customers are increasingly drawn to brands that have good reputations in sustainability related areas. A company considered responsible can benefit from both its enhanced reputation with the customers, as well as its reputation within the business community, increasing a company's ability to attract capital and trading partners.
5. *Increased worker commitment leading to employee loyalty*—Sustainability can act as a very powerful motivator for the staff. Employees that are given the opportunity to express their individuality and give something back to the community tend to be happier with their daily work and are more enthusiastic about their company. Moreover, offering employees the freedom to collaborate on particular projects can promote teamwork and relationships between employees, as well as superiors, and create a healthier and more productive work environment.
6. According to a research conducted by the British corporate think tank, *The Work Foundation*, companies can improve their chances of hiring and retaining talented staff if they make efforts to recognize and support employees' needs and pay more attention to environmental and community concerns.
 - The group's study of more than 1000 British workers concluded that about one in 10 is an «ethical enthusiast», or strong supporter of CSR practices. These groups of workers tend to fall within the age ranges of 18–24 or 45 and over.

- The study also found that employers with no concern for corporate ethics are more likely to lose staff within a 12-month period.
 - The deputy director of research at the Work Foundation, Stephen Bevan, highlights the misconception that most employers assume that the employees resign due to financial reasons. However, according to Bevan, only about 10% of employees leave because they are not sufficiently satisfied with their pay package.
7. *Good relations with government and communities*—Another way sustainability benefits the company is through the establishment of mutual trust and respect between business and local communities. When the company seeks to improve society, and acts not only for its own profit, it gains more privileges. As a result, the bond between communities and the private sector is strengthened.

Possible advantages for the enterprise:

- Gains the community support
- Creates alliances with community sectors
- Gains the political decision-makers' confidence
- Fosters relationships with other businesses that might become clients

The Corporate Social Responsibility (CSR) Debate

After analyzing the various advantages of adopting Sustainable (CSR) strategies, it is still necessary to mention there are arguments concerning the responsibilities of a business, as a balance of power with responsibility. It is essential to manage equilibrium between power and responsibility. Today companies have unprecedented power, which may harm society if those who are in charge do not follow specific steps. However, if firms use their power in a responsible and trustworthy manner, the risks of poor judgment and shortsighted operations shall be eliminated. Businesses committed to CSR are aware that if they abuse the power they have, they might lose it permanently.

Issues for Learning and Discussion

1. Are suppliers considered to be internal stakeholders, external stakeholders or both?
2. In reporting the company's stakeholder engagement, should the frequency of communication be included?

3. If information about a material aspect is omitted from a report, should the company explain the reasons?
4. Should the sustainability report cover aspects which reflect the organization's economic, environmental, and social impacts, aspects which influence the decisions of stakeholders, or both?

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Sustainability (CSR or ESG) Reporting

SUSTAINABILITY (CSR OR ESG) REPORTING

There are two main reasons that make sustainability (CSR or ESG) reporting vital. Firstly, all over the world there are sustainable development challenges, and, in order to bring solutions, companies must play a significant role by promoting economic, environmental, and social reporting. On the other hand, corporate scandals resulted in the loss of trust in the businesses, which can be overcome through a more open and clear communication.

CONTEXT OF REPORTS

The reporting process involves seven topics that corporations are interested in. The first one is called financial analysts. The main point of this topic is to find ways to connect the company with the financial world. Many corporations use the sustainability report in order to gain ground and build a stronger profile among financial analysts. Into this sector of the sustainable reporting companies provide the shareholders with information about the money the company spends for expanding, operating, and protecting the environment.

The second topic that sustainability reports incorporate is called Assurance and Verification. This topic includes detailed reporting process by specific companies—called verifiers—that inform the shareholders about the future goals of the company and recommendations in order to accomplish them.

The topic that follows is called Supply Chain reporting and discusses the complexity of cooperating with the suppliers. Nowadays, there is an increasing percent of stakeholders who take into consideration the supply

Practical Sustainability Strategies: How to Gain a Competitive Advantage,
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chain. For this reason, companies must be very critical when deciding about the suppliers. Ford Motor Company faced a serious problem when more than 100 people had died in accidents due to problems in the tires of the vehicles—supplied by Firestone. Both customers of Ford Motor Company and Firestone lost their trust in these companies.

The fourth topic that is discussed in the sustainability reports is called Emerging and Transition Economies.

The Economic Bottom Line is the fifth issue that the sustainability reports are communicating about. This topic goes beyond financial accounting and discusses about positive or negative economic social and environmental impacts that companies have on their stakeholders. The whole concept is to go beyond balance sheets into Economic Value Added metrics in order to define clearly their true meaning. For example, Rio Tinto in its sustainability report informs the company's stakeholders of its economic contribution to the local communities.

The topic that follows is called Brands and Reputation. This subject discusses how corporate reporting is related to the reputation, the brand name, and the values of the company. Reputation, brand name, and values of the company are considered to be extremely important in order for a company to be successful. The thing that makes brand name essential is the trust that it carries. For example, Nike, McDonald's, and Coca-Cola are companies that people trust because of their commitment to their values.

The last topic that sustainability reports take into account is about Governance. Corporate Governance discusses how the company's management deals with economic and social issues. Corporate governance is concerned with creating a balance between economic and social goals and between individual and communal goals while encouraging efficient use of resources, accountability according to the interests of individual's corporations, and society.

CHANGES OVER THE YEARS

During the earlier years of the 2000s, only a few dozen companies issued sustainability reports. Furthermore, sustainability reports focused rather exclusively on environmental protection and stewardship. Yet as the years progressed, sustainability reports grew more comprehensive and adopted a holistic approach, investigating social and economic concerns as well following global guidelines such as Global Reporting Initiative (GRI).

Reports issued after the new millennium provide shareholders with much more information and insight into a company's operations. This shift gained

traction as companies began to recognize the importance and utility of fostering a socially, economically, and environmentally responsible image for shareholders and stakeholders alike.

The two following case studies are good examples of sustainability reports achieving a high level of completeness and comprehensiveness and reflecting the company's progressiveness and maturity in terms of sustainability issues. They are extracts from Hess and Praxair's sustainability reports (2017 and 2016, respectively) and they are consistent with the GRI guidelines.

HESS CORPORATION 2017 SUSTAINABILITY REPORT

Approach to Reporting

In this report, we provide descriptions of the company's 2017 strategy and performance regarding material economic, environmental, and social issues [1]. Our annual report, Form 10-K filing and proxy statement, detail our financial and governance information and can be found on our website.

Additional sustainability and investor information is available at <http://hess.com/investors>.

Reporting Standards

This report was prepared in accordance with the Core option of the GRI Standards. It was also informed by the Oil and Gas Industry Guidance on Voluntary Sustainability Reporting developed by IPIECA, the American Petroleum Institute, and the International Association of Oil and Gas Producers; the UN Global Compact's (UNGC's) 10 Principles; key environmental, social, and governance (ESG) ratings, including Disclosing the Facts, an investor scorecard developed by As You Sow, Boston Common Asset Management, and the Investor Environmental Health Network; and recommendations from the Task Force on Climate-Related Financial Disclosures.

An index of GRI, IPIECA, and UNGC reporting indicators is available at <http://hess.com/gri-index>.

Materiality

Consistent with GRI's materiality guidance, we identified and prioritized new and emerging issues important to our stakeholders when developing the content for this report. Through a survey of select industry peers and external

stakeholder groups, as well as an annual, document-based assessment of key stakeholder perspectives and Hess' operational and regulatory risks, we identified the 10 most material issues for our company:

- ◆ Regulatory Assurance
- ◆ Water Management
- ◆ Transportation Impacts
- ◆ Emergency Preparedness and Response
- ◆ Process Safety and Spills
- ◆ Community Engagement
- ◆ Climate Change and Greenhouse Gas (GHG) Emissions
- ◆ Stakeholder Engagement
- ◆ Transparency in Business Conduct
- ◆ Human Rights and Security

These material issues have informed our environment, health, safety, and social responsibility strategy and helped to define the boundaries of this report. See reporting boundaries for each of these issues at <http://hess.com/sustainability/approach-to-reporting/boundaries-for-material-issues>.

Boundary Setting

Included within the scope of this report are the principal facilities and assets operated by Hess Corporation and our subsidiaries during calendar year 2017, unless otherwise indicated. Data presented are gross figures from operated facilities, unless specified otherwise. The report includes partial-year data for the following assets that were divested in 2017:

- ◆ West Texas, including the Permian Basin enhanced oil recovery assets. Period of data included: January–July 2017.
- ◆ Equatorial Guinea, including the Okume and Ceiba production operations. Period of data included: January–November 2017.
- ◆ Norway, including the (non-operated) Valhall production operations. Period of data included in GHG equity-share emissions and social investment spend: January–November 2017.

We report GHG emissions on both an operated and equity-share basis in accordance with the GRI G4 Oil and Gas Sector Supplement and the IPIECA Petroleum Industry Guidelines for Reporting GHG Emissions (3rd edition, 2015). We report social investments for our operated assets, joint

ventures, and non-operated facilities in which we hold a significant interest. We include contractors in workforce metrics for those contractors whose hours we track.

See our expanded performance data at <http://hess.com/sustainability/performance-data/key-sustainability-metrics>.

Restatements

We believe our approach to restating data complies with the GRI Standards' principle of comparability and the specific disclosure regarding restatements of information, as well as IPIECA guidance. For GHG emissions, in cases of acquisitions and divestitures and other source ownership and control changes, we adjust our base year emissions if the change exceeds 10% of the original base year emissions total. The exact timing of the adjustment depends on several factors, as described in the Hess GHG Inventory Protocol.

Access the Hess GHG Inventory Protocol at <http://hess.com/sustainability/climate-change-energy>.

Assurance

In order to evaluate accuracy and reliability, we conduct quality assurance/quality control reviews and validation of both aggregated and facility-level data. Individual numbers in the charts, tables, and text may not precisely sum to the total amounts shown due to rounding. All currency in the report is in US dollars.

This report, including our sustainability data and self-declared GRI "in accordance" status, was assured by ERM Certification and Verification Services (ERM/CVS). This external review helps to ensure consistent and objective data collection and reporting of our sustainability performance.

Requests for Information

For copies of our Environment, Health, and Safety Policy, Social Responsibility Policy, or Human Rights Policy, or for more information regarding our operations, please visit our website at <http://hess.com>.

We invite your questions, comments, and suggestions regarding this report. To send us your questions or comments, or to request more information or additional copies of this report, please contact: Hess Corporation. You can also send us an email.

JOHNSON & JOHNSON 2017 HEALTH FOR HUMANITY REPORT

About this Report

Johnson & Johnson's 2017 Health for Humanity Report is our 15th annual report on our progress in citizenship and sustainability [2]. Data in this Report cover the period between January 1, 2017 and December 31, 2017, unless otherwise indicated. The Report is also available in online format at <http://healthforhumanityreport.jnj.com>. It includes an Executive Summary available in five languages.

This Report provides a comprehensive overview of the Company's vision, strategic approach, and performance in the ESG areas relevant to our business. Reporting on other matters specific to financial performance of the Company and its subsidiaries can be found in our 2017 Annual Report.

The content of this Report covers Johnson & Johnson's worldwide operations, including our subsidiaries across three business segments: Consumer, Medical Devices, and Pharmaceutical. Contract manufacturers are excluded from the scope of this Report, unless otherwise noted. There have been no significant changes from our previous reporting period regarding scope, boundaries, or measurement methods applied. From an operational perspective, in June 2017, we completed the largest acquisition in our Company's history, with Actelion becoming part of the Janssen Pharmaceutical Companies of Johnson & Johnson. Pulmonary hypertension is a new therapeutic area that was established within Janssen with the acquisition of Actelion.

The Report has been prepared in accordance with the GRI Standards Core option. Please refer to the GRI Content Index for a complete listing of GRI disclosures included in this Report. Our 2016 Priority Topics Assessment process guided the Report's structure (GRI 102-55).

The Report content is also informed by the UNGC Principles and serves as our annual UNGC Communication on Progress. See the UNGC Index for more information on how we adhere to the 10 UNGC Principles.

This Report includes our annual update on progress against our Health for Humanity 2020 Goals and progress on our commitments to the UN Sustainable Development Goals (UN SDGs). Please see Health for Humanity 2020 Goals Progress Scorecard and UN SDGs Progress Scorecard for more information.

All data in the Report have been subject to various forms of verification and assurance. ERM/CVS conducted independent review and assurance of the following information and data in the report:

- ◆ Progress against our Health for Humanity 2020 Goals, and associated data presented in the 2020 Goals Progress Scorecard.
- ◆ Progress against UN SDG commitments, and associated data presented in the UN SDG Scorecard.
- ◆ Year 2017 global GHG emissions inventory and electricity use generated from renewable energy sources that are also included in our 2018 CDP Climate Change Report.

Click for ERM/CVS independent assurance statements. The financial data and general information about the business in this Report were previously audited for disclosure in our 2017 Annual Report.

We welcome stakeholder feedback on our Report. To share your comments or questions, please contact us at support@jn.com.

B Corporation

B Corporation Certification, which is administered by the non-profit B Lab, is another certification that is introduced [3]. It assesses the impact of a company, through its products and services, in terms of social and environmental performance, public transparency, and legal responsibility to balance profitability and purpose. With a vision of achieving a culture in which businesses will build a more inclusive and sustainable economy that is driven by the benefit of all stakeholders, and not just shareholders, B Corporation Certification aims to build trust and value for certified companies.

Sustainability in the Supply Chain

In order for a product or service to be considered sustainable, it is not enough for it to be managed properly only by its production company, but throughout its life cycle. Therefore, an extremely important factor is the management of environmental, social, and economic impact, as well as the encouragement of good governance practices, of the supply chain.

The objective of Supply Chain Sustainability is to create, protect, and grow long-term environmental, social, and economic value for all stakeholders involved in bringing products and services to market. Potential business risks are related to cooperation in developing countries where poor quality standards relating to social and cultural aspects as defined in the developed world prevail. These markets may lack a legislative structure to protect human rights and the environment, such as the exclusion of child labor and the reduction of GHG emissions, respectively. Other risks may be related to waste management or non-recycling, as well as to social issues such as bribery and labor conditions of suppliers.

The implementation of a Sustainable Supply Chain allows a form of management of the above risks, as well as achievement of operational efficiency, reducing costs without any negative impact on business. Thus, end products can be considered viable through the supply of raw materials with well-considered social and environmental impacts. Such an example is organic plastics that emit relatively fewer GHGs throughout their lifecycle.

Sustainability Reports and Impact Investments

As described, sustainability reporting has exploded over the past decade. While there are a variety of reasons for the growing popularity of sustainability reporting, such as the improvement or maintenance of a desirable public image, these documents also serve to attract and persuade investors. First, investors concern themselves with sustainability reports because the publication of an annual report reflects a strong and organized managerial structure. The United Kingdom's news outlet, *The Guardian*, explains, "The non-publication of a report, or the absence of published policies, targets and performance data, is increasingly likely to be taken as evidence that the company does not recognize social and environmental issues as management priorities, thereby raising wider questions about the quality of the company's risk systems and processes" [4]. This is especially true for Fortune 500 companies. In short, a business that exhibits sound and effective management strategies appears much more attractive to potential investors.

Changes in the values of individual investors have also contributed to the recent explosion of sustainability reporting. As investors grow more sensitive to social and environmental issues, sustainability performance indicators will have a greater impact on investment decisions. Naturally, this is a generalized statement as not all investors concern themselves with social and environmental issues. Nevertheless, the growing public sensitivity to these topics appears to be influencing investment decisions. An increasing number of investors are making commitments to integrate consideration of social and environmental issues into their investment decision-making processes and better quality environmental and social performance data are increasingly available. It is therefore to be hoped that we will, over time, see investors paying greater attention to companies' corporate responsibility performance [5].

Sustainability reports not only record important advancements in corporate environmental and social strategies, but also foster a competitive environment where companies are compelled to match the significant benchmarks of others. This concept is important for investors as increased competition may accelerate financial performance. Companies that choose

to enter this competitive realm stand to gain, while those that neglect sustainability reporting may be left behind.

Sustainability and ESG Ratings

Despite debate, it is clear that many investors do consider sustainability performance when they choose to invest. Issues of corporate governance, social welfare, and environmental development hold a key role in investor decisions. This is due to the desire of investors to maximize the value of their shares and, at the same time, the social welfare by investing in companies that respect and follow international standards and take into account the requirements and needs of stakeholders. It is no coincidence that investment in socially and environmentally responsible businesses has grown significantly over the past 10 years, particularly in Europe and North America. It is, also, noteworthy that, as a whole, UN PRI signatories represent US\$70 trillion of investment funds.

Socially responsible investing (SRI) takes into account the environmental impact, social impact, and the impact of corporate governance (ESG) [6]. However, a socially responsible investment is not just a way for investors to feel satisfied, but it can both lead to better financial results and reduce the risk of business. Many CEOs of international businesses and large organizations have, for example, decided to comply with the Paris Agreement on Climate Change.

Why is this happening? According to Harvard Business Review [7], socially responsible companies show greater profitability and better return on equity than their competitors. Another reason is that a long-term strategy based on the principles of sustainability can ensure the continuity and growth of the company.

It is not just a trend, it is a new investment strategy that is gaining more and more ground, especially over the last 20 years—for the millennial generation, it is almost a prerequisite. According to Bloomberg, about 84% of the millennials are interested in socially responsible investments and this figure is expected to rise as the demand for responsible products grows steadily. In more traditional economics, Deutsche Bank recently made an analysis of more than 2000 empirical studies since 1970 [8] and found that 90% of studies indicate that socially responsible investment had far better returns than non-socially responsible investment.

In addition, good ESG performance has been associated with good financial results and limited business risk. For this reason, as demonstrated by the recent Eurosif SRI Market Study, the ESG criteria are taken into account 60% more than two years ago.

Globally, some of the most important ESG ratings linked to the stock markets are the Dow Jones Sustainability Index (DJSI), the FTSE4Good Index, the MSCI ESG Index, and the CDP, while the United Nations Responsible Investment Initiative (UN PRI) is considered important too. For example, the development of the DJSI is one of the most notable manifestations of significant changes in the ways global investment organizations value and incorporate sustainability metrics. In order to allow investors to make more informed market decisions on sustainability performance, Dow Jones launched the DJSI in 1999. Today, there are many DJSI licenses held by asset managers; these licenses manage more than US\$8 billion [9] worldwide. According to Dow Jones, the utility of this index is twofold; “The DJSI enable investors to integrate sustainability considerations into their portfolios while providing an effective engagement platform for encouraging companies to adopt sustainable best practices” [10].

This DJSI offers further evidence to support the positive relationship between sustainability and financial performance. Dow Jones explains, “A quantitative analysis of the historical sustainability information compiled by SAM over the years, and upon which the DJSI are based, reveals that an investment strategy that selects sustainability leaders and avoids sustainability laggards contributes to superior long-term investment results with improved risk/return profiles” [10]. Such evidence serves to further accelerate the incorporation of sustainability performance into investment decisions and management systems and bodes well for the future of sustainable progress.

CASE STUDY: MARKS AND SPENCER PLAN A REPORT 2018, PLAN A 2025 AND OUR STRATEGY—HELPING TO MAKE M&S SPECIAL AGAIN

Transformation Timeframe

Starting Afresh In November 2017, we launched a new plan to accelerate the transformation of our business with the aim of making M&S special again through sustainable, profitable growth that delivers value for customers, colleagues, shareholders, communities, and planet alike [11]. We have always recognized that our sustainability activities have to be central to how we do business. We have to make sure that Plan A 2025 is aligned with our Business Transformation Plan. We are in the process of ensuring that Plan A is aligned with this new corporate strategy.

Step One: Restoring the Basics Over the next 12–18 months, we want to build on the legacy of over 10 years of Plan A progress. We want

our customers to choose M&S products and services because they are informed about the healthier and more sustainable options we offer. We need our colleagues to be fully involved in developing and promoting the business. And we need to find ways for our operations and supply chains to become even more efficient.

Customers:

- Bringing Plan A product attributes to life. Showcasing the M&S difference of how we source our products.
- A bigger role for Plan A in communication. Both in how we market our brands and drive innovation to meet the needs of our customers.
- Building trust. Reducing risks through robust governance, systems, compliance, and partnerships.

Colleagues:

- Using Plan A for even greater engagement. To build our relationships with the communities around us and contribute to the development of an inclusive business able to attract the best talent.

Efficiency:

- Driving down costs by cutting waste. Through focusing on energy and food waste.
- Reducing risks and increasing productivity. By working with suppliers to address the challenges of plastic waste, packaging, and climate change.

Step Two: Shaping the Future These are actions, which we are already piloting and testing today, ready to roll out in two to three years' time. Together, they will represent the next step to a more digital, technology-enabled business that reflects and values customers' lifestyles. This will be a phase where we take newly founded innovations, efficiencies, and loyalties to create a business for the future.

Customers:

- Plan A attracts new customers. By providing evidence of a contemporary and ethical brand, appealing to a wide range of new and existing customers.
- Transparency. Demonstration of our ethical and environmental standards in action, visible to all.

Colleagues:

- An engaged, skilled, and inclusive workforce. Scaling-up our activities on transforming communities and meeting new benchmarks of equality in the workplace.

Efficiency:

- Leaner supply chains and franchises. Supplier Sustainability Scorecards being extended to all supply chains and efficiency benchmarks to key franchises.
- Sustainably designed products. Designed to give the best customer experience and with the least social and environmental impact.

Step Three: Making M&S Special By 2022, our Plan A 2025 commitments will be a platform of excellence, providing an M&S difference to all our products and services. This will inspire the development of new products and services. As well as addressing today's risks, we want to evolve an approach which automatically responds to any new emerging social and environmental challenges.

Customers:

- Plan A allows us to move into new markets. New products and services meeting the social and environmental needs of our customers.
- Engagement. Engaged customers, clear about the steps they can take to make a positive difference.

Colleagues:

- An employer of choice. Regarded for our values, skills, and innovation.

Efficiency:

- Zero waste, carbon neutral supply chains. Suppliers and business partners adopt M&S's zero waste, carbon neutral culture.
- Sustainable products. Products that exceed customers' expectations while also creating a fairer and sustainable world.

STARS Framework Measuring sustainability performance is an important process that is now being carried out by various entities and not just by businesses. One such set is the academic sector, for which a targeted framework has been created, known as STARS [12]. The Sustainability Tracking, Assessment and Rating System (STARS) is aimed at colleges and universities in order to help them determine their sustainability strategy in a transparent and self-reporting way. In particular, in addition to providing

a framework for sustainability practices and facilitating the exchange of information, a common set of measurements has been developed with the broad involvement of the university community to make meaningful comparisons between institutions over time. At the same time, through participation in the STARS, each organization can earn points for sustainability discrimination (STARS Bronze, Silver, Gold, or Platinum, STARS Reporter). Thus, incentives for continual improvement can be achieved in this area.

Issues for Learning and Discussion

- (1) Is Social Enterprise or B Corporation a legal entity for mission-driven businesses that provide a higher level of legal protection, accountability, and transparency than existing for-profit entities?
- (2) Which is the top driver of Supply Chain Sustainability?
- (3) Are companies becoming increasingly exposed to risks across their supply chain due to disruption resulting from climate change, due to reputational damage or due to both?
- (4) Is the comparison, over time of key sustainability data, or comparison between different companies one of the objectives of the Sustainability?

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Sustainability Metrics for Improving Impact

Sustainability metrics are a combination of economic, social, and environmental indicators related to performance and stakeholder perceptions. The term “metric” encompasses any way of measuring performances, which can be adopted during different stages of the decision-making process, in order to minimize the impact of the decision, as well as after, so as to reduce and improve the actual situation.

It is hard to develop a metric, which can be relevant for all sectors or situations, and therefore a unique metric for sustainability. In the last years particularly there has been a flourishing of different certifications and standards, concerning all possible issues within sustainability. As proof of this there are numerous and different metrics, which can be used to quantify sustainability, and most of the time, just some of its aspects. Some of the most recognized worldwide are the Global Reporting Initiative (GRI) Standard and the ones released by the International Organization for Standardization (ISO 14000, 14046:2018, 14006).

Measuring sustainability is one of the (endless) challenges that companies, as well as other types of organizations, are facing nowadays. Since the concept of sustainability is not yet transferable into quantitative data, the consequence of this is a higher uncertainty concerning the method to apply, and this, until now, has facilitated the proliferation of tools to measure different aspects of sustainability, often with different methodologies.

Since the road to sustainability is a never-ending process, it is important to measure sustainability because this will allow organizations to keep track of the improvement achieved up to a certain point. Indeed, different metrics can be used to evaluate the progress made and set new parameters

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and targets to be achieved; this will also give more credibility of action in front of stakeholders. “Monitoring and metrics have always been essential components of closed physical systems. They are integral to the scientific method. In this context each indicator should have a threshold and a target to guide political and social action” [1].

METRICS IN THE GRI GUIDELINES

One of the most widely adopted global guidelines for sustainability is the GRI’s Sustainability Reporting Guidelines. The GRI Standards version [2], which was launched on October 19, 2016, aims to help organizations shape their sustainability strategy and report their practices on a range of economic, environmental, and social impacts. The guidelines are divided into two parts, the general and specific disclosures.

According to GRI guidelines, the Materiality Principle is a method of determining which *Indicators* should be reported in a GRI Sustainability Report and should be the focus of the organization for managing, measuring, and reporting.

The GRI Guidelines Materiality Test should evaluate the importance of monitoring the Indicators with regard to the organization’s current and future performance.

CASE STUDY: ABM 2017 CORPORATE SUSTAINABILITY REPORT

102-49—Changes in Reporting

On September 1, 2017, we completed the largest transaction in ABM’s history through our acquisition of GCA Services Group, a leading provider of facilities services in the education and commercial industries [3]. We also divested our non-core Government Services business on May 31, 2017. Society Compliance, Product Responsibility Compliance, and Procurement Practices moved up in our assessment and are covered in the 2017 report.

102-50—Reporting Period

Unless otherwise indicated, this report covers our 2017 fiscal year, which ended October 31, 2017.

102-51—Date of Most Recent Report

Fiscal 2016 (November 1, 2015 to October 31, 2016).

102-52—Reporting Cycle

Annual.

102-53—Contact Point for Questions Regarding the Report

Readers can send feedback, comments or questions regarding our sustainability plans and progress to ABM’s Corporate Communication team at sustainability at www.abm.com.

102-54—Claims of Reporting in Accordance with the GRI Standards

This report has been prepared in accordance with the GRI Standards: core option.

102-56—External Assurance

For the seventh year in a row, we submitted our report to the Centre for Sustainability and Excellence (CSE) for an independent audit assessment.

GRI INDEX

The CSE, assessed ABM Sustainability Report in accordance with the instructions of the GRI Standards and confirms that they are in compliance level “In Accordance—Core.”

Disclosure	Description	Reference (page no.)	External assurance
<i>General disclosures—Organizational profile</i>			
102-1	Name of the organization	5	✓
102-2	Activities, brands, products, and services	5–7	✓
102-3	Location of headquarters	7	✓
102-4	Location of operations	8	✓
102-5	Ownership and legal form	8	✓
102-6	Markets served	9	✓
102-7	Scale of the organization	9	✓
102-8	Information on employees and other workers	9	✓

Disclosure	Description	Reference (page no.)	External assurance
102-9	Supply chain	9, 10	✓
102-10	Significant changes to the organization and its supply chain	10, 11	✓
102-11	Precautionary principle or approach	11	✓
102-12	External initiatives	11	✓
102-13	Membership of associations	12, 13	✓
<i>General disclosures—Strategy</i>			
102-14	Statement from senior decision-maker	4	✓
<i>General disclosures—Ethics and integrity</i>			
102-16	Values, principles, standards, and norms of behavior	13	✓
<i>General disclosures—Governance</i>			
102-18	Governance structure	13, 14	✓
<i>General disclosures—Stakeholder engagement</i>			
102-40	List of stakeholder groups	14	✓
102-41	Collective bargaining agreements	14	✓
102-42	Identifying and selecting stakeholders	14	✓
102-43	Approach to stakeholder engagement	14	✓
102-44	Key topics and concerns raised	15	✓
<i>General disclosures—Reporting practice</i>			
102-45	Entities included in the consolidated financial statements	15	✓
102-46	Defining report content and topic boundaries	15–17	✓
102-47	List of material topics	17	✓
102-48	Restatements of information	17	✓
102-49	Changes in reporting	17	✓
102-50	Reporting period	17	✓
102-51	Date of most recent report	17	✓
102-52	Reporting cycle	17	✓
102-53	Contact point for questions regarding the report	17	✓

Disclosure	Description	Reference (page no.)	External assurance
102-54	Claims of reporting in accordance with the GRI Standards	17	✓
102-55	GRI content index	17	✓
102-56	External assurance	17	✓
<i>Specific disclosures—Economic performance (201)</i>			
103	Management approach	18–20	✓
201-1	Direct economic value generated and distributed	18	✓
201-2	Financial implications and other risks and opportunities due to climate change	18, 19	✓
201-3	Defined benefit plan obligations and other retirement plans	20	✓
201-4	Financial assistance received from government	20	✓
<i>Specific disclosures—Procurement practices (204)</i>			
103	Management approach	20	✓
204-1	Proportion of spending on local suppliers	20	✓
<i>Specific disclosures—Anti-corruption (205)</i>			
103	Management approach	20	✓
205-2	Communication and training about anticorruption policies and procedures	20	✓
<i>Specific disclosures—Anti-competitive behavior (206)</i>			
103	Management approach	13, 20	✓
206-1	Legal actions for anti-competitive behavior, anti-trust, and monopoly practices	20	✓
<i>Specific disclosures—Materials (301)</i>			
103	Management approach	21	✓
301-3	Reclaimed products and their packaging materials	21	✓
<i>Specific disclosures—Energy (302)</i>			
103	Management approach	21	✓
302-1	Energy consumption within the organization	21	✓

Disclosure	Description	Reference (page no.)	External assurance
<i>Specific disclosures—Emissions (305)</i>			
103	Management approach	21, 22	✓
305-1	Direct (scope 1) GHG emissions	21	✓
305-2	Energy indirect (scope 2) GHG emissions	21, 22	✓
305-3	Other indirect (Scope 3) GHG emissions	22	✓
305-4	GHG emissions intensity	22	✓
305-5	Reduction of GHG emissions	22	✓
<i>Specific disclosures—Environmental compliance (307)</i>			
103	Management approach	13, 22	✓
307-1	Non-compliance with environmental laws and regulations	22	✓
<i>Specific disclosures—Supplier environmental assessment (308)</i>			
103	Management approach	22	✓
308-1	New suppliers that were screened using environmental criteria	22	✓
<i>Specific disclosures—Employment (401)</i>			
103	Management approach	23–25	✓
401-1	New employee hires and employee turnover	23, 24	✓
401-2	Benefits provided to full-time employees that are not provided to temporary or part-time employees	25	✓
<i>Specific disclosures—Occupational health and safety (403)</i>			
103	Management approach	25–27	✓
403-2	Types of injury and rates of injury, occupational diseases, lost days, absenteeism, and number of work-related fatalities	25–27	✓
<i>Specific disclosures—Training and education (404)</i>			
103	Management approach	27, 28	✓
404-1	Average hours of training per year per employee	27	✓

Disclosure	Description	Reference (page no.)	External assurance
404-2	Programs for updating employee skills and transition assistance programs	27	✓
404-3	Percentage of employees receiving regular performance and career development reviews	27, 28	✓
<i>Specific disclosures—Diversity and equal opportunity (405)</i>			
103	Management approach	28	✓
405-1	Diversity of governance bodies and employees	28	✓
405-2	Ratio of basic salary and remuneration of women to men	28	✓
<i>Specific disclosures—Non-discrimination (406)</i>			
103	Management approach	13, 28	✓
406-1	Incidents of discrimination and corrective actions taken	28	✓
<i>Specific disclosures—Local communities (413)</i>			
103	Management approach	29	✓
413-1	Operations with local community engagement, impact assessments, and development programs	29	✓
<i>Specific disclosures—Customer health and safety (416)</i>			
103	Management approach	13, 29	✓
416-2	Incidents of non-compliance concerning the health and safety impacts of products and services	29	✓
<i>Specific disclosures—Customer privacy (418)</i>			
103	Management approach	13, 29	✓
418-1	Substantiated complaints concerning breaches of customer privacy and losses of customer data	29	✓
<i>Specific disclosures—Socioeconomic compliance (419)</i>			
103	Management approach	13, 29	✓
419-1	Non-compliance with laws and regulations in the social and economic area	29	✓

ISO 26000

In regard to standards, one of the most influential authorities on the topic is the International Standardization Organization [4], which, in November 2010, released a quite expected ISO guideline on Corporate Social Responsibility, the ISO 26000. The aim of such a standard is to provide guidelines to small and big firms, no matter the sector, on how to face and embed social responsibility within the organization [5].

To do so, the document is suggesting addressing seven core subjects and metrics, which are divided into different issues, in order to cover at least the basic issues on social responsibility. Those subjects are the following:

- Human rights
- Labor practice
- Environment
- Fair operating practices
- Consumer issues
- Community involvement and development
- Organizational governance [6]

Ecological Footprint

The ecological footprint is the virtual surface required to produce and dispose the goods demanded by humans. According to the Global Footprint Network already from the middle of the 1980s, humanity overshoot the world capacity, and is virtually consuming more than the world is able to restore [7]. The ecological footprint is a metric that represents how much land surface it is necessary to use in order to produce and assimilate the goods that people (usually grouped by nation) are consuming. This value, which currently is 1.5 times the earth's capacity, is helpful to understand where the main environmental pressure lies in order to reduce our ecological pressure, and, therefore, increase our sustainability [8].

Arguably its usefulness, that is, the benefit coming from the calculation of an ecological (as well as the others) footprint is limited. For this reason the Environmental Department of the European Commission, together with the Institute for Environment and Sustainability had developed a new footprint tool, called Environmental Footprint [9].

The aim of the later is to produce a harmonized methodology to calculate the environmental footprint of products and organizations, also taking into consideration carbon emissions. It is going to be built on the

International Reference Life Cycle Data System (ILCD) Handbook, as well as other existing standards and guidance documents, in order to create as comprehensive and complete a tool as possible [10].

Another relevant guidance, released by the British Standard Institute, is the quite spread out ISO 14000 on environmental management, as well as two interesting ISO, the 14046 and 14067:2018, respectively, on water and carbon footprint.

The ISO 14000 provides guidelines for the implementation of an environmental management system, particularly the 14001 and 14004. The aim of the first is to provide the requirements, while the second provides general guidelines. Furthermore, the other standards focusing on specific environmental issues, including labeling, performance evaluation, life cycle assessment (LCA), communication, auditing, and how we already mention on carbon and water footprint [11]. These last two probably deserve particular attention, about the issues, which they are dealing with. The ISO 14046, or water footprint—requirements and guidelines aims to complement the existing standard on LCA and on carbon footprint, as well as of providing clear guidelines on an issue where there are plenty of methodologies. The purpose of this standard is to contain:

- Principles and guidelines for a water metric of products, processes, and organizations.
- Ways as to how the different types of water (green, blue, and gray) should be considered, as well as the socio-economic and environmental issues.
- Ways of communicating water footprint.
- Compatibility with the other ISO 14000 standards [12].

The ISO 14067:2018 is strongly based on the ISO 14044, which was published in December 2009, and concerns requirements related to the quantification and reporting of the carbon footprint of a product (CFP) by using the LCA. This standard only addresses the impact of the product's life cycle on climate change and does not assess other environmental or socio-economic effects.

Metrics for Carbon Footprint

The Greenhouse Gas Protocol (GHG Protocol) was developed by the World Research Institute together with the World Business Council for Sustainable Development, in October 2010. This document is one of the most adopted

methodologies to calculate the carbon footprint of organizations, including the supply chain. Indeed its main aim is to identify risks, possibility of reduction along the supply chain and define improvements. The methodology used for assessing is the LCA, which is adopted for all the stages (therefore is theoretical applicable for other environmental aspects).

The Protocol is divided mainly in two documents, the corporate accounting and reporting and the project accounting protocol and guidelines. The purpose of the Corporate Accounting and Reporting is to:

- Standardize the approach to a representative account of the GHG emission of the organization.
- Simplify and reduce the costs of the inventory.
- Provide business with suggestion on how to successfully manage and reduce GHG emissions.
- Increase the consistency and the transparency in GHG accounting and reporting.

The key aspects of the Project Accounting Protocol and Guidelines provide specific indications, ideas, and methods on how to quantify, report, and reduce the GHG. Furthermore some addressed protocols per sector, like the one for agriculture and public sector have been released [13].

Life Cycle Assessment (LCA)

The LCA is a holistic methodology, which can be applied to any kind of impact (environmental, economic, and social), since it takes into consideration all the steps of the production cycle as well as the consumption and the end of life. Generally the steps considered are the following:

- Extraction and treatment of raw materials
- Educational tools
- Product manufacturing
- Transport and distribution
- Product use
- End of life

The aim of this methodology is to help identify the main pressure and issues within the life cycle, and help the decision makers to mitigate them. The methodology of the LCA is defined in the ISO 14040 standard, and consists mainly of four phases:

- Setting of goal and scope (define the methodological framework).
- List all the inputs and outputs of the product system.
- Assessment of the potential impacts related to these inputs and outputs.
- Interpretation of the inventory data and impact assessment results related to the goal and scope of the study [14].

Balanced Scorecard

To manage and integrate all these different standards within the organization, it is necessary to have a strong management system. One of the most appreciated is the Balanced Scorecard, which is a strategic planning and management system. It has been created to add the strategic non-financial performances to the more traditional one, in order to give a more “balanced” view to the managers. The balanced scorecard provides guidelines on what should be addressed, and how to measure it, helping the planners to effectively apply their strategies. This management system helps companies to put their vision and strategy into action. At the core of the balanced scorecard system there is the idea that an organization should always be analyzed from four different points of view: the financial, the internal business processes, the learning and growth, and the customer. For each one of these, metrics, data, and targets should be analyzed [15].

How Metrics of Sustainability Can be Used (e.g., Assessments, Audits)

There are different ways in which this broad panorama of different metrics can be used. For instance some of these guidelines and standards are often used to gain certification, which can easily be used to “boost” the green image of the company, or to use them mainly as marketing tools. However, the purpose of implementing measures is far from this; for instance, it would be preferable to see them as a tool to quantify and improve sustainability.

Assuming that the constructive approach is the one overcoming the others, it is interesting to consider how those metrics are used to reach sustainable goals. Once effective key performance indicators (KPIs) are founded, metrics are an extremely useful tool, which can be used to assess activities within sustainability, analyze achievements, and set new goals.

A defined metric allows an organization to have precise parameters to consider tracking progress and applying tailored management practices. The most common way of doing so is through periodic audits within

the organization, as well as the supply chain. Annual Assessment or Auditing is a really powerful tool to gain a deep understanding of the Corporate Sustainability status including sustainability of Supply Chain (e.g., products). Ideally a third party carries out the process, since, clearly, this approach is much more fitting to a transparent and trustworthy behavior which ordinarily characterizes an audit process.

Additionally the results given by the assessment or audit phase of the metric, if its parameters are clear and well explained, can be used to communicate more effectively the achievements of the organization. They can also be used to present an organization's best practices. Furthermore, most of the time the standards required are above the requirements established by the national law; thus, the compliance to such high standards will allow the organization to promote itself as a forerunner and a promoter of best initiatives [16]. As a consequence of this, it is important to choose the right label or right guideline, ones, which have credibility.

Sustainability Assessment or Audits of organization's policies and its products is a holistic approach with many benefits such as

- Selection of actions in order to effectively address the stakeholder needs
- Proper allocation of resources for the implementation of the selected actions
- Identification of opportunities for every stakeholder group
- Use of a comprehensive tool for the evaluation of strong points and areas for improvement

Benchmarking of internal practices against selected guidelines, standards, or labels.

CASE STUDY: THE HARTFORD—2016 SUSTAINABILITY REPORT

Communities and Giving

Corporate programs:

Teaching fire safety—**30% increase** (year over year) in The Hartford's Junior Fire Marshal[®] program kit distribution, helping to educate children about fire safety [17].

*Countrywide impact***2600+ classrooms.****78,000 students.***Fighting hunger:*

Employees in 21 offices participated in Fight Against Hunger events, including:

- **31,000 lives impacted** through the company's annual Fight Against Hunger campaign, helping to provide 735,000+ meals
- **1000+ employees registered** for the Walk Against Hunger in Connecticut
- **1500 pounds of food collected** for a local food pantry in Charlotte, NC.

Diversity and Inclusion*Our workforce:*

- **35% of workforce** works remotely (five days a week).
- Women chaired 15% of audit committees and 11% of compensation committees at S&P 500 companies. At The Hartford, women chair both committees.

Ethics and Governance*Reinforcing core values:*

The Hartford's Code of Ethics and Business Conduct reinforces our core values, and guides our employees to choose the course of action consistent with these values every time.

100% of employees certify annually they have:

- Read the code
- Agreed to comply

Board of Directors:

- ✓ Certifies annually to a Code of Ethics

99% of eligible managers took newly launched experiential courses focused on ethical business practices:

- Managing with Integrity
- Privacy Operations

Environmental Stewardship

Renewable energy:

We offer uniquely designed products that provide end-to-end coverage for the solar, wind, fuel cell, and biomass industries.

US\$500 million (approx.) invested directly in utility-grade solar, wind, and hydroelectric power generation facilities in the US.

US\$7 million+ earned in premiums from our renewable energy practice in 2016.

Recycling and Reusing E-waste Through a Zero-Landfill Process:

- **20,000+** electronic devices = **88 tons** of e-waste not in landfills

Stakeholder Metrics and Sustainability

There is obviously a strong correlation between Key Stakeholder Perceptions and Brand Image, reputation of the organization.

For example, if customer satisfaction feedback shows a negative trend over a period of time then there will definitely be a direct influence on market share, turnover, customer loyalty, prices, and therefore to sustainability. If employee's feedback is also negative over a period of time, then there is a direct influence on employee turnover, productivity, and many other areas that directly affect operational costs. If local community perception is negative over a period of time or after an ad hoc incident then the image of the organization as a good citizen will be damaged and probably authorities, non-governmental organizations (NGOs) will force pressure for change.

Additionally when the organizations implement systemic methodologies to understand and assess the key stakeholder perceptions, it means that they have comprehensive and accurate information but also the flexibility to react in an appropriate manner. If this information does not exist, or is not systematic or is partial, then their ability to react based on real facts and in time is reduced significantly.

The following chart describes direct and indirect correlations between Sustainability Strategies and Systematic Stakeholder Perception Metrics supporting Sustainability Business Case.



ISSUES FOR LEARNING AND DISCUSSION

- (1) What is the scope of GRI reporting guidelines? To provide a framework for mandatory reporting or to act as benchmarking among organizations in the same sector?
- (2) Is the purpose of the GRI to create a common sustainability language for organizations or to be used as a policy framework?
- (3) Does the GRI require only general, qualitative information about the company's profile and governance or does it require quantitative information about specific sustainability topics?
- (4) Does the ISO 26000 focus on environmental protection, social responsibility, or economic development?

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Life Cycle Analysis and Carbon Footprint Reduction

WORLD POPULATION AND ENERGY CONSUMPTION

Today 7.7 billion people live on our planet [1]. Approximately 70% of all human beings are able to cover their basic needs for food, clothes, living space, medical care, and energy for mobility, heat, and coldness. In comparison, at the time of the French Revolution in 1789, almost 1 billion people lived on our planet, and most of them in extreme poverty. These successes have been made possible thanks to the spectacular advances in the field of science and technology and thanks to the continually growing use of raw materials, resources, and energy. But environmental pollution as well as dwindling fuel and energy resources are the other side of the same coin.

Many problems, such as environmental pollution or the use of raw materials, can be solved. The high energy consumption plus climate warming have the potential of being disastrous for the world.

ENERGY CONSUMPTION AND GLOBAL WARMING

Energy is quite special. The availability of an adequate amount of energy is necessary for every aspect of our life. The production of food and the recycling of materials are just two out of many examples. Unfortunately, climate change is strongly linked to the wastage of energy.

In the near future, our planet will be inhabited by more than 9 billion people. The high rate of population growth and the substantial rise in prosperity in newly industrialized countries lead to a strong demand for energy and raw

Practical Sustainability Strategies: How to Gain a Competitive Advantage,
Second Edition. George P. Nassos and Nikos Avlonas.

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Companion website: www.wiley.com/go/Nassos/PracticalSustainabilityStrategies_2e

materials. Due to improved prosperity, there is a higher demand for meat and dairy products. They show a significantly poorer climate balance than cereals. The global problems caused by climate change and the lack of energy sources will continue to grow.

The energy and climate issues may be solved thanks to innovations, which are attained by higher prices/costs. Two opinions should be proceeding parallel:

- Saving energy
- An increased and additional generation of renewable energy

CLIMATE CHANGE AND CARBON FOOTPRINT

The rise of climate change on the international agenda has pushed companies and organizations to find different ways to assess, measure, and reduce their carbon footprint. In this chapter, we will present some concepts on this pressing issue, as well as explanations of how carbon footprint can be measured; and, finally, three relevant case studies from the business world.

Climate change has been defined by the International Panel for Climate Change (IPCC) as: “... a change in the state of the climate that can be identified (e.g. using statistical tests) by changes in the means and/or the variability of its properties, and that persists for an extended period, typically decades or longer. It refers to any change in climate over time, whether due to natural variability or as a result of human activity” [2].

Numerous phenomena have been observed and registered, such as increasing temperature worldwide, rising of the oceans, melting ice, and increasing number and intensity of meteorological phenomena. Whether climate change can be considered a cyclic event in time and modality that has already manifested itself in different geological eras, the current climate change seems to be linked to the high emission of greenhouse gases (GHGs) produced by humans, a process that started with the industrial revolution at the end of the eighteenth century. Furthermore, the IPCC affirms that, for the second half of the twentieth century “The observed patterns of warming, including greater warming over land than over the ocean, and their changes over time, are simulated only by models that include anthropogenic forcing.”

To make projections of the possible trends of climate change the IPCC produced the Special Report on Emissions Scenarios (SRES 2000). According to this study, the temperature in the last decades rose by 0.2 °C per decade, and, according to different projections, within 2100 the surface

temperature can rise between 1.8° and 4.0° . In a similar scenario the consequences, which the IPCC tried to define, concern the ecological resilience to such significant stress; the global reduction of food production, which will occur after a temperature increase above 3°C ; the rise of the sea level, which will increase erosion, floods, and migration flows; the potential enlarging of transmitting diseases, such as malaria; and, probably one of the most important points, the alteration of water cycle, and consequently of the availability of water and its distribution [3].

CARBON EMISSIONS AND CARBON FOOTPRINT

As underlined in the previous chapter, the necessity of measuring carbon emissions has constantly and significantly risen in the last years. Probably one of the most significant inputs that pushed toward the reduction, and therefore quantification, of carbon emission was the Kyoto Protocol. The United Nations Framework Convention on Climate Change released this international amendment in 1997, and it entered into force in 2005, for the reached quorum of 55% of countries ratifying it [4].

The most common way to quantify carbon emissions is the carbon footprint. This footprint is a tool to measure the total amount of GHG emissions caused directly or indirectly by a company or a specific product. It is usually expressed in the equivalent of carbon dioxide (CO_2e), the reason for this being that CO_2 is not the only gas that is considered in this calculation, but, according to the Kyoto Protocol, calculations also include: methane (CH_4), nitrous oxide (N_2O), hydrofluorocarbons (HFCs), perfluorocarbons (PFCs) and sulfur hexafluoride (SF_6). All six of these gases, to facilitate the communication, are transformed into the equivalents of CO_2 that would cause the same warming effect [5].

There are different kinds of footprint that can be calculated, such as individual, organization, or national footprint; however, the most undertaken is usually the one of organizations.

The carbon footprint of organizations is usually divided into organizational and product carbon footprint. The first, which is easier to calculate and therefore the most adopted, contains all the emissions which are related to the energy consumption of directly owned buildings, and sometimes also to waste production and transportation [6].

Product carbon footprint is more complex, indeed it includes the emissions of the whole production chain of the product, going from the extraction of the raw materials to the final disposal (sometime instead of the cradle to grave, the cradle to gate approach is adopted). As one can imagine, this

calculation is far more time and money consuming; however it is also more precise [7].

Once the carbon emissions are quantified, the next step is to try to reach the neutralization or balance of the total emissions produced. Thus the concept of carbon balance or offset was created. This idea pushes the organization, once it has undertaken all the initiatives possible, to reduce its CO₂ emissions, and to balance its remaining ones. In order to do so, a company has to develop projects to reduce or storage CO₂. Examples of such projects may include tree planting, creation of a wind farm, installing solar panels, or carbon storage [8].

However, one of the biggest challenges related to climate change and carbon emissions is how to measure the carbon footprint. Indeed in the last decade there has been a flourishing of different methodologies trying to achieve this. One of the most recognized methodologies is the Greenhouse Protocol based on a Life Cycle Analysis approach.

LCA AND MEASURING CARBON FOOTPRINT

The most adopted methodology for measuring the carbon footprint is Life Cycle Assessment (LCA). It is a methodology for assessing the aspects associated with a product over its life cycle. The purpose of this study is to identify and improve the most pressing issues along the process of production (which are typically environmental, but can also be social and economic). To do so, the stages, which are usually considered, from cradle to grave, are extraction/production of raw materials, manufacturing, packaging and distribution, product use, and final disposal.

The International Organization for Standardization has provided guidelines [9] on how a Life Cycle Analysis should be conducted, and it did so in the ISO 14040 and 14044, describing the four main stages [10]:

- *Goal and scope definition*—Where the product and the methodology of assessment are defined
- *Inventory analysis*—Where all the input and output of extractions and emissions have to be listed
- *Impact assessment*—Which defines impact related to this input and output
- *Interpretation*—Where the results are reported informally and related to the scope and goal of the study

GREENHOUSE GAS PROTOCOL (WRI)

The GHG Protocol is one of the most widely used standards providing a methodology for calculating carbon footprint. It was developed by the World Resources Institute in collaboration with the World Business Council for Sustainable Development, and released in October 2010. The GHG Protocol is comprised of four different, but still related, standards: the Corporate Accounting and Reporting Standards (Corporate Standard), the Project Accounting Protocol and Guidelines, Corporate Value Chain (Scope 3) Accounting and Reporting Standard, and the Product Life Cycle Accounting and Reporting Standard.

The Corporate Standard is a methodology that provides a method to report on the total GHG emissions of the organization. The Project Protocol gives guidelines on how to quantify the improvements achieved by reduction activities (there are specific protocols for Land Use, Land-Use Change and Forestry (LULUCF) and Grid-Connected Electricity Projects). The Corporate Value Chain (Scope 3) Standard provides the tool to quantify the total emissions of the supply chain (which usually are the most significant ones), and how to reduce them. This usually also includes suggestions on how to face the issues with partners and suppliers. Finally, the Product Standard is used to provide a complete overview of the emissions over the entire life cycle of a product (raw materials, manufacturing, transportation, storage, use, and disposal), and identify the biggest reduction chances, in terms of emissions, efficiency, reducing costs, and removing risks.

The GHG Protocol has been successfully implemented worldwide because it can be easily applied to any kind of product or service, and is based on LCA. This theoretically makes it suitable for application on other types of environmental impact (even if the focus stays on GHG). Furthermore, the standard helps companies to identify emission reduction opportunities and risks in the supply-chain, making improvements over time, communicating, engaging partners, etc. [11].

LCA AND PRODUCT LABELS

The LCA also has applications other than being used by international, or national, institutions to release guidelines (such as the GHG Protocol). Indeed it is probably the most widely adopted methodology for the development of standards and labels as well. The LCA of a product has been positively recognized as a powerful tool for companies to understand their environmental impact and costs throughout the supply chain, and to succeed in reducing them.

The LCA of the carbon footprint of a product usually takes into consideration the emissions of carbon dioxide during the whole life cycle, which, if not stated differently, include extraction/production of raw materials, manufacturing, packaging and distribution, product use, and final disposal. The final value is generally presented in CO₂e in kilograms, pounds, or metric tons.

Carbon footprint can be used by business as an environmental indicator. To do so, however, organizations have to find the right balance between the complexity of such analysis and the difficulty of communicating it effectively to the customers (this is where the labels should help).

The Carbon Disclosure Project was one of the first to develop a reporting tool questionnaire for the supply chain, asking companies to look deeper in their direct and indirect GHG emissions and report them. On the other hand, the communication toward the final customer is a complex field, and here labels can have a key role in facilitating the process. The first, and probably one of the most adopted labels, is the Carbon Reduction Label, released by the Carbon Trust together with the UK Department for Environment, Food and Rural Affairs (Defra) and BSI British Standards. This label comprises a black footprint in which the value in grams or kilograms per tons of CO₂e is written. In order for organizations to use this label they should calculate their products' exact carbon footprint through the PAS 2050 Standards [12].

But this label is not the only one created; indeed in the last years there has been a flourishing of different labels both in the United Kingdom and abroad. Some of these are *L'Indice Carbone* adopted by Casino, and developed by BioIntelligence; the Greenext Index developed for E.Leclerc; the *Climatop* carbon label used by Migros (Switzerland); the Certified Carbon Free in the United States, released by the Carbon Fund in collaboration with the Edinburgh Center for Carbon Management on the basis of ISO LCA standards, the GHG Protocol and the UK Carbon Trust's (2007) Carbon Footprint Measurement Methodology. The German government released a "Product Carbon Footprint." In addition, the China Energy Conservation Investment Corporation (CECIC) is creating a label in collaboration with the Carbon Trust; while the European Union has commissioned an Italian consulting, Life Cycle Engineering (LCE), to develop a Carbon Footprint Measurement Toolkit in cooperation with the Swedish Environmental Management Council (MSS); and the list can go on [13].

However, it is interesting to notice that Tesco, the first retailers to develop a program of labeling the carbon emissions of products, was also the first to abandon this path—after four years, after labeling 500, and doing research on 1100 products. The reasons Tesco is providing for this decision are numerous. Firstly, the costs and time for each analysis on the product are high; indeed each good involved "a minimum of several months' work."

Secondly Tesco were expecting that other retailers would also have moved in this direction fast, but they did not [14].

CASES FOR CARBON FOOTPRINT

Aviva (based in the United Kingdom), which is the 11th biggest insurance company worldwide, is working mainly on long-term savings, fund management, and general insurance. Its aim is for its work to reduce carbon emissions and offset all the residual emissions, to become the first insurance company to neutralize its carbon emissions related to its worldwide operations. The decision to go carbon neutral comes in 2006, and rises from the desire of the Group to:

- Be responsible and minimize its contribution to causing climate change and encourage others to do the same
- Maintain its forward thinking reputation and create a market differentiator
- Create competitive advantage to strengthen opportunities to access new markets (e.g., socially responsible investment) or new products with environmental focus (e.g., discounts on hybrid cars)
- Increase employee engagement and trust by motivating staff to take action to reduce energy use and improve energy efficiency in the office and, in turn, at home;
- Identify business risks associated with climate change by taking a proactive approach

Before that announcement Aviva has already reduced its carbon emissions related to its buildings and travels by 54%. In order to reach this ambitious goal, Aviva sets carbon management as part of the corporate objective, and is working on different aspects, which are energy consumption and saving, waste management, paper use, and traveling; then the offset is contemplated.

For the energy efficiency they are focusing on high efficiency in a new building (with low energy lighting, recyclable furniture and carpets, and plenty of natural light), while for the consumption already 55% of their energy comes from renewable energy, and they are encouraging behaviors, which favor the reduction of consumption. They plan to increase the recycling rate for waste, already at 59%, and they have replaced 400 printers with others more energy-efficient and paper saving. Last but not least, Aviva try to reduce its travel emissions by promoting teleconferences within the United Kingdom, and by adopting in-house car-pooling.

For the final offsetting the company has been audited by a third party in 25 countries, which has evaluated the carbon equivalent that the company should offset. Then the projects about whom to offset with have been chosen internally—they are six and they have global focus and are a blend of social and commercial projects [15].

For close to two decades, **Intel**, which is one of the world's largest and highest valued semiconductor chip manufacturers based on revenue, has recognized the great importance of climate change while, at the same time, realizing that this problem can only be addressed by the broad contribution of all sectors of society. For this reason, it has set aggressive targets for reducing its carbon footprint through a series of actions and programs to conserve energy and minimize emissions to the atmosphere. Until 2018, Scopes 1 and 2 emissions decreased by about 40% on an absolute basis, while the 2020 environmental goals include a commitment to reduce direct GHG emissions by 10% on a per unit basis from 2010 levels, while expanding manufacturing capacity.

As the reduction of energy use is of paramount importance for limiting climate change, it is prioritized in the company's strategy, which focuses on investing in renewable energy and other alternative sources. Intel also focuses on cooperations to minimize emissions across the semiconductor industry, with a striking example of abolishing the use of Class 1 ozone-depleting substances in the manufacturing in the 1990s, and significantly reducing the use of fluorinated gases in semiconductor production. An effort is also made to minimize volatile organic compounds (VOCs), hazardous air pollutants (HAPs), nitrogen oxides (NO_x), and carbon monoxide (CO) using thermal oxidants and wet scrubbers. At the same time, year-on-year investments are made to reduce energy use and from 2012 to 2018, more than US\$185 million have been invested to complete 2000 energy-saving projects. In addition, energy efficiency is incorporated into corporate upgrades, with particular success in reducing the requirement for fossil fuels to heat building facilities using large-scale heat recovery chillers or heat pumps [16, 17].

Barclays, which is one of the biggest banking and financial groups worldwide, is trying to minimize the direct impact of its operations through the adoption of a Group-wide environmental management system. Once this is attained the remaining emissions are offset buying carbon credits. Barclays has different offset projects around the world; in 2010 it purchased 1,191,956 carbon credits, from projects in Brazil, China, India, South Korea, Tanzania, Kenya, and Thailand, to offset their energy and travel carbon emissions.

The project undertaken in Brazil addressed over 1000 ha, which were previously run as extensive grazing for cattle, where more than 50 endemic

species have been planted. The adoption of different species guarantees the reconstruction of a more “natural” ecosystem, which facilitates the restoration of biodiversity in the area. The offsetting due to this project in Brazil was 15,000 carbon credits. While the carbon storage was the primary purpose of this project, a wide variety of other benefits can be pointed out within the environmental and social sphere. Indeed, the tree nursery, together with the monitoring and maintenance of the forest, requires that people from the local community are hired and competently trained. From an environmental point of view, this project will help the restoration of a precious ecosystem such as the Brazilian rainforest—which every year loses approximately between 30,000 and 40,000 km².

CARBON OFFSET AND THE GOLD STANDARD

Climate change is a non-local problem, as GHGs are evenly distributed throughout the atmosphere. Therefore, reducing them anywhere contributes to overall climate protection. With this in mind, carbon offsets are being made to reduce emissions of carbon dioxide or other GHGs in a region to compensate for emissions in another region [18]. The offsetting is carried out by a person, a company or an organization, through the funding of climate protection projects, with an amount of money equal to the emissions made by this person, company, or organization. It is measured in tons of CO₂e, that is, 1 ton of carbon offset represents the reduction of one ton of carbon dioxide or its equivalent in other GHGs.

Projects usually include renewable energy initiatives (wind farms, solar panels, etc.) and/or implementing energy-efficient technologies that replace or reduce consumption of fossil energy sources.

In order to ensure carbon-reducing projects, various certification programs have been created, with the most important representative being the Gold Standard. It was founded in 2003 by WWF and other international non-governmental organizations (NGOs) and is published by the Gold Standard Foundation, a non-profit institution, aiming to ensure that carbon credits are not only real and verifiable but that they are measurably contributing to sustainable development worldwide [19].

In particular, the Gold Standard adds branding, with a logo label, to carbon credits already existing or created by projects, which can then be purchased and negotiated by countries that have a binding legal commitment under the Kyoto Protocol. Following its vision of climate security and sustainable development, the Gold Standard has a large number of NGO supporters and many certified projects in over 80 countries.

CARBON-OFFSET-PROJECTS: INDIA, CLEAN BIOGAS FOR 7000 FAMILIES

A Gold Standard Certified Project about carbon offsetting is First Climate's [20, 21] initiative to provide clean biogas to 7000 families in India. In particular, in India's rural areas, there is a serious problem with the continued degradation of timber resources, as most families use wood as a source of domestic energy. Simultaneously, because the firewood collection is traditionally a feminine task, and a time consuming process too, the time spent by women in other important home activities, such as preparing meals and boiling water to make it safe for consumption, is limited. Thus, in Indian families, serious health impacts are identified, both in cases of malnutrition and diseases caused by contaminated water and from the smoke produced by combustion of wood.

This program was carried out with the cooperation of First Climate, the Indian NGO INSEDA and Germany's Governmental Agency for International Development (GIZ) and ultimately aimed at providing a sustainable solution to all the aforementioned problems. Many households have been supplied with their own biogas plant to harness the free and sustainable energy contained in their animals' excrement and its use in both cooking and domestic lighting, as well as in boiling water. As a result, domestic work has become less time-consuming and safer, providing a better quality of life in the Indian community.

ISSUES FOR LEARNING AND DISCUSSION

- (1) According to the GHG Protocol, what scope are the emissions from the production of electricity which was consumed by the company considered to have?
- (2) Which is the first stage of a climate change strategy for carbon reduction?
- (3) Which of the following is part of Carbon Footprint Strategy? Detection of key consuming sources, cost saving related to the energy consuming operations of the organization or both?

Describe what is the scope of the Cradle to Gate analysis, and Cradle to Grave analysis as part of the LCA?

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Water Footprint

OVERVIEW

A.Y. Hoekstra introduced the water footprint concept in 2002; it has been developed as an indicator, used to express the total amount of volume of fresh water consumed to produce a particular good, or to provide a service.

The main objective of assessing the water footprint of goods is to analyze how their production is affecting water resources (and pollution), and how these processes can be more sustainable (from a water perspective).

The water footprint is composed of three components: *green*, *blue*, and *gray water footprint*. The *blue water* represents the volume of water, which is withdrawn from surface (sea, lake, and river) or ground-water, and does not return to the same basin of catchment, or to the sea.

The *green water* is the rainwater, the soil moisture adsorbed, and the evapotranspiration of the plant. As it is easy to imagine this component of water footprint is of particular relevance toward the agriculture sector in comparison with the industry sector (just in the case in which they have green roof, but, even in that case, is usually not significant).

The last component of the water footprint, the *gray water*, represents the amount of water, which is necessary to dilute the load of pollutants/emitted substance from the production chain, in order to make them harmless.

The water footprint can be expressed in a variety of ways, according to what is most relevant to underline in the assessment. For instance, it can be expressed as water volume per piece (when the product is countable), or as water volume per product unit (mass, money, piece, unit of energy, etc.). However, the classical way of its representation is in the form of water volume per unit of time (m^3/ton , or l/kg).

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The water footprint can be calculated on different scales, starting from the individual consumer to companies, products, and even entire countries. Such categories are usually divided into direct and indirect water footprints; for example, an individual is composed by the water consumed at home (direct) and the one comprised from the consumption of products and services (indirect). As for companies the operational represents the direct consumption, while the suppliers/production chains the indirect [1].

The last concept that is worth mentioning about water footprint is that of water neutrality.

This concept is relatively new and was defined for the first time in 2007 by a heterogeneous group of six organizations. The group defined three main criteria characterizing this idea, being:

1. Defining, measuring, and reporting one's "water footprint" (this is not always considered mandatory, especially for actors different from organizations).
2. Taking all actions that are "reasonably possible" to reduce the existing operational water footprint.
3. Reconciling the residual water footprint (amount remaining after a company does as much as possible to reduce the footprint) by making a "reasonable investment" in establishing or supporting projects that focus on the sustainable and equitable use of water.

Unless these points have still some unclear aspects—such as what is reasonable, how to measure improvement and accounting for loss of water in different areas—the water neutrality concept can still be considered a useful tool to decrease the overall water consumption and increase the efficiency of use within the production chain [2].

GUIDELINES FOR MEASURING THE WATER FOOTPRINT

At the contrary of what is the situation for the calculation of the carbon footprint, for which there are plenty of different methodologies to use, for the water footprint the choice is significantly smaller. Indeed, the most used framework is the one provided by the **Water Footprint Network**, which gives guidelines on how to assess the water footprint of a product. According to it a complete water footprint assessment is made of four different phases:

1. *Setting the goals and scope*—In order to define what is to be measured (consumption of water for the supply chain, one stage of production, final consumer, etc.).

The first thing to determine is which type of water footprint to calculate (product, consumer, production step, etc.), and to what level of detail. Then the boundaries should be set, defining what is included in the study and what is not, for example, which component of the water footprint (green, blue, gray), spatial–temporal borders, direct or indirect consumption, etc. The sustainability assessment, which is usually based on geographic perspective, enables the analyst to link the hot spot and issues to a particular place or reality. Finally the response formulation, here is particularly important to individuate who should be involved and participate to the “solution” process started by the water footprint analysis.

2. *Water footprint accounting*—Here the water consumption is quantified and located in space and time.

Its aim is to measure the human appropriation of freshwater in a determinate catchment area—expressed in volume. The water, which composes the water footprint typically, originates from the one catch by evapotranspiration and runoffs. The water footprint of a single process-step is the basic measure for every accounting of water footprint; then the intermediate or final product calculation can also be done, and add to the total accounting. The accounting of water footprint (which is “the water footprint of a product is defined as the total volume of fresh water used directly or indirectly to produce the product”) is therefore calculated by including the consumption and contamination of water in all the stages of the production chain (this process will almost always be the almost the same, no matter the sector).

3. *Water footprint sustainability assessment*—Is where the environmental, social, and economic aspects are evaluated.

The suggestion given by the guideline considers the assessment of the sustainability process of the catchment of water according predominately by two parameters: the location in time and space within a catchment or river basin, and from which kind of process or group it is addressed (i.e., producer or consumer). For the calculation of the sustainability of the water collected, different impacts within environmental, social, and economic spheres are taken into consideration (once relevant criteria are identified). Then precise suggestions on how to calculate the sustainability of processes, products, and consumers are given.

4. *Water footprint response formulation*—In which a strategy is trying to be developed.

The guideline is intended to be a useful tool to understand in which direction solutions should come and go. The main actors which are addressed are consumers, companies, farmers, investors, and governments; and for each of them suggestions are given on how they can act and undertake initiatives to reduce the consumption of water, and therefore the water footprint. Lastly there is also a short mentioning on how it should be preferable to have clear targets of reduction instead of an offsetting program to reach water neutrality [1].

Another interesting tool that is under development and should be released within 2012 is the ISO 14046 or water footprint—requirements and guidelines, developed by the International Organization for Standardization (ISO). The work of ISO is aimed at integrating the standards of Life Cycle Assessment (LCA), and the carbon footprint standard ISO 14067 (which is also under development by ISO). The purpose of this guideline is to include:

- Principles and guidelines for a water footprint metric of products or organizations.
- How different types of water (green, blue, and gray, as well as ground and surface water) should be considered together with the socio economic and environmental issues.
- How to communicate on water footprint.
- Make it compatible with the other standards of the ISO 14000 family [3].

WATER FOOTPRINT AND LCA

At the moment the most commonly used framework to calculate water footprint is the guideline released by the Water Footprint Network, which is assessed mainly through water inventory and accounting.

Indeed water footprint is sometimes criticized for the absence of “characterization factors,” which should weight its components according to their impact, and will be appreciated for an LCA methodology (and perspective). But this, according to the Water Footprint Network, will go to influence the impact on social and environmental impacts, since it will lead to the omission of important key indicators—such as the variability in time and space.

To agree on both the purposes of Water Resources Management and LCA the steps usually undertaken for the calculation of the water footprint should

be approached differently in an LCA perspective. To do so the accounting stage will be part of the inventory of LCA, while the impact assessment and aggregated impact assessment will be part of the life cycle impact assessment. But this, according to the authors, will still invalidate the effectiveness of the water footprint meaning [3].

CASE STUDY: COCA-COLA COMPANY

Coca-Cola is a leading company in the beverage industry with particularly interesting initiatives related to sustainability [4]. Water is one of the most important ingredients for the production of the final product and special attention is paid to its management in the context of the establishment of a sustainable business on a global scale. Most efforts are focused on the areas with the highest impact on people's health, communities, environment, and market places, thus contributing to United Nations Sustainable Development Goal 6: Clean Water and Sanitation.

In particular, the sectors that are the core of the company's strategy are the following: improving water-use efficiency and reuse in bottling plants, managing wastewater, and storm water discharge to prevent pollution, replenishing the water used in finished beverages across communities and watersheds and helping manage water resources in agricultural ingredient supply chain. Global water risk assessments are carried out at the plant-level and each of the 800 Coca-Cola installations is required to assess local vulnerabilities to implement plans to address them. And this is because a clear precondition for water management is the clear understanding of the origin of water, its availability to local communities, current, and expected future pressures on supply and quality, as well as ways of contributing to addressing common challenges in watersheds of operation [5].

REPLENISHING THE WATER USED

Coca-Cola is trying to refresh 100% of its water in the final beverages back to communities and nature, a target it met for the first time in 2015. Community water projects are carried out with the expertise and support of many important partners and focus on improving safe access to water for human consumption and sanitation, the protection of catchment areas by improving the capture, storage, and quality of water, and the supply of water for productive use, such as increased water availability or water efficiency in agriculture.

IMPROVING WATER-USE EFFICIENCY AND REUSE IN BOTTLING PLANTS

A key process for water management is managing the use of water in bottling plants worldwide. The goal is to reduce the water use ratio by 2020, while increasing the volume of units of the company so that profitability improves by 25% compared with 2010 levels. In recent years there has been a significant improvement in the water efficiency, while about US\$1 billion (from 2011 to 2020) can be saved across the water acquisition system, domestic handling, and discharge fees.

WATER IN AGRICULTURE

A fundamental pillar of water stewardship and overall sustainability of water resources is to improve water management in agriculture through integrated farm operations. In this context, support is being given to projects aimed at improving the productive use of water, especially in areas with high risk of water stress or pollution. These projects are aimed at reducing water drainage and improving water efficiency in agriculture or reducing the use of pesticides and runoff in order to reduce water pollution. Partnerships to promote nature assessment work are just as important, as in the case of cooperation with the global partner, WWF, to help rural communities determine which sustainable farming practices provide the greatest benefits for the protection of watersheds. Lastly, suppliers play a key role in addressing farmers' water hazards with global and extensive supply chains. Sustainable Agriculture Guiding Principles (SAGPs) and Supplier Guiding Principles ensure good water management practices.

ENGAGING IN WATER POLICY REFORM

Because the ultimate responsibility for more sustainable and equitable water resource management lies with governments and public authorities, Coca-Cola is available to provide valuable information on relevant discussions and subsequent policies to solve water supply, quality, and access gaps. With great experience on this precious resource, operating in more than 200 countries and territories, the company is working with international policy cooperation platforms such as CEO Water Mandate and 2030 Water Resources Group to help reform water policy.

NESTLE

Nestle has developed a drip irrigation project in Nicaragua, to develop a low cost irrigation system to use in the coffee plantations. This project took place during a three years period (2007–2010), and Nestle developed it in collaboration with ECOM, Rainforest Alliance, and IDE, and it estimated the participation of 1500 small coffee producers.

The purpose of the project was to provide the coffee crops with a supplementary irrigation system during water stress periods (and flowering periods), since this has showed to increase the productivity, the growth of the plant, and the quality of the crop. Indeed, during the critical growth period, a lack of irrigation could have a strong negative impact on coffee performance and quality. To do so Nestle introduced, improved, and recorded a low cost dripping system, as part of a process of embedding sustainability within its value chain.

The project started with the installation of a drip irrigation system in 11 different plantations in Nicaragua. The increase of production was registered the same year, for an amount estimated between 40% and 60% of the yield of the previous year, and at the same time young plants grow faster. These results were confirmed during the following years of the projects, which also allowed observing that irrigated plants of two years were producing similarly to plants that hadn't been irrigated over a period of three years.

Since the successful results of this project, the system of irrigation has been implemented in other plantations in Nicaragua (32 in total), and also in Honduras (9 systems), El Salvador, and Guatemala [6].

The **Dole Food Company**, which is the world's biggest producer of fruit and vegetables, has introduced a new method to reduce the use of water consumed during the packing process of bananas. To allow a sensitive reduction of water consumption, Dole has implemented different types of water recycling systems during the packing process in the last two decades.

The use of water in the packing stage takes place mainly in the following three different phases: to remove dirt and insects from bananas, to hold and carry bananas while they are selected, and for the removal stage. This traditional way of packing bananas requires around 150 l/box (18.14 kg).

A first measure that Dole undertook at the beginning of the nineties was the adoption of sand and other gravel filters inside a partial recirculation system in order to reduce the amount of water per produced box to 100 l. In addition a Mobile Banana Processor was also developed which allowed a reduction of 97% of water use, which distributed in remote areas of the Philippines.

After a few years Dole installed 120 new systems, which completely recycle and re-circulate the water for the packing process in Ecuador, bringing the water consumption for banana boxes down to 18 l. Afterwards, this system was applied in water scarce areas in Honduras, Colombia, and Costa Rica.

The last improvement that Dole designed was part of the “New Millennium Packing Plant.” In this new system, rather than using water pools (as is common practice), most of the activities take place in the banana crop, leading to a 90% decrease of water use (and 50% less energy) in comparison with a traditional system. Dole’s specific initiative has been awarded with an “honorable mention” from the Scientific Committee of the World Water Week in Stockholm in September 2010 [7].

WATER FOOTPRINT VERSUS CARBON FOOTPRINT

The numerous “footprint” indicators that have been born in the last few years have been intended as complementary indicators of the environmental impact of human activities, since they focus on different issues.

Both water and carbon footprint are two really useful tools for the quantification of the emission or consumption of the respective resources, as well as of the improvement that they can lead during their assessment. However, both footprints communicate just the final value of the study. This can be really deceptive (particularly for water), since the amount of water consumption communicated is not related to the hydro geographical area; therefore it is not necessarily synonymous of water stress in the area. Indeed, if we compare with the carbon footprint, where we assume that any quantity of CO₂ emitted (and other greenhouse gas (GHG)) is increasing the greenhouse effect (therefore it has a “negative” impact), we cannot say the same for the water consumption. In fact a high consumption of water in a particular region/basin should not automatically lead to a water stress situation. As a consequence, it is easy to understand how it can be difficult to communicate effectively the assessment of the footprint undertaken.

Regardless of the communication problem, the water footprint and the carbon footprint remain really valuable tools for companies wishing to find their major point of consumption and waste, in order to quantify and try to reduce their water consumption (e.g., through the identification of hot spot). A challenge regarding the future use of different footprints, which is recognized by different experts, is the development of a framework that would be able to integrate all these different analysis in a comprehensive and useful manner.

The last point of difference between these two “footprints” can be found in their way of offsetting. Indeed both of them have developed a system, which allowed offsetting and neutralizing the emissions or consumption through projects, in order to reach the “neutrality” balance. Here the only difference among the two strategies of offsetting is that the one for water has not yet developed an exchange system of quote among nations or companies since water is a local resource.

ISSUES FOR LEARNING AND DISCUSSION

1. Does the product water footprint include the water consumed and the water polluted?
2. Does the water footprint of a product include the use phase of the product as well?
3. Can a water footprint be calculated for a product and an organization?
4. Which are the three components of water footprint?

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Green Marketing and Communication and How to Avoid Green and Blue Washing

GREEN COMMUNICATION AND GREENWASHING

Often green washing is not an outright attempt to be deceptive, but it stems rather from failing to consider environmental impact measures with the same robust attention as is usually given to more established and familiar measures of business performance.

Green washing—making exaggerated environmental—claims in order to curry consumer favor is one of the banes of the sustainability community. When an organization (company, non-governmental organization (NGO), Government), lauded for environmental performance, is revealed to be engaging in environmentally dangerous practices, it provides skeptics with the fodder they need. Even when the claims are found to be inflated (or impossible to substantiate), credibility is lost. For a business model based on the paradigm that transparency leads to credibility, trust and ultimately market advantage, these incidents can be devastating.

For some companies the desire to appeal to “green” consumers and to be perceived as a good corporate citizen is enough to encourage deliberately deceptive claims. However, most companies do in fact embed “green” practices within day to day operations and initiatives; therefore, on average, the main reason for green washing has nothing to do with malfeasance or bad intent. They have to do with the fact that more often we see a fragmented approach to rolling out projects or activities—where one department may be donating toward reforestation while another is measuring energy

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consumptions and verifying their carbon footprint. Both initiatives are green, but do they reflect the core business? Is there substantial reasoning behind why these two departments have taken on these specific activities, and why they are completely independent of each other? How is this reasoning being justified and measured against the financial and non-financial performances of the organization?

The old adage in business—that which is measured gets done—implies the reverse as well; that things that are not measured (or are not measured well) are often prioritized lower. This can create a negative feedback loop, which certainly has too often been the case for sustainability.

Moving within the same framework, with regard to the UN Sustainable Development Goals (SDGs) [1], the risk of SDG-washing emerges. The World Business Council for Sustainable Development found that 79% of the companies analyzed acknowledge the SDGs in some way. However, only 6% have aligned their strategy and targets to specific target-level SDG criteria and measured their contribution to key SDGs. The concept of SDGs with their catchy logo strongly attracts the corporate affairs departments and PR agencies, but often no emphasis is placed on its essence and what it represents. Almost similar to green washing, blue washing is about misleading consumers regarding the corporate social responsibility (CSR) practices of an organization. The information provided by companies is inconsistent with regard to compliance with the rules of CSR and other core values of the United Nations. The blue color is derived from the United Nations' logo, meaning that corporations portray themselves as being compliant with the 10 principles of United Nations Global Compact, while they are actually not.

GREEN MARKETING, COMMUNICATIONS, AND SUSTAINABILITY

What does this have to do with green washing? Everything. Because companies that would never consider having newer, temporary, or inexperienced people produce their quarterly financial reports are completely comfortable with having them produce sustainability reports (sustainability performance measures). When these numbers fall short of standing up to scrutiny or are used to solely generating marketing or public relations materials, organizational reputation is at risk. Failing to recognize the potential dangers of this trend in organizations not only places reputation at risk but also places a chain of other dependent factors—employees, community, shareholders, and partnering organizations—at risk as well.

If embedded effectively, by qualified *Sustainability Practitioners*, in collaboration with key decision makers, sustainability offers solutions to

facilitate greater trust in business by applying fundamental principles to core business values and operations that will generate clear indications of how credible these claims are. As a result there is greater transparency between facts and claims, thus enabling consumers to see an alignment between the brand identity, product characteristics, and acclaimed “responsible”/“good” corporate citizenship initiatives.

The competitive advantage of sustainability is the quality of return on investment (ROI) that emerges as products and services take on green characteristics and communications unite the brand’s core values with consumer passion, stakeholder awareness, and transparent annual reports. Value for money then does not remain stagnant upon the purchase by a consumer, but rather suspends its impacts to include all relationships formed around the brand—both financial and non-financial. Organizations that focus their vision on sustainability and have a clear strategy, which incorporates standards and models, will enable the adoption of best practices, accelerate performance, and ensure transparent communications.

GREEN MARKETING

Green marketing is the marketing of products that are presumed to be environmentally safe or friendly. Thus green marketing must take into consideration a broad range of activities, including product modification, changes to the production process, packaging changes, and modifying advertising according to American Marketing Association.

Consumers worldwide are increasingly admitting that their behavior has changed in the last few years to benefit the environment mainly from their purchasing decisions. Therefore, stakeholders’ purchasing decisions for products worldwide have been affected by green marketing messages. However, organizations should not get too comfortable with this trend in purchasing habits. Statistically, a consumer will spend on average 45 seconds reading a product label before making a “buy-not buy” decision! They are not passive buyers, but rather information seekers and eager to align organizational claims on products with their expectations, knowledge, and perceptions of value for money. Their buying decisions extend beyond their encounters with green products to include access to information regarding environmental claims of the product from trustworthy sources apart from its producer in a timely manner. The breakdown of traditional media channels to all-inclusive, rapid, real-time, divergent commentaries, and news releases also plays a key role in forming opinions on the credibility of these claims.

MATERIALITY AND SUSTAINABILITY

Organizations need to ensure the materiality of their claims. Materiality refers to constituents that are of high concern to stakeholders and of high strategic relevance to organizations.

Materiality is a very important part of a comprehensive sustainability strategy including green marketing. Overstatement of environmental attributes is a very common marketing behavior leading to green washing. An environmental marketing claim should not be presented in a manner that overstates the environmental attribute or benefit, expressly, or by implication. Marketers should avoid implications of significant environmental benefits if the benefits are in fact negligible. For example, if a package is labeled “50% more recycled content than before” based on the increase in recycled content of its packaging from 2% recycled material to 3% recycled material, the claim, while technically accurate, is likely to convey the false impression that the advertiser has significantly increased the use of recycled material.

A comprehensive sustainability strategy is the very first step organizations should design in order to show that they “Walk the Talk.” There are many organizations marketing their green products without having applied fundamental sustainability practices in their operations, thus increasing the risk of being accused as green washers. How *sustainability* is embedded within an organization influences the degree to which it is perceived as a good corporate citizen or green business.

GUIDELINES FOR GREEN MARKETING

In the United States, the Federal Trade Commission (FTC) [2], the government body that regulates and oversees marketing and advertising, has established general principles that contain specific guidance applicable to certain environmental marketing claims. Additionally European Association of Communication Directors (EACD) [3] has released as well a guide for responsible communication.

One of the questions that are frequently raised about “green” products and services is whether or not there is a receptive public that justifies companies modifying their processes and procedures to capture what may be a small niche or transitional market. The very fact that the US FTC developed and released (for public comment) a set of new proposed green marketing guidelines provides an important affirmation that the government of the largest economy in the world believes in the long-term interest and importance of environmental claims in promoting goods and services. The fact that green

marketing was seen as important enough to merit in sustainability eased attention indicates that the environmental impacts associated with goods and services are a long-term prospect.

The FTC has asked for public comment also indicates that they are serious about making sure that the guidelines help consumers to make informed choices—and that they want to make sure that stakeholders have an opportunity to have input into the process. An open, public process is critical to a program with both real and perceived value.

In general the guidelines focus on the importance not only for *accuracy* but also the need for *clarity*.

Accuracy

A recurring phrase that appears repeatedly and throughout the document is the need for claims to be based on “competent and reliable scientific evidence.” This is defined as including “tests, analyses, research, or studies that have been conducted and evaluated in an objective manner by qualified personal and are generally accepted in the profession to yield accurate and reliable results. Such evidence should be sufficient in quality and quantity based on standards generally accepted in the relevant scientific fields, when consider in light of the entire body of relevant and reliable scientific evidence, to substantiate that each of the marketing claims is true.”

This should provide a great deal of comfort for those who worry about the possibility of wasting time and effort focusing on overblown or exaggerated environmental concerns. Clearly the FTC is putting its weight behind the need for scientific consensus about what is being addressed. Hand in hand with this is the natural extension of this requirement; the need for robust and accurate measurement of environmental impacts of the products themselves, all along the supply chain from their sourcing, manufacturing, packaging, distribution through to their use and ultimate disposal.

The guidelines take issue with, and offer very little tolerance for deliberate “green washing” including not only misrepresentation but also the *omission* of salient facts. Stating that doing so is “deceptive if it is likely to mislead consumers acting reasonably” the guidelines clearly are looking at ensuring disclosure as well as offering guidance for claims.

Clarity

Using words like “reasonableness,” the guidelines seek to establish the basis for a “reasonable consumer” standard. They make a credible attempt to cover all the bases by which consumers can be influenced and/or make decisions.

The guidance covers “any environmental claims in labeling, advertising, promotional materials, and all other forms of marketing in any medium, whether asserted directly or by implication, through words, symbols, logos, depictions, product names, or any other means.”

Marketers will be held to a powerful standard; they “must ensure that all reasonable interpretations of their claims are truthful, not misleading and support by a reasonable basis before they make the claims.”

In some cases the guidance is very specific; for a product to be defined as recyclable, for example, the product not only must be recyclable, the infrastructure and means to recycle the product or packaging must be in place for 60% or more of the consumers within the markets where the products are sold.

Another area the guidelines cover is production. If the capture and re-introduction of excess materials is part of the standard production process (i.e., when scraps are routinely re-introduced into the production process) the product cannot take credit for being made with, or including, recycled materials.

The idea here is for those that distinguish themselves by going beyond the industry standard to be the only products that are able to gain the benefit from making the appropriate environmental claims.

The guidelines also seek to define the “rules of the game” when it comes to packaging, marketing, and advertising. Just as the government put an end to the practice of hiding clear glass balls in a bowl to give the illusion that a soup contained more meat and vegetables (by keeping them near the surface) in photographs and television commercials, the new guidelines set a framework for where and how products are positioned and even the use of the familiar triangle three-arrow recycle symbol—regulating that it be placed on the bottom on containers rather than near the name of the product when the container is what can be recycled rather than the product.

What is clear is that the FTC guidelines raise the bar in terms of expectations. As a result, companies may find that their current practices—even those that are industry-leading—only go as far as to conform to the new guidelines, thereby setting a new standard. Leading companies may find that they must either extend their efforts or be willing to accept ceding their leadership position. Consumers may not look favorably on companies that were once seen as leaders when those same enterprises now find themselves engaging in behavior that meets the guidelines—and suddenly their exemplary efforts now are commonplace and common practice.

A recent study showed that physical measurements of carbon in the atmosphere are higher than they should be if all the environmental claims from all companies were added together. Clearly there is some inaccuracy in counting. The FTC guidelines seek to prevent double counting, by prohibiting companies from claiming and sharing environmental results. For example, having solar panels on your roof allows you to claim the *portion of your power generated*, but not if you sell the certificates. In that case, you have also sold the rights to characterize your power use as renewable. And if you have solar panels visible on your facility, unless 100% of the power consumed is generated thus, the company cannot claim “made with solar power” as this, while true, also implies 100% of the power used is from this source. Instead the company must determine the proportion and is limited to making that assertion.

Truly leading companies will develop their own reporting that is more robust than the standards, perhaps using their framework and intent to guide their own actions and messages. A terrific example is the company that decided to apply safety standards not only to its own employees—as was required—but also to all contractors and subcontractors working on its job sites. The immediate result was an accident rate that seemed to “spike” upwards as they began counting all accidents and injuries, not just those from full time employees who earned a paycheck with their company name on it. In response to the immediate questioning of what had suddenly gone wrong, senior management was in the enviable position of pointing out that the accident rate was still the same for full time employees but that what was wrong was, in fact, an industry standard that did not value all people equally. This set a new standard and bar, above regulatory requirements but, by using the same measurement and metrics, the company was faithful to the regulations.

Exceeding and defining industry standards—or even business standards in general—is one way that forward thinking companies will leverage their efforts from the existing guidelines and do more. Doing so will establish that they are not “toeing the line” but rather “raising the bar.” This can also help serve as a preventive (there is no point regulating a problem that does not exist) and help reduce costs. It also can serve as a marker for the company culture, facilitating human resources efforts to promote sustainability and retain top talent. By reducing risks associated not only with potential environmental damage but also associated with defending against complaints due to false marketing claims, companies that engage in proactive efforts also reduce costs due to potential fines for either.

SUSTAINABLE COMMUNICATIONS STRATEGY

Sustainability and Communications symbolize two complementary areas of an organization's mode of operation. Sustainability represents a "holistic" management approach of re-evaluating the overall corporate strategy and operation, as well as monitoring performance via continuously assessing stakeholder engagement. It is an internal approach aiming at instilling sustainability into the overall management and operations of the organization.

Communications, however, represents a function, which aims at communicating and promoting the strategic direction of the overall organization, department, and operations. It represents an external approach to communicate to stakeholders the progress of the overall sustainability strategy. In addition, it aims at convincing stakeholders of the message's purpose, at validating the corporate objectives and at obtaining support for engagement on sustainability issues affecting the organization's economic, social, and environmental impacts.

The main objective of this complementarity is the ability to transmit the right message, in the appropriate amount, through the suitable communication channels.

Sustainable communications is the process of foreseeing stakeholder expectations, and using a pre-determined set of communication tools, being able to articulate the sustainability direction and objectives within the organization's business operations.



Sustainable communications encompasses a new "breed" of actions within the field of communications that enable for a balanced and a transparent promotion of the organization's business and sustainability responsibilities. The process is composed of five phases.

1. DESIGNING THE SUSTAINABILITY COMMUNICATIONS STRATEGY

Regardless of the overall corporate objectives, defining a communications strategy shows attention to planning, an understanding of the situation, an ability to carry out the work, and clear identification of the goal.

2. CONDUCTING AN ASSESSMENT AND DEFINING THE CHALLENGES/ISSUES

Internal assessment—Is the current communications strategy effectively communicating the corporate sustainability strategy and progress in a manner that will effectively achieve to integrate it within the working environment? Stakeholder mapping to determine the sustainability needs of internal stakeholders is the current communications strategy supported by employees? Does it motivate employee to get involved in organization sponsored initiatives and events?

External assessment—External assessment of the organization's impact on the overall society and environment and determining how well the currently implemented communications strategy responds to the needs and expectations of external stakeholders. Stakeholder mapping to determine the sustainability needs of external stakeholders. In this stage, assessment is conducted to determine the level of efficiency achieved using existing communication channels and external partners (e.g., NGOs, Associations). Is there a policy behind partnering with external entities to achieve and implement relative initiatives?

Analyze the results of the assessment and determine what the current strengths of the current communications strategy are and what the areas that need to be improved are.

3. DEVELOPING A SUSTAINABILITY COMMUNICATIONS STRATEGY

For each of the four sustainability pillars of the environment, society, HR, and marketplace, allocate critical internal and external stakeholder groups that need to be targeted in order to assure a sustainable communications approach. Using the results of the assessment, at this stage the organization will already have a viewpoint of their needs and expectation thus in this stage the objective is to design the necessary initiatives and actions that need to be enforced. This step needs to be accomplished in parallel with identifying the type of partnerships that need to be formed in order to design and implement the above initiatives and actions. Again, the results of the assessment regarding the effectiveness and efficiency of the Corporate policies vis-à-vis external partners will work as resources in order to update the terms and condition via which the organization will partner with these external partners. The objective here is to assure that all means of investment

(i.e., monetary, HR) are allocated accordingly, and respectfully of the overall corporate and sustainability strategy.

Setting targets and developing an action plan per stakeholder group and per initiative designed in order to monitor and assess performance.

4. IMPLEMENTING THE COMMUNICATIONS STRATEGY

The ability of the organization to effectively implement and sustain its communications strategy depends on the designed internal processes that enable for a continuous monitoring and management of the internal and external communications initiatives.

Implementing the communications strategy involves closely monitoring the formulated key performance indicators (KPIs) (quantitative and qualitative) and being flexible enough to immediately respond in case an issue comes up.

Organizations are now “evaluated” for their philosophies and what they stand for. In our era of sustainable communications, what you do is more important than what you say!

This is one reason why in the past couple of years organizations have turned into cause related and green marketing initiatives as a means to promote “business as usual,” but in social and/or environmentally conscious way.

The portion of stakeholders (consumers, employees, NGOs, etc.) who care about social and environmental strategy is rapidly increasing, and has a direct effect on purchasing patterns based on personal beliefs. Prior to making a purchase, these groups want to check the company’s sustainability profile to ensure that certain social and environmental standards are respected.

By responding effectively to stakeholder expectations, the organization is able to communicate issues that matter, achieving active engagement, thus creating business value.

A critical element for the implementation of a successful sustainability communications strategy is the ability to engage with stakeholders in a transparent manner, using clear messages, in a joint dialog.

5. REINFORCING STAKEHOLDER INTERACTION

Once the communications strategy has been drafted and implemented, how can we ensure that it will appeal to stakeholders? Most importantly, how will their feedback be assessed in order to update the communications strategy?

There are innovative ways to interest stakeholders of course as per grasping the issues that do in fact matter to them.

The Importance of Social Media

Social media enables real conversation between the organization and its stakeholders. Organizations have the ability to inform stakeholders of news and events, and be able to receive immediate feedback in a collective manner. Furthermore, social media provide the opportunity to stakeholders to engage with one another and be able to collectively provide their opinions and comments, via virtual “town meetings.”

Furthermore, by using the platforms made possible by social media, the organization can provide incentives to stakeholders to access related web links and reports that further promote the corporate message. Stakeholders can be directed to company websites and/or to online news distribution sources where information pertained to the organization is published and ready to be commented on.

The opportunities provided by social media enable organizations to actively interact with a wider stakeholder base by tapping into the limitless potential of the web.

The Importance of the Company Website

The development of a comprehensive, user-friendly website has the ability to transmit messages in an optimum way.

Most importantly, if the website is designed in mind of the stakeholder groups (internal and external) impacting and impacted by the organization’s operations, it can represent the primary platform for real, active, transparent, and bilateral communication.

The web content should be developed to communicate as much information as needed. Providing too little information may leave stakeholder questions and concerns unanswered, exposing the organization to public criticism for lack of transparency.

Having assessed and determined the needs and expectations of stakeholders as they pertain to the sustainability context of operations, the organization can rest assured it will be able to substantially communicate financial, environmental, and social initiatives.

Furthermore, the company website should be developed in such a manner that will transmit information in multiple ways.

- a. Videos presenting the Corporate Responsibility (CR) or CSR strategy.
- b. A visual presentation of the sustainability report is a new trend.
- c. Interviews of executive members.
- d. Advertisements that demonstrate the sustainability communications strategy.

- e. Video documentaries of how initiatives evolved from a plan on paper to a real stakeholder success story.
- f. Testimonials by internal and external stakeholders.
- g. Videos presenting partnerships with third parties (NGOs, Associations, etc.).

Sustainable communication strategy should reflect those issues that are at the core of organizations, the issues that are most important to the stakeholders. While the sustainability concept is rapidly applied to more organizations, stakeholders are becoming more aware, and concerned about how lack of sustainability influence them.

ISSUES FOR LEARNING AND DISCUSSION

1. Which of the two concepts, greenwashing or bluewashing, refers to consumers misleading on human and labor rights?
2. Can using the logo of the United Nations and the info graphics of the SDGs, without presenting how they are linked with the organization's initiative and performance, be considered as a form of green/blue washing?
3. Which attributes of an organization's products/services are highlighted under the concept of green marketing?
4. One of the signs of greenwashing is "just not credible." Which of the following statements best describe it? "A 'label' that looks like third party endorsement ... except it's made up," or "Declaring you are slightly greener than the rest, even if the rest are pretty terrible."?

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CONCLUSION

Epilogue: Where Do We Go Now?

Human life has been on this earth for about 50,000 (or somewhere between 6000 and 200,000 depending on the source) years. If we assume that 50,000 is correct, for the first 49,800 or so years man has lived in the open fields, in the mountains, in the forests, next to lakes, rivers and oceans, or in the savannahs. This is called biophilia from the Greek word meaning “love of nature.” As a result, this kind of life has been imbedded in the human gene as being normal. In the past 100–200 years, however, man has lived in cities consisting of brick, concrete, steel, glass, etc. Genealogically, this is not normal. Consequently, when a person wants to feel good, he/she goes to the ocean, lake, mountains, or open fields. That is why most people have paintings of landscape in their homes, and that is why most pictures in hospitals are that of landscapes. This makes people feel good. Is there a new norm?

Even many of the world’s religions are greatly concerned about protecting the environment. For example, the beginning of the Sunday worship service in the Eastern Orthodox Church starts with, “For favorable weather, an abundance of the fruits of the earth, and for peaceful times, let us pray to the Lord.” Disrupting nature’s life cycle invariably will cause some harm to the environment, something that is opposed by God. Patriarch Bartholomew, spiritual leader of the world’s Eastern Orthodox Church, has verbalized this thought when he said: “If human beings were to treat one another’s personal property the way they treat their environment, we would view that behavior as anti-social and illegal. We would impose judicial measures necessary to restore wrongly appropriated personal possessions. It is, therefore, appropriate for us to seek ethical and even legal recourse where possible, in matters of ecological crimes. It follows that to commit a crime against the natural world is a sin. For human beings to cause species to become extinct

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and to destroy the biological diversity of God's creation; for human beings to degrade the integrity of the earth by causing changes in its climate, by stripping the earth of its natural forests, or by destroying the wetlands; for human beings to injure other human beings with disease; for human beings to contaminate the earth's waters, its land, its air, and its life with poisonous substances—all of these are sins" [1]. The Holy Bible goes further in stating: "If we destroy the earth, then God will destroy us" [2].

As we deplete our natural resources and the population continues to increase, what is in store for human life on this planet? China's economy is growing as is their population while the availability of fresh water is decreasing. This means that this country and other ones may have to import more food. But how are we going to feed this growing population? A new business paradigm is necessary.

In the late 1980s and the early 1990s, a business concept was introduced called Total Quality Management (TQM). This was promoted excessively in journals and by consultants. This eventually led to other business strategies like lean manufacturing, Six Sigma, and just-in-time inventory. These strategies were imbedded in such a way that they became the standard operating procedure (SOP).

Today, more than ever, it is critical to adopt sustainability strategies as described in earlier chapters as well as practices that are considered "green" but are not necessarily sustainable. Included in this latter category of what can be called "less bad" are energy efficiency, pollution prevention, waste reduction and recycling, supply-chain management, and others. All of these can still be considered critically important.

But then again, how does one know whether they are going in the right direction? This is where measuring certain key performance indicators (KPIs) is important. And being able to communicate this information to the business community, the non-governmental organizations (NGOs), the government and, most of all, to the consumer is very important.

Sustainability strategies are needed more and more in organizations today, and the old business model with concentration on shareholder value seems to be replaced slowly by the stakeholder value concept. We need to move on from the short term strategies to long term ones, the core element of which will be more balanced relationships between communities, the environment and economic performance. Perhaps in 10 years we will not be talking about sustainability since all the organizations will have already implemented it just as it happened with the TQM concept. Sustainability should not be the responsibility of a person or a department within an organization, but rather the concept should be imbedded in the culture of the organization so that everyone within will work together to help solve our environmental issues.

Adopting sustainability strategies and measuring and reporting the results has become and will continue to be an important dimension of the business world and, more important, the global environment for the next few decades. While it will become the new norm, the importance of this business philosophy will only diminish after we are assured that the “environmental cliff” is not in the horizon.

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